Transitional Design Histories

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UMEÅ INSTITUTE OF DESIGN
UMEÅ UNIVERSITY

For Barbro & Göran, my parents.

Umeå Institute of Design Research Publications, No. 008

Dissertation for the degree of Doctor of Philosophy in Industrial Design

Umeå Institute of Design
Umeå University, SE-901 87, Sweden.

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ISBN: 978-91-7855-326-6 (print)
ISBN: 978-91-7855-327-3 (pdf)

Design and layout by Marije de Haas.
All photos by author, unless otherwise stated.
Printed by Pantheon, www.pantheondrukkers.nl

Electronic version available at http://umu.diva-portal.org

Umeå, 2020.

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Abstract

Design practices are to a large degree conceptually and methodologically based in ways of designing rooted in the 20th century. Some of the challenges that arise in contemporary design stem from an unawareness of design's historicity, and the discrepancies between what design methods and concepts once were made to handle, and what we presently try to apply them to. This historicity of design, embedded in its methods, tools, and thinking, shape and limit what is possible to do in design. Unless we actively deal with the historicity of design's central concepts, we risk inadvertently reproducing and reinforcing past norms and values in outcomes as well as in practices of design. Bringing history more actively into design can reframe the spaces in which to explore possibilities for how to go about designing differently.

The program of transitional design histories presented in this thesis is formulated on the proposal that the historicity of designing should be made more present in design to support developing approaches and methods for responding to contemporary issues of complexity and sustainment. Design histories therefore need to work differently, taking an outlook in design practices rather than in design outcomes. I propose a methodology for making design histories as prototypes, combining a programmatic approach from practice-based design research with research methods in history that focus on analysing concepts and ideas. These transitional design histories do not provide solid foundations for, or explanations of, what design is or has been. Instead, they aim to make conceptual moves that support developing design practices capable of engaging with a complex 'now' and with uncertain futures. The aim is to support making conceptual moves through using historical perspectives in exploring if it would be possible to see, think, and do design in other ways. By shifting the outlook of design history from product to process – from things to thinking – an ambition is to sketch the contexts in which foundational concepts and central methods in design once came about. This shift of position can provide a provisional and propositional scaffolding that activates an awareness of how – and why – the ways we design have been formed over time.

How transitional design histories could be made is here prototyped in three examples that take a starting point in concepts and themes central to Scandinavian user-centred and participatory design. As prototypes, these histories are constructed in slightly different ways, and aim to explore partially different aspects of mechanisms of design history and designing in relation to each other. The first prototype focuses on the concept of 'participation' related to turn-of-the century 1900 ideas, in the writings of Ellen Key. The second revolves around the concepts of use and users, more specifically the relationship between designed ideal or intended uses, in investigations of 'dwelling habits' in 1940s Sweden. The third prototype works with methods development in user-centered and participatory design, through examples of research into everyday domestic work at the *Hemmens forskningsinstitut* (Home Research Institute) in the 1940s.

Acknowledgements

As a PhD student, it can be challenging to keep in mind that PhD education is not primarily about carrying out a project and writing a book. The production of a thesis is such a substantial part of PhD studies that it is often easy to forget that the primary outcome of third-cycle education is not the thesis, but one's own becoming a researcher. And learning to be a researcher takes time. I have had the privilege not only of having plenty of time to learn, but also of approaching PhD studies as an iterative process. The first time I enrolled in research education, in the history of science and ideas in the mid-1990s, I only had very vague ideas about what PhD studies entailed. The academic trajectory I followed thereafter, led me to the Umeå Institute of Design and quite far away from finishing those once initiated PhD studies. However, with the opportunity to return to research education, this time in industrial design, the past couple years of being a PhD student have been more enjoyable and much more understandable than was the case the first time. Now, being more knowledgeable about the workings of both academia and PhD education, I could approach this second iteration differently than the first time around and actually also present a thesis to round it off.

Bringing it all together as a coherent whole would probably still be a long way off without the support from Johan Redström: colleague, friend and main supervisor. For everything I have learned from and together with Johan during the years we have worked together, I am deeply grateful. His knowledge, generosity and curiosity in combination with a perfect pitch in pedagogics, as main supervisor, have been invaluable for guiding, encouraging, and challenging me to find the ways of taking this thesis from potentiality to actuality. And that is only part of it all: I am so happy that the learning – about design, about what really matters, and about doing things differently – that began during the years of our leading a design school together, is still ongoing. Thank you for always being there!

I am very grateful that Kjetil Fallan came onboard as my secondary supervisor. The depth and width of his knowledge in the field of design history, in combination with his attention to detail in the feed-back on the text drafts I have sent in sporadic intervals, have been immensely valuable in these final two years of thesis work. It has been a pleasure to discuss the making – and writing – of design history with you. Takk!

Throughout my academic ventures, I have had the opportunity to present and discuss my work in many different settings. My sincere thanks to seminar participants, conference discussants, paper and article reviewers and editors who have shared comments on texts and thoughts I have presented along the way. A very special thank you to Thomas Binder, external discussant at my 90% seminar, who provided thoughtful and constructive insights that were very helpful for finalising the thesis.

The UID Research Studio – as a physical space, and during this spring in its digital format – is one of those places where all the best sides of academia seem to converge: there is critical reflection, collegial empathy, and collective concern for developing ideas together. Thank you all Research Studio colleagues for contributing to creating a vibrant design research atmosphere during these past two years: Aditya Pawar, Aina Nilsson Ström, Ambra Trotto, Anja Neidhardt, Anuradha Reddy, Brendon Clark, Catharina Henje, Elisa Giaccardi, Heather Wiltse, Johan Redström, Maja Frögård, Marije de Haas, Monica Lindh Karlsson, Morteza Abdipour, Nicholas Baroncelli Torretta, Sabrina Hauser, Stoffel Kuenen, Suzanne Brink, Søren Rosenbak, Viola Hakkarainen and Xaviera Sanchéz de la Barquera Estrada.

Thank you, all UID colleagues – former and present – for the dedication and enthusiasm that makes Umeå Institute of Design such an inspiring place to work. I owe special thanks to Tapio Alakörkkö for once bringing me onboard full-time at UID, after many years as a visiting teacher. I am also grateful for your inviting me to be part the leadership of the school, which allowed me to develop several other academic skills than those directly associated with research and education. Being part of the leadership teams together with UID rectors Bengt Palmgren, Anna Valtonen, and Johan Redström (then, also with Monica Lindh Karlsson, Demian Horst and Niklas Andersson), has been extraordinarily rewarding. Thank you all!

I owe a particular debt of gratitude to Helena Bergman for the fact that I by now have worked some 25 years in higher education. Without her advice in the early 1990s, I would most likely have given up on university studies altogether. After the first course I took, I was thoroughly convinced that this environment was not for me. Helena – who knew so much more than me about academia – suggested that I should give the history of science and ideas a try, and this indeed turned out to be a discipline that immediately resonated with me. I continued to take one course after another until I suddenly found myself enrolled in PhD education at the Department of Historical Studies. I wish to thank all of my former colleagues there for all the inspiring discussions at the seminar table as well as at the *fika* table. And in particular, my thanks go to Ronny Ambjörnsson and Kerstin Thörn. Not only are they both remarkable pedagogues and storytellers but their research topics – on everyday life, aesthetics, and 'materialised ideas' – lead me towards what became my field of interest. It was also Kerstin Thörn who enthusiastically convinced me to apply to the PhD programme. Many thanks also to my supervisors back then, Peder Aléx and Lena Eskilsson. Peder: You once advised me to go ahead and make my thesis "really theoretical" – I believe I might have done something in that vein.

As the first person in my family to study at university, learning the workings of academia was a challenge. Almost equally challenging was trying to explain what I actually did when conducting PhD studies; especially when perhaps not knowing full well myself what I was actually doing. Nevertheless, my parents Barbro and Göran Olofsson have always unconditionally supported me in everything I have done – practically as well as emotionally. From the bottom of my heart: Thank you! This book is for you.

Contrary to how many PhD students describe the final phases of thesis-writing as stealing them away from their family, for me, the intense immersion in finalising the thesis actually brought me back home. Instead of being the one always rushing away, late for meetings, focusing on PhD education brought about a much-needed slowing down of my overall everyday pace. To my travel companion in exploring both the ways of academia and the roads of life, Rolf Hugoson, and to Evert and Ingrid who have joined us along the way: thank you for all the joy you bring!

During my PhD studies, I have spent quite some time in different archives in search of material. My sincere thanks to all the archivists and librarians who have helped me in the detective work! And heartfelt thanks to Pernilla and Johan Schuber who have always – besides being wonderful friends in every other aspect – provided both housing and company during all of my archival visits to Stockholm.

I am grateful to the Ellen Key foundation for granting me a residency at Ellen Key's home Strand in the autumn of 2017. Not only did those weeks provide time for re-starting PhD studies, but I also had the opportunity to access Ellen Key's library, with the knowledgeable support of Maja Rahm. And to Anna F. Liljekvist, Gun Berger, and Ylva Wallinder, co-residents at Strand in 2017 – Lifvets gäng: Thank you for all the conversations and for the yearly writing meet-ups that have followed!

After stepping down from UID leadership in May 2018, I have been able to conduct PhD studies as part of the 'repatriation time' allowed from the Faculty of Science and Technology. The leadership of Umeå Institute of Design has generously extended that with additional time dedicated to finalising my thesis. For that I thank Thomas Olofsson, Niklas Andersson, Linda Bresäter and Oscar Björk.

Turning my thesis manuscript into a proper book has been a collaborative process, and I am indebted to everyone who took part. First of all: thank you, Marije de Haas, for the round-the-clock work with design and layout of the final manuscript, and with turning my rough illustrations into elegant graphics. Monica Lindh Karlsson provided ice-cream delivery and energy boosts during the final weeks of writing. And a team of proof-readers helped me rid the text of most of the errors and odd formulations: thank you Aditya Pawar, Anja Neidhardt, Brendon Clark, Catharina Henje, Corné de Beer, Heather Wiltse, Maria Lundgren, Sara Eriksson and Suzanne Brink.

Lastly, but in no way least, to all the UID students I have met over the years as a design history teacher: thank you for engaging in discussions on if and how history might matter to designing. Your questions and explorations have often, for me, opened up new paths and perspectives, and I am grateful to you all for allowing me to continue learning together with you.

Umeå, 7 July 2020
Maria Göransdotter

1. Presenting the past

Over the recent decades, it has become obvious that the past ultimately conditions the choices open to us as well as the trajectories plausible to draw towards the future. The latest century of intense global expansion in production, consumption, travel, and communications, continuing the paths set through the previous two centuries of industrial and societal transformation, has given rise to entangled effects of an unprecedented scale. These have by now become difficult or even impossible to amend. What is already out there will continue impacting lives for generations to come as the conditions for human and other living existence are contingent on actions taken (or not) in the near and far past. The changes needed in how we live, work, travel, and make things are enormous, if we wish to achieve more sustainable future ways of life. However, what we can even think of as possible depends on our outlook from the present situations in which we find ourselves. What we experience in the present – what we can see, and how we think – is shaped by the past.

Design has been integral to giving form to things that affect how we live, shaping how we relate to and perceive each other and the world. The material and immaterial compilations of things and environments of our everyday lives have been created over time, building on ideas of what it is to be human, what society should be like, and what things should do for us. Design itself is a practice that has come about over time, in relation and response to the formation of industrial and post-industrial society. How design is carried out, operating with specific methods, tools, and aims, is something that has taken form over the latest one hundred years or so. This means that the present ways of designing rest on conceptual foundations rooted in other times than our own, and have to do with design's emergence in relation to specific societal, industrial, political and economic circumstances.

How a design opportunity or design problem is formulated, what contexts are defined as ones where design takes place, and which types of outcomes might be expected in design, are all linked to the concepts and ideas governing designing. These foundational ideas direct design towards certain practices, certain situations, and certain ways of thinking. With growing insights into the extent of humanity's impact on the world, established ways of doing design have increasingly become both criticised and challenged. In design research as well as in design educations, central questions are nowadays how to transition towards other practices that could support different and inherently sustainable future ways of living, through rethinking how design is done, who is included in designing, and according to which world views the aims of designing are formulated.[1]

Realising that past and present are interdependent in how and what we design means that without an awareness of how historical mechanisms affect

1 Tony Fry, *A New Design Philosophy: An Introduction to Defuturing* (Sydney: UNSW Press, 1999).

and direct designing, it will be difficult to figure out what types of changes to aim for in design.[2] If methods and tools for designing are invented and implemented but the historical foundations for thinking and doing design remain stable, the frameworks and motivations of design will be difficult to challenge. Unless we actively deal with the historicity of design's central concepts, we risk inadvertently reproducing and reinforcing past norms and values in outcomes as well as in practices of design. In other words, with the aim to open up other design spaces than the ones we are already familiar with, and to develop design practices capable of engaging with a complex 'now' and with uncertain futures, history needs to be made present in designing.

This thesis is based on the proposal that presenting – as in making present – the historicity of design can provide perspectives on design's conceptual foundations that support shifts towards doing design differently.[3] In the following, I argue that matters of history are crucial to probing and critically (re- or de-) activating core ideas and concepts in contemporary design. I propose an experimental approach to making design histories, in which a redirection of design history's focus from products to practices could contribute to different outlooks on present challenges in design. Combining approaches from practice-based design research and historical studies in critical analyses, I suggest that such experimental *design histories can be approached as prototypes, addressing the historicity of designing rather than telling histories of design's past outcomes*. These design history prototypes seek to contribute with different ways of seeing, and doing, design through presenting – making present – the historicity of design concepts and methods.

History matters

Design has emerged and evolved in response to specific situations and particular understandings of the world. Doing design in certain ways springs from implicit and explicit definitions, made over time, of what counts as design. As such, design is always something that takes place in history. This taking place can be understood quite literally since design outcomes often are manifest materially as things and environments, and as the infrastructures, impacts, and remains of the processes through which these have come into being. And, of course, design also takes place as actions in the present, in instances of designing.

As we engage with the things that come out of design processes there are historically layered world views, norms and values embedded in these that continue to be in play. In a reciprocal and continuous process we are simultaneously makers of and made by the designed world.[4] But design is also part

2 Clive Dilnot, "History, Design, Futures: Contending with what we have made", in Tony Fry, Clive Dilnot & Susanne Stewart (eds.), *Design and the Question of History* (London/New York: Bloomsbury, 2015).

3 Johan Redström, *Making Design Theory* (Cambridge, MA: MIT Press, 2017), 113-132.

4 Anne-Marie Willis, "Ontological designing", *Design Philosophy Papers* 2006:2 (vol 4), 69-92.

of making the things we think with in designing itself, perpetuating or challenging ideas and concepts that form the foundations and frameworks for doing design. History is a matter of design, and design is a matter of history.

The concepts design relies on have been formative for methods developed to handle specific historical situations of industrial and social change in which it once took form. In design, little (if any) attention is paid to what this means for what design can be expected to do. Even methods or design approaches that explicitly aim to question who designs, what is designed, with which purposes, and towards what sorts of futures, rely on some core ideas and methods that presuppose certain implicit conceptual foundations – and limitations. The design practices that adhere to participatory and user-centric methods, for example, will undoubtedly question how 'participation' best could be supported but not what working with the concept of 'participation' could mean in regard to the historicity of this concept. While a concept such as participation remains central to design, the connection between its history – why, how, and with which purposes it was once brought into design – and its current application in methods and processes of designing has been lost.

In designing, projects and situations are set up and carried out with methods, tools and processes invented or incorporated in design at different points in time. Design methods, and the fundamental concepts these are based on carry over expectations, values, and definitions relating to design that stem from other times and situations than those where design takes place today. While this means that design's ways of working are historical, having come about in times and contexts other than the present, its methods and concepts often seem to be approached as if they were timeless or neutral. When these methods and processes operate, they support ensuring that certain types of design outcomes come about as responses or solutions to problems. At the same time, these ways of working also perpetuate implicit understandings of design's foundations and frameworks: what design is perceived to be, is established through the ways that design is done. When the embedded conceptual foundations in design operate without their historicity being brought to the fore, it is difficult to know what sorts of changes – conceptual as well as methodological – that are needed for actively supporting design's ability to respond to new contexts and complexities.

Industrial beginnings

Design's contexts and materials have changed quite substantially over time. Design practices, and with them also many design educations, are still conceptually and methodologically often based in ways of designing as these came to be defined during the 20th century. The processes, methods and tools applied in design practice, and taught in design education largely rest on understandings of what design once was, rather than on responses to or suggestions for what design could become.

Ways of doing industrial design have come about over approximately a hundred years, with concepts and methods evolving along the way. These ways of designing have been formed, imported, tried, rejected, adopted, discussed, debated, and developed over time. The specific societal contexts, world views, and understandings in which design has taken shape have influenced how its methods have been formed and formulated. These methods and tools were developed to handle design issues out of a 20th century industrial and societal context, mostly from perspectives of a Western and global North, and as such are not always well

suited anymore for supporting and expanding the possibilities of designing in light of the increasing complexities of the post-industrial world of the 21st century.[5]

Industrial design was once called into being in response to massive changes in scale and perspective brought about through the societal transitions towards industrialisation, mechanisation, and expanding modes of communication.[6] New contexts and situations, and new modes of acting and thinking called for different responses than before, in which early formulations of industrial design addressed the separation of form-giving from making, and producing from consuming. Traditional ways of living and working, including craft-based forms of production, were challenged in the shift from an agrarian to an industrial socio-economic system as this affected the lives of more and more people.

The radical transformation that touched and altered practically all aspects of everyday life, from social norms and behaviours to politics and city planning, was increasingly perceived as a state where the previously unthinkable was made possible; where "all that is solid melts into air".[7] While developing design methods and approaches that aimed to deal with how to give form to and how to make things, questions of the role of design in creating a different future society was highly present already from its beginnings.

Now, once again, we seem to experience ourselves as living in a time of unprecedented change in regard to scale, complexity, and social transformation, leading to new demands on and contexts for design. What is it that designing becomes when working with new composite, or 'smart,' or even intangible design materials – as in the case with 'experiences' or 'big data'? And when designed things can be everything from artefacts to AI, or span over networked technologies in fluid assemblages to democratic assemblies, where does design begin and end? [8] These questions also highlight other questions, such as for whom or for what, and with which methods and motivations design is done and what that means for which futures it supports or disables. As design expands and moves into situations that require addressing other materials, relationships and connections than before, the methods and tools at hand seem to become increasingly difficult to apply.

The field of industrial design has continued to change and adapt in response to societal changes and to new understandings of what 'things' could

5 Clive Dilnot, "History, Design, Futures: Contending with what we have made" in Tony Fry, Clive Dilnot & Susanne Stewart (eds.), *Design and the Question of History* (London/New York: Bloomsbury, 2015); Arturo Escobar, *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds* (Durham: Duke University Press, 2018).

6 Clive Dilnot, "The Matter of Design" *Design Philosophy Papers* 2015:2 (vol 13), 116.

7 Karl Marx quoted in Marshall Berman, *All that is solid melts into air: The experience of modernity* (London: Verso, 1983). Cf. Rolf Hugoson, "Clarifying Liquidity: Keynes and Marx, Merchants, and Poets", *Contributions to The History of Concepts*, 2019:2 (vol.14) 46–66.

8 Johan Redström & Heather Wiltse, *Changing Things: The Future of Objects in a Digital World* (London: Bloomsbury Academic, 2018); A. Telier [Thomas Binder, Giorgio De Michelis, Pelle Ehn, Giulio Jacucci, Per Linde & Ina Wagner] *Design Things* (Cambridge, MA.: MIT Press, 2011).

be. The development, testing and incorporation of specific methods, tools and processes for handling new design materials, contexts, situations, and scales have been central to these shifts. Different ways of doing design have gradually taken shape to meet new demands and developments in the environments where industrial design has rooted itself. Design has continued its movement into areas of planning and giving form not only to industrially (mass)manufactured products and environments but to increasingly interconnected systems and services, experiences and interactions, and with increasing attention to how people go about using them.

Changes in the scope and scale of industrial design continue to bring about new practices of designing, whether as a response, reaction, or resistance to the values and logics governing what 'the industrial' over time has been perceived to be in relation to 'design'. If design a hundred or so years ago came about in response to questions of how to give form to physical things *for* people, it nowadays increasingly revolves around processes of how to design things – that need not be neither material nor stable – in collaboration *with* people.

Questions of what industrial design could be, and how designers should work, already from the beginning mainly revolved around figuring out how social life and material things were connected: what sorts of things should be designed, how should they be made, who should design them, what should they look like, how should they work, and what sorts of lives should they support people to live?

With design's coming of age in symbiosis with industrialism, the world views, technologies, economies, societal practices and social norms that took form also shaped the development of design's areas of practice. Ideas and values relating to these contexts have, however, over time, become deeply embedded in designing and continue to impact what is possible to do, and to think, in design. Thus, many of our design methods and processes are based on concepts that carry built-in norms, values and assumptions stemming from times and situations very different the ones we find ourselves in. This means that many of the contemporary values and contexts that shape our present-day understandings, ambitions, and motivations for what we expect design to be able to address, are potentially at odds with the conceptual foundations guiding and shaping design practices.

The ways we think, and the things we think with often tend to change at a slower pace than the material things or technologies we invent and surround ourselves with.[9] What could be opened up as possible for design, when dealing with new contexts, new design materials, and new methods could actually remain unseen due to the limits set by what our ways of thinking allow us to perceive.

As design expanded into situations quite different from those in which it once was called into being, the development of design methods has also shifted the emphasis in order to support handling new types of complexities and constellations in designing. New situations will most certainly continue to call new types of design into being in near and far futures. Therefore, design's methods and tools need to provide resilient ways of adapting to new practices, as well as to support taking action and making choices based on what appears as possible.

The potential futures – or non-futures – we can foresee, give cause to re-examine how we live our lives and to what things or situations we ascribe value.

9 Jaques Le Goff, "Les mentalités: Une histoire ambiguë" (77-94) in Jaques Le Goff & Pierre Nora (eds.) *Faire de l'historie. III. Nouveaux objets* (Gallimard, 1974), 81-82.

This tends to point towards that if we aspire to bring about changes that would lead towards more resilient and sustainable futures, not only would we need to act differently, but we would also need to learn to think differently to do so. The first step towards such a shift in doing and thinking requires acknowledging that how we understand our situation, what is perceived as 'the present', and what we can see from here, is in all aspects a thing shaped by the past.

However, in current industrial design education and practice, the methods and processes taught and applied are seldom considered as having something to do with history. Instead, the processes and methods used in designing are often presented, and taught, as an assembly of tools in the designer's basic tool kit – they are just there, and they serve particular purposes.

Indeed, there are critical discussions about which types of designing different methods or tools support, or not, and how to go about deciding which methods to use in different situations. But these seldom include critically engaging with what it means for designing that the methods, concepts and approaches we use have come about in particular historical contexts to deal with situations specific to the times and places where this happened.[10] The proposal in this thesis is that exploring design histories in terms of design's ideas and world views rather than through its products or objects can contribute to the critical unpacking of central concepts needed if we wish to find ways to go about designing differently.

Activating history in design

From proposing that historical perspectives can contribute to critically revisiting core design concepts, follows that history somehow needs to be activated from *within* design, and not only *in relation to* design. It would, perhaps, seem close at hand to turn to design history for this, since it is the specific historical discipline that aims to engage with design's past. At present, however, it seems that design and design history do not really align in terms of neither attention nor practices. The issues that design grapples with, e.g. how to find methods or tools that support addressing matters of power and agency in collaborative designing, are not typically the issues that design history primarily works with. There, areas of investigation instead tend to revolve around critical approaches to materialities and meanings related to design outcomes or design cultures. With this non-alignment of outlooks between contemporary designing and design history, the relationship between the two is one of polite interest, rather than of critical engagement in matters of mutual concern. So, how could one then explore ways that design history and design might engage more directly with each other?

Design is a practice that in its focus on current and future situations is largely a-historical. In design, the big issue is typically not where we come from, but where we find ourselves today, and where we could be heading next. The past, and how it relates to the present or the future is simply not a part of the picture very often.[11] On a similar note, design history seemingly has quite little to do with present-day practices in design.

Histories of design often tend to begin with a definition – or re-definition – of the type of design that is to be studied. It could be on the basis of a particular variety of design (the history of industrial design), a type of design object (the history of the telephone), or a specific material (the history of steel) and so on. Then, a narrative is built around that framework, tracing – and often also challenging – that phenomenon chronologically either to or from the present-day, highlighting designers, design outcomes, and their contexts. This structure of chronologically tracing a sort of lineage in relation to work, author, and environment has been present in design history since its beginnings.

Design history began to take form as a university discipline in the 1970s, in relation to design education. Aiming to introduce 'theory' in connection to 'practice' in design by focusing on the history of the design profession, design history was brought into studio-based design curricula.[12] Design history, or the history of design, was not introduced in design education with the ambition for it to become an integrated part of designing, but with the aim of explaining, and questioning, the results of design and the formation of professional designing. Readings provided in design history courses would pick up on the few already existing publications, springing from the fields of art and architectural history. In these, the history of design was told as an account of progress, success and prosperity, leading up to a present state of design portrayed as the culmination of a more or less logical and rational development process.

Despite its origin in design educational settings, the relationship between design history and design education has, in a sense, continued to be defined by separation rather than by integration. And despite design history's strong connection to design curricula, the nature of the relationship between design education and design history has come to be questioned in the process of design history becoming a specialised area of academic scholarship. The question of if design history at all should aim to engage with or contribute to design education and practice has been a contested and debated issue since the late 1980s.[13]

As part of the process of academic establishment, the expanding and eclectic development of perspectives and methods in design history has been parallel to boundary work in relation to not only the art historical tradition but also to

10 Examples of studies that have approached designing critically also from a historical point of view are, for example, Andrea Branzi, *The Hot House: Italian New Wave Design* (London: Thames & Hudson, 1984); Alison J. Clarke, "Design for Development, ICSID and UNIDO: The Anthropological Turn in 1970s Design", *Journal of Design* History 2015:1 (vol.29)43-57; D.J. Huppatz, *Design: The Key Concepts* (London & New York: Bloomsbury, 2020); and Daniela K. Rosner, *Critical Fabulations: Reworking the Methods and Margins of Design* (Cambridge: MIT Press, 2018); Molly Wright Steenson, *Architectural intelligence: How Designers and Architects Created the Digital Landscape* (Cambridge, MA: MIT Press, 2017); Alastair Fuad-Luke, *Design activism: Beautiful strangeness for a sustainable world* (London: Earthscan, 2009).

11 Carl DiSalvo, "The need for design history in HCI", *Interactions* 2014:6 (vol.21), 20–21.

12 Joanne Gooding, *Design history in Britain from the 1970s to 2012: Context, formation, and development,* diss. (University of Northumbria: Newcastle, 2012). Victor Margolin, "A Decade of Design History in the United States 1977-87", *Journal of Design History* 1988:1 (vol.1), 51–72.

13 Victor Margolin, "Design History or Design Studies: Subject Matter and Methods", *Design Issues* 1995:1 (vol 11).

the origins of design history as a 'service discipline' to design education. While design historians have pointed out the importance of including critical historical perspectives on design in studio-based design curricula, issues have also been raised as to the limits this sets for design history in terms of scope, aim, and establishment as an independent area of academic research.[14] Though stressing the importance of historical perspectives as a subject of study included in design education to bring about an understanding of the development of the design profession and critical perspectives on design, design schools are not always regarded as settings in which design history as an academic field of research would necessarily benefit from being located.[15]

In light of this, it is perhaps not surprising that design scholarship defining itself as 'design history' generally does not explicitly aim to contribute to design practice today. And design's disinterest in design history is not very surprising either, given that central discussions and critical developments in design practice mainly focus on processes and methods in design doing, and not the things that design history studies: meanings, cultures, and qualities related to the outcomes of design. Design history has seldom engaged in critically investigating design from a perspective of contemporary designing. Such aims tend to instead adhere academically to the broader field of 'design studies', bringing together interdisciplinary perspectives on the meanings, practices and consequences of design in relation to current and future design doing – but not always including historical approaches.[16] To some extent, the differences between these approaches can be seen as related to place and perspective. Design studies' perspectives are often approached from within, or in proximity to, contemporary design practices. Design historical scholarship has instead often turned its attention to design from a perspective of the specialised disciplinary context of history, rather than from a position within design.

Over time, and specifically with the material turn and the influences from science and technology studies, the field of design history has vastly broadened both its scope and its methodological approaches, challenging and questioning canonical grand narratives of design. While continuing to revolve around various sorts of things – outcomes of design – as a central focus or starting point, design history increasingly studies design from a multitude of perspectives, relationships and contexts: societal, political, material and cultural. Also in the practices of designing, issues of socio-political character, of materials, agency, and cultures in design are present in design work and in design research both. But while design history still in many ways takes its starting points in design of and as products, design practice has more shifted its centre of gravity from product to process.

If design history investigates a *what* – what is brought into being through design – it is the *how* – the methods and processes of designing – that has become more focused in design research and design practice. For design history to shift towards the

14 Jonathan M. Woodham, "Resisting Colonization: Design History Has Its Own Identity", *Design Issues* 1995:1 (vol 11). Adrian Forty, "DEBATE: a reply to Victor Margolin" *Design Issues* 1995:1 (vol 11).

15 Kjetil Fallan "De-tooling Design History: To What Purpose and for Whom Do We Write?", *Design and Culture*, 2013:1 (vol.5), 13-19.

16 Hazel Clark & David Brody (eds.), *Design Studies: A Reader* (Berg: Oxford & New York, 2009).

matters that are foregrounded in design, it should make a disciplinary shift towards engaging with the foundations and practices that relate to *how* design operates.

Historical outlooks

In a casual everyday use of the word 'history', people generally tend to refer to a more or less distant past, distinct from 'today'. This can make us prone to think of both as singular instances: there is one today, and there is one history. History in this singular sense can be perceived as something intrinsically linear; a chain of linked causes and effects that can be traced back in time from our present-day to explain how we ended up where we are ('*the* history of television', '*the* history of World War II'). In such a view of history, each link of the chain needs to neatly attach to the previous and following. History becomes the logical back-drop and explanation to where we find ourselves today, which often means that the events, thoughts, practices and complexities that would fail to provide a neat link or coherent route to the present-day are either absent from the story or referred to as unsuccessful paths or failures.

This view is seldom one that historians would take nowadays. That 'history' is always plural, situated, and contested is instead a starting point for most historians.[17] Nevertheless, in relation to other fields of practice, or knowledge, 'the history of' a particular field or entity will often be reduced to a singular story for the sake of making a certain point, or just out of habit. In these cases, the past – neatly ordered in historical accounts – does not come across as once having been a present just as contradictory, multi-faceted and difficult to grasp as the world we find ourselves in today. It is only from a certain distance in time, from a certain perspective, and by tracing trajectories in reverse, that history comes together in a seemingly coherent and straightforward path leading to what is.

With the discipline of history rooted in the cultures and traditions of the humanities, as these were (re)formed in 19th-century European universities, this has provided the predominant perspectives in design history as it has taken shape. This has led to historical research *into* design and many histories *of* design, but to very few design histories that begin *in* design or that propose to do something *for* design as a specific area of knowledge and practice.[18] Design history, as stated by many historians, does indeed both aim and claim to contribute with critical approaches to designing that could be considered useful from the designer's point of view.[19] However, as design histories do not substantially relate to most of the

17 Reinhart Koselleck, "The temporalization of utopia" in Reinhart Koselleck, *The Practice of Conceptual History: Timing History, Spacing Concepts* (Stanford, CA: Stanford University Press, 2002), 99.

18 Bruce Archer, "The Three Rs" [1976], *Design Studies* 1979:1 (vol.1), 17-20; Christopher Frayling, *Research in art and design, Royal College of Art Research Papers* 1993/94:1 (vol.1) (London: Royal College of Art, 1993); Nigel Cross, "Designerly ways of knowing: Design discipline versus design science", *Design Issues* 2001:3 (vol.17), 49-55; C.P. Snow, *The Two Cultures* [1959] (Cambridge: Cambridge University Press, 1993).

19 DJ Huppatz and Grace Lees-Maffei, "Why design history? A multi-national perspective on the state and purpose of the field", *Arts & Humanities in Higher Education* 2012:12 (2–3).

practices and challenges in contemporary designing, design history will not be likely to become something that strongly resonates with contemporary design research and design practice. For that to be the case, design histories' outlook would need to be closer to that of design; approaching history from within design, as well as design from within history.

Besides beginning with a shift in position and outlook, histories also have to be made to work in a slightly different way than the usual in order to actively contribute to addressing matters of concern in design. Histories that work *to activate awareness of the historicity of designing* need to be different from the histories of design outcomes. In either case, the scope is to explore what comes into view in terms of thinking and doing design when questions of history are posed from a position of contemporary designing. Since these questions would be directed towards histories of values, concepts, and ideas in design, one can hardly turn to the past and expect to find very clear-cut answers.

Activating an awareness of the historical dimensions of designing is not only a matter of searching for the first instances, or origins, of concepts or practices in design. As the aim is to support changes in designing through other perspectives on the worlds and the ways of thinking design has created for its practices, the most central things to search for are the things we think with: the underlying ideas relevant to the formation of design in the contexts or situations where its key concepts arose.

As design has evolved and expanded, some of its concepts and methods have gone from being novel and radical to becoming taken for granted and with that also more or less invisible. A concept such as 'user-centered design' has gone from aiming to reframe designing in relation to agency and attention in both process and outcome in a specific point in time of design's trajectory, to becoming deeply embedded in designing. This means that while the practices of designing for or with users might be critically addressed in regard to how, why, and with or by whom such design should be done, the basic idea of 'use' as an integral part of what design is, is relatively stable. As such it has become a basic concept in design so central that it comes across almost as a given. We believe to have a shared understanding of what we mean when we think and act on 'use' in user-centered design. However, concepts do not only provide frameworks for understanding meanings. Concepts also carry agency, as they are made of layerings of historical experiences and expectations that change over time, and that shape and direct certain types of actions. Activating the historicity of designing means re-activating the tensions and dilemmas – and the potentials – inherent in the concepts that form the basis for designing.[20] This can be done through approaching them from positions that could make visible aspects that have been forgotten or that are as of yet unseen.

Through bringing forth other things than the usual in design's history, other types of narratives of the past will potentially also introduce new perspectives on our understandings of the present. Or, to speak with Donna Haraway:

20 Reihart Koselleck, "A response to comments on the *Geschichtliche Grundbergriffe*", in H. Lehmann & M. Richter (eds.) *The Meanings of Historical Terms and Concepts: New Studies on Begriffsgeschichte*. (Washington, DC: German Historical Institute, 1996), 62.

> It matters what matters we use to think other matters with; it matters what stories we tell to tell other stories with; it matters what knots knot knots, what thoughts think thoughts, what descriptions describe descriptions, what ties tie ties. It matters what stories make worlds, what worlds make stories.[21]

How, then, would one go about making design histories that bring other perspectives, that tell other stories, and that address the matters design thinks with? In the following, I suggest a framework for design histories that are explorative and propositional, and that aim to support conceptual shifts that could bring about other practices in designing: making other stories, that these might make other worlds.

Transitional design histories

This research rests on the idea that design does not have one past, but many potential pasts. And following that: the past or pasts that are made visible and present as histories both shape and limit what is possible to do in contemporary designing. Presenting and making visible the historicity of design, aims to show that some of the difficulties and challenges that can arise in doing design stem from discrepancies between what design methods and concepts once were made to handle, and what we presently try to apply them to. For design history to be able to activate an awareness of these things, the outlook from where histories are made, and the questions asked of the past, must be brought closer to the outlooks from where designing begins.

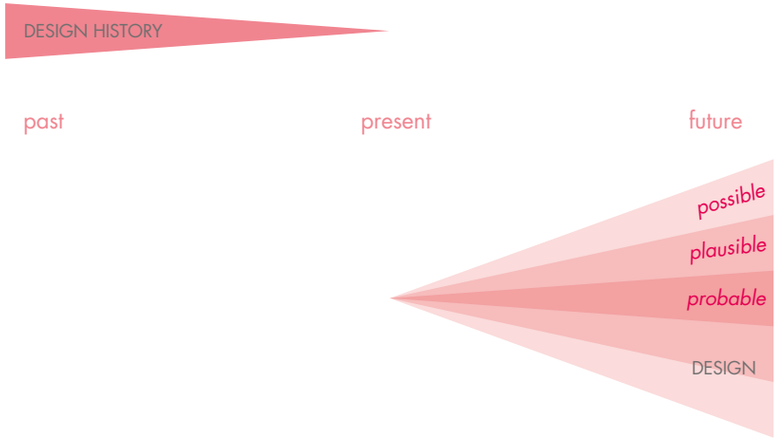
With the risk of being overly simplistic, one could say that design histories tend to start with a 'what' (what type of design is the history about), and designing with a 'how' (how could things be different than they are now?). While design strives towards the future and design history towards the past, they both share a common ground in that their respective queries spring from challenges in the present. History's narratives about what has been and design's proposals of what might become are always situated in the 'now'. However, not only do design and design history gaze in opposite directions; they do so from quite different outlooks in regard to understandings of design. The complexities that contemporary design seeks to deal with are often placed in an entirely different part of the realm of the present than the one where design histories tend to take their starting point.

Design histories mainly work with outcomes of design and propose narratives of the past that support understandings of how these things are meaningful today. Design begins with exploring present situations and proposes different ways that design can bring about intentional change of these – by means not only of making new things but through envisioning how these things can lead to different types of actions, and to creating other future situations. Design's attention is directed towards futures that seem reasonable to project from an a-historical understanding of the present.[22] Just as design seldom looks to the past, design history rarely makes any moves towards positioning itself in relation to

21 Donna Haraway, *Staying with the Trouble: Making Kin in the Chthulucene*, (Durham: Duke University Press, 2016), 12.

22 Harold Nelson & Erik Stolterman, *The Design Way: Intentional Change in an Unpredictable World* (Cambridge, MA: MIT Press, 2012).

present-day, or future, design practices. Attempting to picture this state of things using the 'cone of potential futures', which depicts the range of possible, plausible, and probable, envisioned trajectories of what might become given actions taken from a particular outlook in the present,[23] the disconnect in the positionings of design and design history could look something like this:

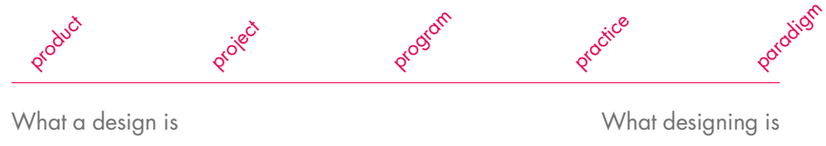


From design's point of view, the line of vision opens up towards a range of futures, more or less probable, but that could be made to come about through different types of choices made in designing. Design history's outlook starts in definitions of design that typically entail establishing or questioning what 'design' is, leading to chronologically linear narratives of, e.g. who can be regarded a designer, what consumers' roles are in design, what can be understood as relevant design outcomes, how ideas of design are mediated, and where designing historically has taken place. While such design histories in many ways critically question both present and past understandings of design, it does not directly engage with issues of how the past connects to present-day and future designing.

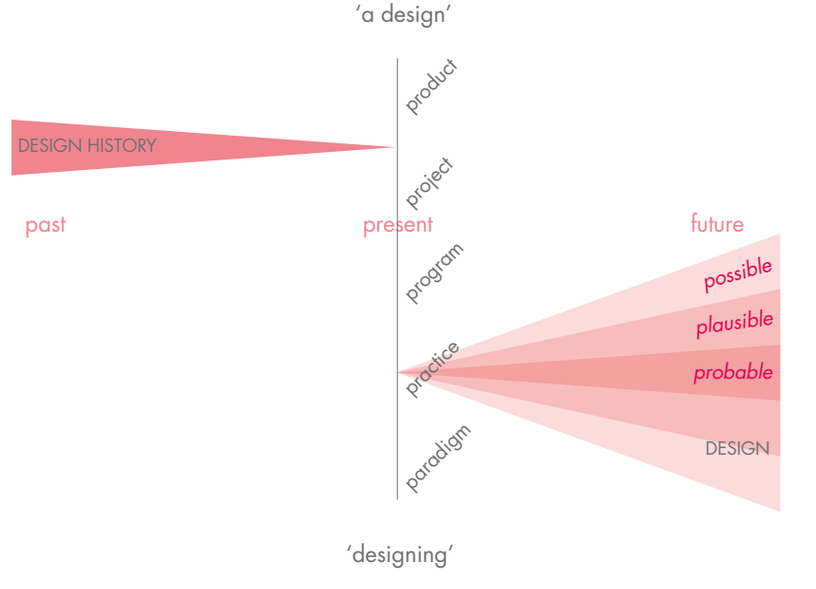
23 Anthony Dunne & Fiona Raby, *Speculative Everything: Design, Fiction and Social Dreaming* (Cambridge, MA: MIT Press, 2013), 3; Redström, *Making Design Theory*, 126-130.
24 Redström, *Making Design Theory*, 30-47.

History from where?

A starting point for providing a different outlook for design history that could connect more to design practices is to avoid beginning historical inquiries with a specific definition of design. Instead of defining instances of design, the starting of such histories would follow Johan Redström's definition of 'design' as a fluid and continuous spectrum spanning between what 'a design' could be (such as products, or projects) to what 'designing' is understood to be (in terms of practices and internal paradigms).[24]



When industrial design was called into being, its attention was quite strongly focused on questioning what 'a design' could be: what things should be like gave impetus to finding out how design should be done in order to achieve this. A wide range of methods, tools and processes for designing has been developed to allow design to take on challenges of proposing not only possible but somehow also preferable, futures. Design's attention has with time increasingly moved towards questions of process and practice, placing the emphasis on how design should be done to achieve certain things – and not explicitly starting with the things themselves. This shift in gravitational pull, from product to design process, that has taken place in designing has not yet come about in design history in relation to how 'design' is defined as an object of research. There, historical narratives are still mostly placed at the end of the spectrum that deals with what 'a design' has been, even if much of design history indeed explores things, projects and situations in terms of production, consumption and mediation, and processes of the creation of meaning and value.



In present-day design practices, challenges addressed and probed in design research instead gravitate towards the *how* of designing, rather than the *what*. How do situations of designing relate to situations of use, and how would open-ended processes of designing work, where there might be no definitive beginnings or endings of design projects or no clear boundaries between 'designers' and 'users'? Processes, not products, are the starting points of many of the issues probed: how can designing be carried out, by which types of design constellations, with what types of methods, given the situations and complexities that form the contexts of design.

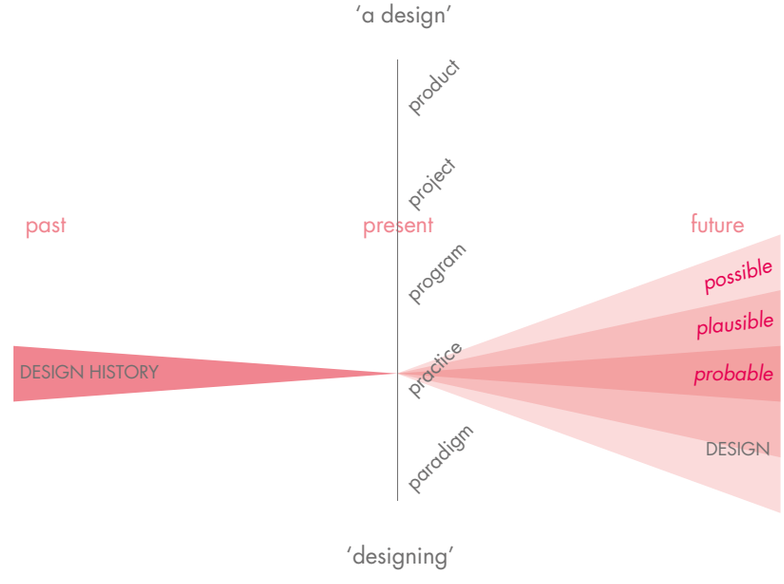
Handling different complexities in various ways in order to find a space from where to aim for a preferable future, is at the core of design. Thus, inherent to design are fluid and changing approaches to its own practice. As design situations change, the ways designing is done also need to change. With design moving into other fields than those from which it once sprang, questions such as what 'form' is if it is intangible, or how design intent should relate to how people use designed things, has led to a continuous development of methods, processes and concepts in designing that are anything but stable over time. But in design, there is little or no attention paid to these from a historical stance. For design histories to contribute to these types of approaches, these would need to more actively engage in questioning the *how* of designing from an understanding of an embedded historicity in ways of thinking and doing in design practices. If design history is to be able to engage with contemporary designing, the first thing to do is to bring these closer to each other.

Instead of contributing to accounting for past practices that could affirm or dispute definitions of design and designing, the scope here is to make histories that contribute to expanding the conceptual spaces in which we go about thinking and doing design[25]. In moving the starting point of such design histories closer to the positions and situations of contemporary designing that deals with how design is approached and carried out, these design histories take their cue from questions regarding how design is done. A focus in this design history approach is to pursue stories that give glimpses of where the ideas underpinning contemporary designing

25 Throughout this book, I speak of histories, and of history, in terms of 'making' rather than 'writing', fully aware that the expression "to make history" usually refers to an individual or collective effort or event that has not been done before, and that is of a character that immediately is recognized as something important to publicly note, record and remember: to include in 'history' as in historical accounts. For example: the moon landing, or Rosa Parks' protest against racial discrimination and segregation. But this is not what I refer to when I speak of 'making history'. While much of history is indeed written, and I certainly also contribute to writing history in the following, I approach history as something that goes beyond written narratives published by historians. History is made by means of many entangled practices – academic and not – and through many more varieties of historical representations than writing alone. Cf. Jaques Le Goff & Pierre Nora (eds.) *Faire de l'histoire. 1. Nouveaux problemes. 2. Nouvelles Approches. 3. Nouveaux objets* (Paris: Gallimard, 1974); Lynn Hunt, *Measuring Time, Making History* (New York: Central European University Press, 2008).

once arose, and how these came to become foundational for design. This is a type of design history that aims to probe what design could become, through making histories that speak to the instability and change inherent to design.

By shifting the outlook of design history from product to process – from things to thinking – an ambition is to sketch the contexts in which foundational concepts and central methods in design came about. The intent of this is to destabilise core understandings of 'design' based on what it has previously been, so that design can move in directions that perhaps have not been discerned as possible before. This shift of position, in which design histories can provide a sort of provisional and propositional scaffolding that activates an awareness of how – and why – the ways we design have been formed over time, is what makes these design histories 'transitional.'



Where we stand, what we see

Transitional design histories aim to engage in a continued re-positioning of perspectives on what is perceived as relevant, and difficult, in present design situations through exploring past assumptions and values embedded in designing. Placing the historical outlook on design in the part of the spectrum of design that deals with practices and ways of thinking that frame how designing is done, these histories engage with both explicit and implicit methods and concepts in design.

These histories are transitional, in that they do not provide solid foundations for, or explanations of, what design is or has been. Transitional design histories aim to work differently, as they instead are constructed to support making conceptual moves that allow for thinking and doing design differently, through using historical perspectives in exploring if it would be possible to see, think, and do design differently not only for the future but in the present. The 'transition' intended is thus not meant to be a passage from one clearly defined state or practice to another, or from a 'now' to a 'then', but something more akin to a quality or a logic in how this sort of history proposes to work.

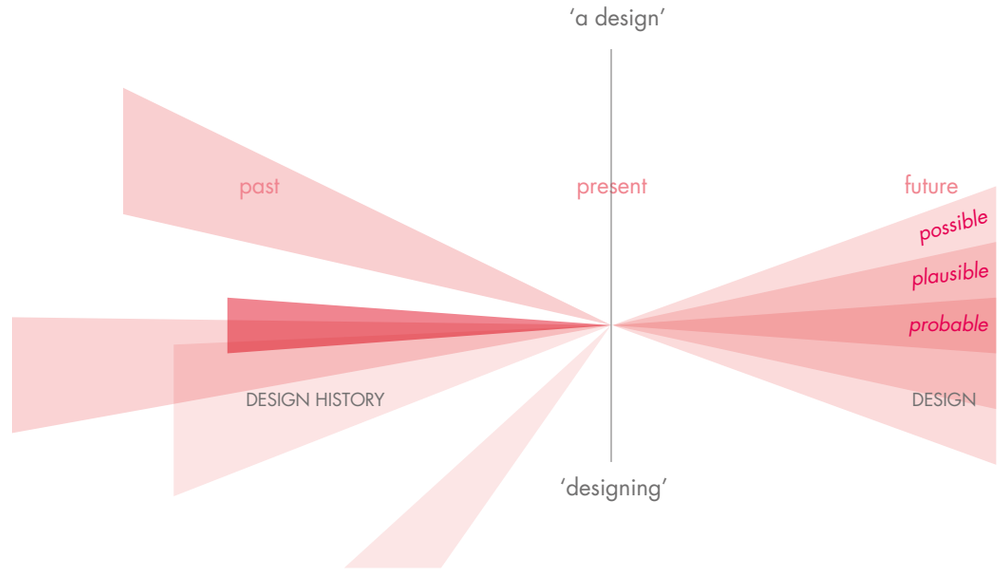
Design can be many things and does, of course, not have one definitive history but many potential histories. Since 'design' is not univocal, neither can 'design history' be thought of as one specific thing. And by proposing transitional design histories as a type of history that might matter to designing, I do not mean that these histories are something that should replace other design histories. On the contrary. Just as there need to be many ways of designing and many voices and knowledges present in design to address different situations and different complexities, we need many varieties of design histories from many perspectives and places. However, what I do suggest is that making design histories that take a starting point in matters of concern in design*ing* can more directly contribute to reframing how and what is possible to think and do in design than the types of design histories that focus on, for example, objects of design.

When these design histories are made from other perspectives, from designing, what seems relevant for us to pay attention to in the past will change as will the methods applied to probe new aspects of making histories. Through making transitional design histories, the perspectives provided will lead to shifts in the position from which design regards itself and what it engages with. As these perspectives lead to new ways of seeing design, and new things are seen in design, the transitional design histories that made this possible will at some point have played out the repositioning role. Then, new outlooks and new challenges in and for design will call for yet other potential histories: transitional design histories themselves are not fixed but fluid and responsive to the changes in practice, position, and perspectives in and on designing.

Taking a perspective on something has to do with several things: Where we place ourselves in order to look at something, what we use to help us look. A perspective, historically, was a sort of telescope – something to look through that made it possible to see distant things up close. What a perspective enables us to see and how we then represent and handle that which we see as well as that which is hidden from sight, varies depending on what types of lenses we apply to look at things through. And all of this – where we stand, what we use to see with, what that guides us to look at, and what then comes into view – depends on what the intention is with going looking for something in the first place.

The above illustrations of the cone of potential futures and its relation to the histories of design are built around the idea of gazing in a certain direction, from a particular point that gives a specific perspective allowing some things and not others to come into view. What is possible or not to see depends on how wide or narrow the frame of vision becomes when applying a perspective, and where the focus point of the perspective as lens lies. As the intention of transitional design histories is to contribute to critically exploring what design could become through activating an awareness of design's historicity, the shift in perspective here consists of applying historical lenses from a position in contemporary designing, shifting both frame of vision and focus in regard to what sorts of histories to go looking for.

From a position in present-day designing, looking to the past through the lenses of core concepts and methods in current design, this will bring into view ideas, practices and contexts within cultural and societal agendas that not only have allowed but perhaps also pushed for certain types of design practices to take form. But we might also see what that means for the limits these ways of doing design carry with them in the situations they are expected to address, and in terms of the norms and values that shaped them and that now might be perpetuated through design.



Making histories of designing, therefore, requires an awareness of the situated practices and knowledges that are integral to design. The proliferation of various design methods or representations, such as the wide-spread adoption of 'design thinking' or the 'double diamond' visualisation of the design process, might give an impression that some ways of doing design are more or less generic.[26] Designing, however, is always a situated practice that deals with the particularities of a specific situation, in a certain context, and for or with individuals that engage in the process of designing in various ways. Making histories of designing, therefore, also necessarily must entail at least some amount of precision in regard to which ways of designing and in which contexts its outlook and perspective is positioned.

In this study, the choice of position and context for exploring how to make transitional design histories is a Nordic context, and more specifically the types of designing that tend to be labelled as 'Scandinavian user-centered and participatory design'. The collaborative methods and approaches applied in designing referred to as the 'Scandinavian tradition' are today very present in design practices emphasising co-designing, within academia in both design research and design education, and in non-academic environments ranging from commercial or industrial design practices to NGO initiatives and community-oriented social

26 Nigel Cross, *Design Thinking: Understanding how Designers Think and Work* (Oxford: Berg, 2011); Tim Brown & Barry Katz, *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation* [2009] Revised and updated edition, (New York: HarperBusiness, 2019); "What is the framework for innovation? Design Council's evolved Double Diamond", https://www.designcouncil.org.uk/news-opinion/what-framework-innovation-design-councils-evolved-double-diamond (accessed 2020-05-30).

designing.[27] In probing how to go about making transitional design histories, I have chosen to position my historical inquiries in relation to Scandinavian participatory approaches to designing in a Swedish context. From the perspective of methods and concepts central to this type of designing, I have looked for potential pasts relevant to these practices, that can have contributed to forming these ways of designing.

Though the primary purpose of this thesis is to explore how history might matter more to designing, and not to write a comprehensive history of Scandinavian participatory and user-centered design, already setting this differently positioned perspective does contribute to highlighting some blind spots in how design histories previously have been made. Design historians have indeed both contributed to establishing and challenging histories of 20th century 'Scandinavian Design,' in terms of what is included or excluded in histories and mythologizations of Nordic design.[28] However, what is referred to, studied and critiqued in terms of a 'Scandinavian design tradition' are certain material aesthetic, discursive, and ideological manifestations of Nordic design. The emphasis is often on the mid-20th century, and apart from the inclusion of select images of ergonomic hand tools or eating utensils for disabled persons in the context of inclusive design, the establishment of user-centric design methods are not very visible in these histories.[29] Almost entirely absent from existing Scandinavian design history overviews, are the collaborative participatory design practices that sprung out of research and union initiatives linked to the introduction of computers into workplaces in the 1970s and 80s. At the time, and perhaps still it seems, these contexts were not defined as 'design,' and as such have not been made part of design history either.

While the positioning and perspectives applied here for studying participatory and user-centered design concepts and design methods as formed historically by design might be a somewhat novel proposal for how to do design history, the ways I have gone about conducting these studies are based in traditional historical methods of text-based inquiry and analysis. The material I have used are the types

27 Pelle Ehn, "Learning in participatory design as I found it (1970-2015)" in Betsy DiSalvo, Jason Yip, Elizabeth Bonsignore & Carl DiSalvo (eds.), *Participatory Design for Learning: Perspectives from Practice and Research* (New York: Routledge, 2017); in Jesper Simonsen & Toni Robertson (eds.), *Routledge International Handbook of Participatory Design* (New York: Routledge, 2013).

28 Kjetil Fallan (ed.), *Scandinavian Design: Alternative Histories* (London: Bloomsbury, 2012); Charlotte Fiell & Peter Fiell, *Scandinavian Design* (Köln & London: Taschen, 2002).

29 Lasse Brunnström, *Swedish Design. A History* (London: Bloomsbury, 2019); Kjetil Fallan, *Designing modern Norway: A history of design discourse* (London: Routledge, 2016); Pekka Korvenmaa, *Finnish design: A concise history* (Helsinki: University of Art and Design Helsinki, 2009); Mark Mussari, *Danish Modern: Between Art and Design* (London: Bloomsbury, 2016); Widar Halén & Kerstin Wickman (eds.), *Scandinavian design beyond the myth: Fifty years of design from the Nordic countries* (Stockholm: Arvinius, 2003); Hedvig Hedqvist, *Svensk form, internationell design* (Stockholm: Bokförlaget DN, 2007); Kerstin Wickman, *Formens rörelse: Svensk form genom 100 år* (Stockholm: Carlsson, 1995).

that historians often work with, in the forms of mediated ideas and events as these come across in the reading and analysis of different types of texts, images, and objects. In the search for material for these studies, I have chosen to mainly focus on published as well as archival material relating to organisations and contexts that to some extent already are present in Swedish design histories. The point with this is to explore if and how the change in position and perspective of making histories that start in designing could also provide new entry points and understandings of already established Swedish design historical narratives.

Histories as prototypes

The studies in this thesis are conducted with the intent of experimenting with how transitional design histories might be made, by means of which materials, which forms they could take, and if the shift in perspective towards designing can make histories that contribute to other design practices. Given this, the examples proposed as transitional design histories in this thesis aim to work as a sort of prototypes, and not as refined and finished final products. In design, prototypes are made and used in various stages of the design process. An early-stage prototype can be made in a very simple way, to test design ideas and concepts that are still relatively openly articulated. In contrast, more refined prototypes can be more detailed and functional in relation to representation and fidelity of form and materials. Whatever the sort of prototype, the whole idea with it is to test something that plays into a continuous process of making choices and proposing directions in an iterative design process.

Besides attempting to bring design and design history closer together, an aim of this thesis is also to explore what happens when bringing together methodological approaches from designing and historiography respectively in the prototyping of transitional design histories. Making histories as prototypes speaks to the idea of presenting histories that change in a reciprocal dialogue with design, since changes in contexts, challenges, concerns, and practices in the one will trigger changes in the other. Design histories that work in this way need to be provisional and open-ended rather than stable and explanatory.

Thinking of design histories as prototypes, and not as definitive accounts, brings together aspects of addressing change and complexity that, in different ways, actually can be discerned as fundamental in both designing and historiography. Designing responds to complexities in the present and proposes changes in and of certain things that could lead to different future situations. As design situations change, the ways designing is done also need to change. Historiography handles complexities in the past, proposing stories about situations and changes in the past that could somehow be meaningful for us in how we handle the present. And as histories are told from different perspectives, views on what is relevant to us in the past will change, as well as the methods applied to probe new aspects of making histories. As prototypes, these histories aim to re-position design's outlook in regard to in what it attends to, in what is valued or questioned, as well as in terms of itself as a discipline, and what that allows us to see and make sense of in the present.

Transitional design histories, made by starting in matters of concern in contemporary design*ing*, and bringing the historicity of current design methods and concepts to the fore, provide glimpses of other possible understandings of design than the ones readily visible to us. If we are to be able to change how design is done to address issues of democracy and justice, global inequalities, distribution and negotiation of

power and resources, sustainment and sustainability, we need to bring into view what is usually not in focus, and see more clearly that which is presently blurred. Transitional design histories can provide lenses to support changes in perspective as well as in position, opening up for other possible stories of the past, which contribute to plural and diverse understandings of our different, potential, and ever-changing 'now.'

Overview

The proposal in this dissertation, as presented in this first chapter, is that design needs a higher degree of awareness of its historicity to be able to alter its own practices. And for this historical awareness to be activated, design histories need to be made differently. The aims are to investigate how to make design histories that can matter to design and also bring relevant perspectives to the field of design history through merging methodological approaches from design research and design history. Can a bringing together of design and design history make visible the historicity of design's conceptual frameworks, supporting a re- or de-activation of these that contribute to changes in designing? And does this shift in outlook lead to histories that matter for designing, making present other historical contexts, actors, and ideas that can also contribute to the field of design history?

Prototyping transitional design histories of core methods and concepts in designing can contribute to altering the outlooks on how the present of design can be understood, and what designing could become. [30] The pasts activated as histories are decisive for the situated presents we are able to perceive and for the futures we thereby make it possible to envision.[31] The overall ambition, however, remains to test ways of doing design history that could be of relevance for understanding and expanding current and future design practices, aiming to "establish a thinking that is capable of thinking past the paradoxical conditions of now and therefore capable of opening our history, which is to say our future, to reflection."[32]

In chapter two, I discuss the disconnect between designing and design history arguing that historical perspectives are needed to make visible how the past conditions the present as well as the futures aimed for in designing. Providing a brief overview of how design history has emerged as an academic discipline in relation to design education and practice, I suggest that design histories could be made with the aim to contribute to shifts in contemporary design practices. For design history to engage with issues in design it would need to reposition its outlook. Instead of focusing outcomes of designing, these histories would inquire into the historicity of designing itself by probing the ideas that once were formative of how core design methods and concepts came about. Making visible what this might mean for contemporary practices, design histories can contribute to critically exploring design's foundations with the aim of supporting emerging design practices.

In chapter three, the propositional approach of transitional design histories is outlined together with the suggestion of making prototypes of histories to explore what

30 Cameron Tonkinwise, "Designing transitions – from and to What?", *Design Philosophy Papers* 2015:1 (vol.13), 85-92; Redström, *Making Design Theory*, 96-99.

31 Ramia Mazé, "Politics of designing visions of the future", *Journal of Futures Studies* 2019:3 (vol.23), 23–38; Tony Fry, *Design Futuring: Sustainability, Ethics, and New Practice,* (Berg: Oxford, 2009);

32 Dilnot, "History, Design, Futures", 211.

these might be like.[33] Narrowing down and situating these studies in relation to a specific type of practice and view on design, Scandinavian user-centered and participatory design is selected as the type of designing investigated in relation to some of its core concepts, methods and themes. This provides the outlook and starting point for the making of the design history prototypes that follow. A very brief account is given of the emergence of user-centered and participatory design in Scandinavia in tracing of some of the core ideas that carry over from these into design practices and research that build on these traditions.

Three prototypes are then presented of how transitional design histories could be made by starting with concepts and themes in Scandinavian user-centered and participatory design. As prototypes, these are constructed in slightly different ways and aim to explore partially different aspects of mechanisms of and in design history and designing in relation to each other. Each takes a different starting point in terms of what sort of perspective is applied, resulting in three separate but interlinked histories rather than a cohesive story.

The first prototype, chapter four, focuses on the concept of 'participation' related to historical conditions of importance for the formation of participatory design in Scandinavia. Through searching for ideas revolving around situations in which ideas of participatory practices and mutual learning took form around the turn of the century 1900, this prototype highlights contexts and ideas of democracy and participation that preceded the appearance of participatory design in the 1970s and 1980s suggesting that ideas and practices unfolding in early 20th century Sweden became formative for a discourse that made possible the emergence of a collaborative approach to designing. The material for this study is mainly related to the thinking of Ellen Key, influential writer in the fields of aesthetics, ethics, education and women's and workers' rights nationally and internationally, in the early 20th century.

The second prototype, chapter five, revolves around the concepts of use and users, and more specifically around the relationship between designed ideal or intended uses and everyday realities of people's actual uses of things. I propose that the conceptual tensions between ideal and real use have been there from the very beginning and that ideas (and certain practices) of the involvement of users in design as these were introduced in Sweden in the 1940s both acknowledged and to some extent sought to deal with these tensions. The material for the study consists of publications and archival material from different investigations into 'dwelling habits' conducted by, or in collaboration with, the design organisation Svenska Slöjdföreningen (SSF) between 1937 and 1943.

The third prototype, chapter six, works with methods development in user-centred and participatory design, proposing that histories of such methods can be traced to the discourse around material and behavioural reform of everyday domestic work in mid-20th century Sweden. Early instances of the introduction of for example iterative prototyping and user studies are discussed based on published material from the research into everyday domestic work carried out at the Hemmens Forskningsinstitut (HFI, The home research institute) in the 1940s and 1950s.

Finally, in chapter seven, these prototypes of transitional design histories are reflected upon and discussed in relation to if and how this approach can provide histories *for* design, rather than histories *of* design. I argue that these types of histories can contribute to opening up views on and potentials for transitions that could support the development of different ways of going about designing, and thus the futures we make possible through design.

33 Redström, *Making Design Theory*, 130

2. Design/History

In the introduction, I put forth the idea that design needs to activate an awareness of its historicity in order to develop other practices. Therefore, design history should be made to work differently and contribute to making the past more present in designing. The question, then, becomes how to go about doing this? In this thesis, I argue that one possibility is to make design histories that are propositional and transitional, and that support fluid and changing understandings of and practices in designing. These transitional design histories would be made to work as prototypes, supporting shifts towards what design could become, rather than as well-rounded stories of what design is or has been, by combining methods from programmatic design research and historical research. In this chapter, I sketch how design history has taken form as an academic field and discuss what that has meant for design history's role in relation to design practice, as well as that this has led to in the relations between design history and designing.

Disconnected outlooks

Ever since it came about in relation to industrialisation design has been instrumental in materialising ways of living that propose to somehow lead towards a better future.[1] How 'better' might be understood depends on how the present is perceived, and on what then seems possible to achieve through design in order to move towards the envisioned future. This future orientation remains in design, focusing what is to come and how to make other things possible than those readily present to us.[2] Design history, instead, usually directs its attention to what has been: and towards the design of things in the past and how these have come to carry cultural and economic significance.[3] So, though design practice and design history both are situated in the present, they are located in different places and approach design from different positions.

The difference in the irrespective outlook makes it difficult to find common ground for design and history to come together – and why should they? The perceived usefulness of history in design seldom goes beyond treating design history as a repository of inspirational or cautionary examples of form-given things, or as a source for understanding the formation of the design profession. Since much of design history has been written and produced along these lines, the limited forms of engagement between designing and design history is not difficult to understand. However, the lack of history in practices of designing can be rather problematic.[4]

1 Ramia Mazé, "Politics of designing visions of the future", *Journal of Future Studies*, 2019:3 (vol.23), 23-28.
2 Johan Redström, "Certain uncertainties and the design of design education", *She Ji: The Journal of Design, Economics, and Innovation* 2020:1 (vol. 6), 83-100.
3 "Editorial policy", *Journal of Design History* 2019:4 (vol.32).
4 Clive Dilnot, "The Matter of Design", *Design Philosophy Papers* 2015:2 (vol. 13), 115-123.

As design makes things that propose to alter present situations, it can unknowingly perpetuate past ways of being and thinking in its outcomes and also through its processes. That the past makings of design are present as material artefacts and environments brought into the world is quite obvious. However, that the very understandings of the world, ingrained in designing as such, are also made in the past and activated in how design presently operates can be more difficult to see.[5] Though aiming for different futures – more just, more sustainable – design might instead inadvertently reinforce precisely some of the mechanisms it seeks to avoid.[6]

While design might critically engage with questioning *how* to design in relation to problematic aspects of its core concepts, it seldom questions the ideas or mechanisms that historically brought these concepts to become central to designing. Without an awareness of how the past is present in design there is a substantial risk that, when addressing the complex challenges of new design situations, design will remain blind to its own historical baggage. This blindness will limit design's potential of finding and inventing other practices.[7]

Design history could engage in exploring how design methods and concepts carry historical assumptions and values into contemporary and emerging design situations and thus provide perspectives on what is made possible, and what is made difficult, given the historicity of designing. Making present the historicity of concepts and world views that govern design, can contribute to de- or reactivating ideas and ideals in design, allowing for broader views on what might be needed to consider in repositioning design to make other future trajectories possible.

However, design history seldom engages much with matters of what design could become but has its attention directed towards what design has been. This outlook stems from a starting point in understandings of *what* design is – often formulated in terms of past design outcomes or products. Contemporary designing, on the other hand, is more interested in questions of *how*: How can design make a difference in changing existing situations into preferred future ones? How would design methods and tools need to be set up, to support the types of designing that current challenges and situations seem to require? The themes and research interests of design history do not readily provide connections that could support design in grappling with questions of how design could be done differently.[8]

Concepts and definitions

The development of methods for handling new design situations is central to design. Methods are, however, not often considered in terms of how, when, or why they once came about. Design is conditioned by history through the thinking that

5 Cameron Tonkinwise, "Always Historicise Design", *Design Philosophy Papers* 2006:1 (vol. 4).

6 Joanna Boehnert, "Design vs. the Design Industry", *Design Philosophy Papers,* 2014:2 (vol.12), 119–136; Tony Fry, *A New Design Philosophy: An Introduction to Defuturing* (Sydney: UNSW Press, 1999).

7 Susan C. Stewart, "And so to another setting", in Tony Fry, Clive Dilnot & Susan C. Stewart, *Design and the Question of History* (Bloomsbury: London & New York, 2015), 275-297.

8 Tony Fry, "Whither design, whether history" in Fry et al., *Design and the Question of History*, 8-9.

governs its outlook on the world, and in the methods and processes that guide its actions and its aims. The contexts that have formed design over time have left traces entangled in how design is carried out, and in the expectations on what design can bring to the situations it engages with. Many of the concepts in design that still form its foundations, setting frameworks for how and what design engages with the world, stem from times other than our own. These conceptual foundations are present in the ways that design is done and in the world views of design , often implicitly taken for granted as a stable ground for design practice.

For example, there is a vast body of design research exploring what open-ended or collaborative design processes means for practices in user-centered and participatory design, in terms of how 'designers' and 'users' might come together in design activities, what this does for emerging design practices, and the roles of 'designers' and 'users.'[9] While the category of 'users' in such approaches tends to be critically questioned, as du understandings of how 'use' might be defined, the fundamental idea that design *does* have something to do with use and users remains unchallenged. These concepts thus provide a seemingly stable foundation for design that aims to be 'user-centered.' However, the concept of 'use' has not always been part of design's foundations, and neither has the idea of designing for – or indeed with – 'users.'

When design emerged, in connection to industrial and societal change, its initial focus was on how material things should be made by certain people (designers) for other people (consumers, or users). The ideas of designing for use, and for or with users, have come about as responses to specific challenges and changes in design and society, in specific times and contexts, and have over time formed into concepts embedded in design. The concept of 'use,' for example, connotes intentional and directed acts that are specifically tied to things that 'users' – not 'consumers' or 'people' – perform in relation to the intentions and actions of 'designers.' While concepts such as these remain both activated and critically discussed in design, in different ways, the connections to the *how* and the *why* that once called them into the realm of designing have been more or less forgotten. The assumptions embedded in these core concepts continue to form how we think, how we talk, and how we take action in design.

9 To give only some examples: Liesbeth Huybrechts, Virginia Tassinari, Barbara Roosen, & Teodora Constantinescu, "Work, labour and action: the role of participatory design in (re) activating the political dimension of work", in *Proceedings of the 15th Participatory Design Conference. i.* (ACM, 2018), 1–11; Åsa Ståhl & Kristina Lindström, "Politics of Inviting: Co-Articulating Issues in Designerly Public Engagement", in Rachel Charlotte Smith, Kasper Tang Vangkilde, Mette Gislev Kjaersgaard, Ton Otto, Joachim Halse, & Thomas Binder (eds.) *Design Anthropological Futures* (London: Bloomsbury Academic, 2016); Thomas Binder, Eva Brandt, Pelle Ehn & Joachim Halse "Democratic design experiments: between parliament and laboratory", *CoDesign*, 2015:3-4 (vol.11), 152-165; Johan Redström, "RE:Definitions of use", *Design Studies* 2008:4 (vol.4), 410–423; Nelly Oudshoorn, Els Rommes, & Marcelle Stienstra, "Configuring the User as Everybody: Gender and Design Cultures in Information and Communication Technologies", *Science, Technology & Human Values* 2004:1 (vol.29), 30–63.

An aim in contemporary design is to find ways to address emerging and complex challenges, bringing plural perspectives and distributed agency into designing. The situations that design seeks to address in the present are the results of choices and actions in the past. If the alliances with industrialism, with the production of consumer goods, and the intentional or unintentional reproduction of these mechanisms are some of the issues that design seeks to revise or address, it is hard to see how this could be done without an awareness of design's historicity. To achieve such a historical awareness in designing, design histories would need to ask questions about what design *does*, rather than what design *is*.

The aim with shifting focus to regard design practices and processes of design in light of the historical contexts in which they arose, is to make way for design histories that take critical views on design's foundational concepts and methods, challenging assumptions that otherwise are either taken for granted or simply invisible. If design history is to engage with the concepts that are central to designing, the positioning from where histories are made needs to change.

Returning to the respective outlooks of design and design history and how these are positioned in different understandings of 'design', this can – as described in the previous chapter – be approached in light of a definition of design suggested by Johan Redström as being fluidly made over a spectrum ranging from 'a design' to 'designing'. Here, a conceptual definition of design spans from the particular of what *a* design (product) is to more general paradigmatic notions of design*ing* (as practices and processes).[10] Much of design history's attention so far could be generically categorised as located in the product and project part of such a spectrum. With its attention focused on the design of things – objects – and their designers, design history has its centre of gravity in the part of the spectrum of design that probes what *a* design can be considered to be in a certain context.

The shift in perspective proposed here builds on moving design history's outlook towards the part of the spectrum of design definitions that revolve around practice and process, to explore what becomes visible from there. This does not mean that instances of design or history that deals with things– the '*a* design' end of the spectrum – would be uninteresting. It means that what these design histories will try to make visible would be more of a *why* and a *where* than a *what*. By situating the outlook in concepts and methods at the core of design*ing* – in practice – the central questions guiding the making of design histories would not primarily be what the outcomes of these ways of designing have been, but how design can be understood as making and defining itself through what it does (products and projects) as well as through how and why it works (processes and practices).

This historical approach would move away from stable definitions of design as things, materials or mediations as starting points for making design histories. Working with an understanding of design defined as a spectrum, the types of design histories that revolve around design practices, would not really have a defined starting point – neither in terms of precise definitions of design nor in terms of historical times or periods pointed out to be investigated. The purpose of moving design history closer to issues in designing is to pose questions, rather than provide answers, about how and in response to what sorts of contexts present ways of designing have come about.

10 Johan Redström, *Making Design Theory* (Cambridge, MA: The MIT Press, 2017), x.

History is always a history *of* something, and at the same time, it is made *for* something. In order to provide causal explanations that make sense for explaining what was before in relation to what is now, history typically relies on ideas of continuity. Approaching histories as chains of events, where one event leads to another, conjures up a sort of idea of intentional directionality, where things are what they are due to these continuous happenings. If we take a closer look at the past, things that seem necessarily connected in a chain of events are made to appear as such through how history is represented. As economic historian Alexander Gerschenkron put it:

> continuity must be regarded as a set of tools forged by the historian rather than as something inherently and invariantly contained in the historical matter. [---] It is the historian who by abstracting from differences and by concentrating on similarities establishes the continuity of events across decades or centuries filled with events that lack all pertinency to the continuity model. It is the historian who decides how far back the causal chain should be pursued and by his (sic) fiat creates its 'beginning' as he creates endogenous and exogenous events.[11]

That history is about narrative, about stories we tell, is apparent already in the name of the discipline. In a sense, all research – not only historical – is about making meaning out of the world by choosing something particular to pay attention to, and finding a way to tell a story of what it means to understand the world and its workings through that perspective. Design differs from many other disciplines in that it aims to approach understandings in terms of handling complexities as meaningful wholes rather than as particulars. To find ways to do that, design methods and processes often aim to find many ways of making change in relationally entangled and diverse ecologies. Thus, design histories also need to support seeing and approaching multiplicity and difference, counteracting ideas of singular worlds and universalist world views.

Historical perspectives on contemporary design practices could provide openings for seeing the present differently, from other angles. Making other design histories can make room for sketching many different potential 'presents' – multiple 'nows', rather than a singular 'now' – that could speak to design's continuous movement between potentiality and actuality. In order to provide design histories that might more fluidly both respond to changes in design practice today, and that might contribute to redefining what designing could become, the move towards designing is needed to shift position and allow us to see things from a different point of view than the usual. This shift in perspective towards histories of design*ing* also builds on the proposal that definitions in design are made in several ways, one of which is *through* design itself.

Thinking of design as something defined through tensions and movements along a spectrum between what it makes, what it does, what it proposes to be, its conceptual spaces and world views, it follows that fluidity and change are at the core of such design definitions. Histories that spring out of such an understanding of design definitions also need to be inherently unstable and transitional, if they are to support processes of continuous unfolding relating to what design can

11 Alexander Gerschenkron, "On the Concept of Continuity in History", *Proceedings of the American Philosophical Society* 1962: 3 (vol. 106), 208.

become, rather than confirming or perpetuating ideas of what design is. Building on this, design histories would need to be transitional, in that they aim to resonate with design understood as something that *becomes* rather than something that *is*:

> Our framework was never meant to account for what is and its relation to what was; it was not meant to be a general theory in the sense other disciplines think of such things. It was the scaffold we needed to make sense of a vast range of possibilities open for us: for navigating the potentiality of what could become, not the actuality of what became. But over time, it shifts from being a transitional and temporary foundation, and when it does, it is no longer a support for development but a framework for making sure things stay the same. It becomes history, not in the sense of a reflective and critical understanding of the past and its influence on the present, but in the sense of a now confined to repetition.[12]

Transitional design histories, then, could provide materials for the temporary scaffoldings that would support provisional foundations that design could build its becoming on. Shifting perspectives to more practice-oriented histories of designing will make visible other things than the ones we usually notice and that have so far not been actively considered or included in histories of design. This also means that we could expect to discover aspects, values and limitations embedded in today's design concepts and methods relating to the designing these will be able to support. This opens up for making design histories that are propositional rather than definitive histories of what design has been or is, in that they contribute to seeing and questioning current concepts and methods in design in light of the historical contexts in which they have emerged and been formed.

Histories, presents, and perspectives

Histories of design, as these have been told so far, have often focused on objects and the socio-cultural contexts in which they have come about, and come to use. And for good reasons. Making things is at the core of designing, and much of what has been designed has left substantial traces in the world and in our lives. The vast body of work that addresses material historical aspects of design outcomes and design culture in a broad sense will, of course, continue to be crucial to deepening and broadening a collective understanding of aspects of design and its outcomes, past and present.

The contemporary practice of research in design history is generally agreed upon to be a pluralistic endeavour of multiple rather than singular stories. Most design historians emphasise that their contribution is only one of many possible histories that could be told. The adoption of different historical perspectives, methods and areas of investigation will not lead to a definitive description of what the past *actually* was like, but will constantly give rise to new questions as "the concerns of the present [are] projected onto the past."[13]

Even if the gradual accumulation of a corpus of design histories does not aim towards a definitive account of the past, it nonetheless makes certain histories more visible than others. These more established stories tend to be told, re-told, and indeed also questioned, in publications, in design museums, and in design history surveys in design curricula. People, periods and places that turn up in most, if not indeed all, of these stories are for example the Great Exhibition, William Morris, the

12 Redström, *Making Design Theory*, 66.
13 Dennis Doordan, "On history", *Design Issues* 1995:1 (vol.11), 76.

Bauhaus, modernism, Raymond Loewy, postmodernism, and so on. In the repeated actualisation of these stories, the questions and topics of design history will continue to be of a certain sort, given the disciplinary practices supported or reacted against. In this sense, there will almost inevitably be some concepts, approaches, and stories that stabilise or even fossilise as a more substantial core narrative of what design is, and has been. Consequently, 'histories' layered upon each other will eventually become 'the history' rather than 'a history' – whether or not this is intentional or even desired. Repeating a certain story enough times will inevitably solidify it as a central part or as a given in design history.

Despite highlighting the importance of multiple perspectives in design history, the histories that become the 'grand narratives' of the field both support and perpetuate certain understandings of what is considered to be design, and of how and why design is done. The problems associated with stable, or fossilised, stories have been core to discussions about canon and narratives internal to the discipline of design history for several decades. However, this is not the only problematic aspect. The ways in which designing happens is built on historical understandings of design, embedded in the ways design is done.

Histories of design have primarily taken their starting points in understandings of design as visual and material form performed by individuals identified as 'designers'. Design history has thus come to revolve around issues of exploring what could count as design things, and in which types of contexts one would find people that do design. As the frame of vision for design history has broadened, more aspects of what can be seen and understood as design have been included in design historical narratives. Given this ongoing reframing, it would be reasonable to assume that there are aspects of the past that have not yet been made visible in design's histories.

Taking a different approach to what history could do in design, as well as how present and past relate to the making of design history, things might emerge that can not only reframe what becomes visible in the past but also in the present. In other words, historical perspectives on present-day designing can shed light on aspects of contemporary practice that otherwise might not be easy for us to see. Making visible some of the assumptions and world views that historically have shaped the processes and methods of designing could enable shifts in the very positions from where designing takes place in the present, and provide openings for seeing other possible ways of designing.

Victor Margolin noted already in the mid-nineties that "'design' does not signify a class of objects that can be pinned down like butterflies. Designing is an activity that is constantly changing. How then can we establish a body of knowledge about something that has no fixed identity?"[14] Instead of trying to pin down definitions of design and designing and exploring the past in terms of which things, practices, uses and situations should be present in histories of design, we could try making design histories that actually speak to the characteristics of change at the core of design.

If we were to sidestep the questions of which objects or classifications of design we might need to define for basing design history on and instead work with

14 Victor Margolin, "Design history or design studies: Subject matter and methods", *Design Issues* 1995:1 (vol.11), 10.

the idea of design as something characterised by constant change, the starting points for making design histories will move away from histories of design defined as things. We might, therefore, want to pay somewhat more attention to not only things made and events played out in the past and presented to us as history, but also to how choices of foci and practices adopted – in designing as well as in historiography – have rendered some potential ways of thinking and doing invisible, rather than impossible.[15]

To make other histories visible, that might be of relevance to contemporary and emerging design practices, we will need to shift the starting points from which we begin our inquiries, and question what the 'design' is that we are looking for. Instead of searching for a 'history of design' – trying to pin down the butterflies, to speak with Victor Margolin – the idea here is to investigate how to instead approach the historicity of designing in relation to the potential tensions between intentional change and unintentional reinforcement of historical values or mechanisms.

Being a practice that has come about in response to particular historical situations, design's processes and methods have been developed based on concepts and modes of thinking specific to these historical contexts. With time, the world views, or proposals, that the development of certain ways of designing was built upon have become so embedded in doing design that these are no longer expressly articulated or even visible. Moreover, when new methods or tools for doing design emerge, these will either build further on or be presented as alternatives to established design practices, proposing that design could be done differently than it is now. While this search for other ways of doing, of finding approaches that could support design in responding to constantly changing societal, technological and disciplinary contexts is central to design, this is seldom something that design history engages with. It is precisely here that design history could provide approaches to historically formed concepts and practices which design would need an increased awareness of if the ambition is to design differently.

By changing the outlook on what could be perceived of as design's history, we could, "shift perspectives – if only slightly – to provide short glimpses of something else" by trying to analyse things that are seemingly familiar and known to us in ways that could perhaps open up for other ways of understanding and using them.[16] This approach, here, is applied to making fluid and transitional histories that mean something to design, out of what is relevant to look for in the present in terms of changing conditions and practices in designing.

Transitional design histories aim to contribute with elements to the expanding, or layering, of many diverse design histories that bring a variety of perspectives, thematics and contributions to the field of design knowledge and design inquiry. This also suggests that transitional design histories could eventually solidify and form parts of more generic or mainstream components of "design history". In the following, let us take a brief look at design history as it has formed academically, and approach the reasons why that type of design history has not yet seemed to resonate with design practice.

15 Tony Fry, "Whither design, whether history", 9.
16 Redström, *Making Design Theory*, 2.

Design history: The formation of a field

An object-focus has been profoundly present from the early days in design historical research and has since continued to be central to the ways design histories have been told. This approach was highly related to not only the contexts in which design history as a practice emerged but also to the early roles that design history played in relation to the emergence of new practices in design itself. Some of the early discussions around what design history could or should be in terms of both an emerging discipline, subject of study, and in relation to the field of design, arose in Great Britain. There, higher education reforms in the 1960s had stipulated that all arts-based education – such as design – must include teaching in historical and contextual subject areas connected to the area of specialisation. In design education, the subject of art and design history came to fill this role.[17]

Many of the people teaching history within design curricula were of the view that the field of design needed histories that spoke to the subject of design, and that more related to critical perspectives on the issues in design practice and education rather than providing art historical foundations and overviews. Therefore, in 1975, a group of scholars gathered in Newcastle for one of the first conferences on design history.[18] The following year in London, the theme for the conference was the concept of leisure in twentieth-century design history. In the proceedings following the conference, the field of design history was described as a new discipline not yet understood, and not accepted, despite already having been incorporated in design education.[19]

The design history conference statements of the mid-seventies bear witness to a reaction against the sort of knowledge and approach to design then present in the discipline of art history. At the design history conference of 1976, the agenda was to establish design history in its own right rather than its being "a poor relation of Art History".[20] The type of history that the new design historians wished to distance themselves from was one that was based on ideas of connoisseurship, such as distinguishing significant works in design on the basis of criteria of aesthetics and excellence. Design history should, instead, deal with objects that were not categorised as objects of art, but industrially produced everyday things:

> In the first place, Design History is concerned with artefacts – things created by man to fulfil a particular function and thus satisfy a social need. It seeks to examine and, eventually, to explain these artefacts by reference to a wide range of criteria – social, technological, psychological, political and economic ones, in particular – within a historical context.[21]

17 Joanne Gooding, *Design history in Britain from the 1970s to 2012: Context, formation, and development,* diss. (University of Northumbria: Newcastle, 2012), 38f.
18 Gooding, *Design history in Britain from the 1970s to 2012*, 58f.
19 Bridget Wilkins, "Introduction", *Leisure in the twentieth century. History of Design. Fourteen papers given at the Second Conference on Twentieth Century Design History 1976* (London: Design Council Publications, 1977), 5.
20 Wilkins, "Introduction", 5.
21 Wilkins, "Introduction", 5.

Not only the "beautiful" or "decorative" artefacts should be studied, but all sorts of objects, that were the results of the work of a designer "immersed in society (…) to understand its needs and attempt to provide solutions to them".[22]

From the beginning of its formation, design history has included a vivid discussion on the methods and areas of study adequate to the subject – the history of design – both of which have been notoriously difficult to define.[23] The first design history conferences aimed to establish frameworks and practices in the history of design were convened in the mid-1970s, gathering many of the academic practitioners active in teaching and researching design history in design educational settings. The discussions of which varieties of histories would be of relevance to design education quite soon veered towards a process of establishing design history as a specialised discipline in its own right. These conferences led to the formation of the Design History Society in 1977, and in the 1980s to the establishment of specific journals dedicated to, or including, design historical scholarship, and to specific educational programs dedicated to the study of history of design in its own right rather than only as a part of studio-based design education.[24]

Design historian Penny Sparke noted, after the third design history conference in 1977, three main approaches to the history of design that could be discerned as forming design history: history of designing, social history with a focus on designed objects, and "a more deliberately 'art historical' track, examining the problem of style and its analysis in objects."[25] While these new directions in design history mainly arose out of a criticism of traditional art history, in relation to categorisations and value judgements of its objects of study, the methods and modes of structuring art history also continued to have a strong presence in design history.

Establishing narratives

A standard narrative of the history of design had initially been set in a book that was a standard textbook in design history for many decades: *Pioneers of the Modern Movement: From William Morris to Walter Gropius* by art historian Nikolaus Pevsner. This book was originally published in 1936 and later revised and reprinted several times with new titles, first *Pioneers of Modern Design* and later *The Sources of Modern Design and Architecture*.[26] Heavily emphasising the built environment and visual form, a history of modern design is told with a starting point in industrialisation and culminating in the emergence of modernism,

22 Wilkins, "Introduction", 5.
23 John A. Walker, *Design History and the History of Design* (London: Pluto, 1989).
24 Gooding, *Design history in Britain from the 1970s to 2012,* 139-174.
25 Penny Sparke, "Design History: Fad or Function? Some Afterthoughts", in *Design History Society Newsletter number three*, December 1978, 14-16.
26 Nikolaus Pevsner, *Pioneers of the Modern Movement: From William Morris to Walter Gropius* (London, 1936); *Pioneers of Modern Design: From William Morris to Walter Gropius* (New York: The Museum of Modern Art, 1949) [new editions 1957, 1960, 1968, rev. 1974, 1986, rev. & exp. 2005; *The sources of modern architecture and design* (London: Thames and Hudson, 1968) [rev. 1975].

with focus on how an understanding of 'good design' and criteria for judging the aesthetics of modern design practice had undergone radical changes during that period. In line with practices in traditional art history, the structure of Pevsner's history was formed around the authorship of individual designers and examples of their works, together with style-based periodisation.

Despite recurring critiques of this narrative, Pevsner's text nonetheless became seminal as one of the early textbooks on design history. It has remained of quite central importance to the discourse of design history, in relation to how its basic structuring of 'the history of design' continues to resonate with the teaching of design history survey courses – albeit with critiques and perspectives added to or challenging the Pevsnerian structure and approach. If we take a closer look at Pevsner's text as a part of the history of design rather than as a design history textbook, the time and context in which it was published can explain its focus on defending and defining the aesthetics of modernism from the point of view of designing.

Pevsner's history of design traced a continuity from William Morris' critique of industrial production in favour of the hand-made to the incorporation of a crafts-based while industrially oriented pedagogy in the socially and industrially oriented Bauhaus education as fronted by Walter Gropius. This was portrayed as a linear and causal evolution in line with the needs of contemporary society. Through this way of doing history, already established within art history (great artists, great works, stylistic evolvement), Pevsner gave the new outcomes of industrial and standardised object production and modernist architecture a solid foundation in a historical chain of important events and styles of art and architecture. In this sense, Pevsner's history of design took its starting point in the service of supporting the radical changes he saw in contemporary (industrial) design practices, by providing legitimacy through history in regard to emerging design ideals that emphasised other objects, other aesthetics, and other practices than before.

We might need to remind ourselves that the Bauhaus, when Pevsner's book was published in 1936, was a relatively small and experimental German design school that had closed only three years earlier. During the time the school had been open, between 1919 and 1933, it had relocated twice, and had continuously developed its curriculum, under three different leaders, aiming to introduce a new type of design education in service of modern industry and society. Exactly *how* that new design education should be shaped and *what* the curriculum should include, was a matter of almost constant debate and experimentation. In the words of Anni Albers, one of the students who went on to engage in teaching and design experimentation at the Bauhaus, the environment was not always perceived as an education in its time:

> (…) well, the Bauhaus today is thought of always as a school, a very adventurous and interesting one, to which you went and were taught something; that it was a readymade spirit. But when I got there in 1922, that wasn't true at all. It was in a great muddle and there was a great searching going on from all sides. And people like Klee and Kandinsky weren't recognised as the great masters. They were starting to find their way. And this kind of general searching was very exciting. And in my little articles this is what I called the creative vacuum. But the word "education" was never mentioned. And the people we think of as the great masters – Klee and Kandinsky – they weren't available for questions. They were the great silent ones who talked among themselves maybe, but never to small little students like me. But we knew that what the Academy was doing

> was wrong and it was exciting that you knew you had the freedom to try out something. And that was fine. But, as I say, it wasn't that you went there and were taking something home from there. You were a contributor.[27]

In Anni Albers' description, the environment at the Bauhaus comes across as an exploratory and experimental environment rather than as an organised design school. The Bauhaus, at the time of Pevsner's publication of his book in the 1930s, was far from the internationally well-known forerunner of modern design education that it later has become established to be.[28] The subsequent, and formative, role the Bauhaus has played in design education can largely be attributed to how design history has been made, and how its narratives were canonised through Pevsner's and others' descriptions. The later affirmation of the Bauhaus as a seminal design education also took form through the establishment of various design educations by former Bauhaus staff and students, explicitly labelling the new design educations as adhering to or reviving 'Bauhaus curricula' or 'Bauhaus ways of teaching'.

Modernism was, in the 1930s, an aesthetic, an idea and an ideology still highly contested – both from 'without' and 'within'. As the narratives of design history grew strong under the auspices of art history, the understanding of and attention to design history as primarily dealing with the visual and material form of objects was introduced as part and parcel of the field. The historical legacy from Morris to Gropius traced by Pevsner can be seen as serving a legitimising and polemical function in advocating the radical and revolutionary modernist agenda and its visual and material form expressions with arguments rooted in history and tradition. This placed the emerging new and industrially based practices of design of the 1930s in a historical lineage traced back to the British Arts and Crafts movement. The historical narrative was thus used to legitimise the – in Pevsner's day – still highly contested expressions and practices of modernist design and the ideals of a new type of society based on industrial production and democratic governance that these were associated with. We can see similar rhetorical practices making use of history to argue for modernism, for example, in the Swedish context when 'functionalism' was introduced at the Stockholm exhibition of 1930. In texts and brochures promoting the exhibition, ample references were made to the "glorious traditions" of Swedish handicraft and arts industries that formed the trajectory leading up to the modern(ist) objects and environments "aimed at the practical needs of modern life".[29]

27 Oral history interview with Anni Albers, 1968 July 5. Smithsonian Archives of American Art. https://www.aaa.si.edu/collections/interviews/oral-history-interview-anni-albers-12134 (accessed 2020-06-10).

28 Laura Forlano, Molly Wright Steenson, Mike Ananny, Ramia Mazé, Fred Turner, Joichi Ito, Natalie Saltiel, Andre Uhl, Carol Strohecker & Stuart Candy, *Bauhaus Futures* (US: The Mit Press, 2019).

29 *Stockholm: Stockholmsutställningen 1930* (broschure), Svenska Trafikförbundet & Stockholmsutställningen 1930 (Stockholm: Centraltryckeriet, 1929).

The history presented by Pevsner, pointing to modernism as the pinnacle of formal and ideal development in design and to the Bauhaus as central to developments in design practice and design education, has to a large extent continued to serve as a model for constructing design history. Specific objects deemed to be of extraordinary quality (thus excluding things made for everyday use) and individual designers, such as Walter Gropius and William Morris, were established as especially significant with reference to their perceived creative and aesthetic genius.[30] This history has been perpetuated through latter narratives and counternarratives – and through the design practices these have supported. The teleological attitude towards industrialism, for example, as the almost 'natural' result of progress and development was still in the 1930s one of projecting a desired and preferred future from the stance of a present that was socially and politically turbulent, to say the least. The idea of a better and more egalitarian future, and the hope of a different material manifestation through design of this future, was integrated into Pevsner's account.

Later, in the following decades, the same basic story of design and industry in a new take on the relationship and practice leading to progress was told from other perspectives critical to Pevsner's history. Sigfried Giedion, for example, in his 1948 book *Mechanisation takes command*: *A contribution to anonymous history,* attempted a social history of design told through the lens of industry and technology rather than in terms of style and authorship. [31] Later, in 1960, Reyner Banham's *Theory and Design in the First Machine Age* brought in issues of function and functionalism – in terms of utility as well as performance – as starting points for writing a history of design.[32] However, all three of these histories of design followed the same overall narrative, thus consolidating a story that has continued to provide a structural foundation for industrial design history: beginning with industrialisation and the arts and crafts movement, emphasising tensions between crafts-based and industrial means of production in relation to form-giving, and tracing works of influential design – mainly understood as visual form – in a linear development towards the present. This type of art-history-based and object-attentive design history contributed to stabilising and legitimising a general understanding of designing as being about giving (visual) form to artefacts. This also largely became the story disseminated through design history classes in design educations with a strong focus on designed products.

Histories of what?

In two articles in the mid-1980s mapping the field of design history and its problems and potentials, design historian Clive Dilnot pointed to the problems associated with design history being "called on to legitimate particular forms of contemporary design practice" in its continued Pevsnerian focus on distinguishing

30 Margolin, "Design history or design studies", 6-9.

31 Sigfried Gideion, *Mechanization takes command: A contribution to anonymous history* (New York: Oxford University Press, 1948).

32 Reyner Banham, *Theory and Design in the First Machine Age (*London: Architectural Press, 1960)

good design and great designers.[33] Design history would need to broaden its understanding of the subject of study, to incorporate designing as understood from current practice, referring to industrial designer Victor Papanek's definition of design as "the conscious attempt to impose meaningful order (…) through the planning and patterning of any act toward a desired foreseeable end".[34]

The question of how to find definitions of 'design' and by consequence also of 'design history' have encompassed a variety of approaches and perspectives. From time to time, discussions have arisen – mainly in the context of debating the direction and focus of design history – where the topic of debate has revolved around whether or not design history should be linked more closely to contexts of design practice and design education. Advocates of the necessity of more collaborative or integrated efforts in bridging historical and design research, as well as bringing history and designing closer together, have been contradicted by those who instead have pushed for the importance of design history's independence from design practice to develop as a free-standing discipline that is still able to contribute to design on its own terms.[35]

Traditional design historical narratives have largely continued to be formed from a Western and global North point of view, and have – in a structural inheritance from art history – mainly been ordered chronologically and thematically around objects, designers, styles and periods. Over time, this way of approaching design's history has been increasingly questioned in relation to how design history is taught within design education as well as in relation to research practices and perspectives within design history as a discipline.[36] Scholarship aiming to expand the history of design in the 1980s and 1990s brought in feminist and gender theory, with contributions leading to broadening perspectives on historical agency in design, and also to questions of which areas of practice that should be included in design.[37] With impulses from feminist theory and cultural

33 Clive Dilnot, 'The State of Design History, Part 2: Problems and Possibilities", *Design Issues,* 1984:2 (vol.1), 12.

34 Victor Papanek quoted in Dilnot, 'The State of Design History, Part 2", 15.

35 Jonathan M. Woodham, *Twentieth Century Design* (Oxford: Oxford University Press, 1997); Kjetil Fallan "De-tooling Design History: To What Purpose and for Whom Do We Write?", *Design and Culture*, 2013:1 (Vol.5), 13-19.

36 Victor Margolin, "A decade of design history in the United States 1977-87" *Journal of Design History* 1988:1 (vol.1), 51-72; DJ Huppatz & Grace Lees-Maffei, "Why design history? A multi-national perspective on the state and purpose of the field", *Arts & Humanities in Higher Education* 2012:12 (vol.2–3) 310–330; Sarah A. Lichtman "Reconsidering the History of Design Survey", *Journal of Design History,* 2009:4 (vol.20), 341-350; Jennifer Kaufmann-Buhler, Victoria Rose Pass & Christopher S. Wilson (eds.) *Design History Beyond the Canon* (London & New York: Bloomsbury, 2019).

37 Cheryl Buckley, 'Made in Patriarchy: Towards a Feminist Analysis of Women and Design', *Design Issues,* 1986:2; Judy Attfield, 'FORM/ female FOLLOWS FUNCTION/male: Feminist Critiques of Design', in John A. Walker, *Design History and the History of Design* (London: Pluto Press, 1989), 199–225.

studies, the themes of consumption and consumers became incorporated into design history during the 1980s and 1990s, taking into account objects and environments from the perspective of people's everyday appropriation of and encounters with these.[38] This also coincided in time with the material turn in design history, in which methods and theories were adopted from anthropology and archaeology, and design outcomes were regarded not only as objects of study for design history but also as source material in its own right and as artefacts shaping and influencing aspects of human life.[39]

Building on material culture studies, and on socio-technically oriented sociology, the Latourian actor-network theory has also substantially added to the design historical toolbox, in working with studies cutting across areas of production, consumption and mediation, in which discourses of design as well as agencies of objects and people are understood and studied relationally.[40] Recent design historical scholarship has also advocated both locally situated as well as global design histories that challenge canonical and hegemonic narratives, not only in terms of geography but also in terms of design as widespread everyday actions carried out by other than 'design professionals'.[41]

38 Adrian Forty, *Objects of Desire: Design and Society from Wedgwood to IBM* (Pantheon: New York, 1986); Maria Göransdotter, "Smakfostran och heminredning: om estetiska diskurser och bildning till bättre boende i Sverige 1930-1955", Johan Söderberg & Lars Magnusson (eds.), *Kultur och konsumtion i Norden 1750-1950* (Helsinki: Finska Historiska Samfundet 1997), 253-274.

39 Judy Attfield, *Wild Things: The Material Culture of Everyday Life* (Berg: Oxford, 2000); Elizabeth Shove, Matthew Watson, Martin Hand & Jack Ingram, *The Design of Everyday Life* (Berg: Oxford & New York, 2007).

40 Grace Lees Maffei, "The Production-Consumption-Mediation Paradigm", *Journal of Design History* 2009:4 (vol.22), 351-376; Kjetil Fallan *Design History. Understanding Theory and Method* (Oxford: Berg, 2010); Alice Twemlow, *Sifting the Trash: A History of Design Criticism* (Cambridge, MA: MIT Press, 2017); Guy Julier, Anders V Munch, Mads Nygaard Folkmann, Hans-Christian Jensen & Niels Peter Skou (eds.), *Design Culture: Objects and Approaches* (London: Bloomsbury Visual Arts, 2019); Maria Göransdotter, "Ting, tecken, text: Om semiotik och smakfostran", in Brita Brenna & Karen M. Fjeldstad (eds.), *Kollektive identiteter, ting og betydninger* (Oslo: Senter for teknologi og menneskelige verdier, 1997), 49-62.

41 Victor Margolin, *World History of Design, Vols. 1&2*. (Bloomsbury: London, 2015); Daniel J. Huppatz, "Globalizing Design History and Global Design History", *Journal of Design History* 2015:2 (vol.28) 182–202; Kjetil Fallan and Grace Lees-Maffei, eds., *Designing Worlds: National Design Histories in an Age of Globalization*. (New York & London: Berghahn Books, 2016). Glenn Adamson, Giorgio Riello and Sarah Teasley, eds., *Global Design History*. (London: Routledge, 2011); Pat Kirkham and Susan Weber, eds., *History of Design: Decorative Arts and Material Culture, 1400–2000*. (New Haven, CT and New York: Yale University Press and Bard Graduate Center, 2013).

The complexities involved in changing and envisioning human practices of sustenance and sustainability are topics also brought to the fore, merging perspectives from various fields of research with design history and environmental history.[42] Many of these stances have also over the latest decades made their way into design research, such as perspectives and methods adopted from material culture studies, gender theory, anthropology, science and technology studies, and Latourian actor-network theory. However, despite the overlapping adoption of some methodological and analytical research tools, design research and design history seem to continue to move more in parallel than to find common ground or points of intersection.

Relating history to practice

The disconnect between design and history, as manifest in the a-historicity of contemporary design practices as well as in the disconnect between research practices in design and design history, has occasionally been acknowledged in design history's formation as an area of research. Questions of how to expand and consolidate design history as a field in its own right have intersected with issues raised around not only how, but also indeed *if*, history could be expected to contribute somehow to design practice.

Debates have arisen in relation to expectations of if and how design history should be made more instrumental to design practice, and if so, why (or why not). One debate often referred to, is the one between 'design history' and 'design studies', as it played out in the journal *Design Issues* in the mid-1990s. Design historian Victor Margolin put forth the argument that pursuing the development of a discipline of design history might not be a viable way to further historical knowledge of more direct relevance to understanding and critically engaging with design. He instead advocated the formation of a field of 'design studies', in which design history should be a part, and that would work from an understanding of studying design as a changing practice that invents itself over time, rather than focusing on categorisable artefacts as the subject of design history:

> But even looking at design from new vantage points we must still ask ourselves whether we are studying a specific class of things that are stabilised in categories such as industrially produced objects or whether the subject matter of design is really much broader. I think the latter is true. The history of design in the 20th century shows us that designers, unlike architects, have not worked with a set of principles and rules that have proscribed the scope of their work. Rather they have invented the subject matter of their profession as they have gone along. [---] The momentous changes the world is currently undergoing are forcing us to revise our old categories of thought and to pose new questions for research. Rather than think of design as we once did, we need to reconsider the way we approach design as a subject for study. I would argue that it is the broad activity of designing, with its multifarious results, that can open up a range of new important questions about design that have not been coherently posed before and at the same time can enable us to consider new possibilities for practice.[43]

42 Kjetil Fallan & Finn Arne Jørgensen "Environmental Histories of Design: Towards a New Research Agenda", *Journal of Design History* 2017:2 (vol.30), 103-121.
43 Margolin, "Design history or design studies", 12-13.

Notably, the tensions and discussions around design history's relationship to design practice seem to have been productive ones, leading to the formation of not only a discipline of design history[44] but also the field of design studies,[45] and more recently also in proposals of design culture,[46] as areas of scholarship. Whereas design history as a free-standing discipline has been described as "the study of designed artefacts, practices and behaviours, and the discourses surrounding these, in order to understand the past, contextualise the present, and map possible trajectories for the future",[47] the domain of design studies is usually situated in nearer proximity to current and future design practice and design education.

Stemming from its emergence out of multidisciplinary attention to theories and criticisms of design outcomes and practices, design studies is something that "encompasses issues of product conception and planning, production, form, distribution, and use. It considers these topics in the present as well as the past. Along with products, it also embraces the web of discourse in which production and use are embedded."[48] Design culture, in turn, proposes to "go beyond the classic dispute" between design history and design studies, instead focusing on design's contemporary manifestations in a broad understanding of 'cultures' related to design's histories, professions, institutions and systems rather than to "the individual objects per se that populate them."[49]

Whether more fluidly or precisely defined as disciplines or fields, and whether labelled 'design history', 'design studies' or 'design cultures', all of these in one way or another lay claims to incorporating historical perspectives and scholarship as significant contributions to understanding design. In many of these approaches, histories of design are still primarily studied in terms of its outcomes as artefacts, around which broader contextualisations and analyses are made. Design, on the other hand, rarely stakes any claims at all to design history as something of relevance to its practices – neither in profession-centred practices of designing nor in the design carried out in practice-oriented design research.

One of the distinguishing features of design is its propositional future-oriented-ness, as "thought and action for solving problems and imagining new futures".[50] Perceived as actions directed towards making things, solving problems, creating possibilities, and initiating change, design brings things, ideas and

44 Grace Lees-Maffei & Rebecca Houze (eds) *The Design History Reader* (Bloomsbury: London & New York, 2010).
45 Penny Sparke & Fiona Fischer (eds.) *The Routledge Companion to Design Studies* Routledge: New York, 2016); Hazel Clark & David Brody (eds.) *Design Studies: A Reader* (Berg: Oxford & New York, 2009).
46 Guy Julier et al. (eds), *Design Culture: Objects and Approaches* (Bloomsbury: London & New York, 2019); Ben Highmore (ed.) *The Design Culture Reader* (Routledge: New York, 2009).
47 Grace Lees-Maffei & DJ Huppatz, "Why design history? A multi-national perspective on the state and purpose of the field", *Arts & Humanities in Higher Education* 2012:2-3 (vo.12), 311.
48 Margolin, "Design History or Design Studies", 14.
49 Julier et al., *Design Culture*, 2.
50 Harold Nelson & Erik Stolterman, series foreword in the Design Thinking, Design Theory MIT Press series, eg. Johan Redström, *Making Design Theory* (Cambridge, MA: The MIT Press, 2017), xii.

actions into the world that propose that life could be different than it is at present. This strong grounding in the experiential now, and the outlook towards the future, seems – at least in the present day – to be connected to a rather a-historical understanding of design.[51]

Design history predominantly continues to revolve around investigating design in terms of visual and material outcomes – their contexts, the processes through which they acquire meaning and hold agency – while emerging design practices have shifted focus from product to process. Questions of what a design thing can be, how it comes about through design and through use, and how collaborative practices of design play out in increasingly open-ended ways have come to the fore, especially in the design practices that stem from a participatory, user-centered tradition.[52]

If design understood as process is not very present in design history, historical contextualisation and reflection is equally rare in these contemporary design practices. Whether in regard to how problems or possibilities are framed or in the methods and processes with which designing is done, history appears to be a non-issue. Similarly, in design history traditions that work with design as objects and with socio-cultural contexts of the consumption, use, and mediation of these, histories that relate to the actual contemporary – and historical – practices of designing are all but invisible.[53]

Victor Margolin already noted this in the mid-nineties, describing design history as not having a successful engagement with aspects of current design practice in terms of "new technologies, innovative collaborative efforts among design professionals, a concern with the impact of complex products on users and the relations between the design of material objects and immaterial processes".[54] If design since the mid-20th century increasingly has become disconnected from a sense of historical context, design history equally appears to have disengaged from many of the issues pertinent to design as practiced in its contemporary setting.

Developing design methods

Discussions on the relationship between design practice and design history have, in various degrees of intensity, been present since the emergence of design history as a field of academic study in the 1970s. The formation of a specific field of study called 'design history' was, in a sense, a double reaction to something already existing that was deemed not sufficiently fulfilling the needs of knowledge in and of design. On the one hand, proposals for a new design history was a reaction against an art historical canon and curriculum taught in the context of design

51 Clive Dilnot "History, Design, Futures: Contending with what we have made" in Tony Fry, Clive Dilnot & Susan C. Stewart *Design and the Question of History* (Bloomsbury: London & New York, 2015), 149f.
52 A. Telier [Thomas Binder, Giorgio De Michelis, Pelle Ehn, Guilio Jacucci, Per Linde, and Ina Wagner], *Design Things* (Cambridge: MIT Press, 2011); Leslie Atzmon & Prasad Boradkar (eds.), *Encountering Things: Design and Theories of Things* (London & Oxford: Bloomsbury, 2017).
53 Dilnot "History, Design, Futures", 151.
54 Margolin, "Design History or Design Studies", 20.

educations. On the other, it was also one of criticising and expanding the narratives of what design was considered to be, formed within this new historical practice.

As design history took form in the 1970s and 1980s, it was initiated from an impetus to propose histories of and historical perspectives on design as part of the curricula aiming to educate future designers. As stated by Clive Dilnot, looking back at the early formation of design history in the 1970s, the hopes were that design history "would play a crucial role in shaping the discourse of design."[55] The topics of design history's – or more generally history's – role in and relevance for design practice have since then occasionally been actualised by design historians advocating the importance of history in and for design.

In the 1960s, practices and educations in design expanded in relation to the contexts and methods of design. During the 1950s and 1960s, for example, design education at the German school HfG Ulm proposed and promoted an approach to design in which processes rather than products would be central to design practice.[56] Moving away from the, until that point, basic understanding that industrial design should take its starting point in issues of how to give form to objects for mass production in terms of relationships between aesthetics and technology, the Ulm approach increasingly emphasised design as an analytical activity for addressing problems and solutions on more systemic levels. Design for the situations that contemporary society and industry stood before could not be resolved only according to principles of artistic form-giving by an individual designer, but were in need of being addressed in team-based and multidisciplinary formats where designers would have the role of 'integrator' or 'coordinator' rather than merely 'form-giver'.[57]

The focus on developing new methods and processes for designing were made in relation to the levels of scale and complexity of situations that design had moved into. Designing on systemic levels entailed addressing situations that were not easily defined or restricted to identifiable problem-solution-relationships, and that could no longer be resolved by the industrial design methods that had developed during the first half of the 20th century.

As John Chris Jones, one of the forerunners of the design methods movement in the early 1960s, stated in his 1970 publication *Design Methods*, the "world-wide dissatisfaction with traditional procedures" in design required a collective seeking of new methods, borrowed from other disciplines as well as developed within design, as well as "new procedures, (…) new aims and a different level of achievement."[58] As areas of knowledge and research from the sciences and social sciences such as psychology, sociology, mathematics, semiotics and economics were incorporated into design, questions were raised and proposals made as to how to develop design methods and practices for handling the new situations in which design was called upon, and the changing knowledge and skills needed in design.

55 Clive Dilnot, "Some Futures for Design History?", *Journal of Design History* 2009:4 (vol. 22), 387.
56 Herbert Lindinger, *Ulm Design: The Morality of Objects* (Cambridge, MA: The MIT Press, 1991).
57 Tomás Maldonado, *Design industriale: Un riesame* (Feltrinelli, 2008).
58 John Chris Jones, *Design Methods* (1970/1981), 2nd edition (New York: Van Nostrand Reinhold, 1992), xviii.

People such as John Chris Jones emphasised that the development of new methods for designing aimed at finding ways to incorporate intuitive and artistic designerly practices and systematic and describable methods in a design process that would aim for something beyond the idea of solving specific problems:

> To think of designing as 'problem-solving' is to use a rather dead metaphor for a lively process and to forget that design is not so much a matter of adjusting the status quo as of realising new possibilities and discovering our reactions to them. To make or invent something new is to change not only one's surroundings but to change oneself and the way one perceives: it is to change reality a little.[59]

However, the strong emphasis on planning, analysis, science and systems that later emerged as one strand from the design methods movement proposed that design could be approached and established in terms of a rational, scientific activity. The still widely cited definition of designing given by Herbert Simon – "[e]veryone designs who devises courses of action aimed at changing existing situations into preferred ones" – stems from his work aiming to formulate a science of design based on rational principles for decision-making in relation problem-solving.[60]

Another aspect of the design methods movement, was the emphasis on the plurality of possibilities in designing that needed to be addressed through the development of new methods. John Chris Jones, distancing himself from the design methods movement that he perceived had turned too far towards searching to stabilise ideas and practices of rational design methodology rather than continuous developments of ways to design collaboratively in complex situations for new ways of living, stated:

> If, as I think now, the main purpose of 'the design process' is collective learning, the deliberate seeking of new ways of living, then we must expect to make changes in our processes and procedures.[61]

As design research evolved out of the design methods movement, and while design education changed, especially in the more technologically oriented design curricula in polytechnic settings veering towards finding systematic and rational methods of a design science, or aiming more to address processual complexities of "wicked problems" through design, the shared idea was that design needed to develop new methods for designing, and new understandings of the situations and contexts of design.[62]

It was in parallel – if not in actual direct relation – to these aims to develop methods, knowledge, forms and frameworks for a different way of designing, and the changes that this entailed for design education, practice and research, that

59 John Chris Jones, "1980 edition: a review of new topics" in Jones, *Design Methods* (1970/1981), 2nd edition (New York: Van Nostrand Reinhold, 1992), xxix.
60 Herbert Simon, *The Sciences of the Artificial* (1968), 3rd edition. (Cambridge, MA: MIT Press, 1996): 111. See also D.J. Huppatz, "Revisiting Herbert Simon's 'Science of Design'", *Design Issues* 2015:2 (vol.31), 29–40.
61 Jones, "1980 edition: a review of new topics", xxvii.
62 Jones, *Design Methods*, 27-44.

design historians also began searching for other ways of doing history than what had arisen out of art history traditions. Depending on the subject of the design curriculum, these design histories would be directed towards material-based or subject-related histories, economic and production-focused history, or leaning towards socio-cultural history.[63] However, in similarity to traditions in art history, the early canons of design history also strongly emphasised aspects of judgment and appreciation, in distinguishing between objects of excellence or importance and attributing these to the work of extraordinary individuals.

By the early 1980s, the focus in design historical narratives centred on distinguishing 'good' or 'important' design from design of lesser importance, as well as the focus on individual designers, came to be both questioned and challenged from different perspectives. While design history by then had started to take form as a discipline of its own – albeit with fluid and contested definitions of boundaries as well as of subject matter and methods – the distance between design history and design practices (designing, design research and design education) remained. This paradoxical situation of distance between design history and designing is all the more interesting to note, since design history's academic beginnings had its impetus from within design education and the perceived needs of integrating history into design and designing.

The relevance of history

A recurring argument in favour of the importance of history in relation to design practice often brought forth by design historians, is that design history would support a specific critical understanding of the objects of design and the situations in which these have come about and come into use. Design history would, from this perspective, allow learning from the past to understand the present, and in order to impact the future. This is usually linked to examples given of how incorporating historical courses and perspectives in studio-based design curricula would contribute to the design student's individual situated understanding of their place in design history, which in turn would inform their future design practice.

The methods applied in providing these historical critical perspectives for design students are often based in exposing students to design historical research methods, such as analysing historical texts and objects, in order for them to better appreciate their own place in a historical trajectory of producing design outcomes. For example, as expressed in a recent publication on design history in the US educational context:

> Engaging students in more active forms of analysis, particularly using diverse primary sources and objects gives them the tools they need to craft their own understanding for design history. And find their place within it.[64]

63 Gooding, *Design history in Britain from the 1970s to 2012*, 75f.
64 Jennifer Kaufmann-Buhler, Victoria Rose Pass & Christopher S. Wilson (eds.), *Design History Beyond the Canon* (Bloomsbury: London & New York, 2019), 3-4.

A similar view, not uncommon as a starting point for including design history in a design school curriculum, could be further exemplified by Swedish design historian Lasse Brunnström:

> It is in my opinion important for prospective designers to train themselves to take a historical-critical approach. A new design is ultimately a small step in a larger context. Studies of older design solutions often yield ideas, demonstrate exemplary role models or by contrast, reveal approaches to distance oneself from. Outdated technology, unsuitable material or another age's aesthetic principles may have formed a temporary dead end. Ensuring that one has critically evaluated previous design efforts becomes, in other words, part of the quality assurance in the design process.[65]

Affirmations such as these of the importance of design history for a critical contextualisation of current design practice are quite common, whether the topic revolves around the need for historical perspectives in design education or a broader take on the ways that knowledge of design history would be relevant to design practitioners. To a substantial extent, though, design historical perspectives in design curricula have built on narratives and approaches perpetuating a Pevsnerian approach to design history as the history of objects and styles. In this way of narrating the history of design, the central elements of study remain objects, their designers, and their form and aesthetics qualities. Since Pevsner's accounts, discussions have arisen on *which* objects, or *who* to consider as a designer, or *how* to understand or appreciate formal and aesthetic aspects of things, especially from perspectives questioning the narratives and ideals of modernism.[66]

In two oft-referenced articles from the mid-1980s, Clive Dilnot pointed out that there was "little explicit consideration of aims, methods, or roles of Design History in relation to its actual or potential audiences" in design practice.[67] If we were to turn to design practice today – whether in professional or academic contexts – and ask for how history might be part of the processes of designing, chances are quite slim that any historical methods or considerations would be an obvious part of the designer's tool box. Design history may be used to commemorate and highlight design practices in the contexts of jubilees of educations or design consultancies,[68] but rarely – if ever – would it be an integrated part of the process of designing.

65 Lasse Brunnström, *Swedish Design. A History* (Bloomsbury: London, 2019), xxiii.
66 Jan Michl, "Taking Down the Bauhaus Wall: Towards Living Design History as a Tool for Better Design", *The Design Journal* 2014:3 (vol.17), 445-453.
67 Clive Dilnot, "The State of Design History, part I: Mapping the Field", *Design Issues* 1984:1, 4-23.
68 Norbert Andersson, *Designed in Umeå: Industrial design at the Umeå Institute of Design,* Sweden (Stockholm: Infobooks, 2009); Susanne Pagold, *Function rules: A Swedish approach to design glamour* (Stockholm: Arvinius, 2006); Kerstin Wickman, *A&E Design: The Book* (Bromma: Business History Publishing, 2018).

Despite its claims of importance to influence the practice of designing, specifically through influence in design education, design history is framed largely as a retrospective area. Through looking 'back' in history, we can contextualise the present and avoid repeating mistakes or repeating what has already been done before in terms of form-giving. Design history, it appears, does not make much of an impact neither on design practice nor on design research, despite its claims to provide critical perspectives or approaches that would be relevant to design.

In regard to design history's relevance to design, Clive Dilnot has consistently argued that contemporary practices of and conditions for design would require a different type of history than what design history has generated so far, since the "historical frameworks developed in the 1970s and 1980s (…) are simply no longer adequate, as history, to comprehending the past from the standpoint of today's present."[69] Additionally, Victor Margolin, as we saw earlier, has pointed out that design history/design studies would need to take more account of design as a contemporary and changing practice. Similarly, Tony Fry has emphatically proposed that history in relation to design should be approached as fundamental for addressing issues of futuring or de-futuring, as history conditions what is possible or not to act upon from the present.[70]

Many design historians have pointed to the relevance of design history for design as a means of critically questioning and contextualising design as a social activity, as well as its outcomes – while the actual ways in which design history could be incorporated in design practice or design research remain unarticulated. As Sarah Teasley notes in her study of design history's relationship to design research, there were statements on the part of early design research as to the importance of including historical aspects in design research as well as in methodologies of designing. However, much of the writing regarding the connections between design research and design history refers to the lack of communication between these fields.[71]

There are examples of design researchers situating design history in the realm of design research, and even noting history as potentially contributing to design research methods. For example, this was the case when Bruce Archer, then RCA professor in design research, in the late 1970s and early 1980s pointed to design history methods as a relevant part of design research in terms of socio-cultural as well as cognitive understanding and contextualisation of design.[72] However, despite this early acknowledgement of the affinity, or at least connections, between design history and research into design practice, neither field has made any substantial attempts to actually bringing history and practice together.

While proponents of design history continue to point to the contributions that it could make to design, the levels of enthusiasm demonstrated from the field of design practice seem low, to say the least. This mutual disinterest, Clive Dilnot has argued, is due to that "in the vast majority of design research the capabilities of design are understood *essentially* ahistorically" while "the cavalier disregard for

69 Dilnot, "Some Futures for Design History?", 384.
70 Tony Fry, *Design Futuring: Sustainability, Ethics and New Practice* (Oxford: Berg, 2009).
71 Sarah Teasley, "Methods of Reasoning and Imagination", in eds. Michael J Kelly & Arthur Rose, *Theories of History: History Read Across the Humanities* (London: Bloomsbury, 2018), 191.
72 Teasley, "Methods of Reasoning and Imagination", 191.

how design is constituted and formed within historical circumstance by design has its echo in the double disdain for the relation to practice and for *thinking* design analytically manifest by design history."[73]

However, if we believe that there is something of relevance that history might bring to designing – what would that something need to be, for it to resonate with design practice?[74] Were we to attempt to bridge the gap between design history and design practice (including practice-oriented design research) through moving design history and design practice closer to each other – what would that move be like? From which starting points could histories of design then be made? Before coming back to these questions, let us first take a moment to consider history and how it relates to pasts, futures, and to the present.

Design pasts, presents, and futures

How we form and take part of histories of design will impact our understanding of what is perceived as important in and for design. How design and designing have been defined is what also defines how histories of design are constructed. Design and history are, in this sense, mutually constitutive. What we today understand as 'design' is a historically contingent construct, that has been shaped over time through different practices having to do with what has been identified as 'design' and 'designing', and the stories told about this. What design is perceived to be, and how and why design is done, is formed by the past conditions in and through which it has come about.

Depending on what is identified as incorporated in 'design', the stories will vary when we account for how this particular 'design' came about and what that might mean to us today. Furthermore, through what is regarded as important in present-day designing, certain historical phenomena are actualised as the trajectories leading to these practices, thereby establishing design histories also through how designing is done. The outlook on design and its capabilities is thus formed by the intertwining of histories and practices of design that result in proposals and outcomes and that all shape the possible trajectories for the design futures we can envision. In his 1970s book on design methods, John Chris Jones pointed to the intrinsic entanglements of past, present and futures in design:

> A great many people will have to lower their belief in the stability of the present before it becomes socially feasible to plan on the basis of what *will be* possible in the future rather than on the basis of what *was* possible in the recent past. The new idea that has to be grasped here is that the details of how the present population has adapted to what exists are of little account – the important thing is to identify the ease or difficulty with which a future population can be expected to break through the thresholds between the way things are now and each of several different ways in which things could be reorganised.[75]

73 Dilnot, "History, Design, Futures", 151.
74 Victor Margolin, "Design in History", *Design Issues* 2009:2 (vol.25), 94–105.
75 Jones, *Design Methods*, 33.

Design is about bringing forth propositions of how things could be otherwise, based on potential and possibility in the present directed towards future conditions. History is similarly about bringing forth proposals of what was, based on possibilities of understandings of that which has been in relation to what is. In this sense, design and history are both centered around potentiality and possibility, proposed through the making of things, whether 'histories' or 'designs'. While in a linear understanding of time, design typically takes off towards a future trajectory, history moves towards a past. Both history and design begin with the starting points and perspectives in the 'here and now', in ideas of what is relevant to us in the present. However, just as 'design' is not the same as 'the future', 'the past' and 'history' are not the same – despite the fact that we in everyday language (and sometimes also in academic-speak) use these latter terms as if they were more or less equivalent to each other.

The past consists of that which has been: people, things, events, actions, thoughts, experiences and environments that have taken place, appeared and disappeared in time. We can, of course, never grasp the past in its entirety, nor can we revisit it or render a comprehensive account of all that it has been. We can only use what is available to us as traces and mediations of the past to create our understandings of it. In order to attempt to at least grasp some aspects of the past, to account for and analyse events, actions and implications of these of relevance to us today, we have 'history'.

History is always a particular and partial story told of the past, from a certain perspective and a specific intent in the present. The past is what it is, and the events that took place can never be fully appreciated or thoroughly made transparent to us. However, what we can make transparent in our histories at least to some degree, is what we choose to go in search of, and how we account for the *why* and *how* we choose that particular perspective of the past for making history. The meanings assigned to the aspects and events chosen to make present – or not – are contingent to the contexts in and methods with which we make history. Debating and probing which topics, interpretations, and contextualisations of the past are relevant to whom, and why, in regard to the present is also a central way in which history is made.

Histories of the past will always entail leaving out more than what can be brought in. Many things, many events, and many people will always be missing. Much of historiographical criticism and debate revolves around these types of issues, and have led to both methodological and theoretical developments that have quite radically changed perspectives on and in history through contestation, critique and negotiations. Marxist history with class perspectives and social history in the Annales tradition opened up perspectives and methods on how to write history "from below", bringing in new methods and types of source materials. Feminist, gender and postcolonial theory brought critical perspectives on power and representation that brought more than merely "complementary" stories to mainstream histories.[76] These perspectives fundamentally challenge not only how histories are told but also periodisation in and perspectives on history as a discipline, and why histories are told in certain ways.

76 Joan W. Scott, "Gender: A Useful Category of Historical Analysis", *The American Historical Review* 1986:5 (vol. 91), 1053-1075; Gayatri Chakravorty Spivak, "Can the subaltern speak?" [1985], in Spivak, *A Critique of Postcolonial Reason: Toward a History of the Vanishing Present* (Cambridge, Mass: Harvard University Press, 1999).

History is to a substantial degree made through these 'historical frictions', where different narratives and interpretations are contested not only among historians but also in broader contexts as history is called upon to support or contradict, empower or disarm.[77] In this sense, we cannot easily define or explain what history *is* in itself, especially considering that the quotidian use of "history" often confuses it as synonymous with "the past". However, history is not the same as the past; it is only the parts of the past that we somehow choose to pay attention to.

History is continuously made in the ways it is manifest, performed and used, through language as well as through other practices and mediations: The roads we travel, the houses we live in, the books we read and the films we watch, the things we encounter in stores and museums, the work we do and the educations we get (or that are not available to us to get), and the traditions and ways of living that are part of our communities.[78]

We will always see different things, find perspectives that open up to new 'facts' or phenomena, and that will resonate – or not – with what we define as that which we need to search for in the past in order to make sense of the present and the future. Similar to design, then, history is something that is formed and made in practice and through use. We make history through our practices, the way we bring things to the fore, the way our considerations today inform what we regard as important to highlight in the past. History is in this sense never given, but always fluid and in many aspects speculative.

In academic contexts, history is mainly made through the practices, methods and narratives associated with historical research in disciplines or fields within or close to the humanities and social sciences. The history made in these contexts is often referred to as something that is 'written' or 'told'. Undoubtedly, language is not only one of the fundamental forms of mediation used in history, as the written material that historical research largely relies upon as carriers of 'primary sources' for accessing different aspects of the past. Language is also the mediating matter itself from which histories are made, as given in forms that allow us to access and relate to parts of the past in the present, in written historical accounts such as books and articles, and narrated orally in stories, lectures, films and the like. So yes, we do both tell stories and write histories, and that is a very substantial way that history is made.

History is also made in and through other materials, tangible and intangible. There is, of course, an abundance of visual material used in history publications as illustrations, and in fields such as art and design history visuals constitute much of the central material used as the subject of study. With the visual turn, and similarly the material turn of the 1990s, visual and material things and practices also gained prominence in broader fields of historical and social studies.[79] In the construction of history, then, the very concrete material manifestations of human settlements,

77 Leslie Witz & Ciraj Rassool, "Making histories", *Kronos: Southern African Histories* 2008:34, 12.

78 Ludwig Wittgenstein, *Philosophical Investigations* (1953), revised 4th edition (Blackwell: Oxford, 2009), 6e.

79 Margarita Dikovitskaya, *Visual culture: The study of the visual after the cultural turn* (Cambridge,MA: MIT Press, 2015); Anne Gerritsen & Giorgio Riello, *Writing material culture history* (London: Bloomsbury Academic, 2015).

from archaeological findings of stone tools and hearths to the contemporary skyrises and congestions of global megacities manifest and embody diverse social, technological and cultural practices of humanity that are present to us now, and that we not only use to make history but that rather materially form and impact the world in which we currently find ourselves.

How we arrange our everyday lives materially and through practices founded in culture, religion or science that shape how we relate to each other and the world around us, are equally bearers of different histories actualised in the ways we behave and think. In the intertwined actuality of our material and behavioural everyday lives, design and history come together. The things – physical or immaterial – brought into the world with an intent to shape something new, something different than before, are in a broad sense all manifestations of design that have come about in specific contexts, and that we in many cases still live with today. History is, in this sense, very much an inescapable aspect of and condition for the present.

When we form histories, we always do so from where we experience ourselves to be situated in the present. But where does the present start, and where does it end? If we attempt to talk about the present as *now*, the time it takes for us to write, speak, or think *now* will inevitably have passed into a *then* by the time it is done. Being in the present is ever a becoming, and never a single stable or static position in time. From such a point of view, we are always heading somewhere, from somewhere – and the potential trajectories of that movement connects the past and the future through the ever-shifting understandings of what the present might be.

Who 'we' perceive ourselves to be, as the collective 'we' that history is made in respect to is a deceptively inclusive thing. In the idea of 'we', used to form a common ground for establishing that there are shared experiences and commonalities that link us together in our understanding of the world we inhabit and of the things that are important to us. Nevertheless, a 'we' – used as I have used it frequently so far in this text – presupposes that there also are others that are significantly different to 'us' and thus not included in our 'we. Consequently, at any time, in any situation, there is a multitude of possible constellations of who 'we' are, of who is excluded and who included, that forms both the situations and the perspectives from which we define what the present is to us, and what is regarded as important. The *now* in which we find ourselves can, from this perspective, never be anything but historical and highly situational, as are 'we'. In the words of historian Reinhart Koselleck:

> all temporal dimensions are always intertwined, and it would contradict experience to define the 'present' as, for instance, one of those moments that accumulate from the past into the future – or, conversely, that slip as intangible points of transient from the future into the past. In a purely rhetorical manner, all history could be defined as a permanent present in which past and future are contained – or as the continuous intertwining of past and future that makes any present constantly disappear.[80]

80 Reinhart Koselleck, *The Practice of Conceptual History. Timing History, Spacing Concepts* (Stanford CA: Stanford University Press, 2002), 30.

The futures that are open to us to imagine and to aim for are conditioned, and in many aspects also limited, by the past. What has been done or not in the past, cannot often be undone or changed in the present – and will, therefore, set the direction for what we see as plausible or possible futures. However, what emerges in terms of possibilities in respect to futures is not only conditioned by what is done or not, but by what is possible for us to discern: to see, think, or even imagine in terms of potential trajectories. This is how history and future are intrinsically entangled: the futures we can envision are dependent upon our understanding of and in the present, the limits of which are construed through the past and conveyed through histories.

If we want to find ways of seeing other potential futures, different from the ones we appear to be heading towards, we would have to make different histories than the ones readily available to us. Furthermore, in order to make alternative histories that could support seeing other futures, the present understandings of who 'we' are, in relation to where our 'now' is located, and what then is important to look for in the past, will have to be a matter of repeated questioning and continuous change.

Following Heidegger, who stated that design and history are both "inescapable for the future" and thus should be of mutual interest to one another,[81] Tony Fry and Clive Dilnot have both pointed to history's relevance to design, and design's to history, in the making of futures of resilience and sustainment:

> We cannot think where we are, we cannot think the transformations that have led us here, we cannot think futurally within frames of thought and practice that eclipse time, if we continue to try to think design only within the corpus and limits of the present.[82]

Be it in the material aspects of our world, or in the cultural expressions that shape our ways of being with each other, or the scientific, religious, or other world views that set the frameworks for what is possible for us to think and imagine in terms of how we understand ourselves and our world – these are all the outcomes of historical processes and constructs. History thus conditions our understandings of and actions in the present, which in turn is only a sequence of transitional 'nows' that, to speak with Hanna Arendt, are "no more than the clash of a past, which is no more, with a future, which is approaching and not yet there".[83] What has come before us, what has been made and unmade, what has been rendered possible and impossible, are all results of previous actions or non-actions, that thoroughly affect what could become of, and in, the present and in the future.

If we, with Clive Dilnot, see the realm of design as composed of (at least) three 'things': that which was and has been actualised as something we understand as design, such as things or practices; that which could have been but was not actualised; and what remains to be actualised as social practices in design, history is something that not only contributes to understanding design but that in itself is *of* and *in* design.[84] As the relationships and boundaries between what we perceive as relevant to us in our pasts, presents and futures shift with our changing notions of what design can be, our histories and how we approach them will need to change as well. History, Dilnot suggests, might then be regarded as prospective rather than retrospective as "the subject matter of design history is the capacity of design as we see it historically from the point of view of our endangered present."[85]

A different type of history, then, does not only become an alternative story told about events past, but supports the opening up of spaces of action in the present – the "most futile and slippery of the tenses"[86] – through addressing the temporality and the historicity of the 'now' in designing. Transitional design histories would in this sense, I propose, be actively engaged in the re-making of the present as well as re-orienting the potential envisioning of meaningful futures, by means of presenting the past in design. Such design histories would be directed towards making histories in which entangled pasts, presents and futures could be perceived differently, rather than with making 'History' that in a causal and linear perception of temporality would highlight events or products as milestones in a view of design-as-progress.[87]

This brings us back to the questions around the positionings and the narratives of design's histories. If we wish to make design histories that are prospective and propositional, that could contribute to designing through making a difference in relation to the things we think with, as well as retrospective in the sense of contributing to histories of design – the stories we use to tell other stories with – then from where and with what perspective would one start making such histories?

Design is not singular as a practice, and methods and processes of designing are diverse and varied depending on the field of design and the specific situation design engages with. Therefore, to experiment with how design histories could activate foundational concepts in design, more precision is needed to narrow down which types of designing, and which key ideas, to explore in making transitional design histories. I have chosen to position a design historical outlook in practices that are central to design research practices, established in commercial industrial design work, and taught in many contemporary design educations; the sort of design doing that is defined as user-centered and participatory. These practices are based in codesigning, through developing design methods and tools supporting designers' and non-designers' collaborative work. In the next chapter, I further outline a proposal for how to go about making transitional design histories – histories *for* design – with examples pertaining to certain core methods and concepts in Scandinavian user-centered and participatory design.

81 "Design and history are, each in its own way, inescapable for the future. The two, however, are also insufficient for the future so long as each busies itself with its own affairs in separation instead of listening to one another. They are able to listen if both – design and history – belong to the future" Heidegger quoted in Dilnot, "History, Design, Futures: Contending with what we have made", 136.

82 Dilnot, "History, Design, Futures", 162.

83 Hannah Arendt, *The Life of the Mind. One/Thinking*, [1971] (Harcourt: San Diego, New York, London, 1981), 205.

84 Dilnot, "Some Futures for Design History?", 378.

85 Dilnot, "Some Futures for Design History?", 393.

86 Arendt, *The Life of the Mind. One/Thinking,* 205.

87 Virginia Tassinari, "The Wunderkammer of the Future: Philosophical Notes on the Idea of History in Speculative Design Practices", *z33Research*, https://z33research.be/2018/04/virginia-tassinari (published 2018-04-13, accessed 2020-05-25).

3. Methodologies and positioning

In making design histories *for* design, what could be relevant as methodological frameworks, positioning, perspectives, and materials? In this chapter, I outline how transitional design histories can be made by combining methods from the fields of programmatic design research and historical research. I further propose that we can investigate how to make histories that take an outlook from designing by means of exploratory examples, through prototyping how these might be built and how they might work.

Before engaging with making prototypes of transitional design histories – which are presented in chapters four, five and six – I discuss the materials and methods applied in this process. Since the prototypes are proposals of design histories positioned in relation to designing in a tradition of Scandinavian user-centered and participatory design, I begin with giving a brief account of how histories of participatory and user-centered design practices so far have been handled in Swedish design historical publications. Following that, I give a brief account of how histories of these design approaches are usually narrated internally within the field of design practice and research. Besides framing and contextualising the historical prototypes, these introductory examples of historical narratives also intend to make visible the current disconnect between design history and design practice. Following this, I sketch the overall frameworks and aims of user-centered and participatory design, highlighting methods and concepts central to these practices. The prototyping of transitional design histories that follows in the subsequent three chapters, then, revolves around three of these core concepts.

Methodologies combined

In the first chapter of this thesis, I proposed that approaches from programmatic design research could be used to bring design history closer to that of design practice, to make histories that could activate an awareness of design's historicity. In the following, I will further introduce these methodological considerations, first from the perspective of programmatic design research, and then from the perspective of historical research methodology. The combining of these methodologies forms the framework for how I then propose that transitional design histories can be prototyped.

A programmatic design research approach

In design research where design practices are an integral part of research processes, the notion of programmatic design research provides a framework for experimenting with what design could be, and what designing could become.[1] Through setting up a program that forms a provisional framework of "a diverse set of inherently unstable and transitional world views",[2] explorations can be made of how design can be done and what its outcomes can be. These explorations can take different forms, as can the questions asked, depending on what the program was set up to investigate. Whether aiming to probe what design practices would be possible or what a particular practice could lead to in terms of products or outcomes, the making of examples and prototypes through design is central to this type of research investigation. Depending on what emerges in the exploratory examples when different ways of designing are prototyped, both program and subsequent explorations will begin to change. In the design work and situations where examples are made – whether these are prototypes, methods, or theories – new questions will emerge. These can then be explored further in relation to the program's frameworks and provisional world view. In sequential experimental moves, the limits and structures of the program can be probed, until the provisional framework set by the program can no longer support the ways of designing that the program set out to make possible, and new programs need to be formulated to address changes in the contexts and practices of designing. The proposal of making transitional design histories builds on this programmatic approach to design research.[3]

Transitional design histories are made as propositional and exploratory examples that aim to shift outlooks on designing in order to support design's ability to respond to increasingly complex and connected present and future challenges. Through making such examples, or experiments, other questions and perspectives will be brought forth within the program, changing the outlooks for further explorations of design histories, from other positions. In a continuous movement, each new position will provide new outlooks and new questions – and new histories.

1 Laurene Vaughan (ed.), *Practice Based Design Research* (New York: Bloomsbury Academic, 2017); Gesche Joost, Katharina Bredies, Michelle Christensen, Florian Conradi & Andreas Unteidig (eds.) *Design as Research: Positions, Arguments, Perspectives* (Basel: Birkhäuse, 2016); Jonas Löwgren, Henrik Svarrer Larsen & Mads Hobye, "Towards programmatic design research", *Designs for Learning* 2013:1-2 (vol.6), 80–101; Ilpo Koskinen et al., *Design Research Through Practice: From the Lab, Field, and Showroom* (Waltham, MA: Morgan Kaufmann, 2011); Thomas Binder & Johan Redström, "Exemplary design research", *Design Research Society Conference,* Lisbon, November 2006.

2 Johan Redström, *Making Design Theory* (Cambridge, MA: The MIT Press, 2017), 95.

3 "Chapter 3 Research Programs" in Ilpo Koskinen et al., *Design Research Through Practice*, 39-50.

These design histories are not meant to serve as genealogies of design methods or concepts. Instead, transitional design histories make propositions that open for questions in a sort of double loop: taking a position from which to ask 'what sort of history can be made from *this* outlook in design?', the history that is made from that point of view is a propositional answer to the question. In turn, this gives rise to another question: 'what could designing become, if *this* were its history? This does not mean, however, that making these types of histories would draw on some sort of radical relativism, in which one could think that almost anything would be possible to propose as a relevant history of and for design.[4] On the contrary, transitional design histories engage with the ideas, concepts, and methods that are so firmly established in designing that they seem almost 'natural' or 'given', forming the very foundations that design builds on.

Making histories of concepts central to the foundations of and practices in design, the intent is not to attempt to define or follow terminologies related to specific ideas or practices. Though tracing core concepts of design etymologically and semantically would most certainly also provide perspectives that expand understandings of design thinking and doing, exploring how and when these have entered into the language and practices of design, this approach does not build on that sort of conceptual history.[5] Instead, transitional design histories aim to explore how concepts central to design can work to support processes of change and fluidity, as well as how these contribute to the fossilising or stabilising of particular understandings of design.

Through making histories that take a perspective positioned in conceptual foundations of designing – the ideas, rather than the words (albeit they do often coincide) – these studies explore and investigate the contexts and preconditions in which different concepts came to emerge, and how they have been acted out and put to use, as well as imbued with meaning, over time.[6] To clarify: this is not a

4 Just to be clear: this is neither a project aiming towards total historical relativity, nor towards making some kind of counterfactual or alternate design history. While these latter methodological approaches are also aimed towards linking future and past narratives through 'what if'-questions, the aim here is not to consider what might have happened in the present or the future if historical events had played out differently, but what the understandings of 'present' and 'future' might become if past events and contexts are actualised in design historical narrative as making a conceptual framework for designing. Cf. Kathleen Singles, *Alternate History: Playing with Contingency and Necessity* (Berlin: De Gruyter, 2013); Richard J. Evans, *Altered Pasts: Counterfactuals in History* (Waltham, MA: Brandeis University Press/Historical Society of Israel, 2013).

5 Reinhart Koselleck & Todd Samuel Presner, *The Practice of Conceptual History: Timing History, Spacing Concepts* (Stanford: Stanford University Press, 2002).

6 Reinhard Koselleck, "Modernity and the planes of historicity", in Reinhard Koselleck, *Futures Past: On the Semantics of Historical Time* (New York: Columbia University Press, 2004), 9-25.

matter of extrapolating recent ideas 'back in time',[7] or to retrospectively construe historical situations into things that fit with contemporary understandings of design.[8] Instead, the ambition is to illuminate some of the ideas that come together in these concepts, and the historical layering of uses and practices that make up their meanings in various ways.

The historical narratives that emerge out of perspectives positioned in contemporary designing should not be understood as a step towards a finished product (*this is* what the history of designing is) but as suggestions made to elicit questions, reactions and responses (what could designing be *if this were* a history of design). Therefore, this way of making histories as examples in relation to a programmatic design research approach has more in common with prototyping than with the production of a final thing.

In practice-oriented design research, prototypes and processes of prototyping hold a central position as a means for representing design ideas and propositions.[9] A prototype is something that is not meant to be completed and finalised, but something that in itself is transitional: it points to that something else will follow. In design, prototypes can be made in various degrees of refinement in different stages of a design process, to test, explain, or explore design ideas or directions, but also to provoke reactions, or enable actions. Prototypes can be everything from very rough and simple representations of an idea, in simple materials, to functional prototypes in a high degree of refinement and detail, elaborated to be very similar to 'the real thing.' Prototyping most often entails an iterative process, making various prototypes of increasing degrees of refinement or complexity for different purposes along the way.

Making prototypes of design histories seeks to probe two things: On the one hand, these prototypes try out a way of shifting perspectives on *how* to go about making design histories that take a starting point in designing to explore how the design histories that emerge might differ from established design history. On the other, these prototypes aim to investigate contemporary concepts and challenges in doing designing, through activating a historical awareness that could contribute to making a difference for transitioning towards sustainable futures through design.

In moving the outlook of design history closer to that of contemporary designing, one methodological issue is the discrepancy between what has been passed down as histories within design practices and what has been made relevant and disseminated as design's history through design historical publications and representations. Stories of the (recent) past from within designing tend to be briefly referenced in design research publications, and shared orally in public talks, or within organisations, offices, and in the fields of practice they 'belong' to, rather than in what could be considered "design history formats". As such, these accounts of pasts highlighted from the perspectives of design practices often are partial,

7 Kjetil Fallan, "'The 'Designer – The Eleventh Plague': Design Discourse from Consumer Activism to Environmentalism in 1960s Norway", *Design Issues* 2011:4 (vol.27), 34.
8 Clive Dilnot, "The State of Design History, Part II" in Victor Margolin (ed.), *Design Discourse: History, Theory, Criticism* (Chicago: University of Chicago Press, 1989), 233–250.
9 Stephan Wensween & Ben Matthews, "Prototypes and prototyping in design research", in Paul A. Rodgers & Joyce Yee (eds.) *The Routledge Companion to Design Research* (New York: Routledge, 2015).

fragmented, and often personal. Additionally, such accounts are often framed in relation to the fact that many of the persons conveying these histories are still active designers or design researchers. Their referencing of history often intends to strengthen a certain perspective in their current practice, rather than primarily aiming to give a historical account as such. Piecing together design histories that begin with issues in current practices, requires some attention both as to how perspectives are set and to the materials from which histories can be made.

Methods and materials for making histories

There is not only one way of making design histories – nor should there be. Since a historical account is always only *a* history in relation to something, and not *the* history of something, we need many different ways of making design histories that bring different formats, methods, materials and perspectives on design in the broadest sense. However, there are certain methodological basics that historical disciplines share, design history included, which have to do with the crafting of histories in terms of materials and methods, as well as practices of interpretation and mediation.[10]

Taking a programmatic design research approach and introducing prototyping as one way to explore making design histories, these prototypes need to be made out of some sort of materials, with some types of methods, that still make them relevant as histories. And – of course – they also need to be made from somewhere – in terms of perspective – that supports the aim of exploring the research program in relation to some sort of more specific context. As mentioned earlier, and as will be further expanded on later in this chapter, the prototypes that will follow after this chapter have all been situated in relation to practices in the tradition of Scandinavian user-centered and participatory design.

Crafting prototypical design histories of Scandinavian user-centered design, in a Swedish 20th-century context, the methods and materials used are those commonly associated with methodologies of making history, as initially formulated in the mid-19th century.[11] In historiography, materials used in research are usually referred to as 'sources' and usually categorised as 'primary' or 'secondary' sources. A secondary source is somehow mediated, such as an account of an event by someone who was not present when it actually happened or consists of previous research made into the field of study. A primary source is some sort of material understood as being 'directly' connected to the event, period, or entity it refers to. Text-based primary sources can be meeting minutes, diaries, lists of inventories, letters, and publications of different sorts, and are often – but, again, not always – stored in public or private archives, with registries listing the archived materials. Primary sources are often text-based but not only: almost anything printed, photographed, recorded, built or human-made can be regarded as a primary source, depending on the questions asked and on the field of historical study. In design history, visual material – photos, sketches, drawings, publicity, films – and material artefacts are also often addressed as primary sources.

10 Kjetil Fallan, *Design History: Understanding Theory and Method* (Oxford: Berg, 2010).
11 For a concise overview of historiographical methodologies in general, and of design history in particular, see D J Huppatz, "Introduction to Methodology: Virtual Special Issue for the Journal of Design History 2018", *Journal of Design History*, 2020:1 (vol.33), e25–e40, https://doi-org.proxy.ub.umu.se/10.1093/jdh/epy021 (accessed 2020-05-31).

A historical research process consists of relating the research question to what could be supposed to be adequate primary source material, locating and making suitable selections of that material, critically analysing it, and constructing a historical narrative. This process is seldom linear but proceeds in iterative loops when new questions and perspectives arise out of the analysis of materials. This leads to new archival explorations and material searches, in a simultaneous weaving together an analysis of past phenomena with accounting for these – most often in written narrative – based on sources that provide legitimacy for these processes of interpretation, presentation, and representation.

In historical research, methodological considerations to a large extent revolve around which sources that can be found, accessed, and analysed in relation to a research question, and which perspectives of analysis are applied. Searching for, finding, and selecting materials, verifying their authenticity and reliability (this is called 'source criticism'), defining the time frames for the study, and refining the research question(s) is a process carried out in close relation to the selection of perspectives and frameworks for the analysis. Which materials one goes looking for, and how they are chosen, searched for, and interpreted, is highly dependent on the analytical perspectives chosen for a study.

The historical methods put to use in this thesis when making prototypes of transitional histories are largely based on methodological stances from the history of ideas.[12] Focus in historical analyses of ideas is on how specific ways of thinking ('ideas') have come to expression within and in relation to diverse contexts – socio-political, academic, scientific, religious, philosophical – and how these move between contexts, relate to each other, and shape understandings of the world. As Arthur O. Lovejoy described it, topics of interest in the history of ideas could, amongst others, be:

> The history of the development and the effects of individual pervasive and widely ramifying ideas or doctrines, such as evolution, progress, primitivism, diverse theories of human motivation and appraisals of human nature, mechanismic and organismic conceptions of nature and society, metaphysical and historical determinism and indeterminism, individualism and collectivism, nationalism and racialism.[13]

With an approach that is quite interdisciplinary, highly attentive to contexts, and focused on how ways of thinking form and are formed by different practices, studies in the history of ideas vary in their encompassing of historical time. From the perspective of examining how ideas travel and change over time, the framing of such studies can span everything from the *longue durée* of how ideas evolve and change slowly over time, spanning centuries or even millennia, to delving deep into the conceptual foundations of a specific period or situation in time.[14]

12 The history of ideas as an academic discipline is quite specific to the Nordic countries. In other geographical and academic settings, these perspectives and research trajectories will be found under academic labels such as intellectual history, history of science, history of philosophy etc.

13 Arthur O. Lovejoy, "Reflections on the History of Ideas", *Journal of the History of Ideas* 1940:1 (vol.1), 7.

14 Fernand Braudel, [1958] "Histoire et Sciences Sociales: La longue durée", *Réseaux* 1987:27 (vol.5), 7–37.

While studies in the history of ideas often will revolve around political, scientific, and philosophical ideas, inspiration from social history has over time also brought attention to practices and ideas of everyday life. With the ambitions of writing histories of everyday life, other historical source materials than the traditional written sources have been sought to highlight other contexts than only those of 'learned history'. Using material things, environments, photographs, film and other visuals as historical sources has also opened up for exploring analytical and methodological approaches from other fields than history. These initiatives aim to take on perspectives of 'ordinary people's' understandings of the world[15], histories of everyday life[16], and histories of marginalised groups or from non-mainstream perspectives.[17] The approaches aiming to make other types of histories also led to historians actively searching for other ways of not only analysing but also of representing histories, with scholars such as Michel Foucault proposing an 'archaeology of knowledge' in which multiplicity, relationality, and complexity emerge as perspectives in the mode of analysis as well as in modes of historical representation.[18]

One ambition with prototyping transitional design histories is to investigate if the shift of perspective through positioning history-making in designing, rather than in design outcomes, can lead to changes in terms of what comes into view. With this in mind, the search for source materials for making these studies has been located to contexts, times, and areas that have already been identified as somehow 'important' to design, and thus already studied from previously adopted design historical stances. The contexts in which the material for these prototypical histories have been located are among those that already – at least in part – are very well-known and researched in 20th-century Swedish design history. This goes for both individuals previously identified as somehow holding influential agency in relation to the formation of design in Sweden, such as Ellen Key, and for organisations such as Svenska Slöjdföreningen (SSF, The Swedish Association of Arts and Crafts). The primary source material used and analysed in these studies, is to a large extent textual: different publications in the form of books, pamphlets, journals and newspapers, and also various collections of archival material such as meeting minutes, registers and various sorts of other administrative documentation found in various archives.

15 Emmanuel Le Roy Ladurie, *Montaillou: Cathars and catholics in a French village 1294-1324* (London: Scolar, 1978); Carlo Ginzburg, *Il formaggio e i vermi: il cosmo di un mugnaio del '500.* (Torino: Einaudi, 1982); Ronny Ambjörnsson, "Om möjligheten av en folkets idéhistoria", *Lychnos: årsbok för idé- och lärdomshistoria. 1983.* (Uppsala: Lärdomshistoriska samfundet, 1983), 158-165.

16 Fernand Braudel, *Civilization and capitalism: 15th-18th century. Vol. 1, The structures of everyday life: The limits of the possible* [New ed.] (London: Collins, 1981).

17 Michel Foucault, [1976] *Madness: The invention of an idea.* (New York: Harper, 2011); Michel Foucault [1975] *Discipline and punish: The birth of the prison* (New York: Vintage Books, 1979).

18 Michel Foucault, *L'archéologie du savoir* (Paris: Gallimard, 1969).

However, since the perspectives chosen and the approach made to prototype different histories have brought other aspects than the usual to the fore, in some cases the texts identified and used in the historical analysis are ones that have not previously been very present in design history. Some of the text-based source material has not in the first hand been approached as 'historical sources' at all in the context of Swedish design history, such as publications in the field of participatory and user-centered design research of the latter decades of the 20th century. Thus, in some cases, these publications are handled as elements of the design research in relation to which this thesis is situated, while in the prototyping examples the same texts are instead approached as objects of historical study or as sources.

Several of the authors of these texts are also still active design researchers and practitioners. From this follows, that the analyses and interpretations, of course, can be contested by these authors and designers – which is not regularly the case in history, as it is a field usually working with 'plucking the dry bones of the dead' as a professor in the history of ideas once described it over a conference lunch[19]. So, any objections that people still active in the field of design research might make as to the interpretations and readings would of course both be reasonable, relevant and highly welcome: they were there and had first-hand experience of what played out from their perspectives.

Why not, then, include first-hand accounts, for example through interviewing people who were engaged in the formation of participatory and user-centered design in the latter decades of the 20th century? That would, of course, be more than relevant, and would be something to do in later iterations of transitional design histories, as well as for histories of design that more specifically aim to go to the bottom with what *actually* happened. Again, the aim with the design history prototypes presented here is *not* to try to provide an accurate and definitive description of *the* history of participatory and user-centered design, but proposals of how histories of central concepts and methods can shed light on embedded ideas that form and inform current design practices. In working with these prototypes, the main aspect to test is if a change in perspective – from design as product to design as practice – could tease out other histories than the usual. Maintaining a fairly strict focus on "basic" primary sources in textual form, and not bringing in interviews and methods of oral history, is then part and parcel of making the moving parts of the prototypes as few as possible.

The prototypes presented later in this thesis, have been given the form of text-based narratives. There are of course several other ways than text-based prose-like narratives could be used to present, represent, and mediate histories: in visual form, through films, historical games, etc.[20] In design history, for example, a narrative format highly present is that of exhibiting things – as design museums do – according to a selection of objects based on perspectives of what

19 This statement was made by the first appointed professor in the History of ideas at Umeå University, Gunnar Eriksson – and there is little I can point to as a source for the quote, since this happened over a lunch during a national History of Ideas conference in Uppsala in the late 1990s.

20 Jaume Aurell, "*Rethinking History's* essential tension: between theoretical reflection and practical experimentation", *Rethinking History: The Journal of Theory and Practice* 2018:4 (vol.22), 439-458.

constitutes 'design' visually, materially, and ideally. In many of these other ways of representing history, however, text-based narratives tend to show up at one point or another. An exhibition, for example, is usually accompanied by a catalogue including texts, and textual signage incorporated in the exhibition space. Using the form of narrative text in making transitional design history prototypes is certainly not the only possible way one could go about experimenting with how these could be made, but – given the centrality of textual narrative in design history mediation – a highly relevant way for at least this first iteration of these prototypical histories.

The narrative adopted in the three prototype chapters moves between a chronological account and a format that is not always clearly linear or chronological, while still attempting to adhere to something that resembles a coherent 'story'. These slightly different ways of building the stories could have been taken further, and perhaps included more diverse experimentations with styles of writing, with fragmented narratives, or other textual formats that even more could have highlighted the aim of making 'different' histories. However, making these first prototypes as part of a doctoral thesis that as a genre in itself sets certain expectations on form and content, posed certain limits to this idea. Additionally, presenting this work in a way that could be meaningful and also make sense in a design research context as well as for a context of academic design history, has led to considerations in terms of how to balance contextualisation, research positioning, accounting for 'established' practices in design and design history, and also in the formats and narratives applied in the thesis.

While the stories that emerge in the design history prototypes are connected in different ways, each chapter is meant to be able to stand, and work, on its own. The three studies are carried out thinking of the operation as a sort of cross-sections through historical time in which different aspects become visible depending on what we look for, and from which perspective. The narratives, therefore, intersect at several points in regard to materials, contexts, and historical time, but also highlight different phenomena in relation to the concepts studied. Thus, partly different, partly similar, histories are made. What is in focus are why and how central ideas and concepts in Scandinavian user-centered and participatory design have come about in specific historical situations (programs), how they have been actualised or materialised (products, projects), and what it means that some of these have transitioned into embedded understandings or foundations for doing design (practices, paradigms).

Positioning perspectives

When selecting materials and formats for the prototypes of design histories that follow, the choices of materials and analytical perspectives have been made with regard to what it is that these explorations set out to do, for design as well as for design history. Since one proposal here, is that shifting the outlook of design history closer to designing as practice, a specific type of practice needs to be selected, and specific perspectives used to analyse it. A general history of design is not only impossible to make, but would also be of little use for probing ways that historical perspectives could contribute to change in practices of designing. In the cases of the prototypes of transitional design histories presented later in this thesis, the perspectives applied in looking for histories are concepts and ideas that

come across as central in design research endeavours and contexts positioned as following a historical trajectory of a (or, *the*, as stated in several papers and articles in the field) Scandinavian user-centered and participatory design tradition.

In terms of trajectories, a user-centered tradition is one that is also strongly present in my everyday context of work. The participatory and user-centered design approaches that came to be influential in the Nordic countries – and beyond – were a strong impetus in setting up the institutional environment in Northern Sweden of which I myself am a part.[21] The new industrial design curriculum introduced at the Umeå Institute of Design (UID) in 1989 was grounded in user-centered and participatory design methods and approaches, and in many ways still remains quite firmly situated in these ideas and practices. I have worked as a teacher in design history and design theory in the studio-based industrial design education at UID since the mid-1990s, with a background in the academic field of the history of science and ideas. Starting as a consultant teacher, visiting once or a couple of times a year to deliver a few weeks of design history teaching, I gradually transitioned to employment as a full-time UID teacher. This meant that I could engage not only in the dedicated design history survey teaching but could work more with integrating historical perspectives in other parts of the curriculum. This transition of my own, from initially teaching design history from an 'outside' perspective (history *of* design) to integrating historical perspectives in user-centered and project-based studio education (histories *for* design), has been significant for considerations of research questions as well as for the selection of methods and materials for the studies. Combining methodologies from design history and programmatic design research, a re-positioning of design histories and design practices in relation to each other, might – I hope – make visible other possible trajectories for emerging design practices.

Now, after having presented the above considerations of methodologies and materials, the background and context of positioning perspectives on and in user-centered and participatory designing will need a more substantial introduction before going into detail of how to make prototypes of transitional design histories.

The way I will be providing background and context for this positioning, is through first searching for accounts of user-centered design in existing design history, and after that piecing together a historical narrative that outlines how a history of Nordic participatory design generally tends to be conveyed from within contemporary design research practice. One could – and rightly so – argue that this is precisely the type of design historical storytelling that I have voiced objections to. Through repeating *the* history of participatory design as it is usually

21 The Umeå Institute of Design at Umeå University was founded in the end of the 1980s (in 1989, to be precise), merging regional university and municipality initiatives with national-level action by design practitioners wishing to reform design education in response to user-centered design methods as well to as a closer connection to industry rather than an art school tradition. Norbert Andersson, *Designed in Umeå: Industrial design at the Umeå Institute of Design, Sweden* (Stockholm: Infobooks, 2009); Lasse Brunnström, *Swedish Design. A History* (Bloomsbury: London, 2019), 179-180.

told, I will probably contribute to a layering or fossilisation of this narrative. Why, then, provide such an overview?

The reason to do so, here, is firstly to sketch the history that is in a sense already made, but not made very explicitly or coherently, through different articulations and representations in design practice of where user-centered and participatory design is perceived to have its origins.

Secondly, through providing this summary of a history of user-centered and participatory design, this will set a reference point, so to speak, for what tends to be seen from existing perspectives on the field. This can then be contrasted and related to the stances that I introduce as possible conceptual perspectives that aim to make other things visible.

Thirdly, this overview also serves as an example of a history of design*ing* as narrated in design from a perspective of design practice, that is all but invisible in existing design histories of Nordic design, thus contributing to the field of design historical knowledge. In contrasting the different outlooks that design history and designing currently have regarding histories of participatory design practices, this disconnect between history and design quite evidently comes into view.

Later, in the final section of this chapter, I will return to discussing some matters of concern that are present in contemporary participatory and user-centered design, with the purpose to take that as the final step towards teasing out some conceptual lenses through which to look for transitional design histories. But before that, let us take a closer look at how design histories relating to Swedish user-centered design have been told so far.

Histories of user-centered design

User-centered in Scandinavia emerged in a historical situation where established design practice and methods fell short of addressing the types of design situations that came about in light of social changes and technological advancements.[22] In the 1960s and 1970s, the role of the designer came to be challenged and criticised, not least from within the design profession itself. There were calls for new design methods and approaches, which could expand the scope of design from a sales-and-production-oriented profession to one that could – and should – address issues of democracy, equality and inclusion through design.[23]

Designers increasingly advocated developing methods that would include knowledge of and dialogue with the people not only buying but also using the outcomes of design. Attention to design for the 'third world', for disabled persons, and for 'professional users' increasingly highlighted the potential for design to make change for people in a specific situation of use. In Scandinavia in the mid-1960s, design students made a joint cause of critiquing the current design

22 Pelle Ehn, "Learning in Participatory Design as I Found It (1970–2015)", in Betsy DiSalvo, Jason Yip, Elizabeth Bonsignore & Carl DiSalvo (eds.), *Participatory Design for Learning: Perspectives from Practice and Research* (New York & London: Routledge, 2017).

23 Anna Valtonen, *Redefining industrial design: Changes in the design practice in Finland*, diss. (Helsinki: University of Art and Design, 2007).

education and calling for other, more socially responsive, approaches to design.[24] The discourse on socially relevant design was shaped in dialogue with designers invited to tutor summer school courses, one of whom was Victor Papanek who after that also published the first version of his book *Design for the Real World* in Swedish as *Miljön och miljonerna* in 1970.[25]

User-centered product design in Scandinavia came to have a strong focus on ergonomics and use that soon carried over into working with concepts of cognitive ergonomics and usability in design for what later became labelled as interaction design. The aims of the new, user-focused, methods proposed were to take on new ways of approaching complex design situations in order to bring about change through design, and also to build new types of research methodologies in order to advance knowledge in and through design.

Developing methods and tools to support shared design processes based on new relations between designers and users in terms of both power and participation, was at the core of the user-centered design that sprang forth. The introduction of computers into working life led to the development of specific methods of participatory design, incorporating the centrality of use and users in design processes. Simultaneously, design methods with a strong emphasis on the work situation and the effects of work procedures on the human body formed a foundation for ergonomic design approaches. These aimed to include users in designing to minimise physical strain, while also incorporating perspectives of production efficiency.

These developments in design are reflected in a design history publication from the late 1990s, documenting the emergence of Swedish industrial design.[26] In the book, historical overviews by art and technology historians as well as accounts by "first generation" Swedish industrial designers map the establishment of the industrial design profession. In a chapter dedicated to the turn towards ergonomic and safety-centred design in post-war Sweden, user-centered (or, rather 'consumer-centred') design is discussed by art and design historian Lasse Brunnström.[27] With a focus on design for increasing workplace safety design, he gives an overview of design work resulting in more ergonomic tools, safer working conditions including safety in cars and tractors, and development of tools and aids

24 Ida Kamilla Lie, "'Make Us More Useful to Society!': The Scandinavian Design Students' Organization (SDO) and Socially Responsible Design, 1967–1973", *Design and Culture* 2016:3 (vol.8); Cilla Robach, *Formens frigörelse: Konsthantverk och design under debatt i 1960-talets Sverige* (Stockholm: Arvinius, 2010).

25 Mateo Kries, Amelia Klein & Alison J. Clarke, (eds.) *Victor Papanek: The Politics of Design* (Weil am Rhein: Vitra Design Museum, 2018);Victor Papanek *Miljön och miljonerna: Design som tjänst eller förtjänst* (Stockholm: Bonnier, 1970); Victor Papanek, *Design for the Real World: Human Ecology and Social Change*, (New York: Pantheon Books, 1971).

26 Lasse Brunnström (ed) *Svensk industridesign: En 1900-talshistoria*, (Stockholm: Norstedt, 1997).

27 Lasse Brunnström "Hjälpmedel för ett säkrare och jämlikare liv" in Lasse Brunnström (ed) *Svensk industridesign: En 1900-talshistoria*, (Stockholm: Norstedt, 1997), 302.

for disabled persons, describing how an impetus towards ergonomics in design was introduced in Swedish industry, almost by accident, in the 1950s.

With a starting point in Taylorism and time studies of work, the consequent establishment of a view that addressed factors of safety and ergonomics became a way to not only increase industrial production efficiency but to view workers as a resource to be maintained and cared for. These practices are traced to the concept of 'human engineering', as practiced and promoted in American industrial design by Henry Dreyfuss, whose book *Designing for people* is pointed to as making a significant impact on early Swedish industrial designers.[28] Brunnström quotes a study trip made to the US by Swedish industrialists and construction technicians in 1955 as the point in time when – due to an error in translation – what was going to be a six-week study program to American technical construction offices, instead became a tour of industrial design offices and consultancies.[29] This is presented as a turning point when the introduction of ergonomics and the 'measures of man' that Dreyfuss had created became incorporated in Swedish industrial design practice.

The main impetus for a shift in design practice, towards ergonomic design for work, is in Brunnström's publication mainly attributed to increasing focus on safety issues in work life, but also to a more inclusive approach to people in society which led to approaches of design for all. In this context, the systematic work initiated by *Arbetarskyddstyrelsen* (The Swedish Work Environment Authority) and the *Handikappinstitutet* (the Handicap Institute) in the late 1960s to promote the design of more inclusive aids for disabled persons was of great importance to the developments within Swedish user-centered design. In the journal *Form,* published by the organisation Svensk Form (the Swedish Society of Crafts and Design), there were recurring calls for designing for disabilities. In 1968, a special issue on the topic was published, advocating designing together with people with expertise in the field in order to provide adequate and inclusive design solutions.[30]

In the late 1960s and early 1970s, constellations of young Swedish designers founded design consultancies such as Ergonomidesign (1969), Designgruppen (1969) and A&E design (1968), all working with ergonomics and user-focused design methods in their consultancy work. From 1979, a merger between two of these, profiled in ergonomic design, resulted in the joint consultancy Ergonomi Design Gruppen. Several of these also applied for funding from government bodies and institutes, setting up collaborative design research projects aiming to improve work environment conditions in specific areas or to focus on disabled

28 Henry Dreyfuss, *Designing for People* [1955] New edition, (New York: Paragraphic, 1967).

29 Brunnström "Hjälpmedel för ett säkrare och jämlikare liv", 302.

30 "Att leva med handikapp" [To Live with Disabilities]. *Form* 1968:10. This was a special issue of the journal Form produced as material aimed for study circles interested in understanding the constructed environment. The study series was themed "Active Perspectives on Environments" (Aktiv miljösyn), and was made up of twenty-four slides, twenty-four text cards, six study pamphlets, and different exercises or tasks. The cards included imagery and presentations relating to disabled persons' living conditions, tools, rehabilitation, and clothes.

persons' situation. At Ergonomi Design Gruppen in the mid-1970s, for example, a collaboration between designers Maria Benktzon and Sven-Eric Juhlin and disabled users focused on the design of everyday utensils – cutlery, knives, pens – based on extensive ergonomic studies of grips and handles together with both medical experts and with users.[31]

The work carried out together with disabled users in the development in different aids and tools, led to the introduction of user-centered methods in designing tools for professional use. In the mid-1970s, a series of screwdrivers was redesigned with a starting point in ergonomic user studies and interviews with people working professionally with these tools. Using video filming, different work situations were studied and analysed, and iterative prototyping then took place together with users in regard to grips, torques, and handle sizes.

This way of working with users at Ergonomi Design Gruppen is described by design historian Lasse Brunnström as a "tangible work method with consumers as co-creators in the design process [that] shall be seen as a further development of the 1940s Swedish tradition of consumer research."[32] While noting this longer historical trajectory of the emergence of new design methods, the shift in design practice brought about in working with users is not further highlighted in this Swedish design history publication, besides stating that it has "given exceptionally good results, but at the price of both time consuming work and high costs." Risks with the process are noted, such as designers possibly nudging "test persons" in desired design directions, or that the methods might entail the designer abdicating from "design responsibility and simply give people what they want".[33]

Overall, what comes across in this historical account is a strong emphasis on the role that work-life ergonomics, safety and security perspectives and design for disabilities have had on Swedish design. This is of course a valid account in many ways. The innovative design and engineering work carried out in this context are undisputable – but in telling the story in this way, a blind eye is turned to what these contexts and design situations have brought in terms of opening up new spaces for design, and new methods and practices in relation to this.

Similarly, design historian Kerstin Wickman also does bring attention to the rise of ergonomics in Swedish industrial design in her history over the design consultancy A&E.[34] Against a background of the crafts-based and traditional Swedish design education of the 1950s and 1960s, she highlights the dissatisfaction and critique among young designers that surfaced as critiques of the roles of designers in relation to social responsibility. While the publication does pay a good deal of attention to design processes from the perspective of form work, and different stages of iterative prototyping of products in relation to ergonomics, materials and production techniques, there is hardly any mention of what the new user-centered methods for designing entailed.

31 Maria Benktzon & Sven-Eric Juhlin, *På lättare sätt: En rapport om utveckling av äta - dricka redskap* (Stockholm: Folksam, 1981).
32 Brunnström, "Hjälpmedel för ett säkrare och jämlikare liv", 321. [my translations, here and in the following]
33 Brunnström, "Hjälpmedel för ett säkrare och jämlikare liv", 321
34 Kerstin Wickman, *A&E Design: The Book* (Stockholm: Business History Publishing, 2018).

The design process behind a series of ergonomic painting tools and brushes for the company Anza in the mid-1970s is presented with examples of innovative detail solutions and materials. However, in relating to the user-centered studies that must reasonably have been part of the process, only very brief references are made to iterative user-inclusive processes. The text about paint brushes mentions that these were "tested with regard to the correctness of grip and the function of the hanging device", which led to modifications and new working models that were "prepared and tested", reworked again and "tested and modified".[35] How these processes played out between designers and users, which methods developed and what that meant for design practices and for how designers and non-designers related to designing together, is not explicitly mentioned or discussed.

Both Wickman and Brunnström have contributed immensely to Swedish design history as a field, and have published extensively on various aspects of histories of Swedish design. However, in these accounts and in those by other design historians, user-centered design is all but completely missing. In a 1990s publication compiling Swedish design histories by both academics and designers that were active in establishing industrial design as a practice in Sweden, a clear focus is on the design of artefacts, and on the role of the individual designer and his (for they are all 'he') career trajectory.[36]

User-centered design methods are often only mentioned in passing, such as in designer Rune Monö's account for the shift from a focus on aesthetics and styling to function, including ergonomics and psychology: "During the 1960s and -70s, they [the industrial designers] increasingly emphasised that it was not the beautiful product that they strove for, but that which best suited the user; implicitly, from all aspects!"[37] Monö references American designer Henry Dreyfuss' work in *Designing for People* in this context of adapting products to human physical demands: "In Sweden the word ergonomics came to the fore in the 1960s, and at the beginning of the 80s the concepts environment and ecology suddenly became important."[38] The consultancy Ergonomi Design Gruppen's 11 step program is presented as emblematic of the change view of industrial design from the 1940s to 1960s "tentative, to a large extent, despite everything, intuitive one-man-theatre to professional scientific and methodically verified team-work", stressing that the work done by Ergonomi Design Gruppen has "to a higher degree than most of our design companies incessantly, and with success, driven this scientific methodology as a condition for a good result."[39]

What comes across in these histories is a strong focus on individual designers and the objects created by them. Any collaborative design work is mainly described as work conducted in pairs or teams of professional designers – such as Sven-Eric Juhlin and Maria Benktzon at Ergonomidesign, or Tom Ahlström and Hans Ehrich at A&E design. Some texts will mention collaborations between designers and other professional expert groups, such as engineers, technicians,

35 Wickman, *A&E Design,* 106
36 Lasse Brunnström (ed) *Svensk industridesign: En 1900-talshistoria*, (Stockholm: Norstedt, 1997).
37 Rune Monö "Produktdesign för arbetslivet" in Lasse Brunnström (ed) *Svensk industridesign: En 1900-talshistoria* (Stockholm: Norstedt, 1997), 281.
38 Monö "Produktdesign för arbetslivet", 283.
39 Monö "Produktdesign för arbetslivet", 295.

and marketing people, but rarely mention neither the users involved in iterative prototyping, testing and designing nor the processes and methods for doing this.

The work of designer Rune Zernell, who set up the design department within Atlas Copco, manufacturer of drilling and pneumatics equipment, can serve as an example. Zernell's work on developing ergonomic hand tools for the company is mentioned, in which he during the late 1950s worked together with a medical expert "for help with the science concerning the hand's anatomy, muscles and nerve centres in connection with his experiments with different grip models in plaster and plasticine".[40] Iterating and testing plaster and wood models with co-workers led to the development of a series of ergonomic hand drills that minimised strain on the body, both in terms of static and dynamic injuries and in terms of drastically reduced noise and thus less risk for hearing impairments.[41] The description of the design work leading to the launch of the ergonomic hand drill is very brief while the actual design work would, of course, be more extensive than what comes across in the historical narrative. This narrative does not include any pointers towards the methods applied, or any mentions of interviews or tests with actual users of hand drills, which might otherwise have been indicative of these early considerations of ergonomics in Swedish user-centered design. Instead, the focus is more on design as a physical object and marketable commodity. These histories map the emergence of industrial design from the perspectives of designers, design contexts and designed things – which undoubtedly contributes to the understanding of key aspects of design discourse in Sweden, but not to the centrality of new methods in the (re) formulation of industrial design practices at that time.

Overviews of Swedish industrial design point to the 1970s turn towards 'design for the disabled' or 'design for all' as important for establishing ergonomics and inclusion as central aspects of Swedish design.[42] Examples presented are mainly everyday utensils such as knives and forks designed for disabled persons, and screwdrivers or other ergonomic hand tools for professionals. Products tend to be described as things in which the aspects of "function" and "aesthetics" came together, for example in "handicap adapted products", which would make these suited to "everyone".[43] Design objects are portrayed as having inherent artistic qualities, where for example cutlery for persons with reduced hand and grip functions is described as "a work of art in plastic", with reference to its public exhibition in museums as well as the sales of these items in museum shops.[44]

With the focus on design as products rather than as process, in the turn towards 'design for all' these are presented as designers' reactions to broader societal issues and discussions on equality, democracy and critiques of consumption. Simultaneously, and perhaps sometimes more explicitly, the formal qualities of these designed object are emphasised from a perspective of their having been "awarded design prizes and

40 Lasse Brunnström, *Swedish Design. A History* (Bloomsbury: London, 2019), 3.
41 Brunnström "Hjälpmedel för ett säkrare och jämlikare liv", 304-305.
42 Lasse Brunnström, *Swedish Design. A History* (Bloomsbury: London, 2019); Kerstin Wickman, *A&E Design: The Book* (Stockholm: Business History Publishing, 2018).
43 Brunnström, "Hjälpmedel för ett säkrare och jämlikare liv", 317.
44 Brunnström, "Hjälpmedel för ett säkrare och jämlikare liv", 321.

are exhibited in design museums around the world, not least because they, besides being ergonomically functional, have had a beautiful form."[45]

Narratives such as these tend to highlight specific objects as design historically important in relation to their artistic qualities from a position of connoisseurship, for example in statements presenting these in terms of that "the functionally simple also is the beautiful."[46] In the focus on design as materiality, concepts, as well as actions of continuity and disruption in design, are approached from a form-giving point of view. In these Swedish design histories, the changes in process and perspective in designing brought about when developing methods for user-centered design is, at best, touched upon in relation to ergonomic design and design for all.

While collaborative and user-centered designing brought about the exploration and invention of new methods and different processes in design, these considerations are relatively invisible in a Swedish design historical context. Even in cases where the "common knowledge" is that the period between 1960 and 1980 was one when designers increasingly begin to work with new methods for understanding and working with users, the processual, conceptual and methodological perspectives on design as design*ing* are rarely present. While the ergonomic or design-for-all-perspectives are included in some sense in Swedish design histories, the processes, approaches, and methods development of collaborative designing that came to be called 'participatory design' are not present at all.

Concepts in participatory design

In place and in time, 'cooperative design' emerged in Scandinavia in the late 1960s to -80s.[47] As this approach eventually merged with developments in user-centered design as well with research initiatives in an Anglo-American context in the 1970s through -90s, it came to be called 'participatory design.' During this time, technological innovation and changes in the organisation of workplaces gradually brought computer-based technologies into the realm of designing.

Between the 1960s and the 1990s, participatory design in Scandinavia grew out of initiatives that strove to include skilled workers and a range of stakeholders into the design of new computer-based work tools. As such, participatory design could be seen as an aspect or a strand of user-centered design, but with a strong connection to its computer-oriented origins. As defined on the very first page of a recent handbook: "Participatory Design is about the direct involvement of people in the co-design of the technologies they use."[48] Perhaps it is this coming about of participatory design in a field – computers and 'IT – not traditionally seen as pertaining to one where design takes place, that has led to its exclusion from design histories. In the type of design history that focuses on design as material outcomes and on the formation of the professional roles of designers,

45 Brunnström, "Hjälpmedel för ett säkrare och jämlikare liv", 321.
46 Brunnström, "Hjälpmedel för ett säkrare och jämlikare liv", 321.
47 Yngve Sundblad, "UTOPIA - Participatory Design from Scandinavia to the World", *History Of Nordic Computing* 2010:3, 176–186.
48 Jesper Simonsen & Toni Robertson (eds.), *Routledge International Handbook of Participatory Design* (Routledge: London & New York, 2013), i.

the ways of designing and the constellations of people doing design in these early participatory design situations would probably not qualify as "design".

Participatory design approaches and methods have with time have moved well beyond the realm of "pure" technology (if such a thing even exists). They are nowadays strongly present in the design of various products, services, of social phenomena, political processes and much more, collaboration between people is at the core of this type of designing. In Norway, Sweden and Denmark, 'cooperative design' evolved in close collaboration with trade unions, inspired by action research, as a method of supporting workplace democracy in developing IT systems in different workplaces. In these initiatives, design methods and processes aimed at bringing designers and skilled professionals together in collaborative processes of designing computer-based tools.

The design approaches formulated in the design methods movement,[49] as well as corporate research into use and usability in IT sector such as the ethnographic methods adopted by the Xerox PARC,[50] eventually merged with Scandinavian cooperative design in formulating research initiatives to further practice and knowledge in the field of 'participatory design', or 'PD' for short. The first Design Research society conference on the topic of "Design participation", held in Great Britain in 1971, is often referred to as a starting point for research on the theme of participatory design, while the year 1990 marks the establishment of a specialised biannual conference on participatory design (PDC) focussing both academic and practice-based methodologies, theories and practices. A heterogeneous 'PD community' has formed, which through design practice and design research seeks to both constitute and challenge what participatory design is, or could be.

Scandinavian participatory design emerged in a historical context where established design practice and methods fell short of addressing the types of design situations called for in light of social changes and technological advancements. Participatory design aimed to take on new ways of approaching complex design situations in order to bring about change through design and to build new types of research methodologies in order to advance knowledge in and through design. Over the almost five decades that participatory design has been practiced, there have been continuous developments of and discussions on its methods, tools, aims and contexts.

49 "Looking back now, at this book, and at what has become of design methods, I think that this is the crux of the matter: the new methods permit collaborative designing whereas the old methods do not. The change the nature of designing, or can if one lets them. (…) The new methods, properly used, release everyone from the tyrrany [sic] of imposed ideas and enable each to contribute to, and to act upon, the best that everyone is capable of imagining and doing." John Chris Jones in the foreword to the 1980 edition of *Design Methods* (1970). In John Chris Jones, *Design Methods: Second edition with new prefaces and additional texts* (New York, 1992).

50 Jeanette Blomberg, Jean Giacomi, Andrea Mosher, Pat Swenton-Wall, "Ethnographic Field Methods and Their Relation to Design" in (eds.) Douglas Schuler and Aki Namioka, *Participatory Design: Principles and Practices* (New York, 1993), 123-156.

As in any emerging field, participatory design has been simultaneously defined and questioned through the theoretical and methodological contributions of its practitioners. Conference papers and publications positioned in participatory design contexts often include an initial or integrated brief history of the emergence of participatory design and highlight the own position in regard to one or several of the aspects that are seen as fundamental to this approach (or field, or community). These include approaching design as a social and collaborative practice of mutual learning that includes designers, users and stakeholders; empowering users as experts of their own experience in the design process as well as in future use contexts; enabling non-designers to have a voice or a say in the design process through introducing tools, methods and techniques that support a shared language and shared design actions; and addressing issues of politics, power, influence, expertise, accountability and agency in design as well as on societal levels.[51] For some, Participatory Design explicitly also must deal with the design of computerised technologies or systems (and is then capitalised when referred to in writing), while others extend participatory design to apply to many other contexts of design.

That participatory design can be understood and practiced in different ways while sharing some fundamental basics, is perhaps embedded in the approach itself.[52] Discussions of what participatory design actually is, should be, or could become are, in a sense, highly reflective of the ethos of participation and prototyping at its core. However, in the discussions and practices aiming to define and develop 'participatory design', there are of course differences in starting points, emphases, conceptual definitions and aims with what is meant with participatory design both as a practice and as a concept.

With a very rough generalisation, we might say that understandings, definitions and suggestions tend to lean towards either descriptions (what participatory design currently is), prescriptions (what participatory design should be) or projections (what participatory design could be becoming). With this follows that some of the ideas, concepts and aspects in participatory design also are approached and understood in partially similar, partially different ways over time and in relation to the types of designing they connect to. What is it, then, that we talk about in participatory design, when it comes to some of its core concepts and practices?

51 Toni Robertson and Jesper Simonsen, "Participatory Design: An introduction" in (eds.) Jesper Simonsen & Toni Robertson, *Routledge International Handbook of Participatory Design* (Routledge: London & New York, 2013).

52 The dual scope of furthering knowledge and understanding as well as bringing about change through design action comes across already in the titles of many publications, not least the anthologies springing from an early 1990s PDC conference initiative and the latter 20 year jubilee of the conference: *Participatory Design: Principles and Practices* (New York, 1993), eds. Douglas Schuler & Aki Namioka, as well as the more recent *Routledge International Handbook of Participatory Design* (2012).

Projects

Practices and developments within participatory design have, since the 1970s, been project-based. To a large extent these projects have, from the very beginning, either originated in or been substantially connected to research. In Scandinavia, the processes of initiating design projects that were also practice-based design research projects came about in a context of changes in workplace legislation in the 1970s.

New regulations governing workers' influence on substantial changes in the work environment and work process resulted in trade unions expanding their roles and range of influence and negotiation into the emerging sphere of new technologies of communication and digitalisation. Unions that had until the early 1970s mainly negotiated issues of wages, physical working conditions and work hours now actively engaged in influencing the introduction and design of new technologies and workflows in professional life. In this process, questions of industrial democracy manifest in power relations between workers, unions and management were linked to needs to develop not only new work practices or designs for implementation at work but also new methods for new types of knowledge development in these processes.[53]

In Norway, the late sixties saw legal reforms aiming towards industrial democracy specifically in relation to socio-technical developments. This brought about trade union initiatives that pushed for the development of knowledge that reflected workers' interests in conjunction with the introduction of computers into working life.[54] With the Swedish Joint Regulation Act in the late 1970s, the legal right was established for workers and unions to be part of codetermining work organisation as well as design and introduction of new technologies.[55] The seminal projects, in a Scandinavian participatory design context, such as the iron and metal workers union NJMF project in Norway, and the later projects inspired by this in Sweden (DEMOS, UTOPIA) and Denmark (DUE) thus largely sprang from changing legal frameworks on national levels, in combination with societal and academic movements.

One of the driving forces for establishing participatory design projects has been that of furthering research in this area, initially mainly within computer and systems related areas, but increasingly also in other contexts. Many of these have initiated as research projects, formulated by design researchers in relation to research interests, as well as to funding opportunities and organisations or government authorities, for example in relation to policy or legislation changes. Research constellations in these settings can include both design researchers active in academic environments, researchers from other disciplines, designers and design researchers based in industrial or commercial contexts, as well as people in the capacities of 'users' and 'stakeholders'.[56] Participatory design projects, in this sense, have been set up by designers/design researchers according

53 Pelle Ehn, *Work-Oriented Design of Computer Artifacts,* diss. (Stockholm: Arbetslivscentrum, 1988), 251-252.
54 Ehn, *Work-Oriented Design of Computer Artifacts,* 272.
55 Ehn, *Work-Oriented Design of Computer Artifacts* 256-58. In Swedish, MBL: Lagen om medbestämmande, enacted in 1977.
56 For example the constellation of HCI researchers from different universities and three Danish companies with usability labs described in Sussanne Bødker and Jacob Buur "The Design Collaboratorium – a Place for Usability Design", *ACM Transactions on Computer-Human Interaction* 2002:2 (vol.9).

to timelines, funding, people, stakeholders and other resources, with emphasis on certain design and research aims of interest for and in participatory design methodology and practice: that of people participating as users in co-designing something for instances of future use.[57]

In the last ten years, researchers in design have been suggesting perspectives in which design could be approached in more open-ended manners. Design processes in traditional participatory design focus designing for use-before-use – in which 'the user' as such does not yet exist – calls into question what (or who) is actually being designed.[58] An increasing focus on design-after-design, and how design could allow for change and adaptation of what is designed in and through use by 'designers-in-use',[59] complicates the idea of the 'project' with a definite start and endpoint of a design process. With strong influences from STS studies and ANT theory, researchers are increasingly instead proposing that design would come about around shared 'matters of concern' and that this could take form as 'design Things' in which people and other entities come together in different ways to both formulate and address issues through design.[60]

Such a broad spectrum in viewing what participatory design could be, from the taking on something to be collaboratively designed between designers, users and stakeholders in a process with a clear start and end, to seeing it as an open-ended design Thing that revolves around matters of concern in which humans and non-humans somehow take part, of course also opens up for substantially different understandings of what 'participation' might be.

Participation

As a concept, 'participation' – the action or state of someone taking part in something – is fundamental to participatory design. There are of course differences in starting points, emphases, conceptual definitions and aims with what is meant with participatory design, and with 'participation'. With this follows that some ideas and aspects in participatory design are also approached and understood in partially similar, partially different ways over time and in relation to the types of designing to which they are connected.

For some, Participatory Design (or PD, capitalised, to mark the specificity of the term) explicitly must deal with the design of complex computerised technologies or systems. Others extend participatory design (in lower case, to signal approach rather than category) to apply to broader contexts of design, including social innovation. A consensus in participatory design is that it as a practice cannot be reduced to limited definitions or formal principles, but that it is characterised by "a commitment to core principles of participation in design" that in their turn connect to a wide range of different methods and tools.[61]

57 Pelle Ehn, "Participation in Design Things", in *Proceedings of Participatory Design Conference* (Indiana University, 2008), 92-101.
58 Johan Redström, "Towards user design? On the shift from object to user as the subject of design", *Design Studies* 2006:27, 123-139.
59 Warwick John Tie, "The subject supposed to design", *CoDesign* 2017:1 (vol. 13), 19.
60 Ehn, "Participation in Design Things", 92-101.
61 *Routledge International Handbook of Participatory Design*, 3.

Participation in Particpatory Design thus from the very beginning has been something fundamentally constitutional, that goes beyond mere involvement by people (such as user testing in which users are objects of study or givers of information), and that involves people not defined as designers becoming active and acknowledged collaborators in the design process as such.[62]

Practices of participation vary substantially in participatory design, in regard to who takes part, where and how collaboration takes place, and how extensive it is in time and engagement in relation to the design process.[63] However, neither as a concept nor as a practice does 'participation' have a generally agreed-upon definition within participatory design.[64] How participation and collaboration are supported, what participation does for designing, what counts as genuine participation, and how much participation is "enough", have been recurring topics in participatory design research.[65] Attention has also been drawn to how decisions of participation and legitimacy of participation come about in projects: who decides who gets to participate – which users, which stakeholders – and how are these processes initiated?

In many participatory design projects, the design researcher or the organisations with which she collaborates in the project would have a strong influence on which users and stakeholders would be considered for inclusion in the process and how power is distributed, negotiated or challenged in the processes of designing. That many papers and articles refer to participants as being "invited" to participate, suggests that someone already holds power over the design domain, and extends the invitation allowing others to step into the process. What does it mean in terms of power relations and ambitions of empowerment to be 'invited' to participate, as opposed to 'come together' around matters of concern, or to see participation as a method for the formation of publics in processes of social innovation?[66]

Since the union-initiated Scandinavian participatory design projects of the 1970s and 1980s came about in a legal-political context of changes in work legislation that by law gave unions and union members the right to have influence

62 *Routledge international handbook of participatory design*, 5.
63 Kim Halskov & Nicolai Brodersen Hansen, "The diversity of participatory design research practice at PDC 2002–2012", *International Journal of Human - Computer Studies*, 2015 (vol.*74)*, 87.
64 Halskov & Hansen, "The diversity of participatory design research practice at PDC 2002–2012", 81–92.
65 Victoria Gerrard & Ricardo Sosa, "Examining Participation", *PDC'14:Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1,* Windhoek 2014, 111-120; Jon Whittle, "How much participation is enough? A comparison of six participatory design projects in terms of outcomes", *PDC'14:Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1,* Windhoek 2014, 121-130; Andrew Drain & Elizabeth B.-N Sanders, "A Collaboration System Model for Planning and Evaluating Participatory Design Projects", *International Journal of Design*2019:3 (vol. 15), 39-52.
66 Christopher A Le Dantec and Carl DiSalvo, "Infrastructuring and the formation of publics in participatory design", *Social Studies of Science* 2013:43(2), 260.

in decision-making processes regarding changes in work organisation and technologies, the question of who should participate in these design projects was fairly given. What was not given, was how a participatory design process in which non-designers took part could or should be set up.

Hence, the strong emphasis in participatory design on developing methods, tools and techniques to support and enable users to, through making, have a voice in the design of technologies, without necessarily having to be able to 'speak the language' of design. Language and knowledge were central to the formulation of practices of participatory design. Bringing designers and users together with the aim of mutual sharing of not only formal knowledge but also experiential know-how was what pushed for other than language-based ways of bringing experience, skill and embodied knowledge to the fore. The purposes of introducing, for example, lo-fi prototyping and workshop methodologies were not only to enable discussion and decision-making in the design process but also aimed to bring forth other types of knowledge in the process through practice.

From the 1980s participatory design projects and onwards, the development of tools to enable and support sharing experiences, engaging in mutual learning between designers, users and stakeholders, and supporting collaborative design practices has been central to participatory design. This design-by-doing approach has, in combination with ethnographic methodologies, brought about a number of tools, techniques and methods used in different stages or aspects of design processes.[67] Whether aiming to support formulation and aims of a design endeavour, sharing of experiences, analysis and reflection, or making other types of design decisions, these methods and tools include working with prototypes, mock-ups, scenarios, probes, design games and various tool-kits.[68] The continued development of tools, theoretical foundations and methods for participation and involvement in design action is core also in the more recent and emerging participatory design initiatives, aiming for more open-ended practices. The question of where design ends and use begins – and thus the questions of how to mediate or support participation between 'designers' and 'users' during 'design time'– also leads to explorations revolving around how to 'design for design after design', when design things continue to change and evolve through use.[69] Infrastructuring, designing for a continuous appropriation and redesign through

67 Ehn, *Work-Oriented Design of Computer Artifacts,* 18.
68 Methods and tools are the "most well-described aspects of PD in the early proceedings" of PDC, between 1990-98, according to Ditte Amund Basballe, Kim Halskov and Nicolai Brodersen Hansen, "The Early Shaping of Participatory Design at PDC", in *PDC '16: Proceedings of the 14th Participatory Design Conference*, Aarhus 2016, 23. Also in later PDC papers, the development of methods has a strong presence both in "traditional" PD contexts and as discussed in papers relating to PD moving into new contexts: Halskov & Hansen, "The diversity of participatory design research practice at PDC 2002-2012", 83.
69 Elisa Giaccardi, "Metadesign as an emergent design culture", *Leonardo* 2005:4 (vol.38), 342-349; A. Telier, *Design Things* (Cambridge, MA.: MIT Press, 2011), 170-181.

and in 'use time' is suggested as ways of opening up for more flexible design processes, that support ongoing changes in design through use as practices, technologies and society evolves.[70]

People

That people – often referred to as 'users' or 'participants' – should come together collaboratively in the design of things that affect their everyday life, is an idea central to participatory design. So is also the ambition that design should happen *with* people rather than *for* people.[71] Design should take its starting point not in considering which objects or products to make, but in questioning how things – situations rather than objects – are used, are experienced by, and affect the people engaging with them. The people labelled 'designers' or 'design researchers' should, therefore, work together with people referred to as 'users', since the aim of early participatory design was to co-design tools and systems that would enhance and strengthen the workers' professional skills in a work-life context. People involved in the design process, who were neither 'users' nor 'designers', would be 'stakeholders', representing, for example, workplace management, other professions, groups, organisations, providers of funds or infrastructures that had some sort of interest in or relation to the design initiative.

As participatory design increasingly moved into new areas of professional practices, users would also be categorised in different types – such as primary or secondary users, end-users, or weak users. Categorisation would depend on different aspects of context such as how directly or indirectly users would work with a designed thing; how professionally skilled or unskilled they were in relation to it, or how likely it was that they would be able to fully participate with equal voice and standing in a design process (in relation to, for example, health, sickness, age and cognition) where designers and often also workplace managers or leadership took part.

Over time, the developments within participatory design practices in conjunction with technological becoming increasingly fluid and digitally interconnected, not only in professional but increasingly also in everyday contexts, has brought about yet other ways of thinking and doing in relation to people involved in co-design processes, which is also reflected in terminology:

70 Per-Anders Hillgren, Anna Seravalli & Anders Emilson, "Prototyping and infrastructuring in design for social innovation", *CoDesign* 2011:3-4, vol.7, 169-183; Christopher A Le Dantec & Carl DiSalvo, "Infrastructuring and the formation of publics in participatory design", *Social Studies of Science* 2013:43(2); Susanne Bødker, Christian Dindler & Ole Sejer Iversen, "Tying Knots: Participatory Infrastructuring at Work", *CSCW* 2017:26, 245-273.

71 "Participatory design is about the direct involvement of people in the co-design processes of the information technologies that they use." Jesper Simonsen & Toni Robertsson (eds.), *Routledge International Handbook of Participatory Design*, xix.

> Many people have become uneasy about terms like *user*, *consumer*, and *client*, which pin people down in narrow and restrictive roles. At this moment, we refer to them as *participants* and *partners*, or just *people*, to express the multiple roles and perspectives that are needed.[72]

Who can define what it is to be a 'user' or a 'stakeholder' of an as of yet non-existent future something, and what does that then mean for who gets to participate – and how – in co-designing? And in projections of future practices where trajectories are drawn towards situations where "everyone designs", what happens to the roles of people hitherto labelled 'designers'?[73]

Questions of which people, in which stages of design initiatives, should be the ones to have a say in how other people are invited, included, engaged, at all considered, or even seen, have increasingly become weighed into considerations of power, bias and perspective on how participatory processes come about and play out.[74] Building on Latourian actor-network perspectives, questions also arise as to whether participatory design should exclusively be about designing with people, or if other entities also should and could be included in participatory processes – and if so, how.[75] Participation then not only becomes a question of which people and in which capacities (such as 'designers', 'users' and 'stakeholders') engage in collaborative practices in design, but also how humans and non-humans could come together around 'matters of concern', and how design comes about in relation to agency and power.[76]

72 Liz Sanders & Pieter Jan Stappers, "From Designing to Co-Designing to Collective Dreaming: Three Slices in Time", *Interactions* Nov.-Dec. 2014, 30.

73 "The people whom we currently call designers are the toolmakers, making the tools and providing the materials the rest of us use to imagine and express our collective ideas about future experience." Liz Sanders & Pieter Jan Stappers, "From Designing to Co-Designing to Collective Dreaming: Three Slices in Time", *Interactions* Nov.-Dec. 2014, 33. See also Ezio Manzini, *Design, When Everybody Designs: An Introduction to Design for Social Innovation* (Cambridge: MIT Press, 2015).

74 Henry Mainsah and Andrew Morrison, "Participatory design through a cultural lens: insights from postcolonial theory", *Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1,* Windhoek 2014.

75 Elisa Giaccardi & Johan Redström, "Technology and More-Than-Human Design", *Design Issues* 2020:4 (vol.36) [forthcoming].

76 Kristina Lindström & Åsa Ståhl, "Becoming response-able stakeholders – Participatory design in times of uncertainties", *PDC '16: Proceedings of the 14th Participatory Design Conference,* Aarhus 2016, 41.; Erling Bjarki Björgvinsson, "Open-ended participatory design as prototypical practice", *CoDesign* 2008:2, vol.4

Politics and power

Inspired by action research, as it had taken form in the social sciences during the 1950s and 60s – not least as formulated in the 'critical pedagogy' by Paulo Freire in *Pedagogy of the Oppressed* – an ideal in early participatory design research was that of making societal and political change not only through design but also through design research. The researcher should not, or could not, be a neutral observer, but should consider how the own practice would play into societal power relations. In early participatory design, taking a political stance was also strongly associated with the role of engaging as a design researcher, and with the conflation of the roles of 'designer' and 'researcher' into one and the same (compared to practice in user-centered design, where the roles of designer and researcher typically would be separate).

In the early Scandinavian participatory design projects, designers/researchers actively sided with unions and workers as opposed to management, expressly with the emancipatory ambition to change the power structures in the workplace by giving a voice to those who were weakest in the decision-making processes that affected their everyday lives.[77] Questions of power relations and empowerment were, then, highly present in participatory design from the very beginning. This strong emphasis on power relations and issues of democracy is also linked to an ethos of accountability, both in regard to the field of design and "the worlds it creates" and in regard to the role of the designer."[78]

How power is negotiated is intrinsically linked to ethical considerations of how roles are defined and ascribed (or described), and how values associated with skills and knowledge are perceived and created. Since people are considered experts at what they do, and of what they experience, they should also have the right to influence how what they do is interpreted and represented in a design situation: personas and scenarios, for example, should be developed by and with the people these claim to represent.[79]

The aims of propelling a democratic development through participatory design have been grounded in the importance placed on mutual learning and negotiation in the design process. That designers and skilled users in early participatory design projects should not only share easily articulated knowledge but also the tacit knowledge associated with their respective practice was the groundwork on which informed collaborative work and relevant negotiations of design decisions could take place. In projects such as UTOPIA, which aimed to develop computer-based tools for newspaper layout processes, knowledge and skill were the criteria by which users gained influence in the process of designing graphical print tools for highly skilled use.

The gradation of different types of users in terms of skill, expertise, or whether they are end-users or not, might be considered as ascribing different values and different possibilities for participation. Again, considering the UTOPIA project, the aim of designing with 'skilled workers' in a graphic design and layout context had the aim not only of creating qualitatively well-designed layout tools for professional use but also the aim of excluding non-skilled workers from

77 *Routledge International Handbook of Participatory Design*, 6.
78 *Routledge International Handbook of Participatory Design*, 5.
79 Tori Robertson and Ina Wagner, "Ethics: Engagment, representation and politics-in-action", *Routledge International Handbook of Participatory Design*, 65.

entering the graphic design or newspaper printing profession, through design.[80] In a broader societal context, these issues of democracy, technology, design and development then can hold different political and participatory power relations depending on how wide or narrow our perspective is.

While issues of power negotiation and democracy have been present in much of participatory design practice, an ideal of consensus rather than conflict has been developed. Pointing out that many participatory design projects have been carried out in the realm of political 'do-goodery',[81] or as a diffuse and general ambition to design for democracy,[82] calls have recently been made for participatory design to re-connect with the political in design and research, returning to an early view of the political as conflictual or agonistic.[83] Differentiating between politics (structures that enable governing) and the political (contestational conditions of society), an increasing emphasis is placed on agonistic design negotiations as fundamental for emerging participatory design practices aiming to promote political design rather than design for politics.[84]

The introduction of concepts of infrastructuring in relation to the formation of publics and 'matters of concern', has been pointed to as a way for participatory design to re-connect with, revitalise, or reimagine, a political stance in relation to issues of democracy and empowerment from which it originated in the late 20th century.[85] The concept of agonistic democracy is then used to frame design-for-future-use in terms of infrastructuring, as well as design-for-use approaches, these design efforts where contestation rather than consensus are seen to form a basis for democratic developments and the formation of publics.[86] Participatory design would in this setting be about bringing forth situations in which a radical pluralism and the ability to dispute become significative for community-based designing that aims to strengthen practices and discourses of design for democracy.[87]

80 Gro Bjerknes & Tone Bratteteig, "User Participation and Democracy: A Discussion of Scandinavian Research on System Development", *Scandinavian Journal of Information Systems*, 1995, 7 (1), p. 78; Peter Botsman, "Rethinking the CLASS struggle: Industrial democracy and the politics of production", *Economic and Industrial Democracy* 1989:1 (vol 10), 132-133.
81 Susanne Bødker and Morten Kyng, "Participatory Design that Matters – Facing the Big Issues", *ACM Transactions on Computer-Human Interaction* 2018:1, vol. 25, 4:8
82 Carl DiSalvo, *Adversarial Design*, (Cambridge: MIT Press, 2012).
83 Bo Westerlund, "The use of the absent and Othering in design and critical analysis of PD activities", *PDC '16: Proceedings of the 14th Participatory Design Conference,* Aarhus 2016, 31.
84 DiSalvo, Carl, *Adversarial Design* (Cambridge: MIT Press, 2012); Alastair Fuad-Luke, "Design(-ing) for radical relationality: 'Relational Design' for Confronting Dangerous, Concurrent, Contingent Realities" in (eds.) Ma Jin & Lou Yongqi, *Emerging Practices: Professions, Values and Approaches in Design* (Shanghai: Tongji University Press, 2014), 42-72.
85 Warwick John Tie, "The subject supposed to design", *CoDesign* 2017:1 (vol.13), 19.
86 Christopher A Le Dantec and Carl DiSalvo, "Infrastructuring and the formation of publics in participatory design", *Social Studies of Science* 2013:43(2), 257.

Products and practices

What is it, then, that would be the 'design Thing', outcome or 'product' which participatory design brings about? Participatory design once came about in relation to the construction of limited, specific systems for particular uses, and grew to encompass more large scales systems and frameworks – still, mainly in work-related settings. With networked technologies and platforms developed increasingly for everyday use – whether professional or personal or private – participatory design has moved into other realms of practice than before, working with issues pertaining to open and social design, multiple users, and open-ended processes.[88] With the shifts in what participatory design focuses, and with the blurring of boundaries between user and designer, the "finishing line" of design is also constantly moving. If the result of a design process previously was a product, refined and worked-through collaboratively between different stakeholders, thoroughly tested and evaluated, and finally released into "use", things are beginning to work differently. The 'end-users' are increasingly also carrying out the final design decisions, whether in forms of personalisation or collaborative open-access formats, or unanticipated uses or re-uses.[89] Therefore, intentionally 'designing for design after design' re-shapes ideas of methods and uses, and also redefines how a 'final product', 'design solution', or 'outcome' of design might be understood.

What is seen as results or outcomes in and of participatory design, ties into the layered and often simultaneously plural views and aims within participatory design practices there since its beginnings, that still are, in different ways, highly present: engaging people in co-designing meaningful things, supporting mutual learning and shared understanding, empowering people within their present and future situations, striving towards the democratisation of workplaces and society in general, and supporting these participatory processes through tangible practices of making and designing. Outcomes in participatory design are also highly connected to its research traditions and contexts: the building and furthering of knowledge in and through design by means of the development of participatory methods, theories, tools and processes.

87 Carl DiSalvo, Andrew Clement and Volkmar Pipek, "Communities: Participatory Design for, with, and by communities" in *Routledge International Handbook of Participatory Design*, 200; Thomas Binder, Eva Brandt, Pelle Ehn & Joachim Halse, "Democratic design experiments: Between parliament and laboratory" *CoDesign* 2015:3-4, vol.11.

88 Elisa Giaccardi, "Metadesign as an emergent design culture", *Leonardo* 2005:4 (vol.38), 342-349; A. Telier, *Design Things* (Cambridge, MA: MIT Press, 2011), 170-181; Monica Lindh Karlsson & Johan Redström, "Design Togetherness" in *Nordes 2015: Design Ecologies* (2015), no 6; Stoffel Kuenen, *Aesthetics of being together*, diss., (Umeå: Umeå University, 2018); Aditya Pawar and Johan Redström, "Publics, Participation and the Making of Umeå Pantry", *International Journal of Design* 2016:1 (Vol.10).

89 Jesper Simonsen & Toni Robertsson (eds.), *Routledge International Handbook of Participatory Design*, 9.

In a participatory design context, there seem to be tensions between practices aiming towards tangible and measurable design project outcomes in terms of things and situations, and more research-oriented practices that are more process-focused and whose outcomes tend to be the publication of research papers.[90] In this context, there are tendencies to see a participatory design result as something veering in a prescriptive direction: Some aspects, themes, practices and outcomes in participatory design are considered more central to the approach than others, and in consequence, a return to the conceptual and political origins should be made in order to (re-)form a participatory design that makes a substantial difference in peoples' lives. This view of also typically includes an understanding of Participatory Design (then often capitalised) as pertaining more exclusively to the realm of advanced technological things, processes and systems, where the aims are "better processes for the direct participants" and "solutions that are controlled by the future users".[91] This also entails a returning to core participatory design foundations in action research, and the conviction that design researchers should actively take a political, and even activist, stance and "engage in some of the more controversial areas where researchers could really help people by taking side with them."[92]

In an emerging participatory design approach that rather instead emphasises process, change, experimentation and deliberation as the fundamentals of politically engaged practice, the focus is not on what participatory design should be, but what it could be becoming. The outcomes in this approach take on the probing of forms of knowledge and research contributions that – rather than aiming to provide stable answers or solutions – instead attempt to further an opening up of questions, proposals and possibilities that "experiment with, dissect, critique, enable and multiply the ways issues and publics are being designed together".[93] The political, rather than politics, thus comes into focus in contexts or aspects relating to the democratic aims of early participatory design. Experimentation and prototyping aim to both propose new tools, methods, theories and contexts of participation to highlight and consciously negotiate mechanisms of power and influence in open-ended design situations and processes, projecting possibilities of doing design differently: "What if there were absolutely no operational selection criteria of who enters the thing?"[94] "When everybody participates but we don't have the hope of clear answers from previous experiences to fall back on or solutions to craft, what is possible?"[95]

90 Morten Kyng and Ellen Balka qutoted in John Whittle, "How much participation is enough?", 121.

91 Bødker & Kyng, "Participatory Design that Matters", 4:1-4:2.

92 Bødker & Kyng, "Participatory Design that Matters", 4:9-4:14.

93 Binder et al., "Democratic design experiments", 161.

94 Binder et al., "Democratic design experiments", 162.

95 Lindström & Ståhl, "Becoming response-able stakeholders", 44.

96 Bødker & Kyng, "Participatory Design that Matters", 4:1-4:31; Binder et al., "Democratic design experiments", 152.

Origins and futures in design

The recent calls made for a return to the origins of participatory design often propose a revitalisation or revision of the field,[96] and for an opening up of new interpretations of what participatory design could be.[97] What revitalisations or openings would potentially be brought to the fore by returning to the 'origins' of participatory design? Depending on which aspects of the perceived beginnings of participatory design practices become activated, these will rely on concepts and ideas that were formative to *how* and *why* these ways of designing came about. If activating such 'original' ideas in participatory practices will re-activate and reformulate conceptual spaces that allow for framing design situations and processes differently, or if this will instead – or also – mean that certain values, power structures, or hierarchies, are perpetuated and re-enacted is difficult to predict. Especially if it is not clear how entanglements of past and present work in regard to design's concepts and practices.

Early participatory design – though critical of many social and disciplinary power structures – still to a large extent worked within a design paradigm in which the process of designing had a distinct starting point, fairly clearly defined (new) roles for designers and users, and an idea of design results being something quite 'finished' in terms of product or concept. Much of today's more recent participatory design work has moved more towards a 'what if – then how?' stance: What if everyone designs? Then how do the roles of 'researcher', 'designer," user', 'participant' or 'stakeholder' shift or change? What does it mean in terms of power relations and ambitions of empowerment to be 'invited' to participate, as opposed to 'come together' around matters of concern?[98] Or to see participation as a method for co-designing towards tangible design outcomes that address power relations in a specific situation or context, versus as a way to support the formation of publics in processes of social innovation aiming to address issues of democracy and deliberation on a societal level?[99] What if not only people but also other entities have agency in design processes – then how would participation come about and what would it mean?[100] How could design be done, and can we say something about how it should be done? What if designing does not start with a brief and end with a finished thing, but is open-ended – then, how would practices of design and use come about?[101] What is it that we mean, when we talk about participatory and user-centered design and could design histories of core ideas to these approaches open up for other ways of thinking and doing in these practices?

User-centered and participatory design – whether understood as potentiality or in more normative ways – basically revolves around how we collaboratively can go about crafting just and sustainable ways of being together through new ways of doing design. Contestations, explorations and proposals of who 'we' are and how 'we' come together, what challenges and contexts can be brought to the fore, and what that means for the knowledge and tools we need to make, will need to be continuously investigated, prototyped and eventually paradigmatically transformed. Whatever design will become, its becoming will relate to, and carry traces of, its historicity. How possible and plausible trajectories of design are drawn between past and future through an unstable and transitional now, will inform and affect the choices we make, whether we are aware of it or not.

97 Lindström & Ståhl, "Becoming response-able stakeholders", 41.

98 Lars Bo Andersen, Peter Danholt, Kim Halskov, Nicolai Brodersen Hansen, & Peter Lauritsen, "Participation as a matter of concern in participatory design", *CoDesign* 2015: 3-4 (Vol.11), 250-261.

99 Le Dantec & Carl DiSalvo, "Infrastructuring and the formation of publics in participatory design", 260.

100 Erling Björgvinsson, Pelle Ehn & Per-Anders Hillgren, "Design Things and Design Thinking: Contemporary Participatory Design Challenges", *Design Issues* 2012:3, vol. 28, 103; Elisa Giaccardi & Johan Redström, "Technology and More-Than-Human Design", *Design Issues* 2020:4 (vol.36) [forthcoming].

101 Erling Bjarki Björgvinsson, "Open-ended participatory design as prototypical practice", *CoDesign* 2008:2, vol.4, 97; Johan Redström, "RE:definitions of use", *Design Studies* 2008:29, 410-423

Moving on

Already in this chapter, even before moving towards the examples of design histories that are about to follow, it has begun to be noticeable that a repositioning of design's and design history's outlooks in relation to each other, does something for what becomes visible in design's past as well as what that means for understandings of its present. From the shift in vantage point of looking towards history from a position of participatory and user-centered design*ing*, some blind spots in design historiography have already become evident, as have some conceptual tensions relating to the historicity of designing. In the following, we will move on to exploring what prototypes made through combining methodologies from programmatic design research and historical research can bring to design and to design history.

Positioning the outlook towards the past from a position of core concepts in contemporary participatory and user-centered designing, the ambition is to bring about a different way of looking at things. Instead of looking straight ahead, at the things that are usually visible from established positions in design, these prototypes attempt to support seeing something different, sort of like trying to look a bit closer at something that one maybe only has glimpsed in the corner of one's eye.

In the following three chapters, three different examples of design histories are prototyped in relation to concepts at the core of a Scandinavian user-centered and participatory tradition. By applying the concepts of 'participation,' 'users' and 'methods' as lenses through which to go looking for design histories, the ambition is to test if these practice-based perspectives on well-established design historical contexts can tease out other histories than the usual. Furthermore, the aim is to explore if these histories could provide glimpses of conceptual openings for thinking differently about design, drawing forth other possible trajectories for designing.

The programmatic approach of making transitional design histories speaks to the potential of doing things differently that is at the core of designing, but that can risk becoming restricted in outlook and practice through the 'fossilisation' of certain methods or perspectives. Transitional design histories made from starting points in concepts perceived as foundational or stable, aim to support change in design. Through activating an increased awareness of the historicity of designing, these design histories strive to contribute to the conceptual spaces in which to think and act differently in design, including how design actively could re-design itself.

Transitional design histories

The program of making transitional design histories begins with the proposal that a historical awareness of design methods and concepts is needed within design. Activating attention to the historicity of design can support shifts towards doing design differently, moving towards more just and more sustainable ways of living. Transitional design histories aim to work as things to think with, and as stories to tell other stories with.

By positioning design histories in issues that matter for contemporary designing, these explore ideas and practices that have come to form the conceptual foundations of design. These histories would be transitional, made to support fluid and changing understandings of designing. They would present – make present – tensions and assumptions embedded in the core concepts that designing builds on, making visible the inherent instability in what is often currently taken for granted.

The program of transitional design histories is a tool for exploring what happens when design histories are made *for* design rather than *of* design. With this follows, that the making of these histories does not ultimately aim to prototype directions for providing more solid or definitive histories of participation, users, or methods. While making contributions to design historical knowledge, the prototypes presented here are not meant to be stable over time, but to be open to change in respect to design histories' outlooks, positions, and directions as design evolves.

Transitional design histories are directed towards exploring if histories can work towards opening up conceptual spaces for thinking and doing design differently. These prototypes investigate the design space of transitional design histories: how could such histories be made, and what would they be able to do?

In this section of the thesis, I present three prototypes that explore the program of transitional design histories. The following prototypes thus present – and are, as such – initial explorations of the program of transitional design histories. In probing the frameworks of this program, the main guiding questions are:

- What becomes visible in design's conceptual frameworks, when a shift in positioning brings design history to the outlook of designing?
- Will histories *for* design be able to activate different understandings of core design concepts, in terms of the historicity of designing?
- What becomes visible as history relevant for design, when perspectives are taken on the past through core concepts in design? Will other historical contexts, actors, and ideas come into view, contributing to design history?

While the sequence of the three prototypes does not represent an iterative process towards gradually more finished or complete designs of histories, the findings and questions that emerged in the process of working with each prototype have been used to form the next iteration. These considerations are presented briefly here below, as are outlines of methods and materials used, before moving on to the prototypes themselves.

Participation

The conceptual lens for this exploration was chosen based on its central and foundational capacity for participatory design practices: as a concept, as a term, and as a name. In making and narrating this historical representation, the prototype relies on the reader having a certain familiarity with participatory design. That understanding would be expected either to relate to experiences of participatory design practices, or to being acquainted with contemporary participatory design research, or with histories of these. Since general histories of participatory design are very few, part of the construction of the prototype consisted of providing a historical narrative as a contextual setting within the prototype itself.

The conceptual lens of 'participation' was then first applied to published material from contexts where the term began to appear in design (DRS conferences in the early 1970s and the UK design methods movement) and to publications relating to situations where the concept usually is considered to make its first appearances in Scandinavian design (union-driven IT projects of the 1970s). In the analytical readings of material relating to Scandinavian participatory design, further attention was given to ideas appearing explicitly and implicitly in the material, e.g. 'democracy', 'mutual learning', 'bridging difference', 'negotiation', etc., as being connected to the formation of the concept of 'participation'.

Based on these readings, and the conceptual layerings added to the lens of 'participation', I turned to focus on a period in time in Sweden when socio-political ideas of participatory democracy were strongly voiced, the turn of the century 1900. This coincides temporally with the emergence of Swedish industrial design discourse – at least as established in previous design historical accounts. Here, the conceptual lens of 'participation' has been introduced and applied to a temporal setting (turn of the century 1900) and actor (Ellen Key) well known in Swedish design history, to test if this would bring other possible design histories into view.

The source material for the analytical readings were selected more broadly from Ellen Key's literary production than is usually the case in Swedish design history. From findings in these texts, original archival material related to participatory situations (Tolfterna), located in the National Library of Sweden, was studied further. In terms of form, the narrative of this prototype does not follow a strictly chronological trajectory, but experiments with a format of moving 'back and forth' in relation to a perceived linear historical time, aiming to place more emphasis on how the prototype works, than on the narrative as a coherent story.

Users

This prototype is made with the intent of being similar to a more established format of historical representations, working with a chronological format of historical narrative covering the period from 1900 to the 1940s. Still, the intent is to test if and how an outlook from the perspective of designing could make a difference for the workings of such a history. The conceptual lenses of 'users' and 'use' chosen here relate to both terms and concepts in user-centered designing. This chapter also incorporates a contextual account of practices (of user-centered design), and an overview of existing design histories as part of the prototype. One emerging question in relation to this, is if and how such a piecing together of 'established' or 'general' design histories as contextual components made to contrast to the more intervention-like parts of the prototyping will diminish the potentially transitional workings of such design histories.

The socio-political framing of a Swedish design discourse in relation to issues of housing and home reform is taken further in this iteration. Framing these in light of well-researched aspects of Swedish modernist design theory and practice of the 1930s (functionalism and the Stockholm exhibition), the readings of original publications applies the concept of 'use' to hone in on attempts to achieve behavioural change through integrating design intent in material things. This is set in relation to a socio-political context of reforms of everyday life on individual as well as societal levels, through studying investigations and interventions into practices of use as well as in practices of designing.

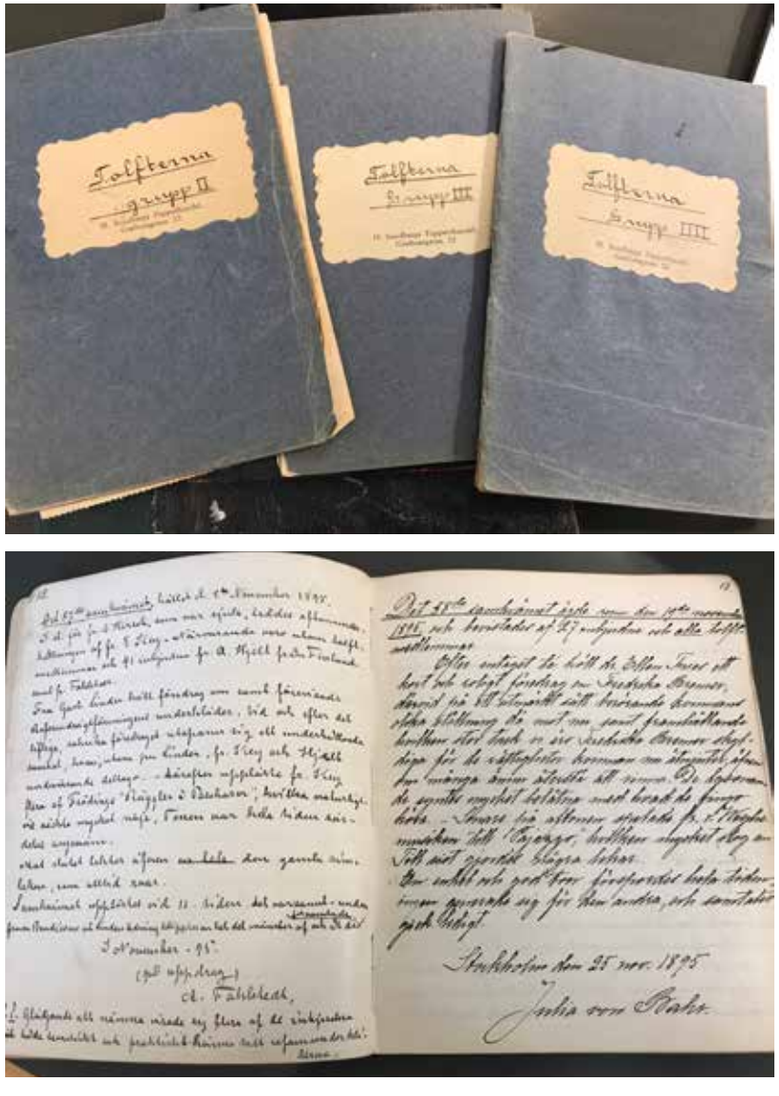
Here, I apply the conceptual lenses of 'users' and 'use' to archival material and publications that I have previously engaged with in other research settings (dwelling investigations from the 1930s and 1940s). Returning to a, for me, well-known archival material from a perspective of designing, is done with the aim of investigating if other things become visible than what was previously noticed. In terms of archival materials, the items and collections studied here are stored in several different archives: the Nordiska Museet Archive, the ArkDes Archive, the Swedish National Archives, and the Centre for Business History. When I first went searching for these materials, the process of locating them took several years of searching, entailing substantial amounts of archival detective work supported by staff working in these different archives.

Methods

In this third iteration, the prototype sets out to investigate design practices through the concept of 'methods'. The exploration here aims to highlight instances of how methods become invented or incorporated into design practices in relation to changing contexts for and expectations on design, and to probe what might count as relevant contexts in the search for histories of designing.

This prototype continues building on the socio-political contexts relating to the formation of design practices and concepts, as presented in the previous two prototypes. The prototype further probes how methods for supporting processes of designing have come about. Picking up on the themes of reforming and redesigning everyday life through interventions in the domestic environment, this study further explores the development of methods for studying and changing everyday domestic practices in the 1940s (the Home Research Institute).

The format of this prototype aligns with the previous two, in that it also includes a quite extensive historical contextual framing, highlighting contemporary as well as past approaches to methods in design. Following this, the part of the prototype where the concept of 'methods' is applied in analysing historical material follows a fairly sequential and chronological historical narrative. In terms of materials, the analysis is based both on published journals and reports, and on a substantial collection of archival documents stemming from the Home Research Institute, placed in the Swedish National Archives. In this prototype, visual material from publications as well as archival items are introduced, to begin extending the analytical framework to other than written materials.



Notebooks with minutes from Tolfterna discussions. Tolfterna archive, Kungliga biblioteket (National Library of Sweden).

4. Participation

Since participatory design was formulated in the latter decades of the 20th century, finding different ways of bringing together designers and non-designers together in collaborative processes has been central to its methods development as well as to its ethical foundations. There are abundant examples of explorations into how to further develop methods, tools and formats for collaborative designing. So many in fact that even starting to point to a selection of 'typical' such projects or programs in a thesis reference is quite difficult.[1] Many of these studies critically question current practices in relation to which types of situations these can be expected to support.[2] However, in this attention to participatory practices in design the *concept* of 'participation' itself is rarely called into question – even when the concept is critically addressed in regard to what it could mean.

As a foundational idea in various approaches to collaborative designing, it is not contested *that* participation is important. What is questioned and challenged is *how* it should happen, *who* should be involved, and *what* participatory designing can bring to situations as diverse as commercial product design or social and political activist endeavours. So, while there are many explorations of how and why participation should or could happen in design, little attention is paid to what 'participation' *means* conceptually to the formation of design practices, or on how it once came to be established as a core concept in designing.

This study probes the concept of 'participation' in design from a historical point of view, with attention to ideas that over time have played into its emergence and incorporation into design discourse and practice. 'Participation' here thus serves as both a starting point and as a lens for sketching a design history from a perspective of designing. This chapter is a prototype, an example, of a what a transitional design history could be like, in that it aims to contribute to opening up ways of seeing the historicity of collaborative design practices through probing the concept of 'participation.'

Attempting to draw forth historical ideas and contexts that made it possible for participatory design to become established, specifically in a Nordic setting, the hope is that a history such as this might give insights into the embedded values and attitudes that still might be in play when activating participatory approaches, methods, and tools. What participation is understood as – actively or not – is

1 For example, the papers presented at the biannual *Participatory Design Conferences* since 1990 extensively explore collaborative methods and considerations https://ojs.ruc.dk/index.php/pdc/issue/archive (accessed 2020-05-29).

2 Liesbeth Huybrechts, Virginia Tassinari, Barbara Roosen, & Teodora Constantinescu, "Work, labour and action: the role of participatory design in (re)activating the political dimension of work", in *PDC'18:Proceedings of the 15th Participatory Design Conference. i.* (ACM, 2018), 1–11.

what draws up the possibilities and the limits for how practices of participatory designing can be made and unmade, and what we can expect these to be able to do for us. *How* design is done responds to the situation it is called into but in doing so, it relies on understandings and on ways of seeing things that are embedded in its concepts and methods.

If existing practices of designing come to the limits of what these can handle in emerging situations, it will in the long run not suffice to try to address this only through developing new design methods. Then, what will be necessary is the bringing about of other ways of thinking in design. The concepts that frame how we see and understand the world need to be presented – made present – and destabilised if we are to be able to question the historicity of concepts and find ways to do design differently. To speak with Hannah Arendt, we would have to unfreeze the frozen thoughts of concepts to find their original embedded meanings, and open up for thinking and acting differently.[3]

This prototype of a transitional design history aims to contribute to begin the thawing of one of the core concepts in participatory designing, making it softer, less stale, and more pliable, and thus possible to perhaps re-shape or handle in other ways than when it comes across as firm and fairly solid. Another aim is to bring design history closer to central issues in contemporary design and make design histories that speak to practices of designing. The other objective with this prototypical design history is to explore if other stories of design come into view when the historical outlook and perspective applied is positioned in relation to current designing.

In this chapter, I direct attention towards what is already a well-established Swedish design historical temporal context, the turn of the century 1900. The approach, however, to events, publications, organisations, and historical actors is positioned in relation to designing, from the perspective of the concept of 'participation' in contemporary design practices. The chapter begins with a brief initial overview of the concept – and the term – of 'participation' as it came into design discourse in the 1970s. I then move on to an exploration of Swedish turn-of-the-century ideas and initiatives revolving around design, society and everyday life, by means of looking for ideas and instances of 'participation' in socio-political contexts relating identified as relevant to Swedish theories and practices in design and aesthetics.

Re-visiting the writings of author, social theorist and design reformist Ellen Key through the conceptual lens of 'participation', examples are given of how practices of mutual understanding and learning became central in negotiating and shifting political and social power structures towards an inclusive democracy in the early 20th century, laying a foundation for the situations in which Scandinavian participatory design later emerged.

3 Hannah Arendt, *Life of the Mind. One/Thinking. Two/Willing.* [1971/1978] (San Diego, New York, London: Harcourt, 1981), 171.

The concept of participation

Participation, in design, entails people being active and acknowledged in a joint collaborative endeavour, in a design process, on equal standing with designers.[4] Terminology varies over time and in various publications: the terms 'participation, 'collaboration', and 'co-design' tend to point to ways of designing understood as characterised by participatory design practices – which as a definition, is a typical example of circular reasoning. However, the 'participatory practices' referred to in such a statement, entails something more than designers and non-designers working together in a design project or design process.[5] The concept of 'participation' in design, includes ideas and practices that go beyond a general understanding of 'collaboration' as working together. Some of these fundamental ideas include approaching design as a social and collaborative practice of mutual learning between designers, users and stakeholders, and enabling non-designers to have a voice or an active role in the design process through introducing tools, methods and techniques that support a shared language and shared design actions. This also builds on understanding the designer's intent as aimed at acknowledging and empowering people as experts in design situations as well as in future use contexts, and on strengthening emancipatory and democratic practices through design processes by addressing issues of politics, power, accountability and agency in design as well as on societal levels.[6]

A general dictionary definition of 'participation' is "the action or state of taking part in something". A legal definition also includes "association with others in a relationship (as a partnership) or an enterprise usually on a formal basis with specified rights and obligations".[7] 'Participation' in more recent participatory design – especially with its transition into other fields of design than strictly computer-related – would lean towards the former definition, while the roots of Scandinavian participatory design can be traced to a definition more veering towards the latter. However, this ambiguity of meaning between participation as generally referring to people somehow taking an active part in design, and a formalisation of what constitutes participation in design, has, it seems, been there already from the start.

4 Jesper Simonsen & Toni Robertson (eds.), *Routledge International Handbook of Participatory Design* (London & New York: Routledge, 2013) 5.

5 Tuuli Mattelmäki & Froukje Sleeswijk Visser, "Lost in Co-X: Interpretations of Co-design and Co-Creation" in *Proceedings of IASDR2011* (Delft, 2011); Elizabeth B.-N Sanders & Pieter Jan Stappers, "Co-creation and the new landscapes of design", *Co-Design* 2008:1 (vol.4), 5-18.

6 Toni Robertson and Jesper Simonsen, "Participatory Design: An introduction" in (eds.) Jesper Simonsen & Toni Robertson, *Routledge International Handbook of Participatory Design* (London & New York: Routledge, 2013)

7 'Participate' in Merriam-Webster dictionary, https://www.merriam-webster.com/dictionary/participate, (accessed 2020-06-10).

DRS Design Participation conference 1971

As a term, 'participation' seems to have made its way into design discourse with the design methods movement and the emerging design research of the mid- to late 1960s. At the Design Research Society (DRS) conference in Manchester in 1971, the theme was "Design Participation". The concept of 'participation' in relation to design was, on the one hand, presented as something new that would alter ways of designing, and, on the other, as a concept that by then was already over-used and at risk of becoming void of actual content. In his opening address, design historian Reyner Banham referred to the "wonder ingredient 'participation'" as having emerged relatively recently, and as having not only become established but already also contested:

> In six short years a concept which had been non-existent had already been narrowed down to a point where it was almost useless. It had gone from something which was absolutely spit-new to all those who heard it for the first time, to a condition where everybody thought they knew what it meant, and were astonished to be reminded that the word can carry a broader spectrum of meaning than they had had in mind.[8]

Nigel Cross, who was director of the conference and editor of its proceedings, stated something similar in his concluding remark:

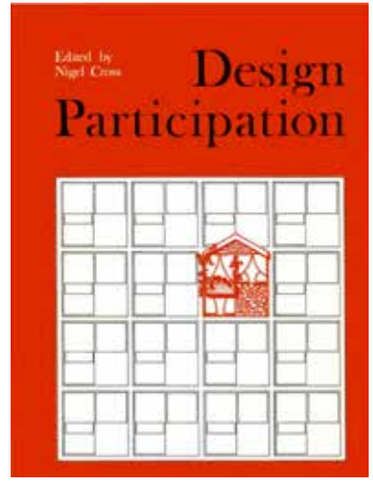
> Finally, I should add that the phrase 'design participation', which I thought I had invented specifically as a title for the conference only nine months earlier, had already become, according to one reviewer soon after the conference, 'an inadequate cliché'.[9]

At the DRS Design Participation conference, the talks and discussions revolved around what would become of the role of the designer if "user participation" were integrated into designing, with examples given mainly from architecture and city planning. What would happen, several of the speakers asked, to design expertise and professionalism when design decisions were made in conjunction with people who had little or no knowledge of design at all? How should designers handle the increasing emphasis on political and social contexts brought to the fore in instances of participatory designing? And what would this mean for the development of methods and processes in design?

In their concluding summaries of the conference, designer John Chris Jones and journalist and futurist Robert Jungk both highlighted the need for designers to work with challenging and changing societal infrastructures and cultures hampering the possibilities for participation between "professionals" and "people", for society as a whole to change. The time frames for achieving design situations in which participatory practices could reasonably be carried out in radically new ways were, they pointed out, probably decades away since the

8 Reyner Banham, "Alternative networks for the alternative culture?" in Nigel Cross (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972), 15.

9 Nigel Cross, "Here comes everyman!", in Nigel Cross (ed.) *Design Participation,* 14.



DRS Conference proceedings 1971, "Design Participation".

education in schools would need to be fundamentally re-worked to prepare new generations for working collectively and creatively. Until such changes had come into effect, new procedures and practices should also gradually be introduced in current design processes as a whole, in matters of participatory idea generation as well as in making participatory design decisions. Jungk remarked:

> As a prognostician, I don't think this change will take place before the end of the century. We will have to suffer first from the lack of foresight of our fathers and grandfathers. After that, something radically different can come, but it won't come on its own; it has to be prepared.[10]

The preparation needed to enable a radical societal change, according to the conference speakers, included developing methods for participatory practices in design. These should, in turn, be based on furthering design knowledge not only among design professionals but among people in general. The development of methods for participation applicable should also be set in relation to initiatives aiming to "undo generations of conditioning" that hindered people from coming together in participatory designing.[11]

For designers, this meant moving into the socio-political realm and becoming more familiar with methodologies and research traditions in the social sciences. This was crucial if designers should better understand "the people" that ought to be brought into participatory decision-making in design. Jungk suggested that anthropologists and psychologists, therefore, must be involved in the work to "open people up". This was work that would take time, and that needed to be carried out on different levels, as a central aspect to this was learning to see and do things in other

10 Robert Jungk's remarks in J. Christopher Jones & Robert Jungk, "Closing comments" in Nigel Cross (ed.), *Design Participation,* 122.

11 Robert Jungk's remarks in Jones & Jungk, "Closing comments", 121.

ways than before. Since this meant the creation of "a new man" (Jungk) and a "lifting of art and design into the area of life rather than hardware, into living patterns and activities, rather than utensils" (Jones), one would need to be patient and work with relatively long perspectives in order for making change come about.

While the lack of knowledge amongst 'the people' (as opposed to 'the designers') involved in design processes could be seen as problematic – or even "appalling" – in relation to participation in design, this unfamiliarity with design was also pointed to as an advantage. While design experts could be trapped in understandings that were the "frozen things of yesterday", ordinary people might "with a lack of knowledge [...] be able to look at things in a more original, more creative way."[12] This idea of bringing people with 'a lack of knowledge' into the design process quite clearly set the framework for where and how expertise was perceived to reside in designing.

Designers, in their professional roles, held certain types of expertise based not only on knowledge about and practice in design but also knowledge about socio-technical frameworks and possibilities that ordinary people did not hold. The eventual contribution that 'people' might make to designing, through methods enabling their participation, was seen as a potential for ideation and creativity through bringing a sort of unconditioned and genuine perspective on what could be possible.[13] One of the speakers, exemplifying with design in a medical setting, highlighted that it was not enough to only bring in 'users' into design situations since the contexts being designed for were very complex. If specialists or experts, together with users and designers, were to truly be able to contribute to finding new solutions, this must be done in the form of participatory teamwork, in which group processes and collaboration would be essential:

> It is also clear that in the team design process we have outlined there is a considerable need for participation at all levels in time and form. But to consider this merely as a reaction between user and designer would be naive to say the least. With major goal-centred activities, participation, including social as well as individual action, becomes the source of design.[14]

In this context of early methods-focused design research that aimed to address increasingly complex design situations, issues of participation were quite tightly linked to questions of how to find adequate ways of bringing people other than designers into design processes. Standpoints differed in that some saw 'users' or

12 Robert Jungk's remarks in Jones & Jungk, "Closing comments", 121.

13 Alberto Feo. "Operational games applied to socio-technical problems", in Cross, Nigel (ed.) *Design Participation,* 42-43: "Our experience suggests that people without the explicit knowledge and experience of a socio-technical problem can be helpful in the exploration of such problems. This should not be interpreted as if they are 'the best' subjects for this kind of exercise, because much better and also more difficult than using 'any' people is using *'the'* people involved in the real problem or situation."

14 E. Matchett and K. G. Williams, "Strategic design for international healthcare", in Nigel Cross (ed.), *Design Participation,* 105.

'people' as needed to participate at specific points in a design process for designers to reach better solutions, while others, as in the above quote, meant that participatory design approaches would fundamentally alter how design is done, and by whom.

While the questions at the 1971 conference revolved around how to include non-designers in designing, and in the situations in which design could take place, the mechanisms bringing design into being were not explicitly questioned. Design was approached as something initiated by companies, organisations, and government or municipal structures. These initiators would engage designers, who in turn could – and, some argued, should – engage 'ordinary people' as participants in designing.

But what was the role of the designer supposed to be, in a situation where non-designers were brought into design? Should designers be increasingly specialised and professional, or should the profession be dismantled in favour of participatory processes? In the discussions, the actual methods and formats through which design would be done were not really in focus at this conference. Participatory design was, at least for some of the contributing design researchers, rather a means for the designer to extend his (for it was 'he', in all of the conference contributions) knowledge about user needs and wishes, rather than a process in which roles of power were collaboratively contested or negotiated. So while 'participation' as a concept was discussed, it was on a level somewhat detached from actually engaging with participatory modes of design practice.

In the Scandinavian context of the early 1970s, the situation was perhaps not inversed but at least quite substantially different. The new ways of collaborative designing that emerged there arose both in response to the opening up of new situations for design, as well as to structural changes on legal and socio-political levels. Labour market dynamics and new legislation in combination with institutional and self-organised socio-political movements, challenged traditional domains of power and knowledge in society in general as well as in designing.

Union-driven design initiatives

In a Scandinavian context, the formation of participatory design practices took its starting point with initiatives from trade unions, specifically in relation to questions of gaining influence in the processes of introducing computer-based technologies and tools in the workplace. The late sixties saw legal reforms in the Nordic countries aiming towards industrial democracy, specifically in relation to socio-technical developments. With the formation of social movements discussing and taking action on issues of democracy and power, new arenas arose for collaboration and negotiation in working life.

Seminal projects acknowledged as such in participatory design contexts were, for example, the Norwegian Iron and Metal Workers Union NJMF project in Norway in the early 1970s, and the later projects in the mid-1970s and early 1980s inspired by this such as the DUE (Demokrati, udvikling og EDB/Democracy, development and Electronic Data Processing) in Denmark, the Swedish DEMOS (Demokratiska styrningssystem/Democratic systems for steering) and UTOPIA (Utbildning, teknik och produkt i arbetskvalitetsperspektiv/Education, technology and product in work quality perspective), and the nursing project Florence in Norway. These were all either initiated directly by trade unions or by researchers in collaboration with unions, with the express intent to gain both knowledge and influence in the introduction of these new tools and technologies.

Between the years 1971 and 1973, the Norwegian Iron and Metal Workers' Union (NJMF) ran a project aiming to incorporate workers' perspectives on the introduction and development of new technology. Kristen Nygaard, who was a computer science researcher engaged in the NJMF project, has described how it came about with a starting point in union-led discussion groups in the late 1960s, revolving around topics related to the consequences of introducing computer technology in various parts of society. The specific discussion group in this case was one "within a broad, democratic movement genuinely representing the interests of the workers" in which the members "came from a wide range of sectors in the society: Job shops, chemical plants, transportation, white-collar work, hotels and restaurants, the public sector."[15] Out of these joint discussions came the conclusion that there was a lack of knowledge about computer technology "based upon the world view of the union members, emphasizing solidarity, industrial democracy, safe employment, safe working conditions, decent wages etc". The decision was then made by the Norwegian Iron and Metal Workers' Union to initiate a research project aiming to address this, with bringing in Nygaard as a researcher based at the Norwegian Computing Center.[16]

The aim of the NJMF project was to increase knowledge about the new technology, in order to be able to take part in influencing how these could be incorporated in working life. Fundamental to the project were the collaborative ways in which workers and researchers together addressed ways of jointly building and sharing knowledge as well as methods for doing development work together. Besides these, results on a national level also included the 1975 Data Agreement between the Norwegian Trade Union Congress and the National Federation of Employers, "stating the right for the trade unions to be informed and participate in the development and introduction of computer-based system impacting upon their working conditions."[17] In these early projects, a strong focus was placed on developing methods for power negotiations between trade unions and company management, targeting on gaining influence in the introduction of new technologies in the workplace.

The formal or legal aspects relating to unions' possibilities to influence the introduction of new technologies into the workplace was decisive for the role that unions came to play in the emergence of participatory design also in Sweden. With the Swedish Joint Regulation Act in the late 1970s, the legal right was established for workers and unions to be part of co-determining work organisation, including changes to work structures stemming from the design and introduction of new technologies. The democratisation of working conditions, in terms of introducing industrial democracy, had been an integral part of the Swedish workers' movement since the 1920s and had aims that went beyond the empowerment of workers' influence in and on their workplace.[18] The legislative

15 Kristen Nygaard, "How Many Choices Do We Make? How Many Are Difficult?", 52-59 in Floyd, C., Züllighoven, H., Budde, R., and Keil-Slawik, R., (eds.), *Software Development and Reality Construction* (Berlin: Springer-Verlag, 1992).
16 Nygaard, "How Many Choices Do We Make?", 52-59.
17 Nygaard, "How Many Choices Do We Make?", 52-59.
18 Anders Kjellberg, "Från industriell demokrati till medbestämmande - fackliga utvecklingslinjer 1917-1980", *Arkiv för studier i arbetarrörelsens historia* 1981:21-22, 53-82.

goals were ultimately to establish shared responsibility for production as well as ownership in industry, and the Swedish approach built on the ambition to avoid social tensions in the form of class conflicts or outright revolution.

Negotiation, collaboration and 'the Saltsjöbaden spirit'

There had been many conflicts regarding working conditions and employment between workers and employers in Sweden during the first decades of the 20th century. Repeated attempts to introduce legislative measures to subdue these tensions were not successful, and instead, the organisations representing employers and those representing trade unions came to an agreement in 1938 that would govern relations and negotiations without state involvement. This so-called Saltsjöbaden agreement was a labour market treaty stating the forms and protocols regulating collaboration and negotiation between the Swedish Trade Union Confederation (LO, *Landsorganisationen*) and the Swedish Employers' Organisation (SAF, *Svenska arbetsgivareföreningen*). Despite being an agreement only between two parties, this became the template for subsequent similar agreements between other unions and employers on the national as well as the local level and is still in force in Sweden today.[19]

Central to the Saltsjöbaden agreement were formulations stating the conditions for worker's representation in issues regarding workplace influence, including agreements of employment security, and forms for handling workplace conflicts and strikes. Unions and industry leaders came together around the joint goal of increasing national production as well as improving workers' satisfaction and motivation in the workplace through cooperation, dialogue and negotiation aiming to reach agreements directly and without interference from state or government.[20] The governing ideas that relationships and negotiations between workers and industry should be based on a collaborative intent, aiming to seek compromise through mutual understanding and respect, came to be called 'the Saltsjöbaden spirit.'

This spirit of deliberation and collaboration profoundly influenced the conditions on the labour market, and also the broader political and societal arena in how cross-organisational and political discourse and practice took form. 'The Swedish model', or the 'middle way' as it was labelled by American journalist Marquis Childs already in the mid-1930s,[21] consisted of a route taken between market-driven capitalism and socio-political aims of egalitarianism that formed a sort of social contract laying the foundation for the Swedish welfare state.[22] Decisive for this was that there was a high degree of enrolment in unions,

19 Christer Lundh, *Nya perspektiv på Saltsjöbadsavtalet* (Stockholm: SNS, 2009). The Saltsjöbad agreement is still in force in Sweden today, with the latest major revision made in 1976 with adaptations made in relation to the change in national legislation.
20 Kjellberg, "Från industriell demokrati till medbestämmande", 63-64.
21 Marquis Childs, *Sweden: The Middle Way* (New Haven: Yale Univ. Press, 1936).
22 Yvonne Hirdman, *Att lägga livet tillrätta: Studier i svensk folkhemspolitik*, Maktutredningens publikationer (Stockholm: Carlsson, 1989); Sven E.O. Hort, *Social policy, welfare state, and civil society in Sweden Vol. 1 History, policies, and institutions 1884-1988*, (Lund: Arkiv, 2014).

by workers, and likewise a high degree of registration in the national industry or employers' organisations by companies, enterprises and agencies, which gave both parties a strong standing in labour market negotiations.

Central to the collaborative ideas at the basis of this 'middle way', was a shared understanding between employers' organisations and trade unions that economic growth and industrial development were the keys to the development of Sweden as a prospering country. Therefore, coming to agreements through direct negotiations between unions and employers would be a smoother way to ensure Swedish financial prosperity than opening these questions for government legislation or control.

During the 1960s, focus from the unions had increasingly turned towards gaining formal rights of actively included in decision-making on a strategic level in the industry, rather than only influencing or having a say in matters of workplace conditions.[23] Towards the end of the 1960s, tensions had again grown stronger between unions and employers, and the shared collaborative aims were replaced with increasing conflicts over issues regarding working conditions and rights. Demands were raised for national legislation that would govern certain fundamental employment security rights, as well as forming a legally binding foundation for how labour market negotiations should be handled.

In the early 1970s, legislation was proposed for governing how negotiations and decision-making should be organised between workers and management. The 'Swedish model' or 'middle way' had begun to lose traction, as negotiation and mutual agreements were increasingly replaced with government level legislation. The Swedish Employer's Organisation went so far in the mid-1970s as to declare the Saltsjöbaden spirit dead, since the labour market power balance, in their perspective, had shifted strongly in favour of the unions.[24]

With the introduction of the Joint Regulation Act of 1976, power relations between unions and employers in determining workplace conditions changed substantially. In connection to this, new formats were needed for setting up processes of integrating union representation in decision-making.[25] In combination with the tradition of collaboration in order to reach agreements, including forms and procedures for negotiation, this new legislation set the now non-negotiable framework for which parties could, and should, take part in all decisions regarding significant changes in work-place conditions. The agonistic perspective, in which the tensions between workers and employers were fundamental to negotiations of power and influence were prominent from the beginning in participatory design projects such as the UTOPIA project, as noted by Pelle Ehn in his description of this in his dissertation:

23 Kjellberg, "Från industriell demokrati till medbestämmande", 74.
24 Gerhard Schnyder, "Like a phoenix from the ashes? Reassessing the transformation of the Swedish political economy since the 1970s", *Journal of European Public Policy* 2012:8 (vol.19), 1126-1145.
25 In Swedish, MBL: Lagen om medbestämmande, enacted in 1976. See *Medbestämmandelagen*, Serien om arbetsrätt 5 (Stockholm: LO, 1977). Pelle Ehn, *Work-Oriented Design of Computer Artifacts,* 256-258.

> Hence, we rejected the harmony view of organizations, according to which conflicts in an organization are regarded as pseudo-conflicts, to be dissolved by good analysis, and consequently we also rejected an understanding of design as a rational decision-making process based on common goals. Instead our research was based on a conflict view of industrial organizations in our society. In the interest of emancipation we deliberately made the choice of working together with workers and their organizations, supporting the development of their resources for a change towards democracy at work. We found it necessary to identify with the 'we-feeling' of the workers' collective, rather than with the overall 'we-feeling' that 'modern management' dedicates its resources to creating, in order to gain more productivity out of the work force.[26]

In the emerging design practices that came about in the context of developing methods for collaboratively envisioning how the new IT-based work tools could be designed, the context was one of agonistic negotiation in which the participating design researchers explicitly "sided" with workers rather than with management. The union-initiated Scandinavian participatory design projects of the 1970s and 1980s came about in a legal-political context of changes in work legislation that, by law, gave unions and union members the right to have an active part in decision-making processes regarding changes in work organisation and technologies. With a decades-long practice of workplace negotiations between unions and employers, a culture of agonistic deliberation had been well established in terms of formats, arenas and representation in relation to workplace democracy.[27]

When the new Swedish legislation was set in motion regarding the new phenomenon of introducing computerised technologies into the workplace, the question of who should participate in the negotiations of these in regard to working conditions and workers' rights to employment was therefore fairly straightforward. What was not a given, though, was how these computerised tools should be set up, what specific qualities of work they should support, or how a process for design and decision-making should be set up in which unions, industry leadership and the developers of these new computerised tools could take active part.

Collaboration in early Swedish participatory design

As the introduction of computer-based technologies into working life gradually spread, the unions' focus on issues related to this also increased. During the first half of the 1970s, the introduction of these new technologies led unions to focus on co-determination processes, and how participation in these could be ensured to secure the quality of working life from a perspective of industrial democracy.[28]

The Scandinavian IT design projects carried out during this period were primarily characterised by attempts to secure influence and power on behalf of the trade unions, in processes highly conditioned by a managerial perspective. The unions' standing in these negotiations was not as strong as the employers', and questions were raised about how co-determination could be extended to the

26 Ehn, *Work-Oriented Design of Computer Artifacts,* 253.
27 Carl DiSalvo, *Adversarial Design* (Cambridge, MA: The MIT Press, 2012).
28 Peter Botsman, "Rethinking the CLASS struggle: Industrial democracy and the politics of production", *Economic and Industrial Democracy* 1989:1 (vol 10), 132.

design of the computerised tools themselves, and not only in the introduction of the tools and their effect on the quality of work performed but on questions of working conditions and power in the workplace.[29]

The 1970s IT design projects in Norway, Sweden and Denmark had influenced "the introduction of the technology, the training, the work environment, and the organization of work only to a certain degree" while "important aspects like the opportunity to further develop skills and increase influence on work organization were limited by the available technology."[30] It is in light of this context that the 'collective resource approach' was introduced as a framework for exploring participatory processes for designing computerised tools, as well as for processes of strengthening workers' power and influence in decision-making on a strategic and managerial level.

Access to knowledge in the realm of technological possibilities and constraints was seen as crucial from the unions' side, which was the main reason for involving researchers, and backing or initiating research projects within this realm. Equally important was the emphasis on finding ways to design and develop tools and situations that safeguarded workers' employment security in a context of increasing computerisation and automation, which from the employers' side reasonably should lead to cutting costs through laying off people or hiring low-cost labour to replace more costly skilled workers.

A focal point of what became labelled as participatory design projects, starting with the 1980s UTOPIA project situated in a Swedish graphic workers' context, was to find methods for designing the technology-based tools with the explicit ambition to support, safeguard and develop the skilled workers' competencies and knowledge. The aim of the project, which ran between 1981 and 1986, was to collectively develop computer-based tools to support skilled work within text and image processing in newspaper publishing. Engaging the graphical workers in the process of developing these new tools, then, required methods to share and build joint knowledge in the process of designing. As described in a publication within the UTOPIA project:

> A major aspect in this first phase of the project was the mutual learning process in which the active participants: graphic workers, computer and social researchers, established a common 'knowledge platform' for the future work.[31]

29 Botsman, "Rethinking the CLASS struggle", 132-133
30 Susanne Bødker, Pelle Ehn, Staffan Romberger & Dan Sjögren (eds.), *The UTOPIA Project: An Alternative in Text and Images. Graffitti: Informationsblad från UTOPIA-projektet* nr. 7 (Aarhus & Stockholm, 1985), 2.
31 Bødker et al., *The UTOPIA Project/Graffitti,* 2.

Democracy and learning together

In the early Scandinavian research projects that came to be influential in proposing and prototyping participatory formats in design, 'participative approaches' alone were not deemed as sufficient to create democratic settings for designing.[32] Inspired by action research, as it had taken shape in the social sciences during the 1950s and 60s – not least as formulated in the 'critical pedagogy' by Paulo Freire in *Pedagogy of the Oppressed* – a strong ideal in early participatory design research was that of making societal and political change not only through design but also through research.[33]

This bringing together of research and design practices in a context of both developing technological tools and simultaneously supporting workplace democracy was initially labelled 'the collective research approach'.[34] The researcher should not, or could not, be a neutral observer but should consider how the own practice would play into societal power relations. In early participatory design, taking a political stance was also strongly associated with the role of engaging as a design researcher, and with the conflation of the roles of 'designer' and 'researcher' into one and the same (compared to practice in user-centered design, where the roles of designer and researcher typically would be separate).

Within the participatory design projects of the 1970s and 1980s, design researchers actively sided with unions and workers as opposed to management, expressly with the emancipatory ambition to change the power structures in the workplace by giving a voice to those who held the weakest position in the decision-making processes that affected their everyday lives.[35] Questions of power relations and empowerment were, then, highly present in participatory design from the very beginning. This strong emphasis on power relations and issues of democracy was also linked to an ethos of accountability, both in regard to the field of design and "the worlds it creates" and in regard to the role of the designer."[36] The aims of propelling a democratic development through participatory design had been grounded in the importance placed on mutual learning and understanding in the design process:

> This way it was thought that the workers and the research group could learn from each other while together building up a new understanding essential for democratization of working life. The understanding acquired in the investigation groups should elucidate and widen the possibilities for action.[37]

32 Ehn, *Work-Oriented Design of Computer Artifacts,* 366: "formal democratic and participative procedures for designing computer artifacts for democracy at work are not sufficient. The design process must also be organized in a way that make it (sic) possible for ordinary users to utilize their practical skill in the design work, and having fun doing this."
33 Ehn, *Work-Oriented Design of Computer Artifacts,* 9; Paolo Freire, *Pedagogy of the Oppressed* (Herder & Herder, 1970).
34 Ehn, *Work-Oriented Design of Computer Artifacts,* 272f.
35 *Routledge International Handbook of Participatory Design*, 6.
36 *Routledge International Handbook of Participatory Design*, 5.
37 Ehn, *Work-Oriented Design of Computer Artifacts,* 283.

In early participatory design projects, there was an ambition that designers and skilled users should not only share easily articulated knowledge but also the types of tacit knowledge associated with their respective practice. Making knowledge accessible, or relateable, through different methods for prototyping and discussing together was seen as the groundwork on which informed collaborative work and relevant negotiations of design decisions could take place. This entailed creating a design space for participatory collaboration, through developing methods such as mock-ups and design games that allowed sharing and building knowledge through doing, at the same time as creating a space from which to have influence in a democratic setting.

In the UTOPIA project, knowledge and skill were the criteria by which users gained influence in the process of designing graphical print tools for highly skilled users working with the layout and graphical production of newspaper printing. The gradation of different types of users, in terms of skill, expertise, or whether they are end-users or not, might be considered as ascribing different values and different possibilities for participation. Again, considering the UTOPIA project, designing with 'skilled workers' had the aim of not only creating qualitatively well-designed tools for professional use but also of excluding non-skilled workers from entering the graphic design or newspaper printing profession.[38]

The ideas and practices aiming towards establishing support for workplace democracy through design, then, became a question of negotiating power and influence for those already included in an established work situation while excluding others that could potentially have been included. The future contexts in which these computerised tools for graphic work should be used were envisioned as situations where new forms of collaborative work would take place. Various professions would engage in work processes based on their respective expertise, and not compete for work on a job market in which computers diminished work possibilities. As a journalist from the Danish newspaper Østlaendingen stated in a debate on the quality of work and product at the UTOPIA project's conference in May 1984:

> The fact that we wish to cooperate obviously means that we wish to continue being journalists and graphic workers. I certainly do not wish to become a graphic worker or anything similar. (…) The new technology has created a borderland for us. We meet there, and our meeting in the borderland requires cooperation, new working practices, new knowledge, new insight, and, to some extent, a new division of labour (…) Training must count more than technology, and for some it will be necessary to do something as central and old-fashioned as improving their general knowledge of the society in which we live and work.[39]

Participation, in terms of deliberation aiming to strengthen unions on both a local and a national level, was also seen as a positive outcome of the UTOPIA project. This was believed to lead to increased union-level cohesion, in joining forces towards employers in the work-related political and technological developments:

38 Gro Bjerknes & Tone Bratteteig, "User Participation and Democracy: A Discussion of Scandinavian Research on System Development", *Scandinavian Journal of Information Systems*, 1995, 7 (1).
39 Bødker, et al., *The UTOPIA Project/Graffitti*, 29.

> If the trade unions involve themselves in research and development projects, and if these projects are successful, they may turn into important and inspiring *demonstration examples* which show the possibilities for influencing the technological development, and which expand the *technological possibilities*. Thus attention will be drawn to the political nature of existing technology, and technological research and development may turn into a political issue, and thus a topic for the unions.[40]

In the context of transitioning to working with computerised technologies in different areas of industry and the public sector, a new way arose for design to come about. In retrospect, the various union initiatives to influence this transition comes across as driving not only design activities but also for new situations that made possible and called for new ways of doing design. At the time, though, the question is if what was being done was considered to be design at all?

In the UTOPIA project, the rationale for inclusion or exclusion in design participation was tied to skill and knowledge. The participatory design methods developed aimed at bringing in skilled graphical workers as experts in the development of computerized tools, strengthening their position in regard to the future use of these profession-based systems in which a certain knowledge and know-how would be required.

This, as participatory design researchers Tone Bratteteig and Gro Bjerknes pointed out already in the mid-nineties, fell back on a sort of knowledge-guarding or gate-keeping tradition firmly embedded in a crafts culture, such as in the guild system, where making and manifesting difference between roles and professions based on skills and knowledge was central.[41] Specific experiences of ways of working, based on knowledge of doing graphical work in relation to the demonstration of specific types of know-how, became the decisive demarcations for inclusion or exclusion in regard to the design process as well as to intended design outcomes.

The context of 'democracy' was tightly connected to a framework of 'knowing' and 'learning' in relation to 'collaborative designing', articulated more or less explicitly and more or less narrowly to a specific situation, or widely to include an overarching societal perspective. In a broader societal context, these issues of democracy, technology, design and development could then hold diverse political and participatory power relations, depending on how wide or narrow the outlook.

Coming back to the attempt to discern from where these entangled ideas of learning and collaboration in relation to democracy might originate in a Scandinavian design context, let us turn to the early decades of the 20th century. It was only during the first two decades of the century that Swedish legislation was passed which allowed men and women of all (or, most) social standings to take part in enacting democratic participation in the form of voting for parliament. The processes towards universal suffrage, in Sweden, were closely linked to initiatives aiming to empower or foster, depending on perspective, people into full citizenship.[42] Issues of learning and participation were central to these processes, and enmeshed in this context were also ideas of the centrality of a material environmental reform as part of the transformation necessary to create a new society.

40 Bødker et al., *The UTOPIA Project/Graffitti*, 43.
41 Bjerknes & Bratteteig, "User Participation and Democracy", 78.
42 Lena Eskilsson, "En arena för möten mellan människor

Let us, therefore, take a closer look at an example drawn from a 'canonical' figure in Swedish turn-of-the-century design history, Ellen Key, who was active in advocating the democratisation of society through design reform. In Swedish design history, Ellen Key's influence is usually described in terms of her engagement in 'everyday aesthetics', as taking a normative approach to teaching people, especially women, the principles of good taste in interior decoration.[43] Her publication *Beauty in the home* [44] has been extensively studied and commented in this respect, with a strong emphasis on her moral and aesthetical judgements of specific objects and environments. Here, we will shift the perspectives on this design historically fairly well-described figure, and apply lenses that enable a search for ideas and situations of participatory practices, such as the coming together with aims of mutual learning connects to ideas of supporting the development of democratic processes.

Ellen Key: Designing a different everyday life

During the 1880s and 1890s, Ellen Key worked in Stockholm as an elementary school teacher. Prior to that, she had also functioned as a political secretary to her father, Emil Key, who was a member of the Swedish parliament since the late 1860s. From 1878, Ellen Key taught in a mixed school open for boys and girls – which was until then quite unusual – that she herself had co-founded with her friend Anna Whitlock. There, Key developed an integrated practice and theory of pedagogics, that later came to influence the founding of both the Montessori and Steiner/Waldorf schools and pedagogies.[45] During these decades, Key also arranged and held open lectures in literature, history and geography at the newly founded Worker's institute in Stockholm (*Stockholms arbetareinstitut*), which was an educational enterprise that was free and openly accessible for the general public interested in furthering their knowledge in the arts and science.[46] Key was an inspiring speaker, and her lectures drew vast numbers, amounting to some ten thousand people per year.

och idéer: kvinnliga medborgarskolan på Fogelstad" in Erland Mårald & Christer Nordlund (eds.), *Topos: essäer om tänkvärda platser och platsbundna tankar* (Stockholm: Carlsson, 2006), 167-182.

43 Brunnström, *Svensk industridesign,* 75; Zandra Ahl & Emma Olsson, *Svensk smak: myter om den moderna formen* (Stockholm: Ordfront, 2001); Linda Rampell, *Designatlas : en resa genom designteori 1845-2002* (Stockholm: Gabor Palotai, 2003).

44 Ellen Key, "Beauty in the Home" [1897], in Lucy Creagh, Helena Kåberg & Barbara Miller Lane (eds.), *Modern Swedish Design: Three Founding Texts* (The Museum of Modern Art: New York, 2008).

45 Ronny Ambjörnsson, *Ellen Key: En europeisk intellektuell* (Stockholm: Bonnier, 2012).

46 Sigfrid Leander, *Folkbildningens födelse: Anton Nyström och Stockholms arbetareinstitut 1880-1980* (Stockholm: Nordiska museet, 1980).

With the public talks, Key established herself as an advocate for social and political change, and her views were even more widely spread through an increasing number of publications. After the turn of the century, she left her teaching career and focused entirely on writing and on giving public talks. Ellen Key demanded access to education as well as to social and political arenas for women and workers, arguing for universal, and especially women's, suffrage. Legal and political rights for women and workers were needed, as well as changes in the relationships between men, women and children, in order to change the dynamics of society as well as those of family life and promote both individual development and the evolution of the human race. [47]

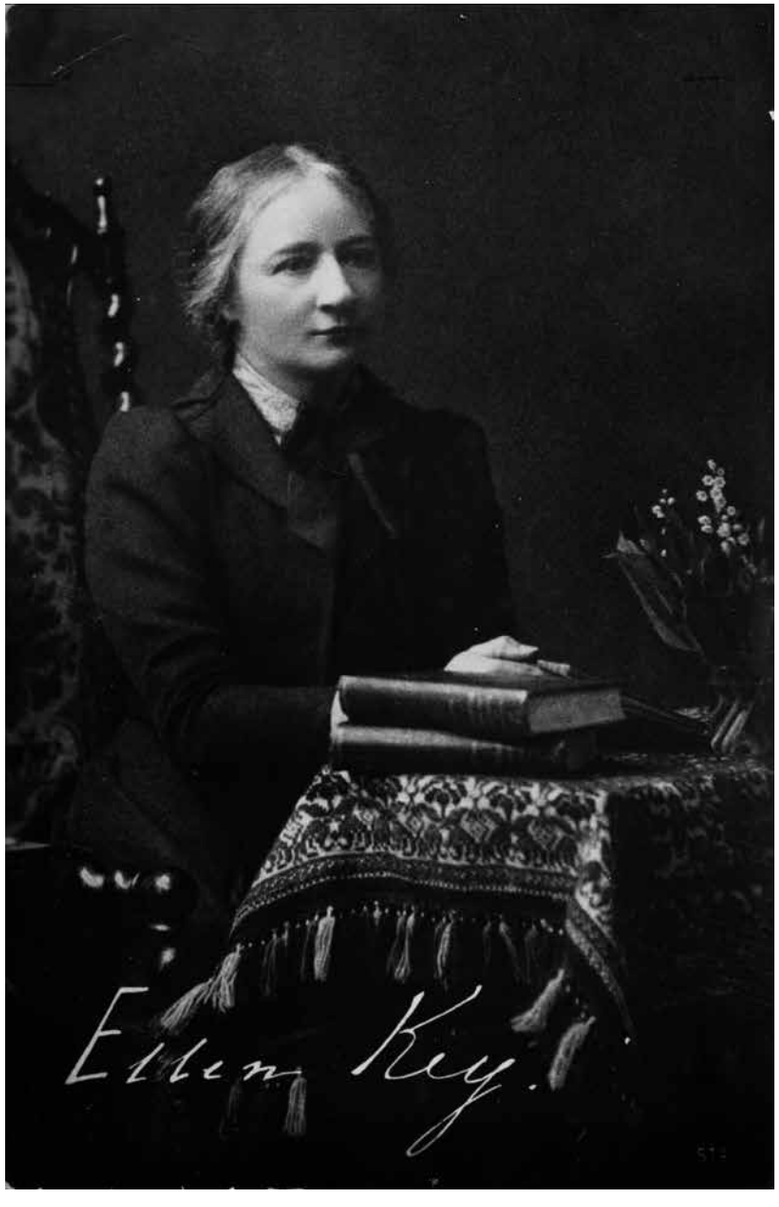
Her later work focused actively on issues of war and peace, envisioning a globally connected world in which a league of nations would work together in ensuring mutual understanding and peaceful cooperation between peoples. By the turn of the century, Ellen Key was a well established radical thinker and public speaker, both in Sweden and abroad. Her publications were translated into several languages, and she regularly toured Western Europe, holding public lectures.[48]

Drawing her pedagogical, philosophical and societal ideas upon the work of Goethe, Rousseau and Montaigne and on more contemporary cultural and political theory – such as the social evolutionism of Herbert Spencer – Key engaged in the socio-political debates of the day in the spirit of the Enlightenment ideals. Learning and knowledge were seen as the foundations for an active and socially engaged citizenship, which in turn was the key to establishing an egalitarian, democratic society. She was one of the leading advocates for pedagogical reforms within the school system as well as for an educational practice aiming towards emancipating the adult population as a whole through access to knowledge as well as to a better material standard of living.

In the years just before the turn of the century 1900, Ellen Key published and lectured widely on the topic of the importance of a new aesthetic of the everyday as a key factor for societal reform. She, as many of her contemporaries, had reflected intensely over how life should and could be lived in the times of profound change and transformation that industrialisation had brought about. For Key, this meant that out-dated norms and roles for women, men and children should be actively challenged, and new ways of living together – in families, as well as in society as a whole – needed to be adapted to other world views and understandings. Society itself was in a state of change, and many were the voices that called for changes also in how life was lived: Class structures, gender norms, education, marriage, sexuality, childhood and religion were issues of discussion, debate and conflict around the turn of the century 1900. And as people increasingly moved from

47 Ellen Key, *Barnets århundrade* (Stockholm: Bonniers, 1900); Ellen Key, *The Century of the Child* (New York: Putnam, 1909). Not all of Ellen Key's publications are translated into English. When I quote Key from a Swedish publication, the English translation is mine, and I provide the Swedish original text in the footnote. When I quote Key from a published English translation, I do not provide the parallel Swedish formulation. And, by the way: her surname is pronounced "Kay".

48 Ronny Ambjörnsson, "Ellen Key and the concept of *Bildung*", *Confero* 2014:1 (vol.2), 133-160.



Ellen Key in 1899. Unknown photographer / Nordiska museet (NMA.0033051)

rural areas to cities and industrialised towns, rented new types of dwellings and furnished them with industrially mass-produced goods rather than home-made or hand-made things, the material conditions of everyday life were changing as rapidly as the social norms and the ways of living.

Key's writings – many translated and spread widely internationally – revolved around pedagogics, socio-political questions, women's rights, and peace issues.[49] Much of her work highlighted the importance of education and learning, for children and adults alike, as the foundation for a citizenship comprising men and women of all social classes. People needed to learn about politics, science, philosophy as well as, art, literature, architecture, music, and 'everyday beauty' – or what later became called 'design'. In the fields of the arts, Ellen Key often made a point of connecting contemporary expressions and developments of content – as in topics or themes – to the broader context of societal transformation.

Just as forms, materials and aesthetics in these various areas were questioned and explored by practitioners in terms of new literary formats, new approaches to the figurative arts, other ways of dancing, etc., the design of everyday objects and environments needed to be actively considered. In a project of constant learning, Key advocated the perspective that truly everyone – including workers, women, children – and not only people of means and money should have access to situations of learning as well as access to beauty and culture in their everyday lives.[50]

This is where Ellen Key brought in ideas of design into the Swedish debate on societal development. She lay forth a proposal, inspired by the arts and crafts ideas of William Morris and John Ruskin, for how design of and for everyday life must be a crucial component in both handling and transcending the impact of industrialisation to create a new society, founded on liberal and democratic principles.[51]

In her work, Key often integrated themes of how aesthetics, taste, ethics and the role of the home in society played crucial roles for human and societal development. Everyday experiences, daily practices and personal choices were highlighted as linked to broader societal and political issues. More specifically, the home and the family were pointed to as central units in the process of transforming practically all aspects of society. Far from being trivial matters, questions of possible

49 To give but some examples of the contemporary English translations, many of which were translated by Mamah Bouton Borthwick and promoted in the US by her and her partner Frank Lloyd Wright: Ellen Key, *The Century of the Child* (New York & London, 1909); Ellen Key, *The Morality of Woman and Other Essays,* (Chicago, 1911); Ellen Key, *Love and Ethics* (New York, 1911); Ellen Key, *The Woman Movement.: Transl. by Mamah Bouton Borthwick. With an introd. by Havelock Ellis* (New York, 1912); Ellen Key, *War, Peace, and the Future: A Consideration of Nationalism and Internationalism, and of the Relation of Women to War. Transl. by Hildegard Norberg,* (New York & London, 1916).

50 Ellen Key, "Skönhet i hemmen"[1897], in *Skönhet för alla* [1913] (Stockholm: Rekolid, 1996); Ellen Key, *Folkbildningsarbetet. Särskildt med hänsyn till skönhetssinnets odling: en återblick och några framtidsönskningar* (Uppsala: Appelberg, 1906).

51 Ellen Key, "Skönhet", in *Tankebilder I* (1898), 3.rd edition (Albert Bonniers förlag: Stockholm, 1922).

everyday individual actions – and especially perhaps those traditionally connected to women's expertise or abilities, such as managing a home and raising children – were not only issues of individual development but were framed as drivers of radical social change that would benefit all people.

Power and political participation

The turn of the century 1900 was a socially turbulent period, in which Sweden was a country initiating a transition from an agrarian to an industrialised society. After years of famine due to failed crops, and in reaction to legislation that did not support religious freedom, large numbers of people had chosen to emigrate – many to the United States. Approximately 1,5 million people left Sweden in search of a better life between the mid-19th century and the 1930s. Within Sweden, living conditions were harsh for many, and both work and housing were difficult to find, in the countryside as well as in urban areas. "The social question", which was extensively debated politically and by philanthropical and other organisations, was formulated as a combination of various societal problem areas including poverty, poor housing conditions, health issues and alcohol abuse, perceived erosion of moral norms, lack of education, and tensions between different social classes.[52]

Providing better living conditions – not least materially – became one of the key areas in focus for organisations and political initiatives with different aims, from halting emigration and promoting traditional forms of social stability to enacting socio-material change towards parliamentary democracy and modernising Sweden.[53] The questions of how material change could enable transitioning into a modern, industrialised, country, without risking social stability and avoiding conflict or even revolution became a uniting element. However, at the core of this also lay fundamental disagreements on which issues should be part of a political solution, and who could and should have the right to participate in influencing political and societal processes.

At this point, large groups of Swedes, based on their class or gender, were excluded from participation in electing parliamentary representatives to the bicameral Swedish Riksdag. Only City and County Councillors – men and women of a certain social standing and with high financial status – could vote for the First Chamber according to a proportional voting system granting persons with land and fortune more influence. At the same time, the Second Chamber members were appointed in direct elections by 'universal' male suffrage. In 1909, any man over 24 who paid taxes, had fulfilled military conscription duty and had not been supported by pauper aid, was legally given the right to vote for the Second Chamber, while women were not at all awarded this right until 1919.

It was not until 1921 that the first general election in Sweden was held, where men and women held equal voting rights to the Second Chamber, and it took yet another twelve years to eradicate the landholding and financial restrictions governing eligibility and proportional votes to the First Chamber. So, at the turn of the century 1900, large groups of the population in Sweden were neither represented nor included in the democratic system.

52 Per Wisselgren, *The Social Scientific Gaze. The Social Question and the Rise of Academic Social Science in Sweden* (Farnham & Burlington: Ashgate, 2015).

53 Anna Lindkvist, *Jorden åt folket: Nationalföreningen mot emigrationen 1907-1925*, diss. (Umeå, 2007).



Ellen Key speaking at a suffrage rally arranged by Föreningen Kvinnans Politiska Rösträtt (FKPR) in June 1915. Calla Helena Sundbeck / Grenna Museum– Andréexpeditionen Polarcenter

The call for new parliamentary forms for representation and influence on an egalitarian and majority foundation was heard from many organisations, political parties, and individual debaters. Counter-voices were fierce, and one of the more substantial arguments against implementing a parliamentary majority principle and extending the right to vote, and thus to legislative as well as societal influence, to both workers and women was that the level of education and knowledge in these groups was far too low to trust them with such power and responsibility. Issues of how to support both formal education and non-school-based situations for learning thus became a central issue to negotiations of how socio-political transformation.

'Bildning' and democracy

Since 1842, legislation in Sweden gave the right for most Swedes to attend six years of *folkskola* (Folk school) free of cost but education beyond this was still a matter for a privileged few, in gender-separated schools, with different curricula for males and females. Many working-class children and rural manual labourers did not even finish the six years of *folkskola* since they were required to help out with work at home or taking on salaried work to help support their families. Many ended their schooling without even basic skills in reading and writing. This was held forth as an argument for *not* extending universal suffrage to masses of people who were perceived as unable to orientate themselves in societal matters of importance. Those who instead advocated the right to claim democratic political influence for all, specifically including women and workers, strongly emphasised that reforms and support were needed to make education accessible for people.

The view that all people could make relevant and essential contributions to society, if only they had the means to formulate themselves and to gain access to both knowledge and arenas for voicing their views led to initiatives to provide arenas outside of the organised educational system to, through access to knowledge, empower men and women as citizens.[54] In this respect, the concept of a comprehensive and holistic knowledge encompassing learning as well as personal experience became formational for the initiating activities of adult education, or '*folkbildning*'.

Central to Ellen Key's thinking, are ideas based on the Humboldtian concept of *Bildung*, in Swedish *bildning*. *Bildning* is a concept that includes both education and learning. It is not exclusively something fact- or knowledge-based but includes subjective experience and formation. The learning processes aiming towards attaining *bildning* comprise many aspects of a person's experience and is a continuous and life-long process.[55] That people of all societal standings have a disposition towards *bildning* – and that some do not – was stressed by Key, in particular with reference to workers:

> the natural disposition towards *bildning* in men and women of the working class is occasionally so strong that they – despite large gaps in their knowledge – have fought their way to a much larger genuine culture, than many so-called learned (*bildade*) which stand blind in front of the times' greatest issues and uncomprehending in front of its' masterpieces in literature and art.[56]

54 Eskilsson, "En arena för möten mellan människor och idéer", 167-182.
55 Sven-Eric Liedman "A lesson for life", *Studies in Philosophy and Ecuation* 2002:21, 328.

Ultimately, a person with the highest degree of *bildning* will reach a state where a holistic overview brings together knowledge, learning, and aesthetic appreciation, in a genuine interest for the world and a "sympathetic understanding" of other people.[57] In a future, where the access to learning and beauty would allow for all people to achieve this, Key envisioned a societal transformation taking shape, which in turn would "build up a higher humanity".[58]

Folkbildning

The concept of *folkbildning* was formulated in Scandinavia at the turn of the century 1900 and largely sprung from within the temperance movement, religious revivalist movements, and within the worker's movement. Within these movements, people would gather in reading groups to share knowledge and learning in order to challenge societal, religious and political norms. Starting with initiatives in Denmark to establish 'folk high schools' on the countryside, *folkbildning* also has a foundation in the concept of *bildning* and the aim to provide access to learning to people (*folk)* in general, and not only to a select few.[59] This spread to the other Nordic countries, and in Sweden became a matter propagated not only by individuals but also widely upheld by non-governmental organisations.

The concept is difficult to translate to English, composed as it is of the words for 'people' and the word 'bildning'. *Folkbildning* is sometimes roughly translated as 'adult education' or 'popular education', but the concept carries a wider meaning than that.[60] Within *folkbildning,* people participate freely and voluntarily in situations of learning together with others, in an egalitarian setting where joint discussion and reflection form the basis for learning together.

The aim of *folkbildning* is on the one hand to support individuals in a life-long learning trajectory, and on the other hand to support 'the people' as a collective in strengthening democratic practices through shared knowledge in forms of public lectures, public libraries, reading rooms and reading groups, evening courses and other types of activities. In Sweden, the activities aiming towards *folkbildning*, and the arenas and material spaces in which these have emerged, have often been emphasized as of importance for the shaping and consolidation of both parliamentary and societal democracy.

Ellen Key was one of Sweden's most avid promoters of *folkbildning*. In her vision, *folkbildning* included integrated learning in literature, philosophy and science as well as in art and aesthetics. Ultimately, Key states, the initiatives aimed at furthering knowledge, learning, taste and other aspects included in the concept of *folkbildning* must serve the broader ambition of bringing about a new society, where material and aesthetic considerations must be integrated and inseparable components.

56 Ellen Key, *Bildning: Några synpunkter*. Studentföreningen Verdandis småskrifter 67 (Bonniers: Stockholm, 1919), 16-17.
57 Key, *Bildning,* 24
58 Key, *Bildning,* 32.
59 Ove Korsgaard, *Kampen om lyset: Dansk voksenoplysning gennem 500 år* (København: Gyldendal, 1997).
60 The Swedish Council for Adult Education, Folkbildningsrådet, still today suggests that the concept be used in Swedish, since its meaning is difficult to convey in translation. https://www.folkbildningsradet.se/om-folkbildningsradet/Oversattningar/English-translations/ (accessed 2019-03-19).

In this socio-political process, universal suffrage as well as access to education were crucial. Without learning and knowledge, people could not become active and responsible citizens. But attaining formal rights was not the only goal. In a process of societal transformation, Key meant, aesthetic qualities were wholly embedded in questions of infrastructure, legality, power and influence and thus had to be explicitly addressed:

> This is the deep societal meaning of beauty, that it teaches us to demand the same fine perfection of man's societal organisations, that already are owned by the mountain's hard crystal and the snow's fast-melting flake – as well as the grand symphony or the cathedral. Only when this is the case, will the power of perfection to create happiness – as in the value of happiness in beauty when morality is accomplished – become fully obvious.[61]

Key's interconnected ideas of societal, aesthetic and pedagogical change built on the premise that spreading and sharing knowledge would serve as a foundation for establishing a democratic societal order. In the process of creating this, *folkbildning* would serve as a means to bridge differences between social classes. To achieve a bridging of the class gap, initiatives towards *bildning* in the realm of learning and aesthetics must aim towards change in the very core and being of people:

> It is first and last a higher *bildning* in terms of being in itself, that the awakening of the sense of beauty should aim towards: a more beautiful will to serve, a quicker sensitivity to others' pleasure and discomfort, warmer words for each other.[62]

For Key, it was in being together and sharing experiences that a true mutual understanding of differences and similarities could take place between people. The understanding between different social classes that such engagements could bring about could form the basis for establishing a different type of society. However, in Sweden at the turn of the century 1900, the places in which people from different social spheres could meet were utterly few. In order to provide places where people together could access knowledge and learning, new arenas were established with the aim of supporting *folkbildning*. Examples included the setting up of the Worker's Institute in the 1880s, initiatives to establish public reading rooms and libraries,

61 Key, *Folkbildningsarbetet*, 92-93. "Detta är skönhetens djupa samhällsbetydelse, att den lär oss kräva samma sköna fullkomligheter av människans samhälleliga organisationer, som redan bergets hårda kristall och snöns snabbsmälta flinga – likaväl som den stora symfonien eller katedralen – äga. När detta blir fallet, då först blir fullkomlighetens lyckodanande makt – liksom skönhetslyckans sedlighetsskapade värde – fullt uppenbar."

62 Key, *Folkbildningsarbetet,* 47. "Det är först och sist en högre bildning i fråga om själva väsendet, som skönhetssinnets väckelse borde åsyfta: en vackrare tjänstvillighet, en snabbare känslighet för andras behag och obehag, vänligare blickar, varmare ord för varandra."

63 Amalia Fahlstedt and Elna Tenow were the co-founders together with Key. Lisbeth Håkansson Petré, *Tisdagar med Tolfterna: Nätverkande kvinnor i sekelskiftets Stockholm* (Stockholm: Stockholmia, 2019).

and open lecture series. These were all primarily aimed towards workers, and some specifically towards women (workers), and thus were not places where diverse social groups would encounter each other and exchange views and experiences.

Tolfterna: Bridging differences

With the aim of creating places where people from different social backgrounds could meet, Ellen Key together with a group of other women in 1892 established an association called *Tolfternas samkväm för kvinnor från olika yrkesområden* ('The Twelfths' social gathering for women from different professional areas'), or just "Tolfterna" for short.[63]

In Tolfterna, twelve groups of women identified as having a good degree of *bildning* (in other words, middle- or upper-class women) each were given the responsibility for arranging two or three events a year, such as lectures or musical evenings, to which women organised in workers' associations were invited. The main aim of Tolfterna was to provide an arena for working-class and middle-class women to meet and talk in a social setting that could increase mutual understanding, and possibly allow for the socialist and liberal women's movements to come together and collaborate more closely.

Another fundamental aim was to provide a setting that would give working-class women access to knowledge and learning, thus laying a foundation for the *bildning* needed in order to fully participate as citizens in society once the suffrage had been gained.[64] This built on an understanding of democracy as a shared endeavour, in which an active search for and discussion based on knowledge of different types is a prerequisite for participation that is not only a system of rules and legislation but a process in which all citizens play an active part.[65]

Linking knowledge, learning, and taking part in a democratic process in the context of *folkbildning* aimed towards overcoming class differences and fostering participatory citizenship bears strong similarities to approaches in later participatory design practices of the 1970s and 1980s. Another similarity is the focus on supporting actions that provide people in a non-privileged position to have or gain a voice through accessing a language tied to a particular cultural, social or professional sphere. In the words of historian Christina Florin, becoming a citizen is to participate in a project of recognition, in which having a language, a voice and a recognised identity is fundamental for participation in society.[66]

This gaining of a language was, in Tolfterna, closely tied to knowledge of and exposure to learned or literary texts. Invited authors or academics would hold a short, maximum 20-minute, lecture summarising a novel or a topic, after which

64 Johanna Ringarp & Karin Carlsson, "Tolfterna – ett bildningsprojekt inför kvinnornas fullständiga medborgarskap?" in Peter Brubech, Ketil Knutsen & Jens Aage Poulsen (eds.) *Historie – didaktik, dannelse og bevisthed. Rapporter til det 29. Nordiske Historikermøde, Bind 1*. (Aalborg, 2017), 9-39.

65 Sven-Eric Liedman, "Democracy, knowledge and imagination", *Studies in Philosophy and Education* 2002:21, 353-359.

66 Christina Florin & Lars Kvarnström, *Kvinnor på gränsen till medborgarskap: Genus, politik och offentlighet 1800-1950* (Stockholm: Atlas, 2001).

sometimes games and musical performances would take place. The idea was to encourage conversation between working-class women and the middle-class women, which would lead to "mutual exchange of different areas of experience promoting mutual understanding".[67]

However, in these sessions, it was often noted that the working-class women were reluctant to talk or contribute to the conversation – and that it was difficult to find topics for the gatherings that were not "political" and potentially generating conflict.[68] Finding an adequate balance between an interesting program for the evenings, and mechanisms that sparked active conversation, was not always easy:

> Because, making both *joint* conversations happen, and individual ones between invitors and the invited, is the wish towards which the endeavors should go, a goal, which the far too rich programmes often counter-act.[69]

So while the ambition was to create arenas bridging class barriers in the aim for shared experiences, mutual understanding, and discussion of social issues, the participation in Tolfterna was highly conditioned by power structures manifest in language as well as materially and through the format of the gatherings. The arranging women were called "the invitors" (*inbjudare*) and the working-class women "the invited" (*inbjudna*). Despite the aim of being a shared participatory conversation, already the ownership manifest by the knowledge-base needed to be an "invitor" created a situation of inequality.

The explicit goal for the Tolfterna gatherings was to create a situation where all participants actively took part in conversation and debate. In a comment to the minutes following the sixth gathering, in 1892, Ellen Key notices that both invited and invitees agree that the purpose of the events was "mutual exchange of different areas of experience with the aim to further mutual understanding; it is so to speak 'in the air' that these gatherings between women from different areas of work is demanded of the times".[70] However, the minutes from the Tolfterna gatherings

67 Ellen Key quoted in Ringarp & Carlsson, "Tolfterna – ett bildningsprojekt inför kvinnornas fullständiga medborgarskap", 25. "...ömsesidigt utbyte af olika erfarenhetsområden till främjande af ömsesidig förstående"

68 Ringarp & Carlsson, "Tolfterna – ett bildningsprojekt inför kvinnornas fullständiga medborgarskap", 24-25.

69 Ellen Key, comment to the meeting minutes of the Tolfterna meeting 16 October 1895. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Första protokollsboken. 1. Samkväm för kvinnor från skilda yrkesområden, 1892-1893. "...ömsesidigt utbyte af olika erfarenhetsområden till främjande af ömsesidigt förstående; det "ligger i luften" s.a.s., att dessa samkväm mellan kvinnor af olika arbetsområden äro ett tidens kraf."

70 Ellen Key, comment to the meeting minutes of the Tolfterna meeting 13 December 1892. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Tredje protokollsboken hösten 1895, 1895-1896, 9. "Ty att få till stånd dels gemensamma samtal, dels enskilda sådana mellan inbjudare och inbjuda, är det önskemål åt hvilket sträfvan bör gå, ett mål, som af de allt för rika programmen ofta motverkar. "

repeatedly pointed to that the working-class women often seemed shy, or were reluctant to fully participate. As one of the Tolfterna 'invitors' noted, the more often the women gathered, and the more familiarity grew with each other and with the venue for the meetings, the easier the "mutual approaching" of each other.[71]

Several of the entries in the minutes bear witness to the underlying tensions between middle- or upper-class women and the women workers that in discussions would sometimes become explicit. In a discussion on the topic of collective kitchens, it was noted that this was something that engaged the working-class women so much, that the evening became more of a heated debate than a casual discussion. Due to this, it was advised that future topics should be chosen with greater caution to avoid "the repressed feeling of resentment over the extensive class difference, that is so commonly spread among the working class, that easily flares up and is expressed in bitter statements."[72]

In the notebooks from the Tolfterna, several rules for the arranging groups of women are added over time, aiming to disarm any potential issues of conflict through subduing some of the all-too-obvious differences between the participating women. Thus, the upper- and middle-class women were reminded not to wear too ostensive clothing, and to address each other as well as the working-class women with the pronoun *Ni* ('You') rather than with the more formal address of title and surname.

These measures were meant to create an atmosphere that emphasised similarities and closeness, rather than differences and distance, as the basis on which to build joint knowledge about each other's' situation. However, in the minutes from the meetings, the comments mainly regard how the working-class women come to understand more about the middle-class women's conditions, rather than the other way around. The revelation that the workers are more envious of the "spiritual pleasures, that are so much more easily accessible" to the middle class, than their finer clothing or better food is noted in a quote, as is the statement from a women saying that the gatherings lead to a new view on many things: "...in one's solitude, one imagines so much about the so-called upper class, but when we meet like this, one can see that one has been mistaken in many preconceptions."[73]

71 Natalia Frölander, meeting minutes of the Tolfterna meeting 7 March 1893. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Första protokollsboken. 1. Samkväm för kvinnor från skilda yrkesområden, 1892-1893.

72 Natalia Frölander, meeting minutes of the Tolfterna meeting 28 November 1893. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Första protokollsboken. 1. Samkväm för kvinnor från skilda yrkesområden, 1892-1893, 67. "...om man ej med försigtighet väljer ämne och noga begränsar detta, de undertryckta känslor af harm öfver den stora klasskillnaden, som äro så allmänt utbredda bland arbetsklassen, lätt flamma upp och taga sig uttryck i bittra uttalanden."

73 Amalia Fahlstedt, meeting minutes of the Tolfterna meeting 3 October 1893. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Första protokollsboken. 1. Samkväm för kvinnor från skilda yrkesområden, 1892-1893, 67. "... man får en ny syn

So while the project of the Tolfterna had both an emancipatory and participatory aim, the formats of the situation and the knowledge and *bildning* promoted were conditioned by the privileged position of the middle-class women.[74] Ellen Key, in the minutes to the meetings, noted that the conversations between the women from different social backgrounds would be difficult, mainly due to the incapacity for some of the middle-class women to "go beyond their own ether and in a natural way be able to socialise with any and all sorts of people."[75]

Despite the attempts to create an inclusive and open environment, that would support the sharing of experiences, the difficulties that often arose in finding ways to ensure that all participants were equally engaged in the conversations come across in the notes and reflections on the gatherings. As one of the Tolfterna members, and 'invitors', Nanna Frölander reflected on after a gathering:

> On the other hand, it was, as usual, difficult to enable conversations in smaller groups and above all to see everyone talk. It is often hard to find common touchpoints: the members of the Tolfterna are hesitant to engage in the questions that are most pressing for the workers, and these do not feel, as would be natural, much for that which interests the former. Would perhaps not this difficulty lessen, if the members of the Tolfterna among the workers and their own questions and interests prepared beforehand by thinking about something, that could be well suited as a topic of conversation?[76]

As the gatherings arranged by the Tolfterna continued, the frameworks surrounding them continued to be tweaked to try to find ways to foster more collective engagement and conversation. The suggestion proposed by Frölander, above, became added as a general rule for the meetings, stating that a specific responsibility lay on the members of the Tolfterna, as organisers and leaders, to ensure that the conversations be energised through not only considering the contents of the program but also to

på så mycket; man inbillar sig i sin ensamhet så mycket om d.s.k. öfverklassen, men när man råkas så här, får man se att man har tagit miste i många fördomar"

74 Johanna Ringarp & Anne Berg, "Borgerlig feminism eller klassolidaritet? Tolfternas bildningssamkväm vid förra sekelskiftet", in Kenneth Abrahamsson, Lisbeth Eriksson, Mats Myrstener, Lena Svensson & Tore Persson, *Folkbildning & forskning: Årsbok 2019* (Stockholm: Förening för folkbildningsforskning, 2019), 17-26

75 Ellen Key, comment to the meeting minutes of the Tolfterna meeting 69, 14 April 1896. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Tredje protokollsboken hösten 1895, 1895-1896, 32. "Mycken urskiljning bör fästas vid dem, som väljas till tolftemedlemmar, ty ett stort fattigdomsbevis för den s.k. bildade kvinnan är hennes stora oförmåga att gå utom sin egen eter och att på ett naturligt sätt kunna umgås med hvad slags menniskor det vara må."

76 Natalia Frölander, meeting minutes of the Tolfterna meeting 9 October 1894. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Första protokollsboken. 1. Samkväm för kvinnor från skilda yrkesområden, 1892-1893, 57. "Däremot var

"somewhat prepare themselves to find topics of conversation. And here over all the invitees' active participation this evening, which the entertainment aims towards."[77]

Gaining a voice, and gaining access to the arenas of societal influence was, for the working-class women, a process of learning the language and forms of argumentation already established in a middle-class setting. For middle-class women, the challenge consisted of acknowledging one's own privilege, in terms of material and cultural power, and trying to find ways of interacting with women who did not share neither the same frames of reference nor the same ways of socialising. The situations created in Tolfterna seemed, on surface level, to be about specific topics – but the substantial content and ambition of these gatherings was to create a place where women would gain new perspectives through reciprocally learning about each other's lives and experiences, to come together in the joint struggle for political power.

Learning together: Study circles

The focus on learning together, on mutual understanding, and on caring for each other, was central to Key's ideas about what constitutes a good society. Participation in building an egalitarian society must, therefore, start with the understanding of one's own situation, abilities and possibilities in relation to others'. For people to be able to begin learning to know themselves, with the aim to understand and contribute to the shared project of societal well-being, school pedagogy must change as well as how knowledge was brought to the masses of adults who had not had the opportunity to go to school. In Key's view, the ability to learn lies with the individual, who through reading and active reflection must take responsibility for the own learning process. Input and impetus for learning can come in different ways, and the main ambition within the *folkbildning* movement was to provide new arenas, situations and methods where this could take place.

Folkbildning was envisioned as a collective project, where the emphasis was on people's active participation in learning together with others rather than being taught or educated.[78] However, in order for this to take place, people

det som vanligt svårt att få till stånd samtal mellan smärre grupper och framför allt att få se alla språka. Det är ofta svårt att finna beröringspunkter: tolftmedlemmarna tveka att inlåta sig på de frågor som äro brännande för arbeterskorna, och dessa känna, som naturligt är, ej mycket till dem som intressera de förra. Skulle månne ej denna svårighet minskas om tolftmedlemmarna bland arbeterskornas och sina egna frågor och intressen på förhand tänkte öfver något, som kunde lämpa sig till samtalsämne?"

77 Rules for the Tolfterna gatherings. Kungliga biblioteket. J10: Tolfternas samkväm för kvinnor ifrån olika yrkesområden. J10:3, Protokollböcker 1892-1964. Tredje protokollsboken hösten 1895 (1895-96). "XII. Samtalets tillgodoseende bör hållas energiskt i sikte genom ett lämpligt och ej allt för rikhaltigt program, samt genom att gärna såväl tolftledare som tolftmedlemmar en smula förbereda sig till att finna samtalsämnen. Och här öfver huvud taget de inbjudnas aktiva deltagande i afton: underhållningen syftar till."

78 Samuel Edqvist, "Det pragmatiska folkbildningbegreppet", in Kerstin Rydbäck & Henrik Nordvall (eds.) *Perspektiv på*

needed access to literature, and to the arts – and also often to basic education or introductions to areas of knowledge. Instating accessible places for learning, open to all, brought about not only folk high schools but also lecture associations, libraries, exhibitions and from the early 20th century through the establishment of adult education associations (*studieförbund*). The temperance movement ran many of these, as did several political organisations – initially mainly the workers', women's and liberal movements.

The aim was that these initiatives should provide the main infrastructures for learning to take place. This also called for people to actively engage in taking action, speaking, and collaborating in other ways than usual. The *folkbildning* idea was envisioned to actively connect to people's everyday experiences, with the aim was to expand the understanding and analysis of current realities:

> The movement must be led by the people's true needs and aimed at the current reality more than on past times, foreign lands and subjects alien to people at the moment.[79]

Learning must, therefore, be grounded in everyday activities and experiences. This was at the basis of Ellen Key's pedagogy, and an idea shared by the many organisations and persons promoting new arenas for learning around the turn of the century 1900.

The system of 'folk high schools' initiated in Denmark by the temperance and religious revivalist movements had, according to Key, been influential in promoting *bildning* but too weak in their foundation in knowledge. In Sweden, on the other hand, the situation was the opposite: The folk high schools were much too focused on imparting knowledge, and lent too little attention to the broader aspects of *bildning*.[80]

The founding of folk high schools had as its aim to provide both education and learning for people who did not have the opportunity or means to attend other forms of (higher) education. The underlying purpose of these schools was to "stand free on the broad foundation of full democracy, uniting youth from the industries and the cities as well as from the countryside."[81] However, the work to achieve *folkbildning* was not limited to schools and lectures. Other phenomena listed by Key as necessary for this work, were open libraries, musical performances and folk music competitions (*spelmanstävlingar*), sports associations, and art exhibitions. Nevertheless, the perhaps most central form of learning support was the study circle.

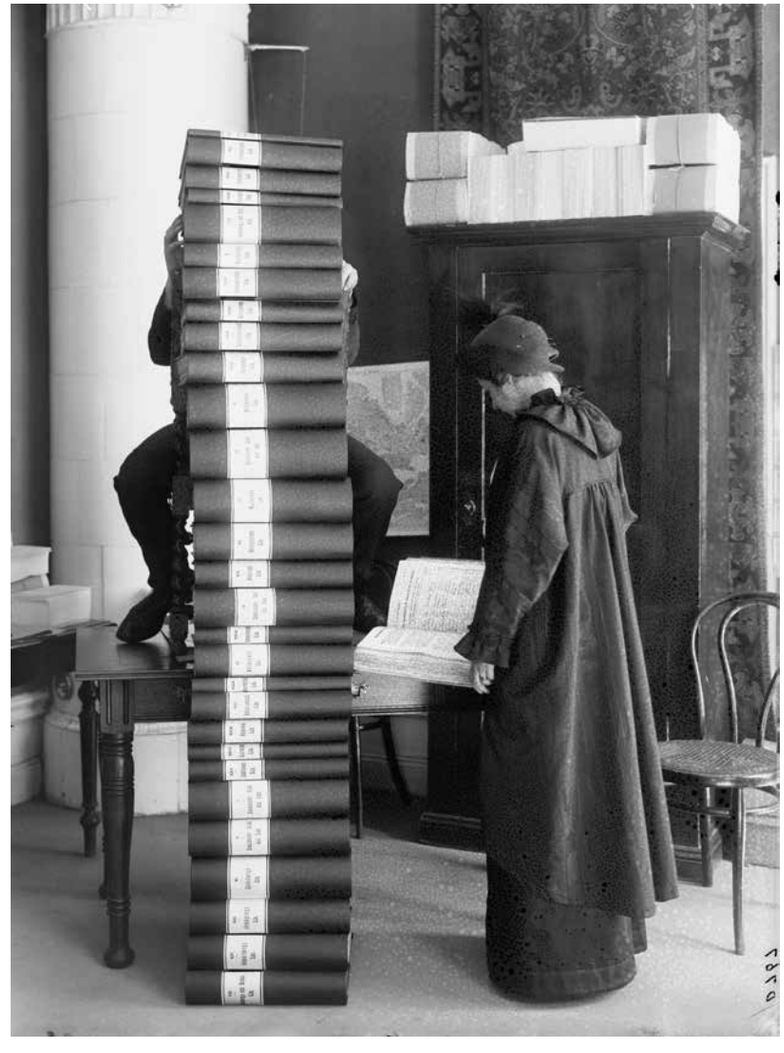
This was a new format, introduced and used above all in the temperance and revivalist movements for studying religious literature as well as readings in more secular and social issues. One of the circle members, or an invited speaker, would hold an introductory talk on a particular theme, followed by a joint discussion

folkbildning. Fyra forskares tankar om folkligt bildningsarbete, Mimer: Nationellt program för folkbildningsforskning (Linköping: Linköping University, 2015) 17.

79 Key, *Folkbildningsarbetet,* 48. "Rörelsen måste bli ledd av folkets äkta behov och inriktad på nutidsverkligheten mer än på forntiden, på främmande land och på för folket tillsvidare främmande ämnen".

80 Key, *Folkbildningsarbetet*, 59.

81 Key, *Folkbildningsarbetet*, 60. "…att folkhögskolan skall stå fri på den fulla demokratiens breda grundval, samlande ungdom från industrien och städerna såväl som från landsbygden."



Feminist theorist and writer Elin Wägner in 1914 next to the binders containing petition lists for women's suffrage. Photographer unknown / Stadsmuseet i Stockholm (D 767)

on the selected topic. Key describes the study circle as a result of "many failed attempts" to find formats suitable for learning together, mainly within the temperance movement.[82]

Already in the mid-19th century, so-called 'circles for *bildning*' (*bildningscirklar*) had been formed with aims to empower workers, while at the same time averting eventual revolutionary tendencies. Through self-reflection and learning together with others, the circles were a response to changing relations in society, where mutual understanding would foster a sense of responsible citizenship among the working class.[83] When study circles later, around the turn of the century, were introduced, these picked up on the format established in the *bildningscirkel*. Ellen Key, however, emphasised that the difference between the previous educational attempts and the format provided through study circles, is that the latter aims towards "joy in knowledge" (*kunskapsglädje*) rather than "school learning" (*skolvetande*).

Only through self-reflection in light of personal relations – between people, and between people and their "spiritual leaders" in science, art and poetry – can a true interest in and joy of learning be sparked. Therefore, fostering egalitarian relations that would allow people to understand each other was crucial to the *folkbildning* ideal promoted by Key. When Ellen Key described the study circle, she explained its format as well as its aims for those who had not encountered it previously:

> The study circle of course has neither 'lessons' nor 'meetings': it has social gatherings, where the members in intimate conversation exchange their thoughts on what the great thinkers and poets have provided them, on what goes on and is done in society, etc. The comings and goings of one's neighbor are here left aside, the self-confident shouting in agitation and discussion is here reduced to its correct proportions, as the issue here is not to be right but to achieve clarity. Interests are broadened and deepened, the tone of intercourse is altered, not only in the circle, but also in daily social activity, which yields more exchange than before.[84]

In a note quoting a 1912 newspaper article, Ellen Key suggested that study circles, when possible, should be located in the home of one of the participants.[85] So doing, the study circle could be a complement to lectures and educational activities held in more public venues. This would make it easier for larger numbers of people to find opportunities to discuss together, without having to find public arenas for this, and the circles would more easily be combined with activities

82 Key, *Folkbildningsarbetet*, 125.

83 Frans Lundgren, *Den isolerade medborgaren: Liberalt styre och uppkomsten av det sociala vid 1800-talets mitt,* diss. (Hedemora: Gidlunds, 2003), 193–243.

84 Key, *Folkbildningsarbetet*, 125–126. "Studiecirkeln har naturligtvis varken 'lektioner' eller 'möten': den har samkväm, där medlemmarna i förtroliga samtal utbyta sina tankar om vad de stora diktarna och tänkarna givit dem, om vad som händer och uträttas i samhället o.s.v. Nästans görande och låtande får här vara i fred, det självsäkra agitations- och diskussionsskränet reduceras här till sina rätta proportioner, då det gäller inte att få rätt utan få klarhet. Intressena vidgas och fördjupas, umgängestonen lägges om, inte bara i cirkeln utan i det dagliga umgänget, som lämnar större utbyte än förr."

85 Key, *Folkbildningsarbetet*, 63-64.



Reading circle at the Olsson family, Marielund. Year unknown. Oscar Färdig / Bohusläns museum. (UMFA54429:4021)

including music, reading together, and playing games. This could also provide a way to increase the already growing number of study circles arranged.

Central to Ellen Key's ideas about learning was that it is born out of genuine curiosity and a personal connection to the situation or the subject. In this respect, the study circle was "the most appropriate form" to work with *bildning* for adult women and men in an organised manner, since it was based on each person's motivation to contribute to furthering knowledge individually as well as collectively.[86]

In arguing for the need for study circles to promote awareness in social issues, that ultimately might lead to the creation of a new societal order, Key quoted several persons central to the Swedish social-democratic political sphere. In the work leading towards a radically different society, where class structures would be upheaved, there has to be a mutual will from both "labourers" and "academics" to understand and trust each other. If not, and if the "new social order would happen through the temporary rawness, ignorance or half-*bildning* temporary victory", this would be the most destructive reaction the world had hitherto seen.[87] Radical change was imperative, but not through outright revolution.

What was needed was a new order realised through mutual understanding between manual labourers and "academically learned", which in turn must be based on knowledge and *bildning*. Key pointed to accessibility to better material conditions in parallel to education and development in everyday aesthetics as central aspects in the mutual learning and understanding that must be fostered to avoid potentially destructive paths:

> And therefore there is no other side of the sense of beauty, of which the awakening would be more indispensable than the *sense of societal beauty*, the sense that carries with it an eye for the beautiful, the valuable, in each age, in each area's particularity and thus awakens the will to mutual *collaboration* instead of mutual *opposition*.[88]

Beauty, bildning and everyday life

Ellen Key, influenced by German philosophy and aesthetics, adhered to the idea that the 'national spirit' (*folksjäl)* was the source or basis of creative practice and appreciation. This spirit was manifest and expressed in different artistic practices – music, literature, arts, crafts – and is renewed through collaborative work. In rural areas, the traditional practices would remain stronger and slower in regard to change, and constitute real and proper styles. In the cities, on the other hand, the fashion would change quickly and people often "satisfied with candy store trinkets, while the strong and stylish handcrafts still ennobles and beautifies the farmer's home."[89] So, in bringing arts-based appreciation and understanding to broader groups of people, different strategies and pedagogics must be applied.

86 Key, *Folkbildningsarbetet,* 125.
87 Key, *Folkbildningsarbetet,* 75.
88 Key, *Folkbildningsarbetet*, 99. "Och därför är det icke någon sida av skönhetssinnet, vars väckelse vore mer oumbärlig än *sinnet för samhällsskönhet*, det sinne, som har med sig blick för det vackra, det värdefulla i varje ålders, varje arbetsområdes egenart och sålunda väcker viljan till ömsesidig *samverkan* i stället för ömsesidigt *bekämpande*."
89 Key, *Folkbildningsarbetet*, 113. "...medan stadsbon snabbt följer modets växlingar och nöjer sig med gottköpsgrannlåt, när det starka och stilfulla handarbetet ännu förädlar och förskönar bondehemmet".

All members of a society need to both practice and access the arts. Thus, the differences in availability to these for people from different social strata needed to be considered and catered for. The democratic ambitions in making art and beauty accessible in everyday life, for Key meant that all people must be granted the opportunity to "rise and develop, so that everything – the largest as well as the smallest, the building for centuries as well as the glass for this day – becomes an expression for the sense of beauty in those who consume as well as in those who bring forth."[90]

In her discussion on how art could be popularised in order to reach everyone and not only a privileged few, Key referred to initiatives by English 'artists' William Morris and Walter Crane who, "permeated by an eagerness to social renewal and responsibility" have tried to make their craft production cheap and available for "the little people".[91] In Key's view, this project had not succeeded in making the arts-and-crafts aesthetic or idea popular, since the great masses did not embrace the art or items produced:

> This mass still probably prefers the 'cute' girls and well-dressed children on the colour prints, before one of Crane's woodcuts; their fake, ugly furniture and ornaments before the simple, genuine and stylish beauty in the wallpapers, textiles and home utensils, that have emerged under the influence of the new English arts direction.[92]

Since the project was to make beauty a part of everyone's everyday life, there was no point, Key meant, in trying to adapt or adjust a certain aesthetic to a certain class of people. The goal was instead to foster a new collective sensitivity to art, music, architecture and everyday items that was ultimately based on an understanding of how beauty and learning are intertwined, in the work of creating a society founded on mutual support and solidarity.

Here, the key to realisation lay in work within the realm of providing access to knowledge and learning. For, as Key stated, "[i]t is not only so, that the uneducated do not have the possibility to appreciate the genuine in art. They often do not even have the ability to understand what it means."[93] Learning, knowledge and social

90 Key, *Folkbildningsarbetet,* 114. "Det betyder, att alla skola höjas och utvecklas, så att allt – det största som det minsta, byggnaden för århundraden som glaset för dagen – blir ett uttryck för skönhetssinnet hos de förbrukande lika väl som hos de frambringande."
91 Key, *Folkbildningsarbetet*, 113. "Man brukar – på tal om konstens popularisering – framhålla, huru de engelska konstnärerna, en William Morris, en Walter Crane, varit genomträngda av social nydaningsiver och ansvarskänsla, och huru de därför strävat att göra sin vackra konstslöjd billig och tillgänglig även för småfolket."
92 Key, *Folkbildningsarbetet,* 113. "Denna massa föredrar nog alltjämt de 'söta' flickorna och välklädda småbarnen på färgtycken framför ett av Crane's träsnitt; sina oäkta, fula möbler och prydnader framför den enkla, gedigna och stilfulla skönheten i de tapeter, tyger och husgeråd, som uppstått under inflytande av den nya engelska konstinriktningen."
93 Key, *Folkbildningsarbetet,* 114. "Det är icke blott så, att de

awareness, therefore, needed to come together in order for people to gain an understanding how these are intrinsically linked: "It takes culture to learn in this way to discern the true value from the worthless, the greater values from the lesser."[94]

The reason for the arts-and-crafts movement's failure in creating societal change through combining socialism and aesthetic reform was, according to Key, due to the insistence on promoting crafts while rejecting the potential of industrial production. This had continued to reinforce the existing inaccessibility to reasonably priced goods of useful and beautiful everyday things for many.

Ellen Key instead saw industrial mass manufacture as a means to further increase the spreading of a more appropriate type of goods to wider groups of people – if only a conscious attempt were made to unite beauty, purpose and quality of production. The late 19th-century view that industry and beauty were unreconcilable was contradicted: "Every day now brings new testimony to the possibility of uniting the beautiful with the purposeful."[95] While some industrial products, such as trains, still "retain their original ugliness", Key referred to attempts to integrate design in the transportation industry: "…an automobile factory already has requested an artist to seek to give the new collective means for transport a form that pleases the sense of beauty."[96]

The introduction of industrially based technologies in the production of everyday goods was, for Key, the decisive factor in the potential reform of not only material things but of social and societal life as a whole. There is a reciprocal relationship between the material and the spiritual or intellectual, which is why questions of form and beauty were crucial to address in a context of *bildning* and empowerment. True liberation and transformation of society would only take place when there had been a reform also of the material world through designing beautiful and useful things that would be industrially produced and made available:

> Only when there is nothing ugly available for sale, when beautiful things are as inexpensive as ugly ones are now, can beauty for everyone be fully realized.[97]

The good life

In Ellen Key's thinking, aesthetic reform must go hand in hand with any attempts to change the social and political situation in Sweden, in order to achieve a truly good life for everyone. Satisfying people's hunger for a life in beauty was

obildade ej ha möjligheter att uppskatta det änkla inom konsten. De ha ofta ej ens förmåga att förstå, vad den betyder."

94 Key, *Folkbildningsarbetet*, 117–118. "Men det fordrar kultur för att lära sig sålunda urskilja det verkliga värdet från det värdelösa, de större värdena från de mindre."

95 Key, *Folkbildningsarbetet,* 86.

96 Key, *Folkbildningsarbetet*, 86. "…en automobilfabrik redan anmodat en konstnär att söka ge det nya samfärdsmedlet en skönhetssinnet tilltalande gestalt".

97 Key, "Beauty in the Home", 35. Swedish original quote from Ellen Key, "Skönhet i hemmen", in *Skönhet för alla* (1899), 5: "Först då intet fult finnes att få köpa; då det vackra är lika billigt som det fula nu är, kan skönhet för alla bli en full värklighet."

comparable in urgency to stilling physical hunger.[98] Through bringing beauty into everyday life also for the poorer people, one could both counter the allure of drunkenness, and contribute to creating a home environment that would be more beneficial to families' well-being both physically and socially.

At the beginning of the 20th century, state initiatives aiming to improve not only the quality of dwellings but also the home environment in general was seen as an essential counter-measure against mass emigration. To propose reforms aiming at changing the living conditions of workers and farmhands therefore fit into the socio-political agenda on a national level, and made possible the inclusion of beauty and the design of everyday things – and everyday life – into this discourse. Besides being the main obstacle to the workers' "self-education and development of the sense of beauty", the poor and cramped housing conditions was one of the areas in which class differences remained very materially present. In diet as well as in clothing, the differences between richer and poorer were perceived to be diminishing, but the standard of the dwelling for the working class was still described to be "in a state of bankruptcy".[99]

Key saw the connection between material manifestation and a person's inner state – in terms of development, learning, ethics and *bildning* – as reciprocal. The "inner relationships are highly affected by the outer", and a person surrounded by or dressed in, taste and harmony would both act and genuinely become more dignified and refined.[100] It was from this perspective of connectedness and reciprocity that design for Key became a crucial element in the re-forming of society into something completely different. Designing for new means of production, in combination with changing values, norms, and social structures could make other ways of living accessible for everyone.

Increased knowledge and sensibility in regard to beauty, and refinement in taste, would lead to improvements in several areas. Not only would "raw ugly acts" become impossible if art were to ennoble people: the level of morality would also be heightened. Additionally, the Swedish economy would be strengthened, people would be able to work harder, and the items produced in industry and crafts would hold a level of refinement that would increase the value of its production.[101]

Things were seen as carriers of an ability to convey both beauty and ethics and should serve as an impetus for developing a higher moral standing on individual level, causing an ennoblement of actions as well as of thinking. This, Key meant, was something that persons with a certain level of knowledge and *bildning* in beauty were already aware of but that people living in a "state of aesthetic wildness" – here, she refers to her critics amongst idealist philosophers and socialists – were as unaware as heathens are of the word of God.[102] Creating a truly egalitarian society where "a sense of community between all members of the

98 Key, *Folkbildningsarbetet*, 122. "They, who so speak, forget that it even in the class that still is constricted to live in need and unsightlyness, there are many whose soul's longing for beauty is as consuming as the bodily hunger." / "De, som så orda, glömma att det även inom den klass, som ännu nödgas leva i nöd och oskönhet, finnes många, hos vilka själens skönhetslängtan är lika sugande som den kroppsliga hungern."

99 Key, *Folkbildningsarbetet*, 123.

100 Key, *Folkbildningsarbetet,* 123.

101 Key, *Folkbildningsarbetet*, 126.

102 Key, *Folkbildningsarbetet*, 127.

people" would have to be founded in a common understanding of spiritual values, where access to and understanding of art and beauty were crucial.[103]

Aesthetics and ethics were, for Ellen Key, inseparable and intertwined in all aspects of everyday life. The new times were defined by changing needs and practices, which in turn should be reflected in new ways of making and using things relating to these new needs:

> The aim of the sense of beauty is and remains to be the ability, through beautiful usefulness, to satisfy the needs that in all times are alike as well as the needs that in every time are new.[104]

Attention to the ways that all material aspects of everyday environments supported and influenced a person's learning and overall abilities for reflection and appreciation would constitute and further *bildning*, and thus the abilities to form the personal opinions and views necessary for an informed contribution to the collective good.

Questions of the material production and consumption of everything from household objects to city planning, therefore, must be considered in terms of reform work needed to create a new social order as well as a fundamentally democratic society. For Ellen Key, the changes in contemporary life and lifestyles were central for bringing about a new society. Among the reforms needed to create this, were the design and production of goods, the use and decoration of homes, the social interaction between people, the forms for living and being together, the rights for children to be taken seriously, and a change in the roles men and women play in society. These were all interconnected, and must organically work together in order for change to come about.

The foundation for change rested in access to knowledge, that would allow people of all social standings to question current orders, and to be able to choose other ways of being than those seen as 'traditional' or seemingly 'natural'. Only when all people had the ability and the possibility to realise new ways of living, materially as well as ideally, could these choices could be made collaboratively – whether in consensus or disagreement. Questions of how everyday things should be designed, manufactured, used, and appreciated were situated in a much broader and more intricate context encompassing ideas of the potential for everyone to be able to make a good life for themselves: morally, aesthetically, and practically.

For Ellen Key, the overarching question was not that of which objects are more beautiful than others, which furniture to buy, or which taste is better than another. The most important issue at hand was that of which type of society one would wish to build, and which practices – social and material – should be instrumental in creating the changes needed to draw up the trajectory that would lead into a preferable future.[105]

103 Key, *Folkbildningsarbetet*, 129.
104 Key, *Folkbildningsarbetet*, 172. "Skönhetssinnets mål är och förblir att genom vacker ändamålsenlighet kunna tillfredsställa dels de i hvarje tid lika med dels äfven de för hvarje tid nya behofven."
105 Claudia Lindén, "Förord av Claudia Lindén: Ellen Keys feministiska estetik", i Ellen Key, *Skönhet för alla* (Ellen Keyinstitutet: Ödeshög, 2006), 9.

Whether or not modernity was preferable to tradition, or crafts to industry, was not the primary issue. Rather, Key posed the questions of *what type* of modernity one would wish to have, and *which types* of practices – aesthetic as well as ethical and social – should be made possible through doing things differently.

Crafts were not for Key inherently better than industrial production since the modes of production and their outcomes in material form could fulfil different needs, both during the process of production and in everyday use. "Our times", she stated, "have of course brought many new needs and many new means to satisfy them."[106] This was something that Key came back to in several of the texts in which she addresses aesthetic, social and material practices in the home.[107] That the aim was to achieve a radical change of society as a whole through new aesthetic practices and material changes in everyday life is perhaps most clearly formulated by Key in the essay "Skönhet" (*Beauty*), first published in 1898, the same year as "Beauty in the home":

> And when one has succeeded in this endeavour [creating beautiful and useful designs of public places and environments], will it not perhaps be the case, that the sense of beauty shall present itself powerful enough to force man to reform the current society itself? [---] No one shall then, for themselves or for others, be able to endure the torment of an ugly legislation, of lethargic working conditions, of ugly social customs. The need for harmony between our surroundings and our living conditions shall then be so demanding, as to that the laws of society finally must be reshaped to follow the laws of beauty. [108]

The intended readers of "Beauty in the home" were primarily women, which is quite evident throughout the text as it directly addresses women as home-makers and carriers of both aesthetical and ethical qualities and capabilities. The main task in creating an environment that integrated usefulness and beauty lay on women. It was

106 Key, "Beauty in the home", 39.
107 For example in the last essay in *Skönhet för alla*, when describing how habits and celebrations must be both safeguard and changed in relation to the changing times: "That some old customs must come to be abandoned, is associated with new views of life, changed means of production, more lively communications and other traits of the time". Ellen Key, "Festvanor", in *Skönhet i hemmen* (1913) (faksimil, Rekolid, 1996), 43. "Att en del gamla seder måste komma att övergivas, sammanhänger med nya livsåskådningar, ändrade produktionsförhållanden, livligare samfärdsel och andra tidens drag."
108 Ellen Key, "Skönhet", in *Tankebilder I* (1898), 3rd edition (Albert Bonniers förlag: Stockholm, 1922), 154. "Och när man lyckats i denna sträfvan, månne då icke skönhetskänslan skall visa sig mäktig nog att tvinga människan att ombilda själfva det nuvarande samhället? [---] Ingen skall då för sig eller andra kunna uthärda plågan af en ful lagstiftning, af förslöande arbetsvillkor och osköna samhällsseder. Behofvet af harmoni mellan vår omgifning och våra lefnadsförhållanden skall då varda så befallande, att samhällslagarna slutligen måste omdanas efter skönhetslagarna."

not only the case that women as part of their nature had – and could further train – a specific aesthetic sensitivity different to that of men: it was also every woman's duty to continually strive for beauty. This, Key meant, was something that most women had not yet understood, as they would approach beauty more as a "pleasurable pursuit" than an essential practical and educational task.

Women needed to understand the power they held when it came to providing a home environment that could not only be nurturing and developing for children, but that would also bring about other forms of relationships between women and men at home as well as in society. Women were to be empowered as agents of socio-aesthetic change in virtue of their specific aptitude for aesthetic judgement, home-making and motherhood in private as well as in public life. Hence, for Key, women became the designers of everyday life and everyday practices.

What was crucial – and what the whole essay "Beauty in the home" worked with – was to in practice show how considering usefulness and beauty in every detail were intrinsically linked to a greater purpose of the home, and ultimately also to making a change in society. This holistic view was necessary to have to be able to make the right judgements, choices and designs, and women, in general, would need more training in this:

> Most women do not yet understand how to treat a room – or an outfit – as a whole, where nothing should exist that is beautiful only by itself, but where colours and shapes should work well together; where the main thing should be emphasized and all secondary things, however beautiful, be subordinate; where all separate entities ought to work together harmoniously; where one should understand how to achieve an effect, one time by filling a void, another time by not filling it.[109]

The approach towards understanding an assembled whole, such as a home, as something that went far beyond its parts in relation to use, ethics and aesthetics, also applied to the individual in relation to others and to society. With the starting point of personal taste and an understanding of the own needs, Ellen Key situated the creation of a home environment distinctly within the realm of experience and utility but with the goal of ultimately transforming both society and people as social beings.

In all planning, purchasing, making, moving, painting, composing, and using, the individual things in a home would come together into a holistic environment, which constituted a thing in itself in a way different than only a constellation of its parts. As such, the home was something more than a house, an apartment or a room. It was also something more than just a compilation of objects that serve different purposes. In the idea of the home was embedded not only its materiality, but also the experience of being in it and the potentiality of what it would allow its' inhabitants to become if the harmony between the useful and the beautiful was achieved:

> By striving for beauty at the same time as observing the need for utility, the woman not only satisfies a legitimate desire in her own nature but also exerts a profound influence on the other members in the home of which she is the soul. The children's senses are educated and refined through the beautiful impressions they receive; the adults experience a peace and a joy that cannot fail to make them more agreeable and often nobler; and the woman herself is made happier through that joy of creativity always accompanying the practice of an art.[110]

109 Key, "Beauty in the home", 38.

Through defining the home – and the realm of material aesthetics in relation to individual and social development – as an area of women's expertise, the space for women to have a voice in a socio-political setting was also created. This was based on an idea of gender-based complementarity, where women's contributions to society would be based on their "collective motherliness" (*samhällsmoderlighet*). On an individual level, women carried the responsibility and ability to create a home environment in which the family could thrive and develop. On a societal level, women were needed to participate in issues of legislation and organisation aiming towards the common good in matters associated with the areas considered of specific female expertise.

Bringing women together as a group in order to claim a presence in the socio-political arena, based on specifically identified aspects of female expertise, therefore became an issue of collective and mutual learning. In order to fully be able to participate in society, women would need to help each other to gain language and learning, and to operate the necessary tools to be able to take place at the table. The arenas identified, and put into play, around the turn of the century 1900 were – for women – the home and the shared experiences women could relate to in regard to their roles in it irrespective of social standing.

Full engagement in the budding Swedish democratic processes needed to be preceded by practicing and preparing for participation. The ambition to bridge differences – of class, education, economy – and create new situations for learning together, led to the establishment of new ways of coming together and finding common ground. The attempts to create situations for facilitating a broad *bildning*, that included aspects of ethics and aesthetics as well as of learning, led to adopting formats such as study circles and shared responsibilities for engagement in building knowledge together in various societal contexts.

The importance of an active and reflective approach to the material aspects of everyday life as crucial for societal change, expressed by Key in her texts, became highly influential in different circles already during Key's lifetime. Amongst Scandinavian artists and architects who strove to promote "a new style" as well as new ways of approaching creative practices, the ideas on how material choices could shape everyday practices in the home as well as form a foundation for societal change were picked up and expanded on.

Ellen Key's writings thus became seminal for the formation of both design discourse and design practice in Sweden during the 20th century. Later design theorists, such as Gregor Paulsson, explicitly point to Ellen Key's ideas of design and everyday life as an impetus for their own thinking.[111] Influenced by, and also in some opposition to, the English arts- and crafts ideas brought forth by John Ruskin

110 Key, "Beauty in the Home", 37.

111 Gregor Paulsson, "Min väg till konstvetenskapen", in *Uppsala universitets konsthistoriska institution femtio år* (Uppsala, 1968), 20. See also Gunnela Ivanov, *Vackrare vardagsvara – design för alla? Gregor Paulsson och Svenska slöjdföreningen 1915-1925*, diss. (Umeå, 2004), 75-110.

and William Morris, Ellen Key thus introduced ideas that came to play a significant role in laying the foundation for two themes of relevance to the formation of the discourse around and the program for a user-centered design that emerged as specifically "Swedish" or "Scandinavian".

Participation, designing, history

This chapter has revolved around a situation central to contemporary participatory design processes: that of people coming together in situations of mutual learning as a starting point for proposing change through collaborative designing. By following some of the fundamental ideas embedded in the formats and approaches in play in these situations, the perspective of 'participation' has been applied to contexts in turn-of-the-century 1900s Sweden. The purpose here has been to explore what a design history of a core concept such as this might be like, and what it might do for addressing or opening up certain issues in designing today. Rather than following, or searching for, a specific terminology or method, I have attempted to draw forth ways of thinking around design, everyday life and society that can be seen as having formed a foundation for ideas about 'participation' in relation to design.

The emergence of participatory design practices in the 1970s and 1980s context of work-place democracy was conditioned by the ways that early 20th-century ideas of egalitarianism and democracy were intertwined with issues of learning and bridging differences as central to making change, not least through design and the material re-formation of the everyday. Another shift in perspective has been made in regard to the design-related ideas of author, social debater and design theorist Ellen Key. She has since long been established in the 'canon' of Swedish design history as an 'arbiter of taste', perceived as providing normative judgements of what makes a thing qualify as 'good design'.

The purpose of choosing to revisit Ellen Key's writings from a different perspective, was to explore if this could also make visible other design histories than the usual. Ideas of 'good design', as promoted and argued by Ellen Key, have been quite extensively studied – and also critiqued – in Swedish design history, focusing on these as expressions of normative aesthetics and culturally oppressive home design reform. Through picking up on concepts foundational to participatory design, such as creating situations where mutual learning forms a basis for collaborative designing with an intent of supporting democratic decision-making, situations and ideas of collective learning in relation to initiatives towards design reforms, is understood as a part of a programmatic approach aiming towards the formation of a democratic and egalitarian society.

It might also be the case, that it was precisely these entanglements of socio-political transformation with matters of how to design – materially and ideally – everyday life that formed the conditions for participatory design to emerge in the latter decades of the 20th century. From this also follows, that some of the conceptual and practical issues sometimes difficult to deal with in participatory design, could stem from the historical situations in which these concepts once emerged and took shape.

In Ellen Key's approach to design in the context of industrialisation, the questions of which things to make and how to provide access to these for everyone, were never only about form, aesthetics or matters of taste. For Key and others at that time, design was about what types of futures could be made through envisioning and enacting change on different societal levels: material, ideal, legal and social.

The period in which industrial design came about – the late 19th and early 20th centuries – was also a period in which Western political ideologies took form, and traditional socio-political power structures were contested. With claims and contestations of the extension of parliamentary democracy to encompass (almost) everyone – women and working-class men – followed that issues were raised of who could, and should, have a say not only decision making on societal level but also of how design could be of fundamental importance to making these changes.

Through design, access to decent housing and useful things, to education and arts, would be granted not to a privileged few but to everyone. However, the concept of 'participation' is in some sense rooted in a situation of privilege, where those holding power or resources for some reason open up the possibility for others to share or take part – to participate.

In many participatory design projects, situated in contexts of design research, the researcher and the partners with which she collaborates in the project, would have a strong influence on who would be defined as potential users and stakeholders, and consequently considered for inclusion in the process. Similarly, since much of participatory design is set up as research-related projects, decisions on inclusion in or exclusion from the process are also influenced by funding schemes, stakeholder affiliations, and other structural and financial bases. That many papers and articles refer to 'participants' as being 'invited' by designers to take part in situations and processes, suggests that someone already holds power over the design domain, and decides to open up the possibility of allowing others to step into a process or situation that has already in some way been established or defined.

As in the turn-of-the-century example of the Tolfterna gatherings, the ambition might well be to establish a situation where collaboration or conversation can take place in a non-hierarchical and egalitarian setting. However, when some are called 'invitors' and some 'invited', or when some are labelled 'designers' and others 'participants', the situation – and indeed the concept itself – carries with it a fundamental inequality in the very initiation of the collaboration expected to take place. In participatory designing, an ambition is often to empower people and to enable people to have a voice and a say in design processes that will affect their situation. This was already in the early 20th century also an explicit aim in the *folkbildning* initiatives that targeted groups of people until then devoid of power or influence – legally and/or socially – over their situation.

Developing methods and establishing a foundation for this empowerment was a project that encompassed issues of providing access to situations of learning, through constructing physical meeting places and of setting up social arenas where conversations could take place between groups of people who would otherwise not meet each other.

Scandinavian participatory design of the 1970s and 1980s came about in a context of national legislation that formally required employers and employees to negotiate conditions and consequences when making major changes in the workplace. The development of new methods and situations for collaborative designing was, then, not so much an initiative from within design but a response to other socio-political forces that – eventually – led to changes in design practices.

Ideas of the importance of collective action as well as of mutual and shared learning were central to the design practices that were advocated in the 1970s and 1980s. Much as the *folkbildning* movements had prototyped different ways of setting up learning situations that were open, reflective and inclusive, the prototyping of different formats for shared understanding and learning as a basis for design-driven transformation, both of specific situations and societal structures became central to participatory design practices.

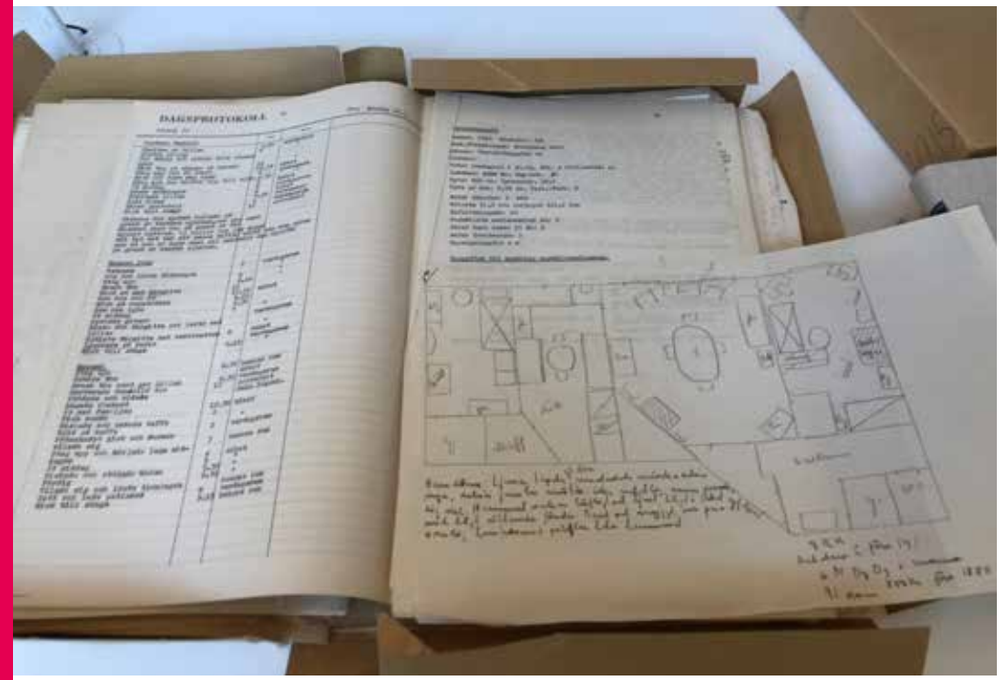
The format of the study circle, which emerged as a suggestion of a more democratic and open format for collective learning in the early 20th century, had by the 1970s and 1980s become a well established format for learning in the context of *folkbildning*. Within different *folkbildning* organisations, many of which were run by worker's organisations and unions, the study circle was an integrated way of disseminating and gaining knowledge. In the UTOPIA project, it was explicitly emphasised that one of the participating researchers had experience of running study circles on new technology together with unions.[112] The study circle thus travels into participatory design through inclusion of idea and format in design situations.

However, here we can also highlight aspects of historical 'baggage' that might be of importance to contemporary design practice, in regard to methods and approaches in participatory design. If we take the ambition in participatory design practice to, as part of the design process, set up situations where every voice should be equally heard and influential in the design process, the formats for doing this will continue to be conditioned by the historicity of the concept of 'participation.' This does, of course, not have to be at all problematic, if the design situation is one where participatory design processes and methods are in tune with the challenge addressed. Whereas if the concept of 'participation' is in play when design is called upon in instances where relationships and complexities go beyond what the concept can allow us to think and do.

The things we use to think with influence the things we do, and the things we make – whether the thing is a material object, a process, or a concept. Approaching the concept of participation from with the aim of making a design history, has as much to do – perhaps more – with exploring what history emerges as what it does for our present-day ways of understanding. Through history, and through what comes into view through this perspective, things we perhaps usually do not actively consider in design practices today come into view as well. A design history such as the one presented here, will not in any way attempt to resolve this, but can possibly provide a shift in perspective that makes visible something which would otherwise have remained unseen or taken for granted.

112 When writing about the persons engaged in the UTOPIA project, it is mentioned that "the engineer Maja-Lisa Perby, [...] had some experience from study circles on new technology together with trade unions." Ehn, *Work-Oriented Design of Computer Artifacts,* 10.

5. Use



Daily protocol of activities and floor plan sketch from the 1937 study for *The Family that Outgrew its Home*. Brita Åkerman's Archive, Umeå University Library.

In the following, I prototype a design history from the perspective of 'use' and 'users,' intending to explore a core concept in the practices of user-centered design. This is, on the one hand, an exploration into the historicity of these concepts in designing and, on the other, a proposal of what a design history of contexts and ideas formative for the emergence of Scandinavian user-centered design might look like.

This transitional design history prototype takes off from an overview of how user-centered design practices are generally described as having come about. Situating the inquiry into user-centered design histories in a Swedish design context, I also give a brief overview of the historical period and theme that the prototypical intervention later in this chapter relates to. I, therefore, sketch a historical background to the formulation of the "housing question" in Sweden in the early 20th century, and the introduction of modernist ideas that directly influenced the national Swedish housing program as part of the broader ambitions of socio-political reforms aiming to turn Sweden into a *Folkhem,* a 'people's home.'

This is a well known and extensively researched historical period in Swedish and Scandinavian design history. It is in relation to these existing narratives that I then prototype a different reading of texts and archival materials from contexts of domestic reform and housing design.

The material studied in this chapter consists of investigations into 'dwelling habits', made by the design promotional organisation Svenska slöjdföreningen (SSF), the National Association of Swedish Architects (SAR) and other actors engaging in issues of reforming everyday life , interior decoration, building standards and design practices. The aim is, on the one hand, to explore if other aspects than those already well known in Swedish design history will emerge when applying a conceptual lens of 'use' to revisit this historical context. And on the other hand – which is the primary aim – the ambition is to investigate the historicity of the concept of 'use' in user-centered design practice.

Users in designing: A brief background

As research and practices in user-centered and participatory design have evolved, one of its core concepts seems to have become increasingly difficult to handle: that of the 'user'.[1] Despite, or perhaps because of, its centrality to many methods and orientations in design, who or what a 'user' is in regard to roles and agency in designing is not always very straightforward.[2] As design moves into situations that are not clearly defined as to when designing starts and ends, the 'use' designed for is perhaps neither easily attributed to a single context, a stable technology, or readily defined type of profession or group of people. Who the 'user' might be, what 'use' will entail, and how it might change over time is, therefore, becoming increasingly hard to say.[3]

In participatory user-centered design, conceptual difficulties also emerge when collaboration in designing take on formats that blur the boundaries between 'designers' and 'users' – not only in terms of roles, power, expertise and

1 Sarah Ahmed, *What's the Use? On the Uses of Use* (Durham & London: Duke University Press, 2019), 21-67; Theodora Vardouli, "*User Design*: Constructions of the 'user' in the history of design research", *Design Research Society 50th Anniversary Conference,* Brighton 2016.

2 Johan Redström, "RE: Definitions of use", *Design Studies* 2008:4 (vol.29), 410-423; Johan Redström, "Towards user design? On the shift from object to user as the subject of design", *Design Studies*. 2006:2 (vol.27), 123–139.

3 Pelle Ehn, Elisabet M. Nilsson & Richard Topgaard (eds.) *Making futures: Marginal notes on innovation, design, and democracy* (Cambridge, MA: The MIT Press, 2014); Kristina Lindström & Åsa Ståhl, *Patchworking publics-in-the-making: Design, media and public engagement*, diss. (Malmö: Malmö University, 2014); Tuuli Mattelmäki, Eva Brandt & Kirsikka Vaajakallio, "On designing open-ended interpretations for collaborative design exploration" *CoDesign: Special Issue: Design and Emotion*, 2011:2 (vol.7), 79–93; Elisabeth Shove, Matthew Watson, Martin Hand, & Jack Ingram, *The Design of Everyday Life* (New York: Berg, 2007), 123.

accountability but also in terms of non-human agency in designing.[4] Design has, over time, come to deal with ever more complex and increasingly less tangible things and materials. Whether in handling the realms of the digital, the areas of experience or service design, or targeting political or social empowerment processes, questions of value, situation, context and scale also affect how 'use' or 'users' in design could be understood.[5]

Design for users

Working with user-centered design can encompass many different things, in terms of what the concept of 'users' is taken to mean in the design process. In some cases, a 'user' is more or less the same thing as a 'consumer' and has often become the preferred synonym for this in descriptions of how a company or consultancy works in relation to the people who are their intended target group. Users, then, are the people who will purchase, live with, and handle a product or service provided for them. A design framework that applies user-centered methods could span everything from doing market research polls that probe people's preferences in consumption, to questionnaires, interviews and observations aiming to understand more of people's everyday situations and behaviours. People might be invited to take part in different stages of the design process, in user tests of prototypes or concepts, or to take part in focus groups or workshops, often dedicated to getting user feedback on usability, ergonomics or product appeal.

In the type of user-centered design that strongly emphasises usability, ways of defining 'use' rely quite substantially on intentionality, direction, and adaptation to functional demands, whether from a cognitive or ergonomic point of view. In pointing to the history of where and how user-centered design took shape, references tend to lead to 1940s methods applied in human factors research, and the adaptation of for example military aircraft to the needs and abilities of fighter pilots.[6] The human factors approach in design incorporated both physical and cognitive ergonomics into industrial design processes and came to have a strong emphasis on developing methods for the design of professional tools in product design, as well as in what later became interaction design.

Design practices based on extensive user involvement grew strong in Scandinavia in the late 1960s and onwards, with emphasis on ergonomic and inclusive design. Users are in this 'classic' Scandinavian user-centered design defined in relation to a particular type of action, quite precisely tied to a

4 Lenneke Kuijer & Elisa Giaccardi, "Co-performance: Conceptualizing the Role of Artificial Agency in the Design of Everyday Life", *Proceedings of the 2018 CHI Conference on human factors in computing systems*. (ACM, 2018), 1–13; Johan Redström & Heather Wiltse, *Changing Things: The Future of Objects in a Digital World* (London: Bloomsbury, 2018).

5 Stoffel Kuenen, *Aesthetics of Being Together*, diss., (Umeå: Umeå University, 2018).

6 Marcia E. Holmes, *Performing proficiency: Applied experimental psychology and the human engineering of air defense, 1940-1965*, diss. (The University of Chicago, 2014); Theodora Vardouli, "*User Design*: Constructions of the 'user' in the history of design research", *Design Research Society 50th Anniversary Conference,* Brighton 2016.

specific context and situation. The future use of the thing being designed is then anticipated, developed and tested in different ways together with people currently engaged in similar situations or types of activities.

Roles are often quite well defined, where designers hold expertise in and responsibility for the process of designing for the people described as 'users'. While these users can be more or less involved in the design process, through everything from user testing to collaborative workshops, the processes in which design comes about are owned and run by designers.[7] Users are subjects of study and providers of information – through surveys, ethnographic studies or user tests – and can also be invited to participate in more collaborative or participatory situations during the process of developing the design, in their capacity of holding a particular experiential expertise in relation to intended future use.

From the 1990s and onwards, as concepts of usability and ideas of designing for rational use moved towards addressing issues of the situated and often less-than-rational way people perceive, adopt and use things, aspects of 'user experience' were increasingly integrated into designing.[8] In terms of user-centered design, this meant that methods and approaches in design would not only focus on ease of operating and handling something, but also on the personal, psychological, emotional and aesthetic experiences of people's engaging with things.[9] In such situations, what the intended 'use' or 'users' of design would be is not necessarily simple to define. As concepts, they can nonetheless seem reasonably straightforward to work with.

In user-centered design, the roles of designers and users are conceptually defined in relation to each other: the designer designs for the user, who in turn will be the one engaging with the final result provided by the designer, and in different ways providing input into the designer's process. However, as issues of how to handle aspects of what use and usability entail in design have been brought to the fore, the attention to ethical implications of user-centered design in terms of roles and responsibilities has also intensified.[10] The contexts and logics that form design also to a high degree shape who (or what) is to be considered as a 'user' in designing, and what that means for the relationships and dynamics between 'users' and 'designers'.

7 Donald A. Norman, *The Design of Everyday Things* (MIT Press: London, 1998).

8 Thomas Mitchell *Redefining Designing: From Form to Experience* (New York: Van Nostrand Reinhold, 1993), 131.

9 Kees Overbeeke, Tom Djajadiningrat, Caroline Hummels, Stephan Wensveen & Joep Frens, "Let's make things engaging" in Mark A Blythe et al., *Funology: From Usability to Enjoyment* (Dordrecht: Springer Science + Business Media, 2005), 7-17.

10 Turkka Keinonen, *Designers, Users and Justice* (Bloomsbury: London & New York, 2017); Michael J. Salvo, "Ethics of Engagement: User-Centered Design and Rhetorical Methodology", *Technical Communication Quarterly* 2001:10 (vol 3), 273-290; Donald A. Norman, *The Design of Everyday Things* (New York: Doubleday, 1990).

Design with users

Turning to participatory design, where the core idea is that designers and users should come together collaboratively in the design process, the concepts of use and the relations between designers and users become more multifaceted. Central to participatory design is the ambition that design should happen *with* people rather than *for* people.[11] The people called 'designers' should, therefore, work together with 'users', or 'participants', in a design process where both parties are involved in making design decisions and influencing the process as well as its outcomes. 'Participants' as a term usually includes the people taking part in designing, based on their being defined as holding different roles, such as 'users', 'experts', or 'stakeholders'.

As participatory design increasingly moved into new areas of professional practices, users would also be categorised in different types – such as primary or secondary users, end-users, expert users and so on. Categorisation would depend on various aspects or contexts such as how directly or indirectly users would work with a designed thing; how professionally skilled or unskilled they were, or how likely it was that they would be able to participate with equal voice or standing in a design process (regarding, for example, health, sickness, age and cognition).

When participatory methods for designing began to be developed, in the context of introducing computers into working life, these rested on concepts of users and usability in relation to systems or software being designed. 'Users' were people in a professional situation, operating a 'tool' or item that should support a specific action in relation to the user's skill sets. In an expanded view on the process and outcomes of design, ideas of 'use' also extended to spheres and processes beyond the direct operational use of the designed thing.[12]

In the early Scandinavian participatory projects, the 'use' that unions would have in power relations and democratic decision-making at the workplace was emphasised, in regard to the knowledge and shifts in positions created through design processes. The difference between 'designers' and 'users' remained upheld in relation to expertise and process. Many of the collaborative methods developed aimed at overcoming or bridging differences between the roles of 'designers' and 'users' in regard to expertise, experience, presence, and decision making in the design process. At the same time, upholding or reinforcing difference was seen as important in regard to other 'stakeholders' such as work-place management,

11 "Participatory design is about the direct involvement of people in the co-design processes of the information technologies that they use." Jesper Simonsen & Toni Robertsson (eds.), *Routledge International Handbook of Participatory Design*, xix.

12 Lucy Suchman, "Working Relations of Technology Production and Use", *Computer Supported Cooperative Work* 1994:1, (vol.2), 21–39; Liam J. Bannon, "From Human Factors to Human Actors: The Role of Psychology and Human-Computer Interaction Studies in System Design" in J. Greenbaum & Morten Kyng (eds.), *Design at Work: Cooperative Design of Computer Systems* (Hillsdale: Lawrence Erlbaum Associates, 1991), 25-44.

unskilled labourers, or colleagues from other professions than those of the intended expert users.[13]

Design of users

Over time, developments of design practices building on user-centered and participatory approaches have brought about new ways of thinking and designing in relation to the non-designers involved in co-design processes. This is reflected in the attention to the terminology concerning 'users':

> Many people have become uneasy about terms like *user*, *consumer*, and *client*, which pin people down in narrow and restrictive roles. At this moment, we refer to them as *participants* and *partners*, or just *people*, to express the multiple roles and perspectives that are needed.[14]

The term 'user' is often replaced by 'stakeholder' or 'participant' to distinguish 'people' in general from 'people' involved in or affected by a design situation or context.[15] While a 'participant' points to someone taking part in the process of designing, the other terms are more complicated to handle: how can we define what it is to be a 'user' of or a 'stakeholder' in an as of yet non-existent future something?

Shifts in what design and designing is about have not only called for developing methods to understand how people go about using or experiencing things. These have also brought about shifts in understandings of the subject matter of design, from focusing on giving form to material things towards shaping users and their experiences. As pointed to by Johan Redström, a 'user' envisioned in the process of designing can only ever be as much a potentiality in the design process as the thing designed. [16]

If design is about projecting and proposing possible futures, the actual thing, its use, and its user, cannot already exist until these have become materialised or actionable in the world. This leads to designing in which not only a final 'thing,' but increasingly also its user, becomes created through design.[17] As such, a person cannot actually exist as a 'user' until entering into a relation to a designed thing, which in turn presupposes an interacting with an actual object or situation of some sort. The type of user created in this understanding of designing is not a real person or group, but an idea or an ideal future intended user. User-centered design then becomes user design "in the sense that it is design where the process through which

13 Gro Bjerknes & Tone Bratteteig, "User Participation and Democracy: A Discussion of Scandinavian Research on System Development", *Scandinavian Journal of Information Systems*, 1995, 7 (1), 78; Peter Botsman, "Rethinking the CLASS struggle: Industrial democracy and the politics of production", *Economic and Industrial Democracy* 1989:1 (vol 10), 132-133.

14 Liz Sanders & Pieter Jan Stappers, "From Designing to Co-Designing to Collective Dreaming: Three Slices in Time", *Interactions* Nov.-Dec. 2014, 30.

15 Jodi Forlizzi. "Moving beyond user-centered design" *Interactions* 2018:5 (vol.25), 22-23.

16 Johan Redström, "Towards user design? On the shift from object to user as the subject of design", *Design Studies* 2006:27, 127.

17 Redström, "Towards user design?, 129.

people turn into users are in focus and where the explicit aim is to work with the results of this process, i.e. how use and user should turn out. We, as designers, turn people into users through our designs, by presenting a thing to be used."[18]

User-centered design brings about proposals of future situations but also implicit or explicit expectations or prescriptions of impacts, limitations or changes on and in people's behaviours. Certain pre-scripted ways of use, as proposed in 'classical' user-centered design but often also in participatory processes, will lead to effects that change current as well as future situations for people. Such changes are frequently highlighted in terms of design improvement, experience, and efficiency.

It is seldom expressly articulated how people would be expected to change in regard to the proposed future use, or in regard to a context broader than the projected framework of the immediate use-situation. Likewise, it is neither easy to predict, nor to prescribe, which potential other consequences the future use of a design thing might open up for. But to propose a way forward in designing, some intended directions need to be regarded, and others disregarded in order to make design decisions. This includes trying to make some ways to handle, react, or relate to something more likely to happen than others, through prescribing certain uses and limiting others. Some futures are thus made more probable, or plausible, than others.

How to negotiate design processes, whether considering initial formulation of a matter of concern or a problem to be addressed, which current and future contexts to consider, whom to include in designing, and how open an outcome could be, are all questions in which the concepts of 'use' and 'users' need to be brought to the fore. In the following, the conceptual perspective of 'use' is turned towards a Swedish 20th-century context of design reform aimed towards the domestic sphere.

Homes and housing in 20th century Sweden

Picking up on these current considerations and challenges in user-centered and participatory design, the time has come to turn to the part of this prototype that explores what a design history seen through the concept of 'use' could be. Turning to the design historically well-established sphere of domesticity and design histories related to the home,[19] and looking at initiatives and mediations aiming towards home reform in Sweden through the conceptual lenses of 'use' and 'users', the aim is to highlight possible histories of designing in relation to ideas of use.

In the following, a brief background is sketched of the context of the 'housing problem' as it was formulated in Sweden in the 1930s. Then, a closer study of a series of 'dwelling investigations' conducted in the late 1930s and early 1940s are analysed

18 Redström, "Towards user design?, 129.

19 For example Grace Lees-Maffei, *Design at Home: Domestic Advice Books in Britain and the USA since 1945* (London: Routledge, 2014); Penny Sparke, *Designing the Modern Interior: From the Victorians to Today* (Oxford: Berg, 2009); Judy Attfield, *Bringing Modernity Home: Writings on Popular Design and Material Culture* (Manchester: Manchester University Press, 2007); Grace Lees-Maffei, *Modern living? Domestic advice literature and design discourse in post-war Britain,* diss. (Portsmouth: University of Portsmouth, 2005).



‘Bostadskök’: a one-room apartment with stove, in Sundsvall ca. 1900. Peter Wide / Sundsvalls museum (SuM-foto017383).

as examples of a turn towards ideas of incorporating user perspectives in design processes aiming to reform housing design as well as the design of everyday things. Applying a lens of ‘use’ to historical material from this period, and looking for early formulations of ideas of user-centered design brings to light aspects of design in relation to use that have not yet been prominent in Scandinavian design history.

This is an attempt to prototype a design history that could contribute to calling out the historicity of core concepts in contemporary designing. Rather than providing an account of a trajectory that led to formulations of user-centered and participatory design, this is *one* example – there could, and should, be many more – of how to draw forth histories that could help us see things differently.

This type of history is more about raising questions than providing answers in relation to designing. Which possible futures could be plausible to move towards through re-articulations and conceptual re-configurations of how, for what or whom, and why design is done? The questions here relate to historically embedded meanings and values in design’s concepts and to articulations of ideas and practices that could be relevant to consider in regard to designing. Attention to the historical contexts in which different concepts emerged could contribute to making moves that support the reframing of what to address as problematic, and what as possibility, when moving further in dealing with concepts of use and users in design.

‘The housing question’

During the years around the turn of the century 1900, Sweden was a nation on the verge of industrialisation. The shift from an agricultural to an industrial and urban way of life had only just begun. Of the five million people living in Sweden, only a fifth lived in cities. With the expansion of Swedish industry, urbanisation and the growth of cities and towns, a steady movement from the countryside to cities began to take place. Many of the people who made their living in the Swedish countryside were very poor and had little or no possibility to own land, a home or earn a salary. Vast numbers of people moved to the growing Swedish towns and cities – or across the Atlantic, emigrating to America – in search of a better life.[20] The flow of people to the more urban areas of Sweden led to difficult housing situations for many. Whether in the countryside or the cities, the systemic changes associated with industrialisation led to markedly worse living conditions than had been the case earlier. The dire situation, especially for working-class families, was perhaps most noticeable in the cities.

In the first decades of the 20th century, Sweden had one of the lowest standards of housing in Europe. In cities and towns, around a third of the inhabitants lived

20 Starting in the 1860s, after several years of droughts and famines, emigration from Sweden to America began and increased progressively during the late 19th century. By the early years of the 20th century, more than a million Swedes had emigrated. Many left for the main reason of sustaining themselves and their family, looking to find a better outcome than was possible in Sweden. For others, the main reasons for leaving their home country were the promises of political, religious and social openness that the new world offered. Anna Lindkvist, *Jorden åt folket: Nationalföreningen mot emigrationen 1907-1925*. Skrifter från institutionen för historiska studier 19 (Umeå University, Umeå: 2007), 40.

five or more persons in small one- or two-room apartments. A general shortage of dwellings in Swedish towns and cities had led to families living in cramped and overcrowded small apartments or *bostadskök*, a one-room apartment with a wood stove in one corner of the room. Since the demand for dwellings was higher than what was available, rents were often high and standards low.

The shortage of dwellings, in combination with a need to reduce the monthly cost often also resulted in already overcrowded working-class families sub-letting a room or a sleeping place to tenants. These cramped living conditions, with low sanitary and material standards, were pointed to as the root of many other evils such as the spread of disease and moral decline. Philanthropic organisations made efforts to both provide low-cost housing for the poor, and to raise the issue of housing to a national political level. Philanthropic initiatives leading to privately funded housing projects aimed at relief for the poor, and also brought about surveys aiming to map the housing situation during the first decades of the 20th century.[21]

During this period, a plethora of social problems and phenomena were formulated as 'questions' (*frågor*) by organisations and politicians. Around the turn of the century 1900, the 'housing question' became highlighted in public debate as a cause for the many other societal problems labelled as 'the social question.'[22] Under the overarching umbrella term of the 'social question, a variety of interconnected problems were identified: poverty, alcohol abuse, the perceived dissolution of families, and poor housing conditions. Other issues formulated over the first decades of the 20th century were the 'women's question' regarding the political and social rights and influence of women; and the 'population question' focusing on the issue of the diminishing number of Swedes. The massive emigration of Swedes during the late 19th and early 20th century in combination with a steady decline in births was defined as a significant threat to the future of Sweden as a nation.[23] A common area of intervention was identified in all of these issues: the pressing need to provide better housing and home conditions for families.

Solving social problems through providing better housing was seen as a key in addressing everything from men's alcohol abuse – since better home conditions would make them more inclined to enjoy the comforts of family life instead of spending time in public drinking houses – to women's rights, in that a different home life would make it possible for women to engage in society in other ways than before. Therefore, a broad political and societal consensus formed around the 'housing issue' as well as around issues of domestic reform, since the two were regarded as interconnected. Resolving issues of how to materially provide adequate housing as well as reasonably priced everyday things for the home thus became a matter of concern which brought together different perspectives and positions.

21 Kerstin Thörn, *En bostad för hemmet: Idéhistoriska studier i bostadsfrågan,* Idéhistoriska skrifter 20, diss. (Umeå, 1997).
22 Daniel Movilla Vega & Dan Hallemar, *99 years of the housing question in Sweden* (Lund: Studentlitteratur, 2017); Per Wisselgren, *Samhällets kartläggare: Lorénska stiftelsen, den sociala frågan och samhällsvetenskapens formering 1830-1920*, diss. (Umeå, 2000).
23 Alva Myrdal & Gunnar Myrdal, *Kris i befolkningsfrågan* (Stockholm: Bonnier, 1934).

For the workers' movement and the politically organised Social Democrats, the housing question was essential to address from an egalitarian perspective. Having access to adequate housing of a decent standard at a reasonable cost was part of the political claims made from the workers' movement. From more conservative political positions, the urgency to find solutions to the housing problems had more to do with providing material conditions that would appease any revolutionary tendencies.

Other associations and movements, such as the Temperance Movement, the worker's movement, and anti-emigration initiatives, also highlighted different societal problems in their respective areas that could be directly or indirectly caused by the poor housing conditions in Sweden. This contributed to establishing the 'housing question' as one of the most crucial social problems to address. Since the situation in the overcrowded homes was seen as a potential breeding ground not only for disease and related social problems – drunkenness, abuse, the disintegration of families – but also for societal unrest, questions of how to build decent, modern, low-cost housing became a matter of national socio-political concern.

Universal suffrage extended to all Swedish citizens in two steps, allowing male workers the right to vote in 1909 and women in 1921, gradually shifting the political weight towards the Social Democratic party gaining increasing numbers of votes. The Social Democrats formed a government from 1920 to 1976, broken off only by a conservative government 1928-32, and broad coalitions during World War II. After the elections of 1932, the Social Democratic Party formed government, with the overall aim to implement national schemes for building, so that the problems of the very low housing standards in Sweden would be addressed.

Until the early 1930s, there had not existed a state policy or even a designated political arena for handling housing issues in Sweden. Providing housing was traditionally seen as a demand-and-supply question for the private market, or the local municipal governments, to solve.[24] It was only with the formation of the 1930s Social Democratic government that a cohesive state housing policy began to be established. This, in turn, was influenced by the international modernist ideas and design principles for city planning and architecture.

The People's Home

The solution to the 'housing problem' was not only perceived to be the production of better housing but for creating better conditions for homes – ideally as well as materially.[25] In the late 1920s, the Social Democrats had proclaimed the intention to turn Sweden into a *Folkhem*, literally a People's Home, in which all citizens were to have equal rights and equal possibilities. If society were to be turned in to a good home for everyone, it had to be reorganised in a more democratic and modern way. Per Albin Hansson, the Social Democrat prime minister from 1932 to 1946, phrased it thus in his seminal Folkhem speech in 1928:

24 Eva Eriksson, *Den moderna staden tar form: Arkitektur och debatt 1910-1935* (Stockholm: Ordfront, 2001).
25 Mikael Hård, "The good apartment: The social (democratic) construction of Swedish homes", *Home Cultures* 2010:2 (vol.7), 117-134.

> The foundations of a home are community and compassion. The good home knows no privileged or neglected, no favourites and no stepchildren. There, the one does not look down upon the other; there, no one tries to benefit at someone else's expense, the strong does not suppress or plunder the weak. In the good home, equality, caring, cooperation, and helpfulness prevail. Applied on the larger home of people and citizens, this would mean the tearing down of all social and economic barriers that now separate citizens in privileged and neglected, in rulers and dependants, in rich and poor, propertied and impoverished, plunderers and plundered.[26]

In Sweden, as in other Nordic countries, the home and its everyday objects have held a very central and symbolic place as carriers of meaning in public discourse in the formation of welfare states based on principles of industrial democracy, intrinsically interlinking 'the big life' and the 'little life'.[27] The centrality given to the housing question in the socio-political agenda was a substantial contributing factor for establishing the home as one of the most important arenas for social and material reform during the 20th century. Family relations, raising children, how the housework was managed and how to design and arrange the furniture all became part of the socio-political agenda to turn Sweden into a *Folkhem*. Specifically, many reform efforts concerned the domestic interior and aimed at promoting a new and modern way of furnishing and decorating the home, spread through advice literature, educational efforts and legislation – similar to initiatives in other Western countries.[28]

The domestic interior became an arena where the vision of the ideal society should be mirrored in the ideal home. This focus on domestic reform brought attention to the issues traditionally connected with the home – housework, homemaking, child-rearing and leisure time – and incorporated these into the political agenda. In this way, the housing question became connected to the quest for home reform, in the ideal sense as well as in a real and material way.

Functionalism and the home

The Svenska slöjdföreningen (SSF, the Swedish Society for Arts and Design) had been founded in 1845, to protect and safeguard Swedish crafts.[29] By the early 20th century, with inspiration from the German Werkbund, the society had turned to promote more radical ideas of design reform in connection to industrial production.[30] The SSF arranged many exhibitions from the turn of the century and during the first decades of the 1900s, to educate and inform general public and producers of industrial goods alike in matters of 'good design', and 'make propaganda' for a better way of living and furnishing the home.[31] The perhaps most influential exhibition arranged by the SSF during the 20th century was the Stockholm exhibition of 1930. This exhibition introduced modernism broadly in Sweden, and also became central to the later formulation of the Swedish government housing policy for decades to come.[32]

In the 1930s, Sweden still had one of Europe's lowest standards of housing. With inspiration from continental design ideas, and impetus from the Le Corbusier exhibit of the Pavillon de l'Ésprit Nouveau at the 1925 Paris exhibition, the Stockholm exhibition of 1930 proposed designing based on standardisation and function-based mass-production of dwellings and furnishings as the way to solve the 'housing question'.[33]

The proposals for building new types of housing at the Stockholm exhibition were heavily influenced by European modernism. The approach to designing should begin in a thing's intended function instead of in issues of form and aesthetics. As architect Uno Åhrén, member of the organising committee of the Stockholm exhibition, exclaimed when explaining what it meant for a thing's form to be an expression of its function, as well as a new aesthetic expression:

26 Per Albin Hansson, "Folkhemmet, medborgarhemmet", in Per Albin Hansson, *Från Fram till Folkhemmet: Per Albin Hansson som tidningsman och talare* (Solna: Metodica, 1982), 227. "'Hemmets grundval är gemensamheten och samkänslan. Det goda hemmet känner icke till några privilegierade eller tillbakasatta, inga kelgrisar och inga styvbarn. Där ser icke den ena ner på den andre, där försöker ingen skaffa sig fördel på andras bekostnad, den starke trycker icke ner och plundrar den svage. I det goda hemmet råder likhet, omtanke, samarbete, hjälpsamhet. Tillämpat på det stora folk- och medborgarhemmet skulle detta betyda nedbrytandet av alla sociala och ekonomiska skrankor, som nu skilja medborgarna i priviligierade och tillbakasatta, i härskande och beroende, i rika och fattiga, besuttne och utarmade, plundrare och utplundrade."

27 Pekka Korvenmaa, *Finnish design: A concise history* (Helsinki, 2009), 83-120; Kjetil Fallan, *Designing modern Norway: A history of design discourse* (New York: Routledge, 2016).

28 Grace Lees-Maffei, *Design at Home: Domestic Advice Books in Britain and the USA since 1945* (London: Routledge, 2014); Judy Attfield, *Utility Reassessed: The Role of Ethics in the Practice of Design* (Manchester: Manchester University Press, 1999); Penny Sparke, Anne Massey, Trevor Keeble & Brenda Martin (eds.), *Designing the Modern Interior: From The Victorians To Today* (London: Berg, 2009).

29 Gunilla Frick, *Svenska slöjdföreningen och konstindustrin före 1905*, Nordiska museets handlingar 91, diss. (Uppsala, 1978).

30 Gunnela Ivanov, *Vackrare vardagsvara – design för alla? Gregor Paulsson och Svenska slöjdföreningen 1915-1925*, diss. (Umeå, 2004).

31 As a genre, exhibitions featuring ideal homes were used as a mediating forum for home reform and modern design in many Western countries. See for example Deborah S.Ryan, *Daily Mail – Ideal Home Exhibition: The Ideal Home Through the 20th Century* (London: Hazar, 1997); Fredie Floré, *Lessen in modern wonen: Bronnenboek over woontentoonstellingen in België 1945-1958* (Gent: WZW, 2004).

32 Eva Rudberg, *The Stockholm exhibition 1930: Modernism's breaktrough in Swedish architecture* (Stockholm: Stockholmia, 1999); Per I. Gedin, *När Sverige blev modernt: Gregor Paulsson, vackrare vardagsvara, funktionalismen och Stockholmsutställningen 1930* (Stockholm: Bonniers, 2018).

33 Uno Åhrén, "Brytningar", in Gregor Paulsson (ed.), *Svenska slöjdföreningens årsbok 1925. Moderna strömningar inom Europas konstindustri* (Stockholm, 1925), 7-36.



The 1930 Stockholm exhibition: Exhibition Hall 36, Rental apartments, by architects Gunnar Sundbärg and Uno Åhrén. Credits: Gustaf W:son Cronquist / ArkDes (ARKM.1990-106-123)

> That is the meaningful element in the modern sense of form. It finds sufficient joy precisely in its capacity of form as made to work. Appearance rather stands as the mere external aspect of work itself, the rational technology, the sensible economy.[34]

In Sweden, the program for reforming the material environment along the lines of modernism came to be called 'functionalism', due to the strong emphasis on function and usability as a basis for designing everything from teacups to city plans. Gregor Paulsson, chair of the 1930 Stockholm exhibition, stated in the exhibition catalogue that the societal revolutions caused by industrialisation and developments in technology craved a new foundation for designing homes – materially as well as culturally:

> We are living through one of the most violent changes in the societal structure that has ever occurred, namely its industrialisation, which from the individual's perspective means an easier and speedier transfer of goods. [---] As certain as it is wrong to try to salvage the industrial society by fleeing to the old, to nature or whatever the saying, it is equally certain that the new homes in the new world must be created on this world's terms.[35]

The dwellings presented at the Stockholm exhibition championed the idea of differentiating between the basic functions of the home, and manifesting these explicitly in the floor plans by labelling rooms on the basis of their function, specifically identifying the 'bedroom' and the 'living room'. Sleeping, cooking, keeping clean, and socialising, were defined as the essential functions the home should support – however small, and for whichever social category of people.

The ideal functionalist dwelling was small and cost-efficient, with large windows to let the daylight in, a balcony for getting fresh air. Cooking would take place in a small but efficient laboratory-style kitchen. Hygiene should be handled in the bathroom with running hot and cold water, and for leisure, there should be a living room for the whole family to gather in. There should preferably be separate bedrooms for adults and children, with separate and proper beds for each member of the family. Different exhibitors at the Stockholm exhibition furnished homes based on these principles, presenting the home as a comfortable place for the family's everyday work and leisure, not a representatively furnished place to manifest one's status or for receiving and entertaining guests.

For the Stockholm exhibition visitors, meeting the whitewashed functionalist exteriors and minimalist interiors with tubular steel furniture, modernist colour schemes and geometric form language became a watershed moment. While some were enthusiastic, others meant that the exhibition promoted an ideology and response to everyday dilemmas that removed any individuality or expressions of comfort from design. Functionalism was much debated both during and after the exhibition, as a style and as an idea.

34 Åhrén, "Brytningar", 9. "Det är det betydelsefulla draget i modern formkänsla. Den har sin tillräckliga glädje i formen just såsom arbetande. Utseendet framstår snarare som den blotta yttre aspekten av själva arbetet, den rationella tekniken, den förnuftiga ekonomin."

35 Gregor Paulsson "Konstindustrien och den nya tiden" in *Huvudkatalog. Stockholmsutställningen 1930.* (Stockholm, 1930), 36-37.

The debate on functionalism caused a big conflict in the SSF between "traditionalists" and "functionalists".[36] In response to the critique, the committee behind the 1930 exhibition jointly wrote and published the book *acceptera* in 1931, which explained and argued the principles of functionalism in relation to the big societal picture as well as to the everyday issues of consumption and interior decoration.[37] To reach a substantial reform towards a democratic society in which people would fully embrace modern and active citizenship, life in the home needed to change. In the 1931 functionalist publication *acceptera*, this was stated as the problem of different aspects of life and society not changing quickly enough in relation to each other:

> With only slight exaggeration one could say that we live,
> or have lived until recently with
> nineteenth-century **upbringing**
> eighteenth-century **domestic life**
> sixteenth-century **religious life**.
> Work and amusements develop most quickly, then come
> upbringing and social issues, followed by domestic
> adaptations to the new circumstances, and finally religion.

The home, therefore, could not continue to be furnished and used as it was in rural Sweden but must be adapted to the modern-day.[38] As living conditions and everyday life changed, the environments and things needed to support work and leisure also must change. The authors of *acceptera* proclaimed, all the radical changes brought on by modern life would bring about a completely new type of family. The responsibilities for child-rearing and housework would be shared in new ways, and women could, therefore, also engage in salaried work outside the home. With the shifting relations within the family, everyday work and leisure would – and should – play out in other ways than earlier. People, therefore, needed to acquire the things that were suited to their own needs, their financial means, and to the times they lived in. One should start with what one had, and piece by piece construct an everyday environment in the home based on needs, functions, and personal preferences. This led the authors of *acceptera* to promote that industrially produced standardised single piece furniture should be used together with heirlooms, instead of buying whole sets and suites of furniture and things copying historical styles.Every choice of furniture or object should be based on fitness to function, individual preference, and personal financial limits:

> If we can afford it, let us give preference to good handicrafts and really outstanding applied art and acquire furniture and household items that also meet our requirements in terms of individual form. If we do not have the money – and this applies to most of us – the let us choose from cheap but nevertheless good and well-designed standardised furniture. A great deal is already available of this kind and there will be more, if only we manage to overcome our prejudice against making use of it. [---]**If we furnish our home with the things we really need, the selection will be an expression of the life in the home as we live it.** In this way the personal home evolves naturally and authentically – just as much if each item is also one in a series of humble, impersonal manufactured pieces of furniture.[39]

36 Allan Pred, *Recognizing European Modernities: A Montage of the Present* (London: Routledge, 1995).
37 Gunnar Asplund, Wolter Gahn, Sven Markelius, Gregor Paulsson, Eskil Sundahl & Uno Åhrén, *acceptera* (1931), in Lucy Creagh, Helena Kåberg & Barbara Lane Miller (eds.), *Modern Swedish Design: Three Founding Texts* (New York: The Museum of Modern Art, 2008).
38 Asplund et al., *acceptera*, 164.

The authors of *acceptera* laid forth arguments that aimed to convince people of the need for changed practices and preferences in the material as well as the behavioural aspects of everyday life. For this change through design to take place the material environment must come about in accordance with the new ways of living that should be supported. Changes were needed in how houses were built, what sorts of things were made, but also in people's understanding of why these material changes were crucial as material infrastructures for modern life.

Promoting the acceptance of 'good design' required educating people, and exemplary exhibitions and information would be needed to bring about the shift that could lead to people appreciating and choosing the 'right' everyday goods for their homes. Following the main principles of modernism, 'good taste' as in the appreciation and acquisition of modern design, was seen as an outcome of a rational function-and need-based way of understanding that the beauty of modern objects and furniture lay in the correlation between what function a thing should serve, and how the thing was designed and made.

In *acceptera,* this was proposed as an entirely new way of approaching the material aspects of human life; a change that in itself was radical, but that the vital thing to understand was that this was a change of principle rather than a change of style or form:

> If we consider this more closely we will become even more certain that we should not view modern building-art as a passing fancy. As we pointed out earlier, its prime characteristic lies in its will to form its materials into an appropriate tool for our lives. If it is to be able to adhere to this principle, its manifestations should not therefore change more frequently or more rapidly than the way in which we live. And, after all, this changes slowly in the more important elements. In addition, modern building-art intends to visibly express the function of things. This too gives us no reason to conclude that the process will be rapid. It is unlikely that our way of seeing and experiencing the expression of function will change significantly in a short period of time.
>
> Finally, we must not overlook the fact that a change in attitude and ways of thinking has taken place in principle, not merely in our taste for forms. Without forcing the comparison, we could say that in aesthetic terms we are now undergoing the same revolution in our very ways of thinking as science did when it liberated itself from its dependence on religion.[40]

Following this functionalist line of thinking, the understanding of modern production, everyday life, and society would form the interconnected basis for a new relationship between form and functionality that would permeate all aspects of life as soon as enough people had understood the logical necessity of this.

39 Asplund et al., *acceptera*, 242.
40 Asplund et al., *acceptera*, 286.

The assumption in the functionalist manifesto *acceptera* was that this shift was imminent and that it was only a matter of time before a standardised production of dwellings and everyday objects would have transformed the last remnants of a pre-industrial taste into a taste for modern life.

The emergence of modern taste would encompass the totality of life, leading people to appreciate new ways of organising their home and family life as well as working conditions. This would include leaving old aesthetics and outdated social norms of the past behind in the transition to a new way of life. Then, people would make the appropriate choices when deciding what to buy for their homes – provided that there were both adequate dwellings and suitable everyday things available and that people actually could afford them. When the Stockholm exhibition opened in 1930, the focus on educating and informing people to make different choices, in regard to needs and use of everyday environments and things, had already been a part of Swedish socio-cultural initiatives for at least a couple of decades.

Changing things

Many Swedish artists, architects and intellectuals had since the late 19th century – inspired by international aesthetic movements and theorists – argued that people needed better living conditions both materially and aesthetically to create a good society. Therefore, how to build better housing as well as how to go about finding ways to push industry towards producing other types of everyday things had become matters of public discussion.

Design theorist and pedagogical reformist Ellen Key had advocated a utility-based home decoration aesthetic of simplicity, influenced by the Arts and Crafts movement and liberalism in combination with social evolutionism. An understanding of real needs and good taste would allow everyone to live a life permeated with beauty.[41] In Sweden, the leading organisation that since the turn of the century 1900 had been actively engaged in trying to "elevate the people's taste" and work towards promoting the industrial production of well-designed things, was Svenska slöjdföreningen (SSF).

From the mid-1910s and onwards, the SSF actively engaged in initiatives to improve the housing situation in Sweden through actions aimed at influencing legislation as well as the material production of consumer goods. In the early decades of 20th century Sweden, this was expressed as an official view of Svenska slöjdföreningen in the seminal publication *Vackrare vardagsvara* (*Better things for everyday life*), in which author Gregor Paulsson built on the ideas of Ellen Key, merged with inspiration from the German Werkbund.[42] Design – or 'art', as it was called in the publication – was expressly called upon to merge with industry in supporting the development of a new social cohesion and a radically different way of life:

41 Ellen Key, *Skönhet för alla: Fyra uppsatser* (Stockholm: Bonniers, 1899).

42 Gregor Paulsson, Better things for everyday life [1919], in Lucy Creagh, Helena Kåberg & Barbara Lane Miller (eds.), Modern Swedish Design: Three Founding Texts (New York: The Museum of Modern Art, 2008). Cf Gunnela Ivanov, *Vackrare vardagsvara: Design för alla? Gregor Paulsson och Svenska slöjdföreningen 1915-1925,* diss., (Umeå, 2004).

> All over the world one social group after another is liberating itself from dependence on such factors as economic individualism, free competition, pointless exploitation – factors that characterised society in the nineteenth century. They are all striving for – if one may venture to group them under a couple of slogans – improved organisation and enhancements of working methods and the quality of the products of their work. As natural consequences there ensue changes in individuals themselves and the social and cultural structure of society. The twentieth century will certainly not permit the nineteenth century's display of such pointless waste of people, time, and raw materials, nor its restriction of culture to a few individuals.
>
> Art organised in a new way will form part of this new society. [---] The new use of art, as it is perceived by the younger generations, has every chance of being able to serve to cement society. There should, however, be no doubt now about the desire of art to undertake tasks that are at least considerably more important for society than hitherto.[43]

For these changes to take place, designers must contribute to the industrial manufacture of everyday objects, and people would need to learn which things to purchase. This dual ambition was expressly formulated in the statutes of the Svenska slöjdföreningen and later came to be of significance for the Swedish emergence of industrial design. Central to this, was the idea of bringing together qualities of usefulness with qualities of beauty in everyday things, and to promote a responsible and active approach to consumption in which people would consider these aspects in all decisions to buy, and use, things. For designers, the task was to understand what were the basic needs that people had, to address these in terms of designing functional and beautiful things.

During the 20th century, utility and usefulness were increasingly incorporated into the requirements for what designers should attend to in the process of designing. In a modernist, or functionalist, understanding of design's ability to bring about a better life, two main obstacles must be overcome: A change in the production of industrial goods, where design expertise is brought in early in the production process, and a shift in people's preferences to promote actively choosing to purchase modern designed things.

Therein lay the implicit idea – sometimes also explicitly formulated – that once a systematic production of well-designed things had come about, and people had learned the basics of making informed choices of well-designed goods, there would be no more demand for 'bad' or 'ugly' goods. Thus, a better life supported by well-designed material environments and things would be realised on an individual as well as on a societal level. However, for this to happen, active educational efforts directed towards producers as well as towards consumers were imperative. These ideas were expressed already at the turn of the century by Ellen Key, who envisioned a future society where the consumption of well-designed

43 Paulsson, *Better things for everyday life* (1919),121-122.

everyday goods would be not only possible for all, but part of the very foundations of an egalitarian society.[44]

From the turn of the century 1900 onwards, the promotion of a rational and conscious consumption began to be viewed as a strategy for shifting towards a democratic and modern society, in a break with not only old modes of production but also with old ways of living.[45] Buying the right things – machine-made and designed with attention to usefulness – was associated with exercising one's power as a consumer, as well as manifesting adherence to modern social ideas. Through conscious consumer choices of well-designed everyday things, change towards a modern life could be brought about on individual as well as societal levels.[46]

Modernist design practice was informed by the idea that function must not only be equally important as form in design but that designing must begin with questions or perspectives related to function rather than aesthetics. In the rhetoric of modernist design theorists and latter-day design historians' accounts of this, a strong emphasis was placed on rational approaches to form through the distinctions made between ornament and function, craft and industry, style and form. The ambition was to find a rational basis for form. That entailed understanding the basic functions or purposes of an object, and the consequences of this for the choice of form, material and means of production:

> The obvious task for everyday things is to be of service. Our aim when we produce them must be that they will not merely serve adequately but irreproachably. We must also recognise that the better something serves, the less fuss it makes. Everyday things should be utilitarian, unpretentious, calm, and appealing.[47]

Designing, in this understanding, is about the creation of objects that aim to serve in a double sense: on the one hand, the object must serve its inherent purpose, and on the other, it must be in service to a person, in terms of fulfilling its essential functions in relation to the person's basic needs. This implies someone using the object with a particular intent, expecting the object to meet certain demands placed on its inherent abilities to support achieving what was intended.

Though the concept of 'use' is implied, it is not articulated as such other than as an inherent quality that supports a person in doing or achieving something. Despite the main focus placed on the thing, and the intrinsic qualities or properties it is perceived to hold including 'fitness to purpose', the idea of 'use' is highly present here. In the strong emphasis on function not only as an inherent quality in a thing but as a starting point for designing things, what a 'function' or

44 Ellen Key, "Beauty in the Home" [1897], in Lucy Creagh, Helena Kåberg & Barbara Miller Lane (eds.), *Modern Swedish Design: Three Founding Texts* (The Museum of Modern Art: New York, 2008).
45 Peder Aléx, *Den rationella konsumenten: KF som folkuppfostrare 1899-1939* diss. (Umeå, 1994)
46 Peder Aléx, *Konsumera rätt - ett svenskt ideal: Behov, hushållning och konsumtion* (Lund: Studentlitteratur, 2003); Paula von Wachenfeldt & Klas Nyberg, D*et svenska begäret: Sekler av lyxkonsumtion* (Stockholm: Carlssons, 2015).
47 Asplund et al., *acceptera*, 256.

a 'functional' thing is considered to become a guiding question for the process of designing. Given that the persons promoting a functionalist design practice also identified a situation of change as the context for designing, the concept of 'functionality' in relation to needs and uses was also changing.

The designer's task would be to ensure that a sensibility to the unknown or the new would guide the process of ensuring that something would fulfil its function in the best possible way. The design of things should not be guided by traditional ways of supporting everyday practices, nor by established understandings of 'beauty'. The designer's task was to make intended future use manifest in the thing designed, through envisioning the future situation in which the thing would be put to use: firstly, to be produced industrially, and secondly to be adopted into someone's everyday environment.

The design intent, and the definition of functions in relation to materials chosen and form given, here, lies with the designer. To achieve changes towards preferred states of societal and individual forms of living, became a matter of rational planning as well as one of speculative envisioning. If only designers were able to translate basic needs into material solutions where the things' functions answered to these needs, design would be able to radically transform society.

Building better homes

That design could bring about substantial change if based on rational principles of design for mass-production of housing, was a standpoint adopted in an attempt to address the housing situation in Sweden. Private, cooperative and state-initiated building intensified in the 1930s, in many cases based on the modernist ideas of creating minimal but highly functional, modern housing. The cooperative building organisation HSB, for example, adopted the modernist ideals of function-based planning and standardisation to provide housing for its members and worked in parallel with producing home furnishings following the same ideals.[48]

Several different types of new solutions were tried out in floor plans. Minimal kitchens with a separate dining space, a larger hallway which could double as a sitting room, and – perhaps one of the most significant changes – indoor plumbing with both bathroom and toilets, all followed the proposed minimum standards suggested at the 1930 Stockholm exhibition.

The most pressing issue was that of homes being overcrowded, which had been noted and addressed through initiatives aimed at providing better dwellings that would allow more space per person. The question if valuable floor space should be used for bathrooms was contested, and in many cases challenged, by those who advocated that working-class families would be better served by larger rooms rather than bathrooms – since they could bathe in public bathhouses, and were used to outdoor toilets. Which technical features and standards a minimum dwelling should hold in relation to the design of the floor plan was also debated – as were the questions of which functions should be provided and designed for collectively versus individually. The building cooperation HSB set a standard that later came to be adopted more widely in formulations of national building norms: bathrooms in all apartments, central heating, collective laundry spaces, and centralised waste handling, to name a few.

48 Kerstin Thörn, *En bostad för hemmet: Idéhistoriska studier i bostadsfrågan,* Idéhistoriska skrifter 20, diss. (Umeå, 1997).

The 1930s saw a variety of design proposals of different types of dwellings ranging from standardised minimum dwellings for families, to collective houses for families or single-family households, sometimes with integrated daycare, restaurant and washing services in the apartment building. The housing aimed for single-person households was based on two main ideas: One, to provide individual and independent housing for everyone who needed it. But this initiative was also pointed towards getting rid of the practice of families' renting out rooms or sleeping space to lodgers – especially to single men. Since this was considered potentially harmful for the private life of the family as well as an unnecessary addition to the number of inhabitants in already cramped households, different varieties of one-person dwellings would also be needed.

But these initiatives were also aimed at single women, employed in salaried positions, to provide independent housing to free them from having to live as lodgers for either economic or social reasons. Since it was considered inappropriate for single women to live alone, the collective houses were also an attempt to break with social gender norms through new ways of living. So, in the attempts to solve the issue of overcrowded homes and housing shortages, different ways of measuring and evaluating person-to-room ratios were tested to try to find suitable ways of constructing dwellings that could provide a minimum reasonable living standard that people could afford. The design of these dwellings also fundamentally challenged normative ideas about gender, family, and social class, which opened up for other ways of arranging everyday life in the home.

Understanding everyday life

By the late 1930s, several initiatives converged in an attempt to bring about changes in the material conditions for everyday life. These included government loans to young couples setting up homes, the building of different types of dwellings, production of new types of everyday goods and furniture, as well as exhibitions and publications to advocate for modern and fitness-to-function-based ways of furnishing the home.[49] The question now was: Had the availability and accessibility to new dwellings and things brought about any changes in people's ways of living their lives in the homes?

Politically, as well as from an industrial market perspective, the material conditions for the individual Swede needed to be stable if the transition from a poor, agricultural nation to a thriving industrial democracy were to be made. Making sure that people saw the possibility for a better life in Sweden, instead of emigrating, would secure a workforce large enough for industry to develop, which in turn was the foundation for the financial change that would enable the transformation of society.

In the mid-1930s, the housing question became associated with 'the population question'. This referred to the diminishing birth rates that in combination with the past decades of substantial emigration of Swedes to North

49 Maria Göransdotter, "Möbleringsfrågan: Om synen på heminredning i 1930- och 1940-talens bostadsvaneundersökningar", *Historisk tidskrift* 1999:3, 444-474;

America were viewed as an imminent threat to the future of the country.[50] In the sociologically influenced book *Kris i befolkningsfrågan* (*Crisis in the population question*) in 1934, the authors Gunnar and Alva Myrdal pointed to the home as the most critical arena for socio-political interventions. They proposed initiatives aiming to promote the birth of more children and better-equipped homes in which these would be brought up.[51] Equally important would be to bring about changes in family relations and social conditions, allowing both parents to work outside the home. At the same time, state-funded daycare would cater for the necessary pedagogical and psychological development of the children.[52]

If women and men were to be able to participate on equal terms in society – in work life, as well as in relation to each other in social life – it was fundamental that legislation and politics changed the overall conditions for this, but also that changes in norms and behaviours came about on individual level. Besides the attention already raised towards material and economic reforms, there should be a focus on actively changing people's way of living. Again, the home was identified as the arena for many of the proposed solutions to societal problems.

The situation for young people in Sweden, with poor housing and difficulties to find work, was identified as the main reason for declining birth rates. Many refrained both from marriage and from having children. In cases where couples both held jobs, having children was also often postponed due to financial reasons: Both incomes were needed, and having a child meant that one of the two (the woman) would need to stop working since child-care was not available.

Picking up on these issues, the state department of social issues initiated a commission, the *Bostadssociala utredningen* (the State Commission on housing), to investigate how to improve housing conditions in order to increase population growth.[53] The commission presented a proposal already in 1935 to not only subsidise the building of more dwellings but also to grant government loans to young couples that would allow them funding for furnishing their first home. The explicit aim of the loans was to provide a substantial sum of money that could be used for purchasing the necessary things to set up home. The considerable cost of acquiring all of the furniture and items needed had been identified as one of the reasons for people not getting married and thus not contributing to bringing children to the world.

50 Ann-Katrin Hatje, *Befolkningsfrågan och välfärden: Debatten om familjepolitik och nativitetsökning under 1930- och 1940-talen*, diss. (Stockholm, 1974)

51 Alva & Gunnar Myrdal *Kris i befolkningsfrågan* (Stockholm, 1934); Allan Carlsson, *The roles of Alva and Gunnar Myrdal in the development of a social democratic response to Europe's "population crisis", 1929-1938*, diss. (Ann Arbor, 1978).

52 Ann-Katrin Hatje, *Från treklang till triangeldrama: barnträdgården som ett kvinnligt samhällsprojekt under 1880-1940-talen* (Lund: Historiska media, 1999).

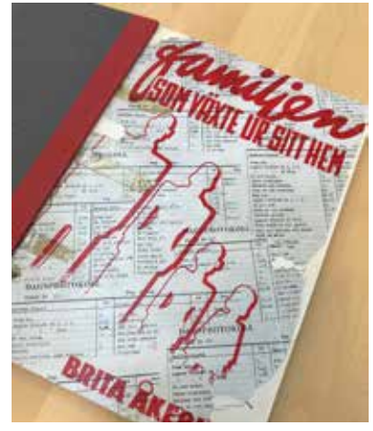
53 The commission worked between 1933 and 1947, and most proposals made were also carried out. *SOU 1935:2 Betänkande med förslag rörande lån och årliga bidrag av statsmedel för främjande av bostadsförsörjning för mindre bemedlade barnrika familjer jämte därtill hörande utredningar.* Bostadssociala utredningen (Stockholm, 1935).



One of the 214 families participating in the study.
Åkerman, *Familjen som växte ur sitt hem,* 6.

During the late 1930s and early 1940s, efforts were made to systematically investigate and understand how people actually used their homes. The State Commission on housing had shown that more than a third of the Swedish population lived in cramped and overcrowded circumstances, with more than two persons per room, in the mid-1930s. The aim with initiating the building of new housing, was primarily to increase both the number of available and reasonably priced dwellings and to provide a wider variety of both larger and smaller apartments to fit different types of households. To achieve an adequate number of new dwellings by the year 1960, which became the aim of the new national housing scheme, the production of housing needed to be standardised and rationalised.

The functionalist ideas adopted into state politics in the 1930s heavily stressed rationalisation and standardisation. Research into engineering and technology was, therefore, promoted and funded to achieve functional solutions. The government housing policy was based on the idea that the new, rational and standardised dwellings should be founded on the actual needs of the average Swede. Thus, a vast array of inquiries into the standards of housing and the practices of everyday life was initiated to lay an objective and scientific foundation for the planning of adequate housing that would support a rational and modern way of life.



Cover of Brita Åkerman's *Familjen som växte ur sitt hem (The Family that Outgrew its Home*), 1941.

The Family that Outgrew its Home

In 1937, an investigation into everyday life in Swedish homes came to trigger a sequence of 'dwelling investigations' over many years to come. Brita Åkerman – whose husband Alf Johansson was an economist and part of the State Commission on Housing as well as a manager in the department of social services – planned and conducted a sociological study of families' everyday life in modern HSB apartments.

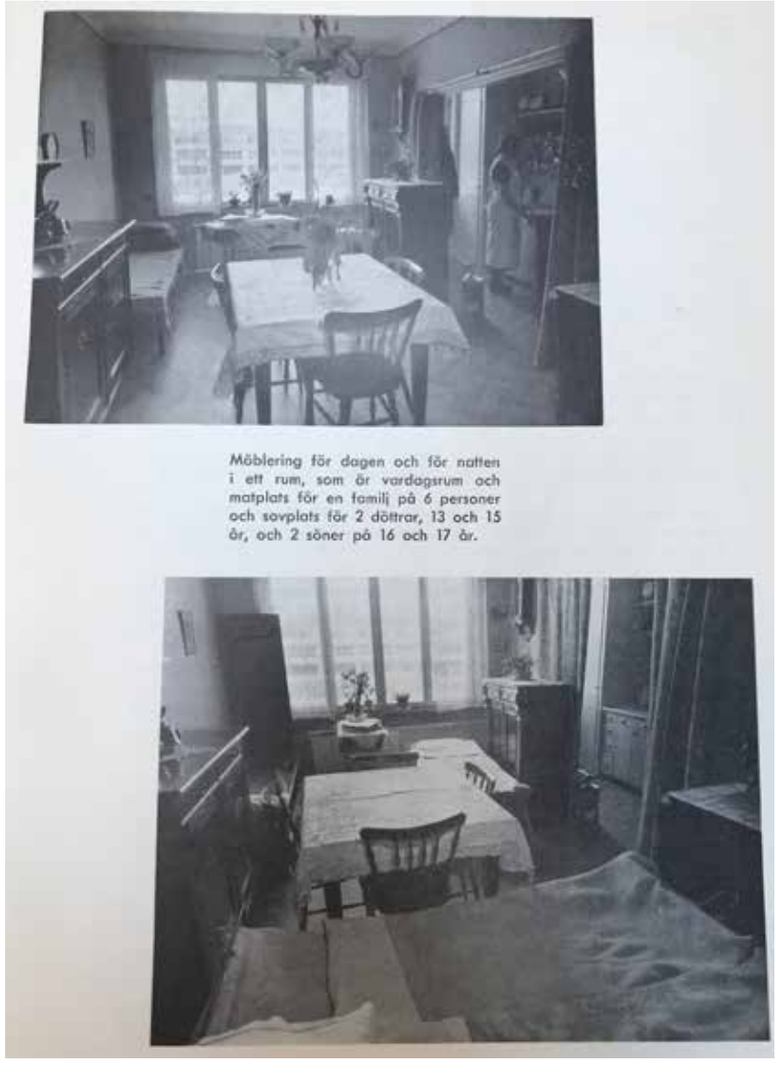
To investigate if social issues and problems for families had been addressed in relevant ways in relation to how their apartments were planned, new sociological methods were applied in the study. A total of 214 families, living in both older and newer apartments were interviewed and asked to describe how life in the home was led, what they appreciated with the apartment and what they would like to change.

The furnishing was documented both by drawing placements of furniture on a floor plan and, in some of the homes, by taking photographs.[54] In 1941, the study was published as *Familjen som växte ur sitt hem* (*The Family that Outgrew its Home*).[55]

The study was a thorough analysis of family life in relation to the dwelling, its furnishings, and the daily work conducted there. A strong emphasis in the book was on women's working conditions in the home, tracking and mapping all the types of work carried out, where, when, and under what circumstances it was done. Brita Åkerman herself was active in different organisations advocating a professionalisation of women's work in the home, and later became one of the founders of Hemmens Forskningsinstitut – which will be discussed more in-depth in the following chapter.

For the process of establishing women's expertise concerning domestic work as a social and political issue, *The Family that Outgrew its Home* became highly influential. But the publication was also seminal in regard to its methodology for studying the impact dwelling design and furnishing had on people's everyday life. In vivid detail, Åkerman described the challenges families handled to make place for all aspects of daily life in the small and overpopulated apartments: how

54 Floor plans and protocols. Brita Åkermans samling. Ämnesordnade handlingar. Vol.7. Handlingar ang. boken Familjen som växte ur sitt hem. Undersökningsprotokoll.
55 Brita Åkerman, *Familjen som växte ur sitt hem* (Stockholm, 1941).



Day-time and night-time furnishing in a room, that is the living room and dining place for a family of 6 and a sleeping place for 2 daughters, 13 and 15 years, and 2 sons of 16 and 17 years. Åkerman, *Familjen som växte ur sitt hem,* 31.

and where family members ate and slept, where the children were allowed to be and to play, where clothes were stored and how housework was carried out. The interviewed women often expressed frustration over the minuscule kitchens in the modern apartments, and the lack of space to cook and bake.

Another aspect of Åkerman's study, which brought new methods and new insights alike, was the direct inclusion of 'ordinary people's' – and specifically women's – opinions in the final publication of the investigation. Besides quoting the observations of the interviewers, Åkerman brought forth the opinions, experiences, and comments of the persons participating in the study. While the direct quotes from the participants could, of course, have been chosen to support the findings of the team making the investigation, they nonetheless provided insights into the everyday life of Swedish families. With incorporating these quotes, the voices of the inhabitants of apartments, the users of furniture and homes, were given a central place in the publication.

The alternating quotes from people sharing their everyday experiences and the interviewers' comments, together with photographs of both women, men and children in their everyday environments, gave glimpses into what life could be like in the small, often overcrowded, homes. Brita Åkerman concluded that planners, architects and furniture designers must actively listen to the expertise of people in general, and of housewives in particular, to design dwellings and things that would make home life and housework easier and more efficient. The ideas that families moving into apartments planned according to functionalist principles would be able to lead lives more adapted to the modern times – including furnishing their homes in accordance with their needs now that industrially produced things were available – was demonstrated to not be the case.

Åkerman noted that no matter if they lived in older or newer apartments, people would continue to live as they had been used to, and to buy the same sorts of furniture and things that had been the norm so far. This was usually suites of furniture consisting of a set number of 6-8 pieces, in a historical or contemporary cohesive style, and placed in the home according to conventional practices. In these parlour-furnished rooms, the children were not allowed to play, and the parents rarely used the room daily either. Families living in larger apartments would often double the parlour furnishing in more than one room, and hardly anyone had a regular bed for sleeping but used different variations of sofa beds or stow-away mattresses:

> As naturally as the families submit to that the walls around them have their given measurements and given placements, they submit equally naturally and without any discussion to a given set of furniture and a given placement of these. One generation after another of newlyweds who set up home follow the same norm for what a home should look like.[56]

56 Åkerman, *Familjen som växte ur sitt hem,* 19. "Lika självklart som familjerna fogar sig i att väggarna omkring dem har sina givna mått och sin givna placering, lika självklart och diskussionslös infogar de sig i en given uppsättning möbler och en given placering av dessa. Den ena årskullen av nygifta efter den andra följer vid sin bosättning samma norm för hur ett hem ska se ut."

People would buy furniture when they were married and moved into their first home together, committing to monthly down-payments of the very expensive pieces for years to come. In Åkerman's study, these consumption habits were seen as detrimental to people's ability to change furniture as the living conditions changed, since they were stuck with the furniture payment scheme, and couldn't afford to buy new things. But this was also difficult from the perspective of the use of the home, especially for children. The 'over-furnished' parlours and small kitchens left children without spaces that were theirs in the home, and Åkerman – echoing the message in the Myrdal's publication *Crisis in the population issue* – connected the low birth rates to the traditional norms that guided the general ways of furnishing homes. People needed to learn how to use their homes in more flexible ways, suited to other possible ways of life, instead of conforming to established norms and habits:

> In hundreds of small instances, we have been able to study this peaceful – and a bit monotonous – family life, where the children are guided forth from diapers, over the sandbox and the homework, to communion, and the first employment. And the question is: is there not something missing? Does not this existence entail an inadequate use of resources, that would be available for people if they only took charge of their powers? Is it not a pity that husband and wife become so settled, would they not be able to continue playing a little also after the time of youthful erotic stimulation has passed? Wouldn't there be a need for a little more imagination and carelessness, a little less shiny furniture surfaces, and frankly, a bit more messy curtains?"[57]

With a more carefree attitude towards the upkeep of the home, there would be more time for husband and wife to explore possibilities to continue developing themselves, and also spend more time with each other, and with their children. In this context, Åkerman also stated that designers would need to be more active in "exploding the old system of furniture and create a new one, that is more flexible, simple and stimulating for the imagination" by designing things that people *actually* needed and could use in a decent way, rather than design what the one *thinks* that people need.[58] The gap between design intent and people's actual ways of living pointed to other foundations, that took into account the actual use of things and homes, needed for the planning of modern housing, as well as for the design of furniture and other everyday objects.

Even before its official publication, the investigation conducted by Brita Åkerman met a great deal of interest. In the first three years of the 1940s, three subsequent studies into people's 'dwelling habits' were carried out, all with connections to Svenska

57 Åkerman, *Familjen som växte ur sitt hem*, 237. "I hundratals smådrag har vi kunnat studera detta fridsamma – och en smula enformiga – familjeliv, där barnen lotsas fram från blöjorna över sandhögen och läxboken till konfirmationen och den första platsen. Och frågan ställer sig: är det inte något som saknas? Innebär inte denna tillvaro ett bristfälligt utnyttjande av resurser, som skulle finnas tillgängliga för människorna, om de spände sina krafter? Är det inte synd att man och hustru blir så stadiga, skulle de inte kunna fortsätta att leka litet också sedan ungdomserotikens stimulans är förbi? Skulle det inte behövs lite mera fantasi och sorglöshet, litet mindre blanka möbelytor och, rent ut sagt, lite slarvigare gardiner."

58 Åkerman, *Familjen som växte ur sitt hem*, 20.

slöjdföreningen and all based on the questions raised and methods developed by Brita Åkerman. The aims of the different dwelling habit surveys were not only to describe *how* but also to find out *why* different practices came about in the home.

A 'dwelling habit' was understood as the way people actually used their homes and how everyday practices played out in relation to the intended functions of both floor plan and furnishings. In all of these three dwelling habit investigations, the motives for the studies differed. Still, the intent of what to study was similar: the design of both dwellings and furniture must take into account how people really live in and use them, to better design the objects and environments that allow people to match their home environment to their real everyday needs. Therefore, design should not only incorporate attention to people's actual practices and wishes. Designers must also consider how to go about shaping people's way of using both apartments and things through design, and through explicit explanation and education aiming to form preferred behaviours.

The SAR and SSF dwelling investigation

In 1938, the SSF received a joint invitation together with the Association of Swedish Architects (SAR) to participate in a housing exhibition in Helsinki, Finland, the following year. The Swedish contribution to the exhibition was asked to target the housing situation for families with many children. This became the starting point for a more substantial joint SAR and SSF investigation that would not only provide material for the exhibition but also form a basis for developing norms and standards for future Swedish dwelling design.

Inspired by Åkerman's study, the SAR and SSF investigation set out to explore the relationship between dwelling and furnishing in regard to actual use. The plan was to make an even more systematic analysis of the functions of the dwelling in relation to how it was used. The project initially consisted of three interlinked studies: "a theoretical inquiry into plans, a special investigation and practical tests regarding housework and child care, investigation of current dwelling habits".[59] The Swedish association for furniture retailers was brought into the project to conduct a fourth study, mapping types of furniture for sale in Sweden in combination with interviews with retailers about their experiences of sales and customer contacts. This aimed to provide knowledge that would lead to producing "the right furniture for the right dwellings."[60] There was, however, no time to do all this before the Helsinki exhibition, so the actual work in the project began first in 1940.[61]

59 *Möbelmarknaden. SAR:s och SSF:s bostadsutredning*, (Stockholm, 1941), 5. "...en teoretisk planutredning, specialutredning och praktiska prov rörande hushållsarbete och barnavård, undersökning av rådande bostadsvanor."

60 *Möbelmarknaden. SAR:s och SSF:s bostadsutredning*, (Stockholm, 1941), 6. "...de rätta möblerna för de rätta bostäderna."

61 For the Helsinki exhibition, the SAR and SSF focused on presenting the housing issues and plans for the investigation. The exhibition was only open three days, during the autumn of 1939: it was shut down prematurely due to the outbreak of the war. Centrum för Näringslivshistoria; Föreningen Svensk Form; F1:96 Registraturen 1938-1970; Vol. 598a Möbelinventering i samband med Statens bosättningslån 1939-1940 "Redogörelse för deltagandet i Bostadsutställningen 39 i Helsingfors och Program för fortsatt utredning"

In the autumn of 1942, the SAR and SSF initiated a survey of dwelling habits. This was led by Gotthard Johansson who was a member of the SSF board, and as editor and art critic actively engaged in promoting functionalism since the early 1930s. Interior architect and furniture designer Lena Larsson was hired to make visits to one hundred families in apartments in Stockholm, built after 1935, in which she documented the dwelling habits in relation to floor plans and furnishings.[62] Larsson would interview the inhabitants about how they used the different rooms and the furnishings, and make detailed descriptions of the interiors, schemes of furniture placement, and finally, the whole apartment was photographed by her husband Mårten Larsson.[63] The outspoken ambition was to establish, through these interviews, a "collaboration between professionals and the public" so that the families would be able to influence future housing production through their "wishes, comments and own experiences".[64]

The analysis of the material was presented in journal articles, and also at the SSF 100-year jubilee exhibition held in Gothenburg in 1945. The final publication of the dwelling habit survey, however, was not made until 1964 due to lack of funding and changes in staff at the SSF.[65] Results of the full SAR & SSF study later led to the establishment of Swedish building norms and standards for housing, including kitchen and storage standards. Additionally, the findings in the dwelling habit survey were later to be decisive in bringing together a variety of popular movements, political parties, and other organisations in establishing education aiming to promote different ways of furnishing and using the home, under the coordination of the SSF.[66]

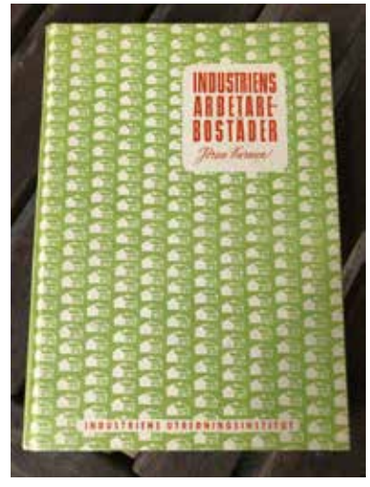
62 Letter: "Vill Ni hjälpa oss skaffa bättre bostäder?" Nordiska museets arkiv. Lena Larssons arkiv. F3 Handlingar rörande Svenska slöjdföreningen 1933-1944; Vol. 6 Bostadsvaneundersökningar. Arbetsmaterial 1942.

63 Arkitekturmuseets arkiv. SAR & SSF bostadsutredning. Vo.l 63 100 fam i 100 läg: Familjer och lägenheter; Vol. 79 100 fam i 100 läg.; and 100 fam i 100 läg, Enkät 1-50 and Enkät 51-100.

64 Lena Larsson "På visit hos 100 familjer", *Form* 1945:1, 12-16

65 Gotthard Johansson (ed.), *Bostadsvanor och bostadsnormer. Svenska Arkitekters Riksförbunds och Svenska Slöjdföreningens Bostadsutredning. Dwelling Habits and Housing Norms. The Housing Investigation of the National Association of Swedish Architects and The Swedish Society of Industrial Design* (Stockholm: Kooperativa förbundets bokförlag, 1964), 11. The inclusion of English summaries in each chapter of the publication, as well as having an English translation of the title of the book on its title page suggests that the aim of the publication was to contribute with methodologies and findings in housing research to not only a Swedish, but als to an international research context. When quoting the book, I sometimes use the original English text, and sometimes translate the Swedish text (which often gives more detail). When I have translated text myself, I include the Swedish original text in the footnote.

66 Göransdotter, Maria, "Möbleringsfrågan: Om synen på heminredning i 1930- och 1940-talens bostadsvaneundersökningar", *Historisk tidskrift* 1999:3, 444-474.



Cover of Jöran Curman's *Industriens arbetarebostäder,* 1944.

The Industries' housing investigation

In the same period as the SAR and SSF investigation was conducted, a research unit funded jointly by Swedish private industries, *Industriens utredningsinstitut,* decided to initiate a housing investigation of their own, also with inspiration from Åkerman's work.[67] In smaller cities and rural areas without any municipal or state-funded housing, it was the responsibility of employers to provide adequate housing for their employees. Around half of the Swedish industrial workers lived in smaller cities or communities, where one single employer provided both work and housing.

As Swedish industry expanded rapidly in the 1930s and 1940s, the housing situation had become a complicated issue to address also for industry, since construction was costly, demand for housing high, and workers' wages low. In this context, an extensive investigation into conditions specific to the production of low-cost, high-standard, workers' housing was needed. Part of this entailed a dwelling habit investigation aiming to understand if industrial workers' needs and practices differed compared to those of city dwellers.

Therefore, four dwelling habit surveys were carried out by architect Jöran Curman. He had worked at the cooperative building organisation HSB and served as secretary in the State Commission on Housing and was, therefore, well acquainted with Brita Åkerman's investigation, and followed her methodology. The surveys Curman made among 263 families, living in different parts of Sweden were also analysed in comparison to the dwelling habit investigations made by Åkerman as well as those made by SAR and SSF. Curman's study also included interviews and questionnaires accompanied by photo documentation of furnishing as well as detailed descriptions of placement and types of furniture.

The study aimed at finding rational ways of providing enough adequate housing for workers, so that their living situation would have positive effects on their work conditions, and thereby the expansion of Swedish industry. The conclusion was that industry needed to invest in standardised mass production of

67 Decided in 1941, the investigation was carried out in 1942-43.

larger and better dwellings to secure better conditions for the workforce. Through building new types of houses, people's "opinions, habits and also thereby their needs" would gradually change as they began living in these environments. But since people were not used to living in more spacious apartments, they would also need support and education on how to rationally use the new, larger, apartments:

> People must, then, in each case *learn* to live in 2 or 3 rooms and kitchen, since this is something new to them. This will entail a very substantial information and PR campaign to influence people to adapt their views and habits in the desired direction, but that is the case for every new item, that is taken to market. If this information can be built on a trustworthy investigation material rather than base subjective suppositions and ideas, the expectations for success should be the best.[68]

Government dwelling loans

Many of the social reforms brought about by the Swedish social democratic government in the 1930s aimed at creating better conditions for families, not least materially through providing means for more spacious and modern housing. The *Bostadssociala utredningen* (the State Commission on Housing) presented several proposals, but also the *Befolkningskommissionen* (the Population Commission) came to identify reforms and legislation regarding housing and homes as necessary.[69]

For the Population Commission, the main goal was to initiate state-funded measures that would ensure that the population in Sweden increased: More children needed to be born, and creating better homes was identified as a crucial part of this. One of the initiatives implemented to increase the number of couples marrying and setting up home – and having children – was to initiate government loans that could be used for purchasing furniture and household items.

In 1941, the Population Inquiry was formed to follow up and evaluate the results of the reforms implemented during the 1930s on the suggestion from the Population commission and also to push the implementation of the previous commission's other proposals. As part of this work, an investigation into dwelling habits was made to analyse the outcomes of the more than 50.000 government

68 Jörgen Curman *Industriens arbetarebostäder* (Stockholm: Industriens utredningsinstitut, 1944), 134. "Folk måste alltså i varje fall *lära* sig bo i 2 à 3 rum och kök, vilket för dem är något nytt. Det fordras visserligen en mycket kraftig propaganda- och reklamkampanj för att påverka folk till att anpassa sina åsikter och vanor i önskad riktning, men så är förhållandet även för varje ny vara som skall föras ut på marknaden. Om denna propaganda kan bygga på ett vederhäftigt utredningsmaterial i stället för på lösa subjektiva förmodanden och föreställningar, synas utsikterna till framgång vara de bästa."

69 The Befolkningskommissionen (the Population commission) was active between 1935 and 1938. Later, the Befolkningsutredningen (the Population inquiry) was initiated and between 1941 and 1947 was tasked with implementing the proposals from the Befolkningskommissionen. As part of the 1941 Population inquiry, a specific Women's delegation was formed to address issues of home life, housework, and women's rights.

loans granted to couples setting up home since 1938. Commissioned to conduct this investigation were Åke H Huldt, interior architect and vice president of the SSF, and Ingeborg Waern Bugge, architect and member of the SAR. Both had first-hand insights into both Brita Åkerman's investigation, and of course into the dwelling investigation jointly carried out by SAR and SSF.

Through questionnaires and interviews with 164 families in eight different cities, the investigation focused on how the dwelling loans had been used, and if that had resulted in the "desired family development".[70] The conclusion presented in the official state report in 1943, was that in practically all cases, the loans had indeed been used in the intended way: for buying home furnishings and making it possible to set up a first joint home. On the other hand, the study also showed that "the loans often have been used wrongly or impractically in regard to the distribution of financial means available between different categories of furnishing on the one hand, and in regard to the real needs and economic resources of the clients on the other."[71] Therefore, guidance was suggested as necessary for couples applying for dwelling loans to support them in buying appropriate things suited to their needs, and that a counselling service should be set up for this.

Questions of use: Functions, rooms and furniture

These dwelling investigations all took a starting point in attempts to map how homes were used on a daily basis. In the SAR and SSF survey, "dwelling habits" were defined as "nothing more than the *way in which the dwelling is* used, regardless of whether this is determined by habits in the stricter sense or by more deliberative acts."[72] The initial ambitions had been to map functions identified in relation to the intended use of rooms as these were defined in the floor plans: "*The living room* is primarily meant for the family's daily being together, *the bedroom* to sleep in, *the dining room or dinette* to eat in, *the kitchen* to cook in."[73]

Many of the apartments and houses built during the 1930s were designed in ways that more corresponded to ideal ideas of living than the actual possibilities for – or interest in – these new ways of life. The attempts to "counteract or if possible hinder" a "wrongful use" of the dwelling through floor plans and fittings had been made in housing design without any real insights into what effects these initiatives would have:

70 *SOU 1943:18 Utredning och förslag angående planmässigt sparande för familjebildning och statens bosättningslån* (Stockholm, 1943), 15.

71 *SOU 1943:18 Utredning och förslag angående planmässigt sparande för familjebildning och statens bosättningslån* (Stockholm, 1943), 13.

72 Johansson, *Bostadsvanor och bostadsnormer*, 33.

73 Johansson, *Bostadsvanor och bostadsnormer*, 109. "*Vardagsrummet* är främst avsett för familjens dagliga umgänge, *sovrummet* att sova i, *matrummet* eller *matvrån* att äta i, *köket* att laga mat i."

> But one also has through the very design of the dwelling intended to counteract or, if possible, prevent a perceived inaccurate use of it. Aiming to exterminate the so-called living kitchen and with it the associated parlour system, kitchens were for a long time so small, that there was no space in them for other than the pure functions of the kitchen – and sometimes not even for these. Through the differentiation of rooms and the very naming of the rooms – "living room", "bedroom" – one has tried to stimulate to a certain use of these, but without assuring oneself that what was intended has become realised.[74]

The questions guiding these studies revolved around creating a solid understanding of what could be seen in terms of causal effects, and what in more value-based ideas or traditions, that governed how a dwelling was actually used. Was the floor plan, and the way an apartment was fitted a decisive factor? What roles did the choice and placement of furniture play? And did social norms and conventions steer how people behaved in the dwelling, or were other factors more influential?

In order to investigate this, questions were asked of the daily activities of the inhabitants; where one slept and ate, and where social activities took place, and where, how and by whom work – housework, homework and other work – took place. Therefore, the surveys were planned as in-depth inquiries rather than only quantitative questionnaires: "The investigators were not so much interested in discovering the incidence of dwelling habits, as their cause-effect relationship, above all the correlation between dwelling design and use."[75] One of the conclusions was that while efforts should continue in mass-producing housing of higher standards, there must also be efforts made to promote coherence between the design of these apartments and of how they were used. The processes of designing modern dwellings had thus far been all too loosely founded on unsubstantiated ideas of what people were doing in their homes:

> The significance of dwelling habits for actual housing standards has long been appreciated, and many different efforts have been organised to combat habits that are felt to deteriorate these standards. Various kinds of publicity and information activities in the form of exhibitions, study circles, etc. in Sweden (with the Society of Industrial Design as the prime mover) have been and are being conducted to interest the public in a more rational and effective use of the dwelling based on actual needs. In the course of these activities, the negative aspect of dwelling habits has been emphasised to the point where their positive aspect is readily overlooked. The very concept of dwelling habit has come to connote something irrational, a conventional view of things, or a routine pattern of behaviour. All dwelling habits, of course, need not be poor dwelling habits. They may also be justified by objective realities. Ideas on how a dwelling *should* be used may be just as conventional as those which, consciously or unconsciously, motivate how it *is* used.[76]

74 Johansson, *Bostadsvanor och bostadsnormer,* 27. "Men man har också genom själva utformningen av bostaden sökt motverka eller om möjligt förhindra en som felaktig ansedd användning av densamma. I avsikt att utrota det s.k. bostadsköket och det därmed förbundna finrumssystemet gjorde man sålunda länge köken så små, att det inte blev plats i dem för annat än de rena köksfunktionerna – ibland knappast ens för dem. Genom rumsuppdelningen och själva beteckningen på rummen – "vardagsrum", "sovrum" – har man sökt stimulera till en viss användning av desamma, utan att dock förvissa sig om att den i verkligheten blivit den avsedda."

75 Johansson, *Bostadsvanor och bostadsnormer,* 49.

An example given of such conflicts between intended and actual use was the planning of the kitchen and dining areas in the 1920s and 1930s that had followed modernist ideals of differentiating between and physically trying to separate, functions in the home. It had been taken for granted that the kitchen should not be a place for eating and that the planning of the dwelling should ensure that this was not the case. But after analysing the dwelling investigations, first, the one made in 1937 by Brita Åkerman and subsequent studies after that, the different investigators identified new norms and practices around meals that seemed to have emerged in relation to changes in different social strata.

While the upper and middle classes earlier had found it more or less unthinkable to eat in a kitchen – a practice associated with servants or housekeeping staff, which most of these families employed – eating in the kitchen common in rural homes and working-class families. By the end of the 1930s, the workers' way of taking meals in the kitchen had largely become adopted also in middle-class households.[77] Attempting to separate the function of eating from that of cooking, a small dining area in connection to but separated from the kitchen was introduced in many small apartments designed in the 1930s and early 1940s. In the different dwelling studies, the room called 'the dinette' (*matvrå*) came to be extensively studied as the perhaps most striking example of how design intent and actual use differed.

Using dinettes and living rooms

One aim of the SSF and SAR survey was to find out "where dwelling functions took place, and how the different rooms were used and furnished."[78] When making these connections, the investigators assumed that matching room to function, or vice versa, would be reasonably easy both by observing the furniture itself and through making follow-up interviews. However, this seemed to be more complicated than planned, since the intended room functions only rarely matched the actual use:

> There are living rooms, that are not used for daily interactions, bedrooms that are not slept in, and dinettes that are not eaten in. On the other hand, the use of the rooms on the floor plan marked as bedrooms were not the only ones used but also living rooms and dinettes were used as bedrooms, the real dining room can be the dinette, living room, bedroom or kitchen, and the daily being together can take place either in the living room, dinette, or a bedroom.[79]

76 Johansson, *Bostadsvanor och bostadsnormer*, 33.

77 Johansson, *Bostadsvanor och bostadsnormer,* 27. "The latter development, not only in Sweden but also in other countries, has above all through the lack of servants led to, that many also in the middle class nowadays have changed idea and found it connected to practical advantages to place the dining area in the kitchen."

78 Johansson, *Bostadsvanor och bostadsnormer*, 49.

79 Johansson, *Bostadsvanor och bostadsnormer*, 109. "Det finns vardagsrum, som inte användas för vardagsumgänget, sovrum, som det inte soves i, och matvråar, som det inte ätes i."

Tab. 86. *Funktioner och möblering. / Functions and furnishing.*

Verkliga funktioner[1]	Möblerat för[2]: Umgänge[3] sömn[4] måltid[5] skrivarb.[6]	Umgänge sömn måltid	Umgänge sömn skrivarb.	Umgänge måltid skrivarb.	Umgänge sömn	Umgänge måltid	Umgänge skrivarb.	Umgänge	Måltid skrivarb.	Antal rum[7]
Umgänge[3] sömn[4] måltid[5] skrivarbete[6]	7	2				1				10
Umgänge sömn måltid	1	3								4
Umgänge sömn skrivarb.	11	7	2							10
Umgänge måltid skrivarb.	2			4						6
Umgänge sömn	1	7			3					9
Umgänge måltid	1									1
Umgänge skrivarb.	1		2	2		1	1			7
Umgänge	1	3	3	2	2	2		3		16
Sömn måltid skrivarb.	2	3								5
Sömn måltid					1					1
Sömn skrivarb.	1	1								2
Sömn	2	1			2				1	5
Måltid		2				1				3
—		4				1	1			6
Antal rum[7]	31	34	7	4	8	6	1	3	1	100

[1] Actual functions. — [2] Furnished for. — [3] Gathering. — [4] Sleeping. — [5] Eating. — [6] Writing. — [7] Number of rooms.

Mapping of living room functions in the SAR and SSF study.
Johansson, *Bostadsvanor och bostadsnormer*, 174.

In schematic overviews in the SAR and SSF investigation, the various uses of rooms were mapped for each family, depicting how the use of different parts of the apartment would lead to several functions co-existing in the same space. Only one room, the living room, could in some families completely lack any daytime functions or uses, while an average across all of the 100 apartments showed that the dinette was the room which housed the greatest number of uses in regard to the different functions. Several of these resulted in what was named 'function conflicts', which were regarded as equally important for the actual use of the apartment as conventional dwelling habits were.

What the three investigations showed, was that many families who by any statistical or numerical analyses of square meters or number of rooms per person could have ample living space, in reality, were to be regarded as living in overcrowded conditions. Trying to get to the root of the causes of this was not easy: bad dwelling design, irrational dwelling habits, unsuitable furniture, or inherent function conflicts were entangled in a way that made analyses difficult.

An example of how the design of the apartment had made a negative impact on how people managed to use their homes, was the dinette. The ambitions to differentiate functions had led to the introduction of Frankfurt-style small laboratory-like kitchens, with a separate small adjoining room that was intended to be used as a dining space.

This was not only an attempt to separate the functions of cooking and eating but also of ensuring that neither sleeping places nor playing children would be able to fit in the kitchen since this was deemed unhygienic. This was a design that people – especially women – already in Brita Åkerman's study had criticised thoroughly. Many families stated that they would have preferred a larger kitchen that was also more spacious to work in, and that allowed room for a dining table.

At the same time, the dwelling survey analyses detected that it was, in fact, the dinette that, despite being the apartment's smallest room was the one that used the most. In the Industry dwelling investigation, the intended dinette was used for so many diverse functions, while the 'living room' was often left unused, that the investigator Jöran Curman suggested a change in both denominations and functions of the rooms to better match how people in practice wanted and needed to use them.[80]

80 Curman, *Industriens arbetarebostäder*, 169-177.

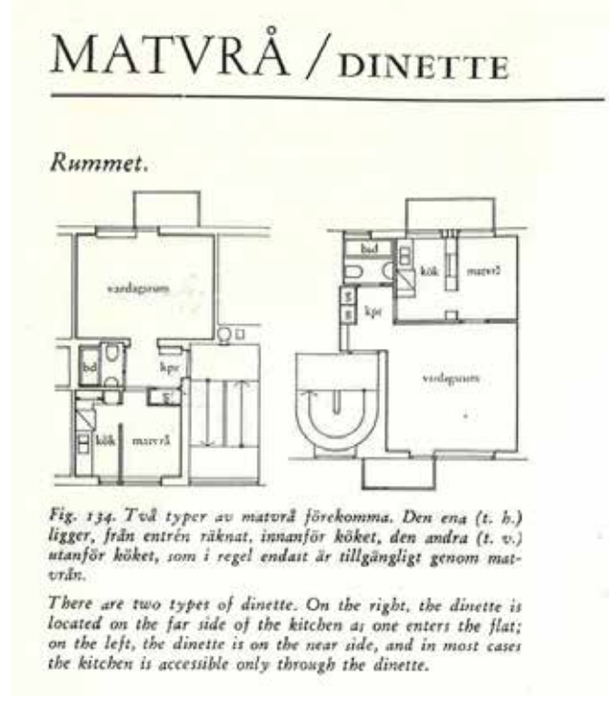


Fig. 134. Två typer av matvrå förekomma. Den ena (t. h.) ligger, från entrén räknat, innanför köket, den andra (t. v.) utanför köket, som i regel endast är tillgängligt genom matvrån.

There are two types of dinette. On the right, the dinette is located on the far side of the kitchen as one enters the flat; on the left, the dinette is on the near side, and in most cases the kitchen is accessible only through the dinette.

Examples of dinettes in the SAR and SSF study.
Johansson, *Bostadsvanor och bostadsnormer*, 200.

Seventy-two of the apartments in the SAR and SSF study held a dinette. Of these, 59 families took all their meals there. But the room was also used for so much more. On the one hand, it was in most homes used exactly in the way it was intended – for eating – but on the other, it was at the same time used for a vast array of functions for which it was neither intended nor designed. Two-thirds of the dinettes held one or more sleeping places, and it was also the main daily gathering place for some 20 families. In many cases, activities like sewing, reading, doing homework, and children's play also took place there. For a room that on average was only about 5,5 square meters, and sometimes even smaller than 4 m², it housed a fair share of many families' daily activities. This led the investigators to proclaim the dinette a "functional abnormality" often holding a "congestion of furniture and functions [that] often becomes fairly grotesque".[81] The causes for this were people's limited understanding of how space should be used properly, in combination with designers being out of touch with the reality of people's actual ways of living. The dinette, therefore, was pointed to as an idea that had been based on designers' and architects' making a "bad guess about people's living habits".[82]

The living room was another example where design intent did not at all align with everyday use. Since the turn of the century 1900, home reform had aimed at establishing the 'living room' as the place where families should gather together for daily activities rather than using it as a 'parlour'.[83] In Swedish, this was explicitly made clear in the naming of the room on the floor plan, as the term is *vardagsrum*, which means "every-day room". This term was contrasted to the terms *finrum* (a room with finery) or *salong* (parlour), which were the uses of the room that were to be discouraged. As noted in Brita Åkerman's study, and confirmed the subsequent dwelling surveys, the practice of using the living room on an everyday basis had not at all been adopted. The room, the SAR and SSF report suggested, could instead be called a "Sunday room", if one were to name it after how it was actually used:

> It was striking how the social and recreational functions were given a disproportionally large place at the expense of the space for sleep and rest. The parlour in its true meaning – a fine, un-used room, whose doors are only opened for important or long-distance guests – hardly exists in big city apartments, but what has come in its place is what we could call a "Sunday room". "Would like to have a nice room if someone visits" is a very common statement – usually said as an excuse, for the propaganda about the abhorrence of the parlour has at least as a rallying cry come through quite well.[84]

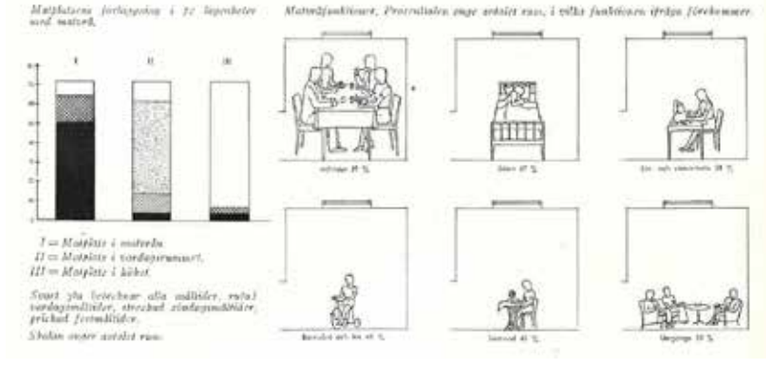
81 Johansson, *Bostadsvanor och bostadsnormer*, 206.
82 Johansson, *Bostadsvanor och bostadsnormer*, 206.
83 Kerstin Thörn, *En bostad för hemmet: Idéhistoriska studier i bostadsfrågan,* Idéhistoriska skrifter 20, diss. (Umeå, 1997).
84 Lena Larsson "På visit hos 100 familjer" Form 1945:1, 16. "Frapperande var hur funktionen sällskap och rekreation fick en alldeles oproportionerligt stor plats på bekostnad av utrymmet för söm och vila. Finrummet i dess egentliga betydelse – ett fint, oanvänt rum, vars dörrar slås upp endast för storfrämmande och långväga gäster – existerar väl knappast i storstadslägenheterna, men i dess ställe har kommit något som skulle kunna kallas "söndagsrum". "Vill gärna ha ett rum snyggt, om det kommer någon" är ett mycket vanligt yttrande – vanligen sagt som ursäkt, ty propagandan om finrummets styggelse har åtminstone som slagord trängt igenom ganska bra."



Examples of dinette furnishing arrangements.
Johansson, *Bostadsvanor och bostadsnormer,* 205



Dinette with 'overloaded functions.'
Johansson, *Bostadsvanor och bostadsnormer,* 201



Dinette functions: Staple diagrams over where meals are taken, and different uses of the dinette in percent.
Johansson, "Hur bor vårt folk?", 10.

In the study made of how dwelling loans were used, similar conclusions had been made. People often spent most of the loan buying furnishings for the living room: "An especially prominent position in the overview of the situation, that the investigation gives, is taken by the general presence of the 'fine room' which with its costly, stereotype and unpleasant furnishing does not fill any real function."[85] While the habit of furnishing the living room in such a way was understandable from a perspective of tradition and ambition to keep a home atmosphere that was based in the sanctity of home life, this practice was nonetheless something to try to work against:

> In small apartments, the parlour becomes a parasite, that prevents crucial tasks from being carried out. The intent may be laudable, but the result becomes grotesque, when one in these small homes finds that most of the space and the main share of the resources when setting up home is used for a room that does not support the family's daily life, indeed, that through its furnishing has become such that it cannot even do so. In its consequences it will therefore lead to directly counteracting the intended family development. If one has reason to talk about dwellings that are unfriendly to children, one can as easily talk about furniture and furnishings that are directly hostile towards children.[86]

The wide-spread dwelling habit of furnishing the larger room of the home with upholstered and polished furniture in a representative setting was a dwelling habit that – when probed in the surveys – at first seemed to be completely irrational, but upon closer investigation might have an at least partially rational foundation. The fact that many chose to keep the parlour furnishing rather than adapt the room to needs of socialising as well as providing sleeping places, seemed to perhaps also be grounded in conflicting needs and functions, and not only in conforming to outdated social norms.

In the investigation of workers' dwelling habits, Jöran Curman noted that the daily non-use of the parlour or living room had quite rational reasons. The fact that people didn't use the room on a daily basis, he meant, was based on the well-founded ambition to keep a nice room where dirty work-clothes would not soil

85 *SOU 1943:18 Utredning och förslag angående planmässigt sparande för familjebildning och statens bosättningslån,* 14. "En särskilt framskjuten position i den bild av läget, som undersökningen ger, intar det allmänt förekommande 'finrummet', som med sin påkostade, stereotypa och otrivsamma inredning inte fyller någon verklig funktion."

86 *SOU 1943:18 Utredning och förslag angående planmässigt sparande för familjebildning och statens bosättningslån*, 14. "I småläghenheter blir finrummet en parasit, som hindrar väsentliga uppgifter att kunna tillgodoses. Avsikten må vara lovvärd, men resultatet blir groteskt, när man i dessa små hem finner, att det mesta utrymmet och större delen av hemmets tillgångar vid bosättningen utnyttjas för ett rum, som icke tjänar familjens dagliga liv, ja, som genom sin möblering blivit sådant, att det icke ens kan göra det. I sina konsekvenser kommer det därför att direkt motverka en önskvärd familjeutveckling. Om man har skäl att tala om barnovänliga bostäder, kan man med lika stor rätt tala om möbler och möblering som äro direkt barnfientliga."

the furniture. If one were to get people to use their finely decorated living room, the consequence would be that "the housewife would have a hefty workload with constantly cleaning and replacing things in the living room, minding the children, changing and washing covers on the upholstered furniture, etc., or that upholstered and polished furniture pieces, certain decorative objects etc. were not allowed in the home."[87] Taking away the possibility for people to have a nicely decorated room for festive occasions and Sunday use would just be meaningless:

> This leads to conclude that the dwelling habits here discussed basically are very well-founded. They are in their main traits neither built on habit nor delusion, but are based on clear needs and practical necessity, and must, therefore, be assumed to continue in the future, as long as the purely technical conditions are the same.[88]

Designers would, then, have to consider these practices of use when designing future dwellings. At least for working-class families in the industrial regions of Sweden, the living room ought to be allowed to be the Sunday room it had turned into. At the same time, the dinette should be given an equal size as the living room, in its capacity of a daily room – and be renamed as such.

Necessary and voluntary overcrowding

One of the core problems in the housing question had been that of homes being too small for the families living in them. An aim was, therefore, to find out if the newer and more spacious apartments built in the 1930s had brought about improvements in the families' living conditions. One of the most crucial functions to fit in the dwelling was that of providing enough space for beds and sleeping furniture for the whole family. Mapping the function of 'sleep', therefore, was a central aspect of all of the different surveys.

Since 'sleep' as a function was deemed quite easy to define, as would be the appropriate furnishing for this with different pieces of furniture, the use of the home for sleeping in was thought to be perhaps easiest to reach substantial knowledge and conclusions about. However, finding out where people slept and how they arranged the furniture to support this brought about methodological difficulties in all of the three investigations. There seemed to be some correlation between the size of the apartment and the practice of trying to allow family members as separate sleeping places as possible. However, the practice of

87 Curman, *Industriens arbetarebostäder,* 175. "…att husmodern fick ett mycket betungande arbete med att ständigt städa och återställa föremål i vardagsrummet, passa på barnen, byta och tvätta överdrag över de stoppade möblerna m.m., eller att stoppade och polerade möbler, vissa prydnadssaker m.m. icke finge förekomma i hemmet."

88 Curman, *Industriens arbetarebostäder,* 175. "Det vill alltså synas som om de bostadsvanor, vilka här tagits till utgångspunkt för resonemanget, i grund och botten äro mycket välgrundade. De äro i sina huvuddrag icke byggda på slentrian och vanföreställningar utan grunda sig på klara behov och praktisk nödvändighet och måste därför antas att komma att bestå i framtiden, så länge de rent tekniska förutsättningarna i huvudsak äro desamma."

"clustering" beds – placing many sleeping places closely together in the same room – seemed to have little to do with the actual layout of the home:

> If sleeping habits follow a certain pattern, this seems to be fairly complicated and often confusing. Even in the most tightly clustered families, white areas tend to appear: rooms, that completely lack sleeping places. When the web dissolves, when overcrowding diminishes or completely disappears, these areas tend to increase, but are here also here fairly distributed and are often combined with clusters in other parts of the apartment – in order to keep one or several rooms free from sleeping places one has instead more than filled others, and thereby submitted oneself to voluntary overcrowding.[89]

In the SAR and SSF investigation, the measurements of floor plans in combination with the findings from the furniture retail overview had shown that the rooms were often too small for the sizes of the beds on the market. This was a reason for people getting other types of sleeping furniture than proper beds. The market survey of furniture had also highlighted this issue, so synchronisation of standard sizes of beds and bedrooms seemed like an adequate design change based on these insights. But investigations had shown that the lack of space was not the only reason for people's difficulties in making room for places to sleep. Some families would indeed be overcrowded due to too many people and too little space. At the same time, others would use the apartment in such a way that while it on paper seemed spacious enough, in practice it became overcrowded, and created difficulties to arrange reasonable sleeping solutions for all family members.

One example was that of how sleeping places were distributed in the 30 families with many children, who lived in so-called *barnrikehus* with apartments specially designed large families. It turned out that even in apartments such as these, with several rooms, people tended to not distribute sleeping spaces in different rooms. Instead, many slept together in one or two rooms. So what on paper looked like an improvement in terms of spaciousness for the families with many children, in practice was still a situation where living conditions were very cramped:

> If one started only with statistical data, i.e. the ratio between family size and apartment size, one would find that only two of these families were overcrowded according to the usual norms of overcrowding. But if one went into the apartments and found out how they in reality were used, it showed that only two of the families were *not* overcrowded.[90]

89 Johansson, *Bostadsvanor och bostadsnormer*, 113. "Även i de tätast sammanklungade familjerna framträda stundom vita fläckar: rum, som sakna sovplatser. När nätet glesnar, trångboddheten minskas eller helt försvinner, bli de flera men ligga även här ganska spridda och förenas ofta med sammanklungningar på andra punkter i lägenheten – för att hålla ett eller flera rum fria från sovplatser har man is stället överfyllt andra och därmed underkastat sig en frivillig trångboddhet."

90 Gotthard Johansson, "Hur bor vårt folk? Bostadsvanorna utforskas" *Form* 1945:1, 8-16. "Utgick man enbart från statistiska data, dvs. förhållandet mellan familjens och lägenhetens storlek, fann man, att av dessa familjer voro enligt vanliga trångboddhetsnormer endast 2 trångbodda. Gick man däremot in i lägenheterna och tog reda på, hur de i verkligheten användes, visade det sig, att ensat 2 familjer *icke* voro trångbodda. Felet låg alltså inte i första hand i bostaden utan i det sätt, varpå den användes.", 9.

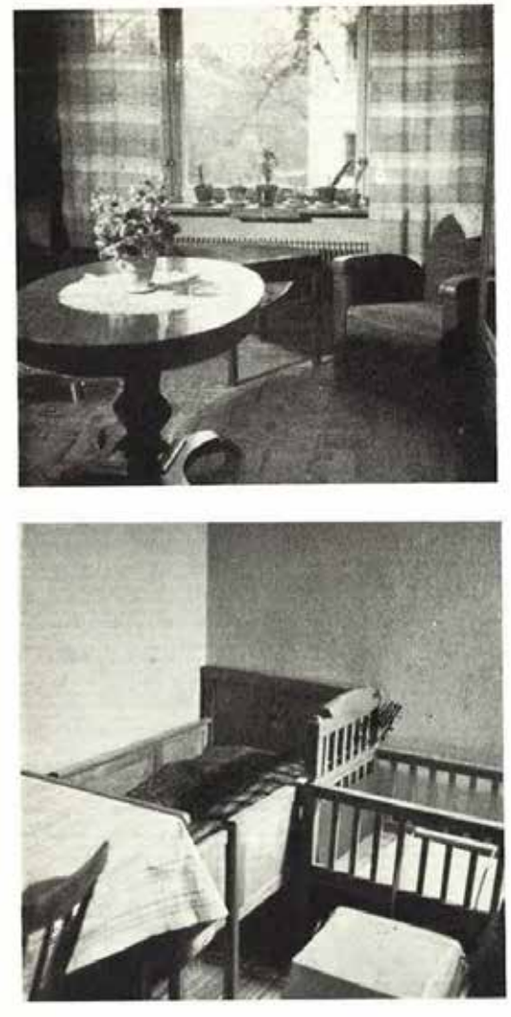


Fig. 66. Frivillig trångboddhet i överbefolkad lägenhet. En lägenhet om 2 rk, där det mindre rummet endast är något över 8 m², bebos av en familj på fyra personer, man, hustru och två barn i resp. 2 och 4 års ålder. Alla ligga i det lilla rummet, som även är familjens matrum, ingen i vardagsrummet, som har karaktären av "finrum". Fam. 19.

Voluntary overcrowding in small dwelling. In a 2-room apartment, where the smaller room is only about 90 sq. ft. lives a family of four persons, man, wife and two children, 2 and 4 years of age. The whole family sleeps and eats in the small room. The living-room is the "elegant room". Fam. 19.

Voluntary overcrowding: A family of four in a 2-room apartment all sleep in the dinette. Johansson, *Bostadsvanor och bostadsnormer,* 114.

Overcrowding would, in these cases, be induced by people on themselves by not spreading out the sleeping places over the different rooms in the apartment, but by sleeping more than two people per room – in several cases as many as 4-5 persons per room. The reason for people's difficult living conditions was, in other words, primarily caused by how the apartment was used – not how it was designed. It seemed to the investigators that this voluntary overcrowding would not be solved by providing more spacious apartments since it was the way people chose to use the different rooms, that was the problem.

A general conclusion, made early on in the process of analysis, was that "(i)t is *not* enough to provide enough space. One must make sure that it is used correctly. That is the first learning from this dwelling investigation."[91] So, not only would apartments need a redesign – but also the people inhabiting these dwellings needed to learn how to use them in a rational manner deemed more appropriate to their actual needs.

The reluctance of people, in general, to use the living room for sleeping in, was often referred to in home reform literature as an expression of an outdated way of furnishing the largest room of the home as a parlour, to be used only for guests and festive occasions, and many attempts were made to correct this practice. In interviews targeting the question of why that was often not the case, parents often raised the need for having a place in the apartment to spend time after putting the children to bed. If all rooms were used for sleeping in, there would be no place for the adults to sit and talk, listen to the radio, or read, after the children's bedtime:

> The ambition to keep one room free from sleeping places needs thus not necessarily be an expression for a despicable parlour system but can have its origin in a conflict of functions, impossible to avoid in a dwelling providing insufficient space.[92]

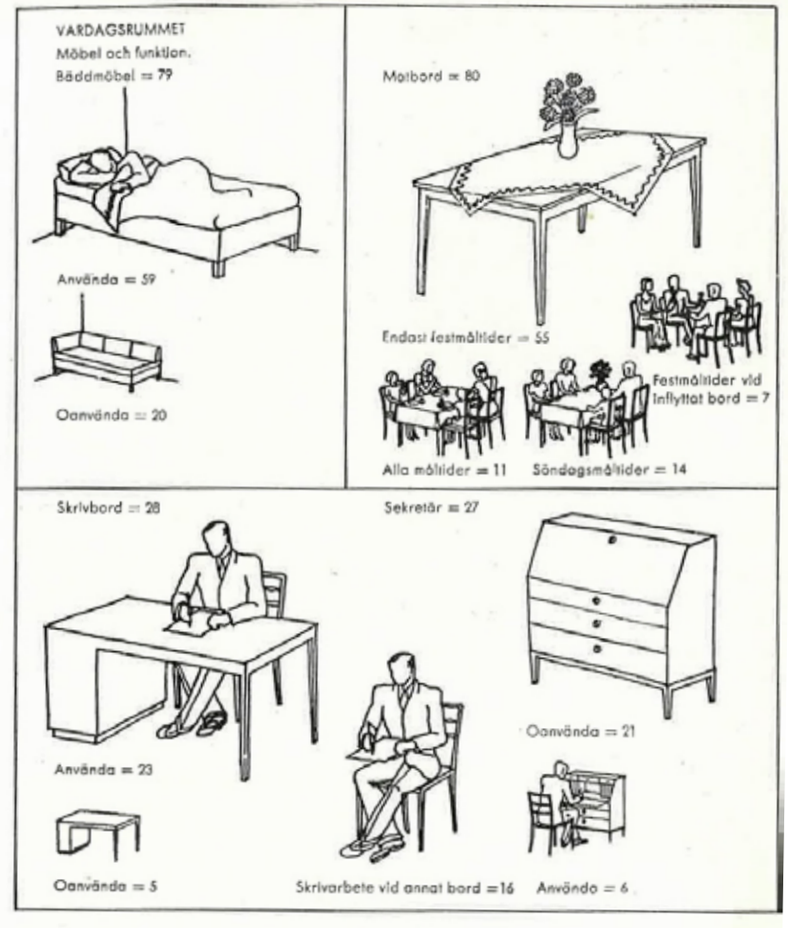
This mismatch between design intent and actual use led to methodological difficulties for the mapping of room functions in connection to the furniture in the home, and the actual functions covered by the use of these in combination:

> The existence of a dining table in the living room does thus not necessarily mean that one eats there. Of the 100 investigated apartments, 80 had dining tables in the living room, but no less than 55 of these stood unused except at festive occasions, and only 11 were used daily. But not even the absence of a dining table has to mean that one does not eat in the living room. It happens, that one moves a table from a room or the attic.[93]

91 Johansson, "Hur bor vårt folk?", 9. "Det räcker *icke* att skaffa tillräckligt utrymme. Man måste också se till att det användes på ett riktigt sätt. Det är den första lärdomen av denna bostadsvaneundersökning."

92 Johansson, *Bostadsvanor och bostadsnormer,* 27-28. "Önskan att hålla ett rum fritt från sovplatser behöver sålunda inte med nödvändighet vara uttryck för ett förkastligt finrumssystem utan kan ha sitt upphov i en funktionskonflikt, omöjlig att undvika i en bostad med otillräckligt utrymme."

93 Johansson "Hur bor vårt folk?", 11. "Förekomsten av ett matbord i vardagsrummet betyder sålunda inte med nödvändighet, att man äter där. Av de 100 undersökta lägenheterna hade 80 matbord i vardagsrummet, men av dessa bord stodo inte mindre än 55 oanvända utom vid festliga tillfällen, och endast 11 voro i dagligt bruk. Men inte ens frånvaron av ett matbord behöver betyda att man inte äter i vardagsrummet. Det händer, att man flyttar in ett bord från ett annat rum eller från vinden."



Möblerna och deras funktioner. Diagrammet visar förekomsten och användningen av olika möbeltyper i 100 vardagsrum.

Furniture and functions. The diagram shows the presence and use of different types of furniture in 100 living rooms. Johansson, "Hur bor vårt folk?", 11.

Similarly, both writing desks and sofa beds, that many had in the living room, were mostly unused, while activities of reading, writing and sleeping happened in other parts of the home. Understanding how the rooms of the apartment were used in relation to the functions identified, required a closer study of how furniture was both placed and used.

The influence of furniture on use

During the SAR and SSF dwelling investigation, and even more so in the later analysis of the material, the specific types of furniture people had was identified as playing a decisive role in the way apartments were initially set up, and then used. An idea that had a strong presence in modernist design theory, as well as in the writings of proponents of home reform in general, was that if people were given the opportunity to live in well-planned dwellings produced according to rational principles, this would not only address the issues of overcrowded and cramped living conditions but also induce a substantial change in how everyday life was led. If people were given enough space, together with information and education on other ways of setting up the home rather than the 'traditional' or 'conventional' way, they would eventually embrace the possibility to furnish according to the needs of the family. This, in turn, would provide the basis for different ways of life.

For people engaged in attempting to reform housing and furnishing, these issues were seen as intrinsically linked to making changes on a societal level: A different home life, labour-saving and function-based, could support a more egalitarian way of life in the home as well as in society. How people could engage in societal democratic processes, and what was seen as the range of potential for personal development of men, women, and children was substantially different from the conditions of previous generations: this should also be materially manifested in the home.

That things – furniture, fittings and everyday objects – were essential to consider in relation to making a different life possible had been on the reform agenda in Sweden since the turn of the century 1900. However, the main emphasis had then been on bringing about a substantial production and distribution of everyday goods, making it possible for people as consumers to buy the 'right' things. Education and information directed towards promoting guidance for what types of things would be preferable to purchase were to a substantial extent based on what was imaginable at that time.

The inquiries into 'dwelling habits' showed that most people in Sweden, during the early decades of the 20th century, would invest in buying the bulk of their furniture and household objects when they were married and set up their first home. After that, the home would not change much in terms of which material things were added to it. Therefore, much of the attention in home reform came to focus on how to affect decisions about which items to buy. If people only buy things – substantial things, like furniture – on one occasion in their life, that very first act of purchasing something became the most critical situation to influence.

Many strategies and initiatives aiming at home reform came to target that specific moment of consumption, the one and almost only occasion when people would buy their costly furniture, committing to a down-payment plan that would last for decades to come. It became important to convince people to buy 'appropriate' things based on their 'actual' needs. Another practice that came under scrutiny was the buying of expensive multi-piece sets of 'suite furniture' that

would then be part of the household for many years to come. What was promoted instead, was buying single-piece furniture, cash on the barrel with no down-payments, only acquiring the pieces needed at that specific moment and with the money one had available. Adding things and furnishings to the household as the family grew and needs changed was something that should come later. This was the theme for many publications and information initiatives from a variety of sources, e.g. the cooperative consumer unions as well as in government subsidies and interventions connected to housing and finances. But did these messages come across, and had peoples' habits of buying and furnishing actually changed?

In the dwelling survey conducted by Waern Bugge and Huldt as part of the evaluation of the use of government dwelling loans and published in 1943, a primary task was to find out if the loans in practice had any effect on the types of furniture purchased, and to the quality of home life achieved. The loans had been successful in supporting more young couples to set up home – there had noticeably been an increase in marriages. However, the results showed that difficulties for young couples in finding both adequate housing *and* suitable furniture had led to "the young home in our country has developed a form, that only to a very low degree responds to the demands of everyday life and that is very poorly suited for a family with children."[94]

In choosing which things to buy, young people satisfied their "imaginary needs" due to lack of knowledge and misguided ambitions in combination with a lack of adequate housing. Young homes in Sweden were primarily dominated by "child-unfriendly dwellings with child-unfriendly dwelling habits", since the lack of bedrooms in combination with the parlour system made the home ill-suited for child-rearing.[95] The dangers were that the habits of buying representative furniture or – even worse radios or bar cabinets – instead of things that were actually needed, was too firmly rooted. In the report, this was exemplified with a couple in northern Sweden who had spent more than half of their loan on buying an expensive suite of furniture for the living room and then only having 25 crowns left to spend on a bed, they bought a one-person camping bed, which they both shared.[96]

The choice of furniture and its placement in the home seemed to be a decisive factor for how the home was used. In many cases studied by Waern Bugge and Huldt, this was neither in line with the intended nor the real functions of either furniture or room, and it was practically impossible to understand how space was used based on how it was furnished:

> One cannot, in other words, directly from a room's furnishing deduce its functions. The existence of a bed does not necessarily mean that someone sleeps in the room, a dining table that one eats there, or a desk that one writes there. On the other hand, the lack of a piece of furniture does not mean that the associated function does not take place in the room. A bed or a table can at the moment of use be moved from another room or a special storage place, writing work can just as well happen at a dining table or any other table as at a desk or

94 "Ungdomen behöver vägledning vid hembildningen" *Form* 1943:1, 22.
95 "Ungdomen behöver vägledning vid hembildningen" *Form* 1943:1, 24.
96 *Waern Bugge & Huldt's rapport,* 26. Arkitekturmuseets arkiv, SAR & SSF bostadsutredning vol 50, Andra utredningar.

> a secretaire; yes – there are school children who have to do their homework at a sewing machine or a chest of drawers.[97]

Why, then, was there such a discrepancy between functions, uses, and furniture? In some cases, the difficulties lay in the mismatch between being stuck with a certain collection of furniture when moving into a new apartment and not being able to afford, or not wanting, to buy new things. The fault could also lie with the apartment, which could sometimes be designed so that a specific type of furniture could not be used. And, in some cases, "conventional ideas of how a room should be furnished" could be as conservative as the dwelling habits.

When studying furniture, some pieces were defined as tools, connected to a specific function, and were analysed in relation to this. Other pieces of furniture – such as the "suites of furniture", sold in fixed sets – needed to be analysed in relation to each other since these were "not only functionally motivated but create, especially for certain rooms, an independent, more or less conventionally combined unit".[98] It became clear to the investigators that the relations between rooms, use, and furniture were not at all straightforward to map or understand. How people lived with and used their everyday environments defied the ideas of rational and well-defined use in relation to function, that had been the starting point for both designing modern housing as well as modern things.

Home reform ideas, from Ellen Key's turn-of-the-century writings to those of the 1930s functionalists, had built on the conviction that people's ways of using objects and environments could be altered through designing other types of things than before. Better and more beautiful things would directly influence people's ways of living and behaving. In a sense, this meant that what was expected from design outcomes, was that people would use things *as intended* by designers – also when that included changing behaviours as a consequence of how things were designed.

The dwelling surveys indicated that the multifaceted uses of things could not always be understood in terms of rational intentions or expected behaviours. The reasons for how and why people used things could not be pinned down in ways that were easily overviewed or understood. This, in turn, led to difficulties in deciding on which courses of action to take to provide better housing, and how to achieve behavioural changes in everyday life.

97 Johansson, *Bostadsvanor och bostadsnormer*, 109. "Man kan med andra ord inte utan vidare av ett rums möblering sluta sig till dess funktioner. Förekomsten av en bäddmöbel betyder inte med nödvändighet, att någon sover i rummet, ett matbord, att man äter där, eller ett skrivbord, att man skriver där. Å andra sidan behöver avsaknaden av en möbel inte betyda, att den därmed förbundna funktionen inte förekommer i rummet. En bädd eller ett matbord kan vid användningen flyttas in från ett annat rum eller från en särskild förvaringsplats, skrivarbete kan lika väl försiggå vid ett matbord eller vilket annat bod som helst som vid ett skrivbord eller en sekretär, ja det finns skolbarn, som få göra sitt läxarbete vid en symaskin eller en byrå."

98 Johansson, *Bostadsvanor och bostadsnormer*, 110. "Men möblerna äro, som redan antytts, inte bara funktionellt betingade utan bilda, särskilt för vissa rum, en självständig, mer eller mindre konventionellt sammansatt enhet, möblemanget."

The issue of overcrowded homes, for example, seemed like a reasonably straightforward problem to solve. If one were to provide new housing according to standards of minimum room sizes in relation to defined functions and peoples' real needs, it would probably improve the situation of overcrowding. But the problems with designing dwellings adapted to many different people's needs, and also to a range of diverse instances of use, was a significant issue. How was one to make sense out of which were rational dwelling habits that pointed to the need for designing differently? What was instead an irrational way of behaviour, that ought to be addressed by designing apartments and furniture in ways that promoted or limited certain practices? How could one direct the future use of things through information and education that aimed at changing people's understanding of their own needs as well as their appreciation of certain types of things above others? And how could these perspectives be brought into the work with establishing norms and standards for building and other areas of design?

One of the aims of the dwelling habit investigations had been to lay the foundations for establishing norms and standards for housing production on a national level, as in the SAR and SSF investigation, and in private initiatives, as the Industry investigation. This was compared to the similar way in which standardisation of sizes and qualities had been carried out in industrial clothing manufacture. But defining norms for dwellings would be much more difficult than standardising clothes sizes:

> But how are we to find the right standards? Aren't all standards subjective to begin with? The attainment of complete objectivity is naturally impossible, but that should be no excuse for not trying to attain it within reason. The mass-produced dwelling cannot be given as individual a fit as the ready-made suit – the housing problem contains too many factors to permit such a solution. But some of these factors can be objectively charted and measured. To begin with the kitchen, it requieres (sic) certain work-places and storage units, whose correct dimensions and locations can be determined by study. A bedroom may be large or small, but should have enough room for one or two beds of given dimensions. The space needed for a dining place can be calculated if one knows how many persons are to make use of it, etc. The dwelling can only be properly standardised through careful study of its functions and the needs of occupants.[99]

The difficulties in design when incorporating aspects of actual use came to the fore in the dilemma of how to bring in the complex realities of everyday life in the home into the design work of defining general standards and requirements for housing. In the attempts to make a different way of living possible through design interventions in dwellings and the design of everyday objects, the tensions between what was predictable and what was not, in terms of future use, were brought to the fore. Everyday life would most likely always include practices that were hard or impossible to preconceive or understand in design. However, this did not mean that design should not provide certain frameworks that would prescribe certain uses of the dwelling and its objects. What was most difficult, was distinguishing which sorts of use should bring input for designing, and which use should be eradicated through design outcomes.

99 Johansson, *Bostadsvanor och bostadsnormer*, 23.

Bringing 'use' into design

In the SAR and SSF study, the reasons for studying everyday life in the dwelling were explicitly discussed in relation to its design. Since a fundamental issue with mass-produced modern housing was the dilemma that the "owner-occupier" in the majority of cases had no possibility to "influence the layout and design through architects and contractors", it was necessary to find other ways of bringing in aspects of actual use into the process of designing dwellings. The methodology applied in the dwelling investigations was an attempt to lay the foundations for a user-centered design approach in which both observations of and the active input from people was aimed to influence the outcome of new forms of dwellings, and also new types of furnishings.

People's experiences were needed in the process of designing dwellings, but to both understand and evaluate this input, the actual ways of using the dwelling had to be systematically and objectively assessed. With methods and general theoretical assumptions borrowed from the newly introduced modern American sociology, finding out how people really used homes could serve as the starting point for reconsidering how dwellings should be designed. Earlier attempts to provide modern housing, in the aftermath of the 1930 Stockholm exhibition, had arrived at solutions that by the 1940s had proven "to be illusory, mainly because the architects have underestimated the significance of dwelling habits."[100]

In an article in the journal *Form* from 1945, the project leader for the dwelling investigation Gotthard Johansson summarised and discussed the main findings of the SAR and SSF investigation of dwelling habits. "A real knowledge", he stated, "of how the Swedish people live is the first condition for creating better dwellings and for getting people to use them in a better way."[101] In handling issues of how to improve the living situation for people, both architects and inhabitants often have gotten things wrong, since they disregarded the real conditions under which people lived and instead promoted ideal ways of living:

> When it comes to the relation between the dwelling and dwelling habits, I have often especially in radical circles of architects met the opinion: 'What interests us is not how people live but how they should live'. On the other hand, I have often in public discussions heard those who represent or consider themselves to represent the dwellers state that: 'Just give us enough space, and we'll surely use it. We don't need to learn how to live!' Both standpoints are equally wrong.[102]

100 Johansson, *Bostadsvanor och bostadsnormer,* 24.
101 Gotthard Johansson, "Hur bor vårt folk? Bostadsvanorna utforskas", *Form* 1945:1, 8. "En verklig kännedom om hur svenska folket bor är den första förutsättningen både för att skapa bättre bostäder och få folk att använda bostäderna på ett bättre sätt."
102 Johansson, "Hur bor vårt folk?" "När det gäller förhållandet mellan bostaden och bostadsvanorna, har jag särkilt i radikala arkitektkretsar ofta stött på den uppfattningen: "Det som intresserar oss är inte hur folk bor utan hur de *bör* bo." Å andra sidan har jag ofta i offentliga diskussioner från dem som representerat eller ansett sig representera de boende själva, fått höra: "Skaffa oss bara tillräckligt utrymme, så kan vi nog använda det. Vi behöver inte *lära* oss att bo!" Bägge inställningarna äro lika felaktiga."

Designing new apartments should, instead, take the starting point in both actual use and in the wishes and suggestions from people. Designers should make well-informed decisions in matters where the design intent in regard to defined needs, functions, and peoples' wishes should come first:

> By studying how the dwelling is actually used and recording the observations and desiderata of the occupants, housing researchers try to get an idea of how a dwelling should be designed in order to fulfil its functions and to meet the needs of the occupants.[103]

The designer/researcher must, therefore, find a way to balance the quantitative study with the investigations into people's behaviour and wishes and analyse how to lay a foundation for further research and experimentation. The important thing was to find out was how to distinguish between dwelling habits that might seem strange, but that were caused by different circumstances that could be more than reasonable, and practices that were not founded in needs or intentions, but were irrational remnants of traditional ways of living perpetuated without consideration. This was important to separate since any bad habits caused by understandable difficulties could be addressed through designing things differently to support the unknown needs, while bad habits without rational foundation would need to be addressed in other ways:

> It is not necessarily the dwellers', but can also be the dwelling's fault, that this is used in a way that one has become accustomed to regard as bad dwelling habits.[104]

In order to design for both current and future needs of people, the designer would have to balance the input from people regarding how they actually used their homes, with design constraints as well as the foreseeable futures made possible through design. In the analyses of the dwelling surveys, warnings were raised concerning taking people's input too literally, since that would entail a risk of designing too much based on current use than for future, desired, use:

> The pursuit of a function analysis along purely logical lines is liable to belittle the actual dwelling habits; but if we proceed from these habits, we run the risk of becoming too involved in existing conditions and thus tempted to preserve the bad dwelling habits along with the good.[105]

A certain measure of prescriptive or normative constraints in the dwelling plan and furnishing needed to be actualised, if a break with old social and formal norms were to be made in favour of new ways of living that enabled more egalitarian and emancipatory lifestyles. How would one find a way to balance prescriptive approaches in relation to openness for including various potential types of future use in designing? And to what extent should current input from users weigh into designing?

103 Johansson, *Bostadsvanor och bostadsnormer*, 24.
104 Johansson, *Bostadsvanor och bostadsnormer,* 27. "Det behöver inte heller alltid vara de boendes utan kan också vara bostadens fel, att denna användes på ett sätt, som man vant sig vid att betrakta som dåliga bostadsvanor."
105 Johansson, *Bostadsvanor och bostadsnormer,* 24.

What should change – use or design?

While the combination of interviews, observations, and diaries would provide a certain insight into what people did in, and thought about, their home, this gave rise to issues relating to how to handle people's input as users of things and environments in the process of designing. Methodologically, it was stressed, the questions of *what* one wants to know need to be tightly linked to *how* one would go about planning and conducting the research. Given that the intent with dwelling investigations differed, the views on use were also approached in slightly different ways.

In the case of the investigation tied to dwelling loans, the main issue at hand was to find ways to directly influence people to buy truly useful things for their homes. The conclusion in that study was that people needed to be educated in which things to buy to match their "true" needs. The ways people used their homes, and their things, was wrong and needed to be corrected through ways that would push actual use towards the design intent. The need for education was also highlighted in the industry survey, but there rather in relation to the use of the dwelling as such, and not so much in regard to things or furniture. People needed guidance as they moved from smaller to larger apartments, but not necessarily in the sense of prescribing certain furnishings above others. Instead, the functions of the rooms needed to be clarified, so that people would understand, e.g. how to use a living room in the best way.

The intent with the SAR and SSF study was to form a solid foundation for the design of better dwellings, highlighting the connections between "floor plan, fittings and furnishings [are] of value to be able to pinpoint the connection between dwelling and dwelling habits, between the apartment and its use, between what one has and what one wishes."[106] Since the questions asked in the initial stages of the SAR and SSF project were fairly easy to formulate in relation to the very tangible aim of creating "a concrete basis for the theoretical study of the dwelling's functions and spatial needs", it had also been relatively easy to pose the questions of where one eats, sleeps, what activities go on in the living room and so forth.

But upon trying to analyse the answers to these questions of everyday use, neither the methods nor the results were straightforward to handle and understand. The investigators quite soon discovered that only very rarely were the answers direct and univocal: "Completely contrary to what one earlier, in general, had believed and often had assured, the dwelling habits were very varied, disparate, and unstable."[107] People's ways of using and relating to their homes differed immensely, in ways that could not be captured, neither systematic quantitative study nor attributed to any specific social factors:

106 Johansson, *Bostadsvanor och bostadsnormer*, 30. "...en beskrivning av de undersökta bostädernas planlösning, inredning och möblering [är] av värde för att kunna fastställa sambandet mellan bostad och bostadsvanor, mellan lägenheten och dess användning, mellan vad man har och vad man önskar sig."

107 Johansson, *Bostadsvanor och bostadsnormer*, 47. "Alldeles tvärtemot vad man tidigare i allmänhet föreställt sig och ofta försäkrat, visade sig bostadsvanorna vara mycket varierande, oenhetliga och instabila."

> There are well situated middle-class families where the wife's interest in "interior decoration" has resulted in the amassment of all sorts of knickknacks, and there are others where one with simple means has made do in a rational and practical manner. There are grumblers who complain about everything, and there are those who are happy with almost anything. There is, overall, a scale of human attitudes and reactions, of shifts in atmosphere and environment, that cannot be drawn in any furnishing plan or registered in any table.[108]

The complex and often contradicting ways in which people lived with things in the home was something that was at odds with most previous understandings of what a rational use of the home ought to be like. A general idea, widespread in home reform and architecture circles, had been that one issue to come to terms with, were stereotypical home environments. The general idea was that most people used and furnished their apartments in more or less the same – problematic – way, and that this was the main issue to be solved:

> How often had one not heard from people, who thought themselves to have experience of these things, that all modern small apartments looked exactly the same, that all families made their lives in the same way, and that the use and furnishing of the dwelling was so bound to a conventional scheme, that the monotony and lack of individual variation was seen to be the main problem.[109]

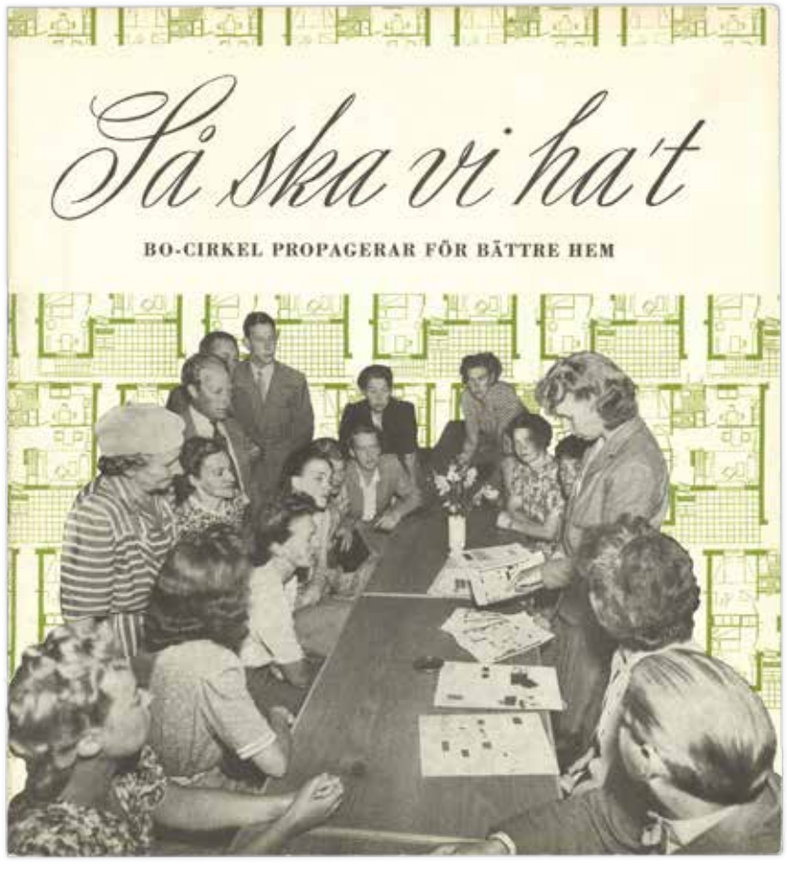
On the surface, as documented in furnishing plans, types and placements of furniture were indeed very similar in many homes. Schematic illustrations in the dwelling surveys would outline all the possible combinations of arrangements of furniture in the different types of apartments included in the studies. Some 'irrational' ways of placing furniture were attributed to the way the apartments were planned, that only allowed for certain furniture placement.

In the dwelling loan survey, for example, the evolutionary stages of parlour furnishing were mapped both in relation to the placement of furniture and to the types and styles of objects. From 'bourgeois' with the dining table in the middle of the room to the more modern version where the furnishing is made according to "clearly marked function groups" of undecorated furniture.[110] But when more in-depth analyses of the material were made, it became clear that almost no two

108 Johansson, *Bostadsvanor och bostadsnormer*, 46. "Där finns välsituerade medelklassfamiljer, i vilka fruns intresse för 'heminredning' resulterat i en anhopning av allsköns krimskrams, och där finns andra, där man med enkla medel inrättat sig praktiskt och rationellt. Där finns över huvud en atmosfär och miljö, som inte kunna inritas på någon möbleringsplan eller registreras i någon tabell."

109 Johansson, *Bostadsvanor och bostadsnormer*, 47-48. "Hur ofta hade man inte fått höra av personer, som ansågo sig äga erfarenhet av dessa ting, att alla moderna smålägenheter sågo precis likadana ut, att alla familjer inrättade sig på samma sätt och i fråga om bostadens användning och möblering voro så bundna av ett konventionellt schema, att själva enformigheten och bristen på individuella variationer syntes utgöra det egentliga problemet."

110 "Ungdomen behöver vägledning vid hembildningen," *Form* 1943:1, 22-26.



That's the way we want it. Broschure based on the first study circle in dwelling knowledge in Gustafsberg, 1944.

families would *use* their home in the same way, despite having furnished them in similar ways. Similarities between the homes could be attributed only to the styles and types of furniture, and in many cases to how these were placed in the dwelling. But concerning the use of rooms and things, the variations in use were more substantial than the similarities.[111]

The conclusion was that more research was needed to distinguish between poor dwelling habits with a rational foundation, caused by faults in planning and design of dwellings, and those that were caused by routines inherited from outdated social norms. Some findings, both in regard to people's feedback and to the observations made, already showed that certain dwelling habits were caused by faulty dwelling design, such as the tiny kitchens and dinettes. These needed to be addressed in new standards and different designs for housing. But equally important would be to pinpoint dwelling habits that were built on convention, and that caused "hygienic, social and practical problems". These must be eradicated through designing different ways of behaviour in the home, through the design of material things and environments. Additionally, education and information must be intensified and also adapted to how people actually live, and how the functional needs, as well as the more socially grounded wishes of people, could be catered to:

> If the dwelling in this way is adapted to the dwelling habits, there must also be an interaction in the opposite direction. One must consider if it is more rational to adapt the dwelling plan to the dwelling habits, or vice versa. A habit, founded in convention and lack of knowledge should not be made the norm for a dwelling plan. It will of course take a very substantial information campaign to influence people to change their views and habits, but on the other hand there have hardly been any general dwelling habits developed in modern two-room apartments, since the industry workers mainly live in outdated apartments of one room and kitchen.[112]

In the material gathered from the surveys of dwelling habits, the ambition to give voice to people's opinions, needs and practices in the process of designing comes across clearly. However, equally apparent as the intent to bring in a greater understanding of the how and the why people use their homes in certain ways, was the intent to shape specific future practices in the home to align better with the design intent. To work with this forming of future uses – and users – one must not make the same mistakes that previous "radical architects" and "societal improvers" had made, of paying too little attention to reality and over-emphasising the ideal intentions while being:

111 Johansson, *Bostadsvanor och bostadsnormer*, 48.
112 Jöran Curman "Industriens bostadsutredning" *Form* 1943:7-8, 120. "Om bostadsplatsen på detta sätt anpassas till bostadsvanorna, måste även i motsatt riktning en växelverkan äga rum. Man måste överväga om det är rationellare att anpassa bostadsplanen efter bostadsvanorna eller tvärt om. En vana, som grundar sig på slentrian och bristande kunskap bör ju ej göras normgivande före en bostadsplan. Det fordras visserligen en mycket kraftig propagandakampanj för att påverka människor till att ändra på sina åsikter och vanor, men å andra sidan ha några allmänna bostadsvanor knappast hunnit utbildas i moderna två rum och kökslägenheter, då industriarbetarna för närvarande mest bo i omoderna lägenheter på ett rum och kök."

> so dedicated towards how the dwelling *should* be used – i.e. in regard to one's own opinion of its use – that one has lacked interest for the question of how it is actually used and regarded opinions and wishes from them, whose dwelling situation one has wanted to improve, as irrelevant for the solution of the problem. Such an attitude is not uncommon amongst those who wish to improve society, but it is nonetheless out of touch with reality.[113]

Educating users

The results of the dwelling surveys came to provide arguments that legitimised the home as an arena for socio-political intervention and made possible the incorporation of issues regarding life in the home as well as questions of interior decoration into public debate and government policy. Conclusions in the different dwelling surveys indicated that it would not be enough to focus only on the design and production of new types of dwellings and things. Education and information were needed to shape people's understanding and use of the home. This came to unite organisations and associations of different ideological and political views in efforts to, in a sense, 'form users' that would adapt to handling modern ways of life.

The housing issues, and questions of how to address issues of people's everyday behaviour in the home, came to be established as a matter of concern that resulted in different actions aiming towards home reform as well as the design of new sorts of things and dwellings. A low standard of living, which during the early 20th century above all was seen as a moral and hygienic problem, towards mid-century more and more was defined as something that prevented the individual from evolving into a modern and democratic citizen.

Home reform during the 1940s, therefore, aimed at changing the lifestyles not only of the working class but also of the middle class, as part of a more extensive societal program of empowerment and emancipation. What people needed to be liberated from, in this view, were norms and conventions that stood in the way of people being able to fully embrace the advantages of full citizenship in a modern, democratic, society. In order to enlighten people, education in matters of everyday home life and interior furnishing was deemed necessary.

Questions of how to promote a modern use and furnishing of the home had been brought up on the political agenda in relation to the dwelling loans in 1943. Then, the Slöjdföreningen was asked by the state department of social affairs and the Swedish Riksbank to produce a brochure on rational interior decoration, that could be distributed to everyone who applied for a dwelling loan. When the brochure *Bosättning* (*Setting up home*) was published in 1944, it demonstrated how to create a modern home within the financial limits of the dwelling loans, and how to choose the appropriate furniture for ones' needs as individuals as well as from the perspective of a family. The same year, another of the suggestions proposed by

113 Johansson, *Bostadsvanor och bostadsnormer*, 28. "…så inriktad på hur bostaden *bör* användas – d.v.s. hur man själv ansett, att den bör användas – att man saknat intresse för frågan hur den i verkligheten användes och betraktat synpunkter och önskemål hos dem, vilkas bostadsförhållanden man velat förbättra, som problemets lösning ovidkommande. En sådan attityd är inte ovanlig hos samhällsförbättrare, men den är inte desto mindre verklighetsfrämmande."

the dwelling investigators became a reality, when the city of Stockholm instated the function of a "dwelling counsellor" that would advise young couples asking for dwelling loans in regard to matching their planned purchases to both their needs and their financial situation.[114]

The educational efforts that formed the basis for the *Bosättning* brochure, as well as for the counselling service, were based in similar ideas as those that had been promoted by Ellen Key and others at the turn of the century. Educating people in matters of taste was seen as a needed training of the faculty of judgement, that people needed to be able to apply in relation to a broad understanding of social issues as well as their individual needs and personal preferences.

Taste, then, was seen as something broader and deeper than subjective opinions on what was pleasing or what things looked like. The faculty of taste was important when choosing the appropriate material environment for ones' daily life, since that was seen as formative for the practices that could – and should – take place in the home, as well as a manifestation of adherence to a particular view on society. The explicit ambition was to strengthen democracy and equal opportunities for people:

> that the young home is created in closer conformity to the demands and possibilities of the times. It has been produced with the conscious knowledge that the classless home, the type home of democracy, can only be formed in a state of larger freedom from conventional and traditional delusions through a simpler and natural selection of everyday things and increased knowledge of fitness to purpose and quality.[115]

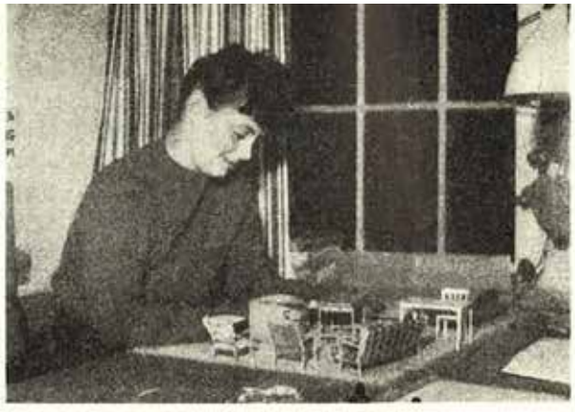
Designing democracy

In December of 1943, when the brochure *Bosättning* was presented, the representatives of a variety of national popular movements were invited by the Svenska slöjdföreningen to a conference on the theme of housing and interior decoration. The aim for the SSF was to involve all of the Swedish national movements such as the Temperance Movement, and various educational associations, women's organisations, youth movements, unions and the consumer cooperation in the practice of educating people in dwelling knowledge and interior decoration. At this conference, the support from these organisations for the home reform project presented by the SSF was unanimous.

A decision was made to form a committee, headed by the SSF, to work with the practical arrangements of how to teach 'dwelling knowledge'. The *Bokommittén* (the Dwelling Committee) was formed in 1944. The SSF was the organisational centre, and the funding of the committee's work was given in equal amounts both from *LO*, the Swedish Trade Union Confederation, and from the *Industriförbundet*, th Federation of Swedish Industries. Representatives from different national movements took part in the committee. They united their efforts

114 Centrum för Näringslivshistoria, Föreningen Svensk Form 7069:1, F1:104, mapp 629, "Motioner i andra kammaren, nr 161. Av fru Humla m.fl. angående anslag till Svenska slöjdföreningen för anordnande av en utbildningskurs för bosättningskonsulenter m.m. [ort, år,]", 8-10.

115 Åke H Huldt & Ingeborg Waern Bugge *Bosättning* (Stockholm, 1944).



Bocirkelverksamhetens ledare arkitekt *Lena Larsson* förbereder en demonstration av möblering. Hennes åskådningsmaterial är SSF:s modellmöbler



Studieledare arkitekt *Erik Berglund* framställer i papp nya typer av modellmöbler

Course leaders Lena Larsson and Erik Berglund prepare generative materials for dwelling knowledge study circles. *Så ska vi ha't, Gustafsbergscirkel propagerar för bättre hem.*

to educate people in matters of taste and everyday practices under the motto "Better homes – better society".[116]

The first study circle in dwelling knowledge was held in the town Gustavsberg, in cooperation with the porcelain factory Gustafsberg not far from Stockholm, in 1944 with SSF's Lena Larsson as one of the circle leaders.

The participants gathered once or twice a week, during six to eight weeks, for introductory lectures by circle leaders from the SSF. After the talk, there would be discussions and exercises so that the participants could analyse and practically exercise the theme for the evening.

Keeping to a democratic and participatory study format throughout the course was central to the pedagogical foundation of the study circle, which was also expressed by the chair of the Gustafsberg study circle, Mrs Alice Pettersson: "We are thankful to our leaders from Svenska slöjdföreningen, who have come to us not as teachers but as good comrades with empathy for our problems while generously sharing their expertise".[117] At the end of the whole course, the study group arranged a public exhibition of one or more apartments that they themselves furnished and decorated. Following the model of the Gustafsberg study circle, the Dwelling committee supported adult education classes and courses in interior decoration and dwelling knowledge all over Sweden. Over ten years, from 1944 to 1954, more than 300 courses in interior decoration were held with the support or participation of the SSF.[118]

These courses aiming to teach people to form their everyday material environment differently became a great success for the Dwelling committee. One reason for this was that the program of rational home reform was not identified as an end in itself, but as a means towards a goal that many different organisations could unite around: the creation of a democratic society. Different groups engaged in education and home reform had diverse motives for supporting the SSF courses and other initiatives taken to "elevate taste" and teach people how to use their homes. But all of them could unite around the idea that changing uses and practices in the home was central as a means of building a different, and better, society.

116 After two years, the financial support from the unions and industrial leadership ceased, but there were still representatives from them in the committee. Eventually, in 1946, the committee was transformed into the SSF:s Study Department. But the work initiated by the Dwelling committee in the first couple of years set the standards for how teaching 'dwelling knowledge' was to be done through study circles and exhibitions all over Sweden during the whole 1940s and 50s, until it later entered the public-school curriculum in the late 1950s.

117 *Så ska vi ha't: Bo-cirkel propagerar för bättre hem* (Stockholm 1946), 31.

118 These courses were commissioned from the SSF, and targeted towards couples – and even if participant lists show that a majority of attendees probably were women, both men and women did join these. Maria Göransdotter, "Möbleringsfrågan: Om synen på heminredning i 1930- och 1940-talens bostadsvaneundersökningar", *Historisk tidskrift* 1999:3, 444-474.



Course literature for study circle in dwelling knowledge.
Så ska vi ha't, Gustafsbergscirkel propagerar för bättre hem.

In these educational endeavours, the main message was that the most important thing to learn was the methods and principles of modern living – understanding ones' own needs, aspirations and potentials – rather than being given pointers on what things to buy. The way things were used was more important than what they looked like. As Gregor Paulsson, the director of the SSF, said in the inaugural speech when the Dwelling committee was formed in 1944, the ambition to bring about material, as well as behavioural reforms in the home, was ultimately a way of laying the foundation for a democratic society:

> Democracy is not only a constitution, but also a way of life. It is the nature of the society we have built for ourselves, and even of society in its most concrete sense. History teaches us, without a doubt, that the social and political structure of society manifests itself in its exterior expression. [---] The popular movements are democracy's primary organ, and work towards realising certain cultural and political goals. Together, they not only build democratic society, but democratic man. Svenska Slöjdföreningen sets the final stone, so to say, in this construction through endeavouring to give democracy its adequate environment.[119]

In histories of the Swedish welfare state, these initiatives have been described as social engineering. Interventions aimed at reforming people's private domain, the home, has been interpreted as a civilising and normative project of social engineering and control over the individual. And from one perspective, that is hardly the case to dispute. But if we take another perspective, and regard these initiatives in the light of emerging practices of user-centered design, another interpretation could also be made.

This coming together around interior decoration and material everyday practices could just as well be seen as a sort of participatory articulation of drawing things together in relation to matters of concern. With the wide variety of organisations and associations represented in the Dwelling committee – from women's rights organisations and the temperance movement to industry representatives – we could also argue that this is an instance of the coming together of different stakeholders, of which many were struggling to give voice to issues that were not yet on the political agenda.

In this context, inviting people to take part in study circles in interior decoration to share the knowledge that would empower them in their roles as citizens, as well as contributing to making material changes in their everyday life, was understood by those engaged in it as a fundamentally emancipatory and empowering project:

119 Gregor Paulsson, "Demokratisk miljö. Hälsningsanförande vid S.S.F.:s konferens på 'Vår Gård' den 7 december", *Form* 1944:1, 2. "Demokrati är inte blott en författning, utan också ett levnadssätt. Den är beskaffenheten av det samhälle vi byggt upp åt oss, och även samhället i dess mest konkreta mening. Historien lär oss otvetydigt, att samhället sociala och politiska struktur visar sig i dess yttre beskaffenhet. [---] Folkrörelserna äro demokratiens primära organ, som arbeta för realiserandet av vissa kulturella och politiska mål. Tillsammans bygga de inte blott ett demokratiskt samhälle utan också den demokratiska människan. Svenska Slöjdföreningen lägger s.a.s. slutstenen i denna byggnad genom att arbeta för att demokratien får en adekvat miljö."

> It is a – unfortunately much too common – misunderstanding that the problem of interior design is to give an answer to the question if green is more beautiful than red or the opposite in a certain context, and similar trivialities. The problem of interior decoration is instead primarily a question of fitness to function and quality, of a correct way of life and of our homes' adherence to it. What one means with good taste is nothing else than the final consequences of utility. "The beautiful is the brilliance of the true" as a philosopher so correctly put it.
>
> *Svenska Slöjdföreningen does not strive to implement an aesthetic program, but it announces – and to its best ability seeks to solve – a cultural program.*
>
> The association strives to realise this program along two lines. Through collaboration with production, it aims to achieve an improvement of items of furnishing, in quality as well as in form, wherewith the most important item, the dwelling itself, shall not be forgotten. Through education among the consumers, it seeks to distribute knowledge about all the factors, that form part of the setting up and the use of a dwelling.[120]

A historical understanding of 'use'

This prototypical history has provided a backdrop, or foundation, to the coming about of Scandinavian user-centered and participatory design, as located in domestic everyday settings rather than the military or professional work arenas usually referred to as where the 'user' comes into focus with the study of human factors and ergonomics. Articulations of ideas of 'use' and 'users' in design emerged in activities aiming to address housing issues and reforms of everyday practices that different organisations engaged in during the first half of the 20th century. In this process, the tensions between design intent and real use came to be explicitly considered and addressed, in ways that might also provide entry points to reflecting on how to negotiate the inherent tensions between ideal and real, potentiality and actuality, in designing.

120 Gregor Paulsson, "Demokratisk miljö", 4. "Det är ett – tyvärr alltför ofta återkommande – missförstånd, att det problem heminredningen sysslar med är att ge svar på frågan, om grönt är vackrare än rött i ett visst sammanhang, eller motsatsen, och liknande oväsentligheter. Heminredningsproblemet är i stället i första hand en fråga om ändamålsenlighet och kvalitet, om en riktig livsföring och våra hems anslutning till densamma. Vad man menar med god smak är ingenting annat än de sista konsekvenserna av det funktionsenliga. 'Det sköna är glansen över det sanna,' har en filosof riktigt sagt. *Svenska Slöjdföreningen kämpar därför inte för realiserandet av ett estetiskt program, utan den förkunnar – och söker efter bästa förmåga lösa – ett kulturprogram.* Detta program strävar föreningen att realisera efter två linjer. Genom samarbete med produktionen strävar den att åstadkomma en förbättring av bohagsvarorna, såväl i kvalitet som i form, varvid den viktigaste varan, bostaden själv, icke glömmes. Genom upplysningsarbete bland konsumenterna söker den bibringa kunskap om samtliga de faktorer, som ingå i bildandet och begagnadet av en bostad."

The concept of use, as approached in the context of dwelling surveys and home reform, was formed in a historical situation where the explicit intention was to enable certain ways of using the home, while disabling others, through design. Simultaneously, active efforts were made to shape the ways people lived on an everyday basis by initiating broad educational programs that targeted the consumption of certain things and specific ways to use them. This goes both for the instrumental or rational use tied to enabling or fostering a particular individual behaviour in relation to specific things or environments, and for a more collective and systemic design program aiming to bring about new norms, practices and socio-material (infra)structures that would support new ways of life.

As the different dwelling surveys showed, envisioning a particular use when designing does not necessarily mean that design intention is what will be played out in actual use. What also came across already in the 1940s, in the attention to real versus ideal use, was that even if design intent and people's appropriation and use of the design thing did not align, change of some sort did still come about. It might not take the expected trajectory, but it would make a difference – that then, as a next step could form a new framework for future design proposals. If we were to look at a couple of the aspects targeted in the dwelling design and educational efforts of the mid-20th century Sweden, some of these brought about actual and intended change fairly fast, while others were less successful.

The way design was approached in these initiatives speaks to an awareness of negotiating the complicated relationship between design intent based on definitions and precisions of functional needs and performances with the unpredictable ways that people make design outcomes a part of their everyday life. While design decisions based on understandings of people's current ways of using things could lead towards certain possible futures, they were also seen as open for negotiations and other potential interpretations though how people used them. As the dwelling investigations showed in the 1940s, designing with attention to use is about providing proposals and possibilities that in the end are difficult to control from a position of design intent, since these can always be actualised in other ways than expected – or not actualised at all – through use.

The new methods applied in the sphere of re-shaping the material and behavioural practices of everyday life in Sweden of the 1930s and 1940s targeted both how people acquire things, and the continued use and appropriation of things and the environments these formed as part of something that needed to be considered in designing.

With the introduction of the idea of incorporating studies of actual use in order to design for future situations, the shift from seeing things as imbued with qualities of usefulness and purposefulness to seeing use as an emerging relational practice emerged in the 1940s. In this context, we can also see the beginning of the formulation of 'the user' as necessary to have included in the process of designing, not least through the ambition to shape correct use through education. This was both explicitly expressed and linked to design of material things as well as to work towards political and social arenas to influence legislation and societal discourses.

The attention to how the dwelling was used in practice opened up for discussing and re-visioning how life could be lived together – not only on a family level in the home, but on a societal level. Satisfying material needs with a focus on the practical, rational, use of the dwelling and its objects was perhaps not so much a goal in itself, as it was a means to both formulate and bring about ways of doing things differently.

Historians have pointed to the processes in which scientific methods and "social engineering" were applied to change the everyday practices in the Swedish homes in normative and oppressive ways, aesthetically as well as socially.[121] From our contemporary perspective this is, of course, one very valid interpretation. But when does a norm become a norm, and when is a challenge a challenge towards "power" from a "marginalised" group? Where do processes shift, and when do power structures solidify, so what was a challenge of an existing hierarchy or state of things, in turn, becomes the new hegemony?

In 1940s Sweden, full participatory democracy was only two decades old. The workers' movement had turned the tables on traditional political governance through pressing forth radical reforms allowing both workers and women fundamental political rights that had shifted the weight of decision-making power in society. Reforms and activist movements opened up and advocated new forms of relationships between men and women, both in public and in private life. Traditional norms and conventions governing social life were questioned, and radical new ways of living seemed not only possible to strive for, but actually also plausible to achieve.

Bringing about radical change, politically and socially, was formulated as a program aiming to transform Sweden into a democratic, classless and genuinely modern country. In this, the project of reforming material conditions linked to ideological and ideal visions of what it meant to be human in a modern, industrial world. Design was not only about making material things but about creating new ways of thinking and acting. It was not enough to address issues of working life and industrial production, or secure that democratic principles governed institutions and gave everyone the right to take part in processes of influence and governance. Formal rights to vote, to work, to higher education, to medical care and so on were the infrastructures that could make a different way of life possible. Equally important was the transformation on the individual level that was seen as each person's responsibility for actually *using* the rights to take part in and to influence society – which was linked to the ability to also *use* all aspects of everyday life to not only make that empowerment possible, but to actually become realised.

Embedded deeply in notions and the practices of Scandinavian user-centered design, lies the concept of not only designing things for a particular 'use', whether for or together with people. In this idea also lies implicitly that a particular design outcome should be adapted to a specifc sort of future user – and as such, the 'future user' is also part of what becomes envisioned and designed. This is not to say that designing users is necessarily problematic. Quite the contrary: In many instances of design, a 'tight fit' between design intent and future use is both desired and required.

A future user of specialised technologies or tools – be it in professional fields or everyday situations – would not always be supported by leaving too many flexible options for alternative uses open. But in instances where design

121 Yvonne Hirdman "Utopia in the Home: An Essay" *International Journal of Political Economy* 1992:22, 5-99; Yvonne Hirdman, *Att lägga livet tillrätta. Studier i svensk folkhemspolitik* (1989) (Stockholm: Carlssons, 2000); Zandra Ahl & Emma Olsson, *Svensk smak: Myter om den moderna formen* (Stockholm: Ordfront, 2001); Linda Rampell, *En kritisk undersökning av det modernistiska projektet för design i Sverige*, diss. (Lund, 2004).

aims to open up processes and methods of designing for complex contexts, evolving relations and fluid constellations of people and non-humans, the concept of 'users' becomes more problematic.[122] If the concept of 'the user' is in play without attention to the implications of design intent embedded in the methods and processes applied in design, or even the notions or perceptions carried by language and terminology, the directions and potential consequences of promoting a certain design intent become obscured.

Projections of possible futures, and the potential future uses or situations aimed for, are perhaps not expressly identified as always instrumentally pointing towards a readily identifiable 'user'. Nonetheless, since design inherently carries the ambition to do something for someone, all designing will inevitably not only lead towards what the 'something' might be but also towards which sort of 'someone' that thing is directed.[123] People defined as 'users' are made and shaped in relation to the current situations and contexts where designing takes its starting points.

Increasing attention in design research has been turned towards the development and prototyping of methods and approaches that would avoid limiting preconceptions or definitions for what and with whom design would need to engage.[124] Neither does design intent only reside with people identified as 'designers' in a design process. In situations of co-designing, the intent is often both formulated and given form in collaborations where notions of the futures of the design things being proposed and created are collectively imagined. As more open and collaborative spaces for articulating matters of concern are searched and probed, the roles and responsibilities (and response-abilities) identified in designing are undergoing significant shifts in articulation.[125]

The inherent tensions between the ideal and the real, between intent and adoption, embedded in the concept of 'use' since its emergence, does not necessarily have to be something negative. Nor does it have to be restrictive in terms of supporting innovative or critical emerging design practices. It does, however, need to be present and visible in terms of accountability, ethics and considerations of power relations in designing as well as what the outcomes of 'user'-oriented designing can afford.

There are always values embedded in regard to what is considered appropriate or possible in relation to use, such as in participatory design practices characterised as building further on a Scandinavian tradition.[126] And as what we value changes over time, what in a certain historical context was perceived as important for

122 Johan Redström & Heather Wiltse, *Changing Things: The Future of Objects in a Digital World* (London: Bloomsbury, 2018).

123 Harold G. Nelson & Erik Stolterman, *The Design Way: Intentional Change in an Unpredictable World* (Cambridge, MA: MIT Press, 2012), 47f.

124 Kristina Lindström & Åsa Ståhl, "Politics of inviting: Co-articulating issues in designerly public engagement", in *Design Anthropological Futures: Exploring Emergence, Intervention and Formation* (London & New York: Bloomsbury Academic, 2016), 183-197.

125 "The Lancaster Care Charter", *Design Issues* 2019:1 (vol.35), 73–77.

126 J. Gregory, "Scandinavian approaches to participatory design", *International Journal of Engineering Education* 2003:1 (vol 19), 66.

liberating or reconfiguring practices, in regard to certain norms and values, will in another time be interpreted as restricting, normative and limiting.

Applying the concept of 'use' in design not only requires attention to the process and future proposed situations of design in regard to the expectations or limitations envisioned in future use. Going back to the historical contexts in which ideas of 'use' became important to address in the process of designing, we could argue that the concepts of 'use' and 'user' will most likely always come with embedded understandings and mechanisms of intentionality and limitation in regard to what 'use' can be.

To use something is to have an intent, a direction, an aim. It might align with the intent, direction or aim formulated in the designing of the thing, or it might be in conflict, resistance, or opposition to it.[127] In any case, 'use' can really only consist in an active engagement with something. For this reason, 'user experience' will also always point to the experience of intentionally handling something for some purpose, rather than other types of experiences that are not aimed or directed towards achieving a specific result: "…what happens when we reduce the space for personal interpretation of 'use' in our ambition to design for efficiency when our understanding – and indeed the limits of what knowledge we can have access to during the design process – of 'use' cannot match the complexities and subtleties of actual use?"[128]

The importance lies in being aware of what is activated in terms of implicit and explicit consequences when we talk of 'use' in design. Often, we want to create the tight fit between design intent and use that this implies. But in processes aiming to open up, or even question, relationships between people in design situations, the notion of 'users' is likely to be more problematic. Rather than trying to find ways to upheave or dismantle perceived problems associated with dichotomies such as designer-user, or intended use-actual use, we might approach these inherent conflicts historically embedded within the concept of 'use' as that which might make it possible to create openings for thinking and doing things differently.

127 Mahmoud Keshavarz & Ramia Mazé, "Design and Dissensus: Framing and Staging Participation in Design Research", *Design Philosophy Papers* 2013:1 (vol.11), 7–29; Carl DiSalvo, *Adversarial design* (Cambridge, MA; MIT Press, 2012).

128 Redström, "Towards user design", 134.



Archival material from the Hemmens forskningsinstitut. Riksarkivet, Hemmens forskningsinstituts arkiv.



Publications by Hemmens forskningsinstitut. In 1948-49 these covered small utensils for the home, processes of conservation, and children's situation in the home.

6. Methods

The attention to methods is central in design practice. Developing design methods through experimentation and critical reflection is fundamental to both commercial design and design research in how to address the continuously changing situations for, and expectations on, design. However, the attention to the historicity of design's methods is minimal – if it even exists at all. I argue that an awareness of design's historicity needs to be part of critical inquiries for approaching existing and emerging design methods, questioning what it means for processes and outcomes of design its methods have come about in particular historical situations.

In the following, I prototype a design history made from a perspective of 'methods', with the dual aim of making present historical ideas embedded in designing, and of presenting a design history of Swedish user-centered design.[1] This example takes a starting point in exploring historical aspects of how methods in user-centered design have emerged in response to new contexts and challenges for design. This prototype of a transitional design history is constructed in a way that begins with establishing the conceptual lens for the study – in this case by giving an outline of how the concept of 'methods' is central to design. After that, the context is set through outlining a historical situation for the study, which here builds both on the prototype presented in the previous chapter and on previous historical research. This is then the context in relation to which the conceptual lens of 'methods' is applied to a more specific location and situation. A brief glimpse of the 1960s emergence of the British design methods movement is given, as this is often identified as a springboard for initiatives aiming to develop new methods for designing. The focus thereafter is on Swedish design-related organisations of the 1930s and 1940s. Here, the source material springs from the *Hemmens forskningsinstitut* (HFI, the Home Research Institute), and consists of different types of written and visual documentation of the work carried out in the 1940s intending to investigate and change domestic work practices.

I propose that design methods and approaches that came to the fore in Scandinavian user-centered and participatory design in work-related contexts in the latter decades of the 20th century, were to some extent already present in the contexts of domestic research and home reform in 1940s Sweden. In the development of the systematic studies of 'dwelling habits' discussed in the previous chapter, continuing with deepened studies of domestic work practices and environments, iterative processes, ergonomic studies, prototyping, and new methods for user participation, were introduced in a context of designing. This precedes by about two decades what is usually considered as the coming-to-be of user-centered and iterative methods in design, and calls into question what counts as relevant contexts for genealogies of designing in design history and design practice.

As design has evolved, various methods and tools have been developed to support practices of designing. Some of these, such as sketching, have been around even since before industrialism brought about the split between designing

and making that called industrial design into being. Others, like brainstorming, function analysis, user testing, and iterative prototyping have been developed or brought in from other fields of practice along the way to address new design situations and new practices in designing. Most – if not all – of these methods are still actively present in design.

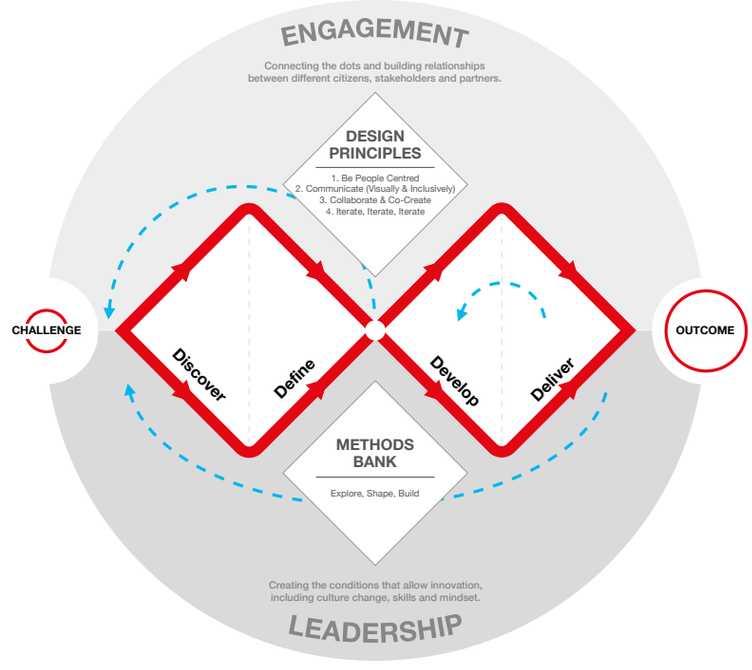
The situations in which these methods came about and the things they were made to handle and address are no longer present, but the ways of designing that they gave rise to are still around. Embedded in how methods, tools, and processes work and what they can do, are remnants of what the world was like, and what designing was like when they came into being. Making visible the historical dimensions of contemporary designing, aims to re-position the outlook of design doing in the present: What is it that we can expect our current – historical – design methods to help us deal with, and what would it take to handle historicity when aiming to develop other ways of thinking and doing design?

Methods in design

In the autumn of 2019, the British Design Council celebrated that 15 years had passed since its launch of the 'double diamond' visualisation of the design process.[2] On its website, articles described how a group of people at the Design Council in the early years of the 21st century developed the 'double diamond' as a general description aiming to cover the fundamental commonalities of all design and innovation processes.

The 'double diamond' visualises a design process trajectory building on steps of divergence and convergence, labelled 'discover', 'define', 'develop' and deliver', that in turn include several different methods aiming to support these steps. The Design Council promotes its own "methods bank" which is described as a "portfolio of design methods which help our clients to identify and address their challenges and achieve successful outcomes", sorting the methods in categories

1 Maria Göransdotter & Johan Redström, "Design methods and critical historiography: An example from Swedish user-centered design", *Design Issues* 2018:2 (vol.34), 20-30.
2 Cat Drew, "The Double Diamond: 15 years on", https://www.designcouncil.org.uk/news-opinion/double-diamond-15-years (accessed 2020-06-10).
3 Jonathan Ball, "The Double Diamond: A universally accepted depiction of the design process" https://www.designcouncil.org.uk/news-opinion/double-diamond-universally-accepted-depiction-design-process (accessed 2020-06-10).
4 Jonathan Ball, "The Double Diamond: A universally accepted depiction of the design process" https://www.designcouncil.org.uk/news-opinion/double-diamond-universally-accepted-depiction-design-process (accessed 2020-06-10).
5 IDEO "Design thinking defined", https://designthinking.ideo.com Retreived 2020-06-10. For collections of methods, see for example IDEO's Human Centered Design Kit https://www.ideo.com/post/design-kit (accessed 2020-06-10), or Nahman's Service design tool kit https://www.servicedesigntoolkit.org Retrieved 2020-06-10.

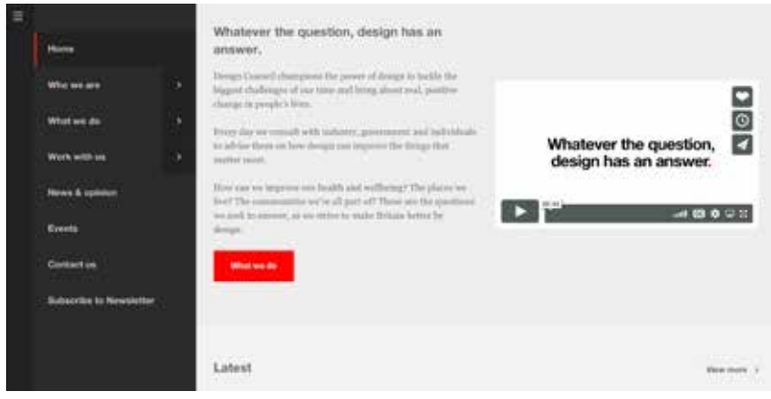


The double diamond, depicting the design process as visualised by the Design Council. www.designcouncil.org.uk/news-opinion/double-diamond-15-years

of those that explore challenges and needs, that shape prototypes and visions, and that build ideas and expertise.[3] These methods can be used by designers and non-designers alike to support opening up and narrowing down an iterative and collaborative process applied in order to find design solutions to problems or challenges, with the message that "whatever the question, design has an answer".

In the online description of its history, the Design Council points to the 1960s as when design methods and a process focus emerged in designing but states that there was then no commonly understood or shared model of the design process. In passing, names of eight men are listed as people who between the 1960s and 1990s worked with "models containing elements of divergence and convergence, cycles and iterative structures" of importance to developing the double diamond.[4] Besides this, the only references to past explorations carried out within design research and design practice to develop the methods and process described by the Design Council are to design consultancy IDEO, referred to as in practice running a double diamond design process already in the 1990s, but without visualising or labelling it as such.

This way of presenting *the* design process and associated methods is relatively common, not least in commercial design. Several companies and organisations highlight 'design methods' or 'design thinking' in easy-access formats as toolkits or method cards for "people who aren't trained as designers to use creative tools to address a vast range of challenges".[5] One could argue that these simplified and



On the first page of the British Design Council's web site, it states that "Whatever the question, design has an answer". https://www.designcouncil.org.uk (2020-06-10).

easily accessible entry points to designing have spread insights in society in general and in both private and public sectors, which strengthen the presence of design in many areas. One could also argue, that these understandings of designing as based on easy-access, universal, methods and tools, reduce design to a procedure of producing problem-solving outcomes rather than seeing it as something that addresses complex situations through proposing other possible futures.

Design methods and processes seem to implicitly be perceived as recent, neutral, and a-historical sets of pre-defined tools and techniques that can handle a variety of situations. Design methods tend to simply be there, as something that can easily be picked up from others' explorations or toolkits be taught and learned in design education, and practiced and further developed through own design experiences.

There is little awareness, it seems, of the where and how different methods and processes have come about, and how they have come to be adopted in designing. There is perhaps even less appreciation of the time frames and the efforts required, that have made significant contributions to methodological changes in designing. In other words, neither the historicity nor histories of designing tend to be present in current practice and research in design.

While methods, on the one hand, are seen as 'just being there', the attempts to alter the ways design is done has become increasingly focused in design practices during the latter half of the 20th century. Indeed, there are many examples of design research that emphatically emphasise that designing needs to change radically, which would require substantial shifts in positioning and performing design.

Over time, as new tools and methods are introduced and others continue to be incrementally tweaked and altered, some ways of doing design tend to stabilise as core methods that become more or less taken for granted. These methods once came about as radically new, in specific historical contexts where traditional methods no longer provided the support needed for addressing new design situations.

The conscious development of new design methods aims to support exploring ways of opening up for different types of knowledge and experiences to be brought into designing, aiming to address change and complexity.[6] However, little attention is given to what implications for present-day and emerging design practices that might come into view if one were to explore the historicity of designing. When historical perspectives are introduced in relation to designing, making a case for the importance of temporal contextualisation, these temporal views seldom stretch more than a decade or two from the near present.[7] The temporal aspect – historical and futural – of the design methods that new proposals for methods build on, reject or relate to, is seldom addressed.

Seeing the present, the temporal 'now', as the springboard for design's movement towards making possible futures, requires paying attention to where the present comes from: how we come to perceive *this* specific state of things as *the* present. Approaching the questions of why these methods came about, and how that formed not only new tools, techniques and processes in design will also need to highlight the actual ways in which these developments took place.

Scandinavian user-centered and participatory design

Prototyping tools and methods for user-centered and participatory designing has been central to the establishment of a Scandinavian design tradition. Tools for exploratory and generative practices, methods for bringing people together in codesign situations, and infrastructures that allow for open-ended design processes, has had a strong presence in commercial design as well as in community contexts and research settings[8].

The centrality of methods can, for example, be seen in the Participatory Design Conferences initiated in 1990, where papers discussing and proposing methods and tools were the most frequent contributions throughout the first

6 Paul Gardien, Tom Djajadiningrat, Caroline Hummels, & Aarnout Brombacher, "Changing your Hammer: The Implications of Paradigmatic Innovation for Design Practice", *International Journal of Design* 2014:2 (vol.8), 119-139; Seyram Avle, Silvia Lindtner, & Kaiton Williams, "How Methods Make Designers", in *Proceedings of the 2017 CHI Conference on human factors in computing systems* (ACM, 2017), 472–483.

7 To give but two examples: İdil A. Gaziulusoy & Chris Ryan, "Shifting Conversations for Sustainability Transitions Using Participatory Design Visioning" *The Design Journal.* 2017 (vol.20, sup1), S1916–S1926; Liesbeth Huybrechts, Niels Hendriks & Sarah Martens, "Counterfactual scripting: challenging the temporality of participation" *CoDesign: Special issue: Participatory Design in an Era of Participation.* Guest editors: Rachel Charlotte Smith, Claus Bossen and Anne Marie Kanstrup. 2017:2 (vol.13), 96–109.

8 Susanne Pagold, *Function rules: En bok om Ergonomidesign* (Stockholm: Arvinius, 2006); Kerstin Wikström, *A&E design: The book* (Stockholm, 2018); Liam J. Bannon & Pelle Ehn, "Design: Design Matters in Participatory Design" in Jesper Simonsen & Toni Robertson (eds.) *Routledge International Handbook of Participatory Design* (London: Routledge, 2012).

two decades of the conferences.[9] Developing tools, theoretical foundations and methods for participation and involvement in design action is highly present also in recent and emerging design practices, aiming towards finding methods for codesign and other more open-ended practices.[10] The abundance of techniques and tools developed to support participatory design practices have in turn led to the compilation of handbooks, toolkits and frameworks aiming to systematically describe and group different methods in regard to the aims and contexts these could be used in, as well as where tools and methods are lacking.[11]

Whether intending to support formulation and scope of a design endeavour, sharing of experiences, analysis and reflection, or making joint design decisions, the focus on methods and tools is prominent in user-centered and participatory design. These often include working with different ways of making people come together, through introducing prototypes, mock-ups, scenarios, design games and various tool-kits. The purposes of introducing for example lo-fi prototyping and workshop methodologies were not only to enable discussion and decision-making in the design process but fundamentally also to bring forth other types of knowledge in the process through practice. This design-by-doing approach has, in combination with the introduction of ethnographic methodologies that aim to gain both broader and deeper insights into people's actions and thinking,

9 Methods and tools are the "most well-described aspects of PD in the early proceedings" of PDC, between 1990-98, according to Ditte Amund Basballe, Kim Halskov and Nicolai Brodersen Hansen, "The Early Shaping of Participatory Design at PDC", in *PDC'16* (Aarhus, 2016), 23; Kim Halskov and Nicolai Brodersen Hansen, "The diversity of participatory design research practice at PDC 2002-2012", *International Journal of Human-Computer Studies* 2015:74, p. 83.

10 For example, Kristina Lindström & Åsa Ståhl, *Patchworking publics-in-the-making: Design, media and public engagement,* (PhD diss., Malmö, 2014); Ezio Manzini, *Design, When Everybody Designs: An Introduction to Design for Social Innovation* (Cambridge: MIT Press, 2015). Per-Anders Hillgren, Anna Seravalli & Anders Emilson, "Prototyping and infrastructuring in design for social innovation", *CoDesign* 2011:3-4, vol.7, pp. 169-183; Christopher A Le Dantec and Carl DiSalvo, "Infrastructuring and the formation of publics in participatory design", *Social Studies of Science* 2013:43(2); Susanne Bødker, Christian Dindler & Ole Sejer Iversen, "Tying Knots: Participatory Infrastructuring at Work", *CSCW* 2017:26, 245-273.

11 Elizabeth B.-N. Sanders, Eva Brandt & Thomas Binder, "A framework for organizing the tools and techniques of participatory design" *Proceedings of the 11th Biennial Participatory Design Conference* (Sydney, 2010), p. 195; Jesper Simonsen & Toni Robertson (eds.), *Routledge Handbook of Participatory Design* (New York: Routledge, 2012); Elizabeth B.-N. Sanders & Jan Pieter Stappers, *Convivial Toolbox*: *Generative Research for the Front End of Design* (Amsterdam: BIS, 2012); Mike Kuniavsky, Elizabeth Goodman & Andrea Moed, *Observing the User Experience: A Practitioner's Guide to User Research* (Waltham, MA: Morgan Kaufmann, 2012).

brought about many tools, techniques and methods used in different stages of design processes, with different aims.[12]

As participatory and user-centered design practices have evolved, initiatives towards expanding the range of methods for user insights and collaborative design work have to a large degree over the latest decades sprung from initatives "within" design. Approaches, procedures and methods from other fields also continue to be bought into design, often through inter- and multidisciplinary projects and programs that aim to develop new ways of handling design situations and participatory practices. At the same time as ways of doing design continue to be explored and expanded, there are certain formats, terminologies and methods that over time more or less have become "standard" to these design practices. This might include the format of the 'workshop', the practices of 'prototyping', the use of 'ethnographic methods' such as observations and probes, working with 'generative tools', and so on.

The conceptual foundations of these terms, tools and methods have become so embedded in the way design is thought and done, that they are hardly thought of as once having been new and introduced into designing at some point. It would be quite difficult – and frankly not even very relevant – for designers to be knowledgeable of the history of each and every method or technique applied in every design situation. Nevertheless, it would be necessary for designers to be aware of the fact that the tools and methods in their toolbox are historically shaped and carry with them certain implications for designing. Overcoming the a-historicity of designing could, then, begin with questioning what it means for thinking and practicing design that its methods have emerged and become established in specific historical contexts.

Acknowledging that the material things that enter our lives through design will affect how we live and what we can do with them, in terms of their historicity, is not very hard. In our daily lives, we can all probably relate to experiences of how designed environments, things, and systems sometimes manifest their historicity in terms that make us acutely aware of their outdated assumptions about who we are and how we wish to live: the scarcity of power outlets in older houses, the difficulty of biking with thin tyres on cobblestoned streets, the way many societal institutions are built on heteronormative presumptions, or how 'skin-coloured' band-aids or nylon stockings tend to be light beige in colour.

It will often take a bit more of an effort to grasp that the *way* these things have come about – the methods and processes of designing – are themselves also designed things that carry a similar type of historicity that affects us daily. The concepts design methods, tools, processes and ways of thinking are founded in have set the limits for what can be expected for design to take on, and what it can present as possible design outcomes. What becomes in a design process, and what comes out of it, is not only dependant on the situations in which the designing takes place but on the situations in which the ways designing is done once have come about. The methods applied in design, the premises on which they were built when they were invented or introduced in design will undoubtedly influence what design can and cannot do, and also what remains hidden and what is made visible in design.

12 Sanders et al., "A framework for organizing the tools and techniques of participatory design", 195-198.

Historical methods and considerations

Despite the centrality and attention to matters of methods and process in most user-centered and participatory design practices today, the historicity of the methods, tools and techniques applied in designing is seldom considered. If it is, it is mainly with reference to developments in the last decade or so, or – if the historical time scale stretches beyond that – with a more sweeping pointing to 'tradition'. In participatory design research, for example, a quite common formulation that points to the historical development of methods or perspectives in the field is that of 'the Scandinavian participatory design tradition'. Here, methods referenced include tools and techniques for mutual learning in designing, as well as methods aiming to provide ways for non-designers to actively take part in design work and processes. Only very rarely, it seems, are questions raised of when and why these design methods once came about, and what preceded the formulation of participatory design, initially called the "collective resource approach", in 1970s Scandinavia.

Historicity is neither considered in terms of the Nordic historical context leading up to the formulation of participatory design in the late 20th century, nor in what it could mean that this sort of design did not just 'emerge', but that it came about in a situation where existing ideas and practices of past situations made its formation possible. Considering that designing comes from *somewhere,* not from *nowhere,* should probably mean something for designing, in terms of the embedded historical values or affordances at play when applied in contemporary design situations.

Similarly, investigations of histories of design methods have had very little presence in design history. While design historical scholarship has studied many layers of historical contexts of design, the question of where, how, and why various design methods came about is all but lacking in the history of design. In accounts of the early formation of industrial design in the US, attention is often drawn to the way that the formation of design consultancies incorporated methods from the emerging field of advertising and marketing that led to the adoption of specific types of user research, such as consumer surveys and market segmentation aiming for understanding consumer preferences.[13] Alison Clarke has pointed to the incorporation of ethnographic methods that came with the anthropological turn in design of the 1960s and 1970s, and the need for design historiographies that would include material outcomes of the 1970s 'design for development' practices as well as the methodological and political foundations for these practices.[14] However, when design history pays attention to methods, it is not usually the methods of designing but its own historiographical methodologies which are in focus.

Transitional design histories that explore the historicity of design methods address the relations between stability and change in design practices. When new methods have come about in design, these have been developed or imported to support design's expansion into areas where traditional ways of designing have not been

13 See for example, Nic Maffei, *Norman Bel Geddes: American Design Visionary* (London: Bloomsbury, 2018)

14 Alison Clarke, "Design for Development, ICSID and UNIDO: The Anthropological Turn in 1970s Design", *Journal of Design History* 2016:1, vol.29, 43-57.

able to deal with the situation at hand. As many of these methods have been tried and tested enough to be deemed appropriate for designing, they have become perceived as stable and belonging to core design skills and practices. Approaching the historicity of these methods aims to make visible ideas and assumptions that can be questioned or called out in relation to which types of design situations these methods now are expected to address.

As design develops, the once-new methods that previously could support design's move into new areas, could perhaps nowadays instead make desired ways of contemporary or future designing difficult. Making histories of design methods – of designing – contributes to probing the conceptual spaces in which designing takes place, with the intent to shift perspectives and points of departure for design's becoming otherwise. What we consider design to be, influences the histories told about what designing has been, and through that also project trajectories of what one could imagine designing to become.

Other potential histories will make visible other possible futures. How we perceive the present and futures made possible in and through designing both conditions and is conditioned by what we acknowledge as design's past. The ways in which design is done in the present is formed and shaped by the past, and not only then by the histories we make of where design comes from. The historical situations in which our ways of understanding and making the world has come about will be re-enforced and re-produced in the ways and procedures that set the directions for what types of things we make in design.

Through making design histories that are transitional in the sense that they are provisional, situated and changing as the understandings and practices of designing changes, the aim is not to provide objective or definitive accounts of the past as a stabilising account of what design is. Instead, the aim with transitional histories is to support a shift in perspective that might also open up for other ways of thinking in design, by questioning things that are perhaps often considered non-issues or that are taken for granted.

The historicity of many methods used in design is, I propose, such a blind spot that could be investigated and questioned in order to make visible ways of doing and thinking that can be re-considered if we aspire to do design differently. What if there were histories of the development of design methods that would enable us to see our, often invisible, assumptions of what we regard as design today, and therefore also what we go about searching for as 'its' history? How do we recognise design when we go searching for it in the past?

If the places and situations where new ways of designing were developed, were not spoken of as 'design' at the time, and have not later appeared to be relevant when searching for 'design history'– how can we then begin to see the blind spots of where our doing and thinking might stem from? How have design methods been developed historically, and what could that speak to in terms of the aims to develop yet new ways of designing for current and future potentialities?

In making this prototype of a design history of methods, I have, of course, applied methods of my own in doing so, which requires some considerations and clarifications. To start with terminology: A method, in a standard definition, is a plan of how to go about doing something, often for a particular purpose or aim. A method implies a systematic and practical approach to the type of action or doing aimed for, but can also refer to a combination of several linked ways of doing, or methods, brought together in a process.[15]

When I state that this historical study takes its starting point in contemporary methods of designing, I refer both to combinations of methods spoken of as "design process", categories of methods gathered under a specific label such as "methods for creativity", and specific methods like "brainstorming" or "lo-fi prototyping". The closer we move towards specific methods, the more likely it is that other terms can be used more or less as synonyms – which is often the case in texts dealing with design methods. In reference to different ways of supporting actions and perspectives in design, the terms 'techniques' and 'tools' are sometimes used in parallel to that of 'method'. While these words are sometimes used almost interchangeably, I use 'tool' as something that is applied or enacted in order to allow or enable the doing of something. A technique is a pre-defined procedure that one can follow in order to achieve a specific result, and combine in sets of procedures. While these terms are linked, the differences between considering a way of doing something a 'method' or a 'technique' will imply different levels of openness or directedness in what could be expected.

In positioning the outlook towards Swedish mid-20th century practices aiming for a design reform of everyday domestic life, I have searched for ideas and practices that preceded the emergence of what became user-centered and participatory Scandinavian design in the 1960s to 1980s, in regard to 'methods'. When looking for material for this study, I have stayed quite close to the design contexts around the Svenska slöjdföreningen (SSF, Swedish Arts and Crafts Association), since that organisation has been a central node for many design initiatives and policy proposals and in addition, is an actor that is highly present in much of the established Swedish design history narrative.

Approaching the SSF through the studies made into housing and everyday dwelling practices of the early 1940s, I trace the formation and use of methods in design to contexts of the rationalisation of domestic work that linked different persons and organisations to the SSF. In the archival and published material used for this study, I have searched for descriptions and discussions, implicit as well as explicit, of practices and concepts relating to 'methods', 'tools' and 'techniques' aiming to influence design processes and practices.

While presenting a design history that is also complementary to existing design historical narratives, this is mainly a proposal of how transitional design history could contribute to shifting perspectives on current issues in design. Methods are not neutral. Where and when these were developed will have an impact on the values embedded in these, as well as on the types of design they can support.

In applying various tools and methods, there needs to be an awareness of the historical situatedness from which these have sprung. If not, the world views, norms and assumptions these methods are based on, as well as the historical traditions these carry, will be put into play in contemporary situations without reflection on what this means for the trajectories we draw. With a historical perspective on how methods come about, we might also see other possibilities in the present for different fields from which design could borrow and build processes, practices and tools that make sense for the sorts of open-ended designing that could open up towards resilient and sustainable futures.

15 Merriam-Webster dictionary, "method"; "tools"; "technique", https://www.merriam-webster.com (accessed 2019-10-29).

Working with the material from which to draw out a history of design methods in a Scandinavian context, I have used archival and published material connected to several organisations and projects: the Svenska slöjdföreningen (SSF) and the Swedish Architects' association's (SAR) investigation into dwelling habits, the home life investigation by Brita Åkerman that had inspired that investigation, both discussed in the previous chapter, and the research institute HFI, Hemmens forskningsinstitut (The home research institute) that picked up on themes from the previous two investigations.

However, I have chosen to begin with a brief overview from an Anglo-American and context: the conference proceedings on 'design methods' from the early 1960s and statements on methods as these appeared in publications from the design school HfG Ulm in that period. The reason for beginning there – despite the theme being Scandinavian user-centered design – is that these settings are the ones most often pointed to in histories of where the concepts of "design methods" were initially formulated and developed. The later emergence of participatory and user centered design in a Scandinavian context, in the 1970s and 1980s, and the subsequent developments of methods based in this tradition, is usually understood to build on the approaches formulated in the design methods movement in regard to how collaborative designing can be supported (see chapter 3).

Building on the outlines of what in the period 1960-1980 was considered to be central to the new methods for designing that shifted the focus from product to process in design work, the analysis of the Swedish material revolves around the search for earlier instances of the development and use of similar concepts or methods in the context of changing design practices.

Needs for new methods

In the process of industrial design's becoming a field of practice, responding to societal, economic and industrial transformations, a core element has been the development of different methods for designing. In the transition from crafts to industrial production, methods were developed to support the separation of designing from production. Techniques and tools that honed in on creative practices and ways of working with exploring and defining qualities of visual and tangible form were developed based on crafts traditions as well as in relation to emerging societal and technical contests.

For example, the separation of making from production in designing led to the Bauhaus formulation of the prototype as something not only representing a potential final, actual object to produce but as a support for the iterative process of exploring of different possibilities and limitations in designing.[16] Similarly, and reacting to the move of design into societal situations and levels of complexity that could not be handled or reduced to issues of visual or three-dimensional form-giving, the design methods of the 1960s and 1970s proposed to introduce a plethora of methods, rationally described, that could form tools for designing in response to wicked problems and systems-level approaches.

The coming-about of the design methods movement built on expansions in practices of designing during the 1950s, that built on bringing in different

16 Walter Gropius, "Principles of Bauhaus production [Dessau]" in Ulrich Conrads (ed.), *Programs and Manifestoes on 20th Century Architecture* (Cambridge: MIT Press, 1971), 96.

methods and perspectives from different disciplines into designing. At the Hochschule für Gestaltung in Ulm (HfG Ulm), for example, the explicit aim was to break with an older educational format that emphasised the designer's artistic individuality.[17] Instead, the curriculum at the HfG Ulm from the early 1950s to late 1960s emphasised teamwork and brought in teaching in semiotics, sociology, mathematics, ergonomics, cybernetics and a range of other fields to both introduce new ways of thinking in design, as well as tools and methods for working with formulating and addressing design problems.

In industrial design education, director Tomás Maldonado argued, knowledge of different types was necessary in order to tackle the increasingly complex situations of design, which meant that design must find ways to assimilate

> the new types of relations between theory and practice, engendered by the most recent scientific developments. We know now that theory must be impregnated with practice, practice with theory. It is impossible to act without knowledge, or to know without doing.[18]

The incorporation of new requirements for designing needed new methods to match the developments in industrial production as well as in societal contexts, which was emphasised in the description of the aims of the industrial design programme:

> The purpose of the department is to train designers of industrial products. The development of new kinds of methods of fabrication has faced the designer with problems which can no longer be mastered from previous artistically-based standpoints. The education of the designer must give greater emphasis to the scientific and technological disciplines which now, as they will in the future, guide the operational processes in industrial production, and which ever more critically determine the product. Today, the designer of industrial products must be capable of using the foundations of professionally-based knowledge for his work, which will be in close collaboration with constructors, production engineers, and economists. Above all, he must be in a position of awareness of the cultural and sociological context in which his activity takes place.[19]

A systematic approach to designing was introduced to students in all programmes at the HfG Ulm already from the first semester's foundation course, in order to "guide students to have, as far as possible, a systematic approach – rather than merely to work intuitively; thus to acquire some knowledge of method."[20]

The HfG Ulm published a quarterly magazine, in which the school's philosophy was shared through articles and project examples, all in parallel text covering German, English and (initially also) French. These ideas about designing

17 René Spitz, *HfG Ulm: The View Behind the Foreground. The Political History of the Ulm School of Design (1953-1968)* (Stuttgart & London: Edition Axel Menges, 2002), 17.
18 Tomás Maldonado, "New Developments in Industry and the Training of the Designer", *ulm 2, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, October 1958, 39.
19 "Industrial Design Department", *ulm 1, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, 1958:1, 6.
20 Anthony Fröshaug, "Visual Methodology", *ulm 4, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, 1959:4, 57.

connected to other initiatives in the same vein in other places. By the early 1960s, a critical conversation had arisen on the theme of design methods, and in one of the Ulm quarterlies from 1962, director Tomás Maldonado refers to an established tension in relation to design methods:

> The necessity and possibility of a particular methodology for industrial design has recently become – above all in Europe – a polemical object found quite often among industrial designers and educators dedicated to the training of industrial designers. On the one hand, there are those who believe that all design problems can only be solved by mathematical formalisation. On the other hand, there are those who assume that all problems in this field can only be solved with common sense. [---] Both positions suffer from a very alarming lack of realism, and I believe that this results from the fact that both groups understand radically different things a [sic] industrial design. The main problem today is that we constantly operate with the notion 'industrial design,' as if it meant one and only one reality. There exist [sic] not only a single but many realities of industrial design.[21]

In the same autumn, of 1962, a conference on the topic of Design Methods was convened in London. It brought together people who, in different locations, were exploring new ways of going about designing. The participants, active in design and design education in the UK and the USA, discussed and proposed different ways in which new methods could bring a more transparent and systematic process to designing. Several of the presenters were active in engineering, architecture and polytechnic institutions, and many professed an interest in methods from fields such as mathematics, cybernetics, operational research, and "digital computers".[22] In the proceedings from the conference, one of the members of the organising committee, Peter Slann, describes the background to arranging the event:

> The Conference, believed to be the first to be concerned with the methods, processes and psychology of the Design act, was the outcome of a discussion which took place a few years ago, when I first met Christopher Jones. Then, as now, we were particularly keen to seek out and establish systematic methods of problem solving, especially those problems associated with design. We also sought a means by which design could be taught as a creative process that could be aided by a systematic process of conscious thought, integrating experience with academic knowledge whilst at the same time keeping the imagination free from inhibitions.[23]

21 Tomás Maldonado, "Preliminary Note", *ulm 6, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, October 1962, 3.
22 Jones, J. Christopher & Thornley, D.G., *Conference on design methods. Papers presented at the Conference on Systematic and Intuitive Methods in Engineering, Industrial Design, Architecture and Communications. London, September 1962.* (London: Pergamon Press, 1963), ix-x.
23 Peter A. Slann, "Foreword" in Jones, J. Christopher & Thornley, D.G., *Conference on design methods. Papers presented at the Conference on Systematic and Intuitive Methods in Engineering, Industrial Design, Architecture and Communications. London, September 1962.* (London: Pergamon Press, 1963).

The focus on bringing systematic approaches to design through the incorporation of methods from other disciplines was, at this first conference, highly connected to questions of what could be said to constitute, or bring about, creativity in design. These designers and design educators working in various fields of design suggested that the process of designing could, and should, be made as comprehensible as possible through developing joint terminologies that described how designing happened, in order for new methods to be able to be developed and adopted. Many pointed to the increasingly complex situations of designing required other tools, other ways of designing, and new ways of approaching design problems than what traditional ways of designing could provide.

Following the 1962 event, a series of conferences on the topic of design methods were held during the 1960s, which later led to the formation of the Design Research Society.[24] In the USA, the Design Methods Group was formed at UC Berkeley in 1963 by Horst Rittel, formerly a teacher at the HfG Ulm. Several books focusing design methods were published during the 1960s, including, towards the end of the decade, two publications that became highly influential on developments in the field: Herbert Simon's *Sciences of the Artificial* and John Chris Jones' *Design Methods: Seeds of human futures.*[25]

Inventing design methods

The aim with the book *Design Methods* was to present different methods and techniques that would allow designers to go about defining and exploring design situations in terms of systematically setting and analysing design problems, which then would need to be approached in designerly ways that built on a combination of experience, intuition, and systematic methods. In a process of alternating divergent and convergent steps, diverse approaches could be opened up and leading to designers being able to choose between different tools and techniques appropriate to the situation at hand.

In *Design Methods,* the methods, or techniques, exemplified and categorised in the second part of the book relate to the different moments of divergence, transformation and convergence that Jones ascribes to the process of designing. This bears strong similarities to the example of the double diamond design process launched by the Design Council some 35 years later. A significant difference, however, consists of the views on the approach towards methods in relation to design. John Chris Jones, in *Design Methods,* emphasised that the ways of working in

24 'The teaching of engineering design' in 1964, 'The Design Method' in 1965, 'Design Methods in Architecture' in 1967. Design Research Society, Publications, Conference proceedings, https://www.designresearchsociety.org/cpages/publications-1, (accessed 2019-10-28). Nigel Cross, "A History of Design Methodology", in M.J. de Vries et al. (eds.), *Design Methodology and Relationships with Science* (Kluwer Academic Publishers, 1993), 16; John Chris Jones (1974), "how my thoughts about design methods have changed during the years" in *Designing Designing* (London: Architecture design and technology press, 1991), 15-16.

25 Herbert Simon's *Sciences of the Artificial* (Cambridge, MA.: MIT Press, 1969); John Chris Jones *Design Methods: Seeds of Human Futures* (London: John Wiley & Sons, 1970).

design were increasingly moving towards handling the unknown and the unstable in proposing different futures, which required other methods than before:

> We can see that the designers of the future can expect to find few fixed points of departure. Their job will be to give substance to new ideas while taking away the physical and organisational foundations of old ones. In this situation it is nonsense to think of designing as the satisfying of existing requirements. New needs grow and old needs decay in response to the changing pattern of facilities available. To design is no longer to increase the stability of the man-made world: it is to alter, for good or ill, things that determine the course of its development.[26]

Developing systematic methods in design was, for Jones, not a matter of providing procedures for reaching design solutions to problems, but fundamentally re-designing how design as a process could be approached more flexibly, depending on the scale and context of the design situation. While systematic and analytical methods could be used during stages of the design process, the aim with the methods was not to conduct a systematic analysis to reach objective decisions:

> the purpose of divergence is to seek questions, not answers. The effect of divergence is to deliberately create muddle and confusion in one's mind, to upset one's own assumptions so that one can become more sensitive to existing realities and new possibilities.[27]

Which method to use in a particular situation must always be a matter of choice made by the designer or design team, based on intuition and experience as well as broad and systematic knowledge, and cannot be reduced to a matter of rational procedure. The making of new methods for designing was necessary in order to create a systematic and transparent way to both be aware of and to handle complex design situations that went far beyond giving form to products. Designers could no longer rely on traditional, individual and drawing-based design methods. New ways of working and thinking were required if design were to be able to address systems-level issues in designing.

In order to approach systemic levels of designing, the tools and processes that had sufficed earlier could no longer support the new types of design. In these more complex processes, Jones and others stressed, the designer could no longer work on their own but would need to collaborate and coordinate with multiple other disciplines, and with users. Opening up design processes for the participation of others than trained designers, therefore, required ways of both talking about and questioning what happens in designing. For Jones, this was one of the main differences compared to the traditional ways of designing. As he wrote in the preface to the 1980s edition of *Design Methods*:

26 John Chris Jones *Design Methods,* (1970) 2nd edition (London: John Wiley & Sons, 1992), 33.

27 John Chris Jones, [1973] "Love, hate and architecture. or what is a method? Memories of a seminar in Bologna" in Johan Chris Jones, *Designing Designing* (London: Architecture design and technology press, 1991), 8.

> Looking back now, at this book, and at what has become of design methods, I think that this is the crux of the matter: the new methods permit collaborative designing whereas the old methods do not. They change the nature of designing, or can if one lets them. [---] The new methods, properly used, release everyone from the tyrrany [sic] of imposed ideas and enable each to contribute to, and to act upon, the best that everyone is capable of imagining and doing. This is not easy. It requires not only new methods but a new conception of the self.[28]

That other ways of thinking, as well as other methods of designing, were needed in order to approach new contexts for design was also stressed in the emerging Scandinavian participatory and user-centered design contexts. Notably, early proponents of the design methods movement such as John Chris Jones and Christopher Alexander had publicly denounced the movement they had been part of initiating already in the early 1970s, with reference to methods being adopted in too instrumental and procedural ways.

The critique towards these developments in the first generation of design methods was also highlighted in the Scandinavian context of making participatory methods that aimed to counter the rational and science-oriented route that had grown strong in the design methods movement. For example, in UTOPIA project researcher Pelle Ehn's dissertation, where he stated that:

> Today, participative and creative approaches to design are championed as candidates to replace the systematic and rationalistic design in the movement of design methodology. This is a critique that comes from the inside, from some of the most important contributors to a Cartesian design approach. Theory and practice in architectural and industrial design have moved away from the early ideas of design as decomposition of complex hierarchical systems, and towards a qualitative understanding of what good design is.[29]

Towards the end of the above quote, Ehn mentions the ambition to shift understandings of the concept of 'good design'. In design historical studies of Swedish design, this concept has been highly present, and many established narratives and critiques of 20th-century Swedish design have related to this. Many of these design histories have pointed to the centrality of the private sphere of the home in reform work aimed to promote 'good design' through influencing both production and consumption towards asserting a particular aesthetic and formal expressions.[30] In contrast to this, the context where the development of participatory methods for 'good design' was the professional sphere of the work-place.

28 John Chris Jones "1980 edition. A review of new topics" in *Design Methods,* (1970/1980) 2nd edition (London: John Wiley & Sons, 1992), xxxiii.

29 Pelle Ehn, *Work-Oriented Design of Computer Artifacts* (Stockholm: Arbetslivscentrum/Almqvist & Wiksell, 1988), 214-215.

30 Maria Göransdotter, "Från moral till modernitet: Hemmet i 1930-talets svenska funktionalism", *Nordisk arkitekturforskning* 1999:2, 45-54; Christina Zetterlund, "Just Decoration? Ideology and Design in Early-Twentieth-Century Sweden", in Kjetil Fallan (ed.) *Scandinavian Design: Alternative Histories* (London: Berg, 2012), 103-116; Zandra Ahl & Emma Olsson, *Svensk smak: Myter om den moderna formen* (Stockholm: Ordfront, 2001).

In the search for new histories of how and where design methods aiming for iterative and participatory design practices arose in Sweden, the position taken here is that of looking towards the domain of everyday life, and the enmeshed categories of "private" and "professional". Historians have pointed to the domestic sphere as a location where we can find entry points to histories of design that go beyond the narratives of the professional design of things. Relevant calls have been made for considering women's engagement in the perceived private and domestic domains of, for example, interior decoration, homemaking, crafts, and the mediation of advice or ideals in these fields, as a relevant part of design history.[31]

Furthermore, while not opposing this type of expanded design history approach based in feminist theory and perspectives of power and interpretation – quite the contrary – why not also consider the possibility that design methods applied in "public" or "professional" design contexts might also have origins in "private" or domestic settings?

Investigating everyday life

In 1964 – during the same period that design methods were intensely discussed in an Anglo-American setting – the Swedish design organisation Svenska slöjdföreningen (SSF, the Swedish Arts and Crafts organisation) published a book called *Bostadsvanor och bostadsnormer* (*Dwelling habits and housing norms*).[32] The publication described and analysed a study of how 100 families in 1940s Stockholm used their apartments, in regard to what types of furniture they had, and what sorts of work and social activities took place in the home.

This study had been part of a more extensive investigation into housing and design issues, run by the Svenska slöjdföreningen and the Svenska Arkitekters Riksförbund (SAR, the Swedish Architects' National Association) in the 1940s. Even though the study was already 20 years old at the time, the SSF and the SAR obviously saw it worth publishing anyway. In the introduction, the editor Gotthard Johansson discussed the time gap between research and final publication, that was attributed to lack of funding and staff at the SSF and SAR for publishing the complete study. However, since substantial parts of the study and the analysis of the material had already been disseminated in the 1940s, through journal articles

31 Grace Lees-Maffei, "From service to self-service: advice literature as design discourse, 1920-1970", *Journal of Design History* 2001: 3 (Vol.14), 187-206; Penny Sparke, *As Long as it's Pink: The Sexual Politics of Taste* (London: Pandora, 1995); Cheryl Buckley, "Made in patriarchy: towards a feminist analysis of women and design", *Design Issues* 1986:2 (Vol.3), 3-14; Cheryl Buckley, "Made in Patriarchy: Theories of women and design - a reworking" in Hazel Clark (ed.) *Design Studies: A Reader* (Oxford: Berg, 2009), 233-238; Alison J. Clarke, *Tupperware: The Promise of Plastic in 1950s America* (Washington: Smithsonian Inst. Press, 1999).

32 Gotthard Johansson (ed.), *Bostadsvanor och bostadsnormer: Bostadsvanor i Stockholm under 1940-talet*. Svenska Arkitekters Riksförbund och Svenska Slöjdföreningens Bostadsutredning (Stockholm: Kooperativa förbundets förlag, 1964).

and partial analyses publications, many of the findings had already contributed to establishing housing norms and national policies, as well as to impacts on the design of things as well as information campaigns aimed towards households.

Why, then, publish the book in the mid-1960s? Johansson stated that he hoped and believed the publication to still be highly relevant: even though the living conditions have changed substantially in the two decades that had passed, the intent with the book was to contribute to a methodological body of knowledge that could support further housing research – and design. That the book included summaries in English of the main points of the material also suggests that the intent was to contribute not only to a Scandinavian but to an international context. What was it then, in terms of methods, that the study aimed to contribute to?

The impetus for the investigation of dwelling habits described in the publication came from a Finnish invitation to SAR and SSF to participate in a housing exhibition in Helsinki, Finland, in the autumn of 1939.[33] The Swedish contribution to the exhibition was asked to target the housing situation for families with many children. The exhibition did open but was closed prematurely due to the outbreak of the war between Finland and the Soviet Union in November. It later reopened again after the end of the war, in 1940, but by then the SSF and SAR with the exhibition had extended into an extensive research program that could provide a solid foundation for dwelling building in the Nordic countries after the second world war. As there was little or no material to base the design of new types of dwellings for large families on, the idea came up to first conduct a thorough investigation of the current Swedish situation. A scheme was set up, that would target the material conditions set by the design of the dwelling, in terms of the floor plan, and in relation to practices of interior furnishings.

As part of this joint initiative between SSF and SAR, a committee of three architects, representing SAR, and three persons from SSF first organised the exhibition, and after its closing in 1940 began the planning of the research investigation. The scope was extended to also include small apartments in larger cities, and the Swedish Association for Furniture Retailers was brought into the project that would come to comprise four different, but interlinked, parts: A general study of dwelling functions and rooms, a specific kitchen investigation, a survey of dwelling habits, and an investigation of the furniture market. In the 1964 publication, it is only the dwelling habit investigation that is presented as a methodological contribution.

In the book from 1964, Gotthard Johansson who headed the SAR and SSF investigation, several times mentions that the whole investigation was based on the 'groundbreaking' and 'seminal' methods developed by Brita Åkerman for investigating everyday life in the home, published in 1941 as *Familjen som växte ur sitt hem* (*The Family that Outgrew its Home*). The decision to conduct a similar study in the SAR and SSF investigation was based on methodology applied by Åkerman. The aim was to further develop the methods she had used so that these would provide even more in-depth material for understanding the relationship between furniture, people's everyday habits, and the design of dwellings: "Our immediate problem will, therefore, be to develop a methodology, first for the collection of data

33 "Program för svenskt deltagande i 'Bostadsutställningen 39' i Helsingfors". Arkitekturmuseets arkiv. SAR:s och SSF:s bostadsutredning: Vol. A10.

and, second, for the tabulation and preparation of data."[34] Before coming back to the study conducted by Åkerman, let us first take a closer look at which the methods were, that Johansson pointed to as innovative and groundbreaking.

Methods for investigating use

The SSF and SAR survey aimed to find out "where dwelling functions took place, and how the different rooms were used and furnished" by people on an everyday basis.[35] The overall aim of the research was targeted explicitly towards how to understand use in relation to apartments and objects. The dwelling habits as such were interesting as an entry point to discover "their cause-effect relationship, above all, the correlation between dwelling design and use."[36]

The questions guiding the SAR and SSF study revolved around understanding what were the causal effects, and what the more value-based ideas or relationships that governed how a dwelling was used. Was the floor plan, and the way an apartment was fitted a decisive factor? What roles did the choice of furniture play? Did social norms and conventions steer how people behaved in the dwelling or were other factors more influential? In order to investigate this, questions were asked of how and where the inhabitants spent time together, where one slept and ate, and where housework, homework and social activities took place. Therefore, the survey was planned as an in-depth inquiry rather than a survey of statistic and quantitative character:

> Every particular concerned with the use of a dwelling could not, however, be inserted in the furnishing plan; the observations, statements and wishes of the interviewees, and the observations and impressions of the interviewer (covering both family and dwelling) had to be given in an adjoining text. The question here was whether to set up this test point by point as a questionnaire or to write it more freely in unrestrained form; for several reasons, the investigators finally decided in favour of the latter method. The result is a series of small monographs on the 100 families and their dwellings. While containing all the required facts, they also describe in clear and concrete terms the relationship between the dwelling and the persons living in it, between the family and its environment.[37]

This study of the dwelling habits was conducted from November 1942 to May of 1943. The planning and framework of the study had been set up by Gotthard Johansson, writer, editor and member of SSF, together with interior architect Elias Svedberg after discussions with Brita Åkerman.[38] All interviews and visits to the 100 Stockholm families participating in the study were made by furniture designer Lena Larsson, accompanied by her husband Mårten Larsson, architect,

34 Johansson, *Bostadsvanor och bostadsnormer,* 33. Cf. Arkitekturmuseets arkiv. SAR:s och SSF:s bostadsutredning. Vol. 57 "Plan för bostadsvaneundersökning 7/10 1941".

35 Johansson, *Bostadsvanor och bostadsnormer,* 49.

36 Johansson, *Bostadsvanor och bostadsnormer,* 49.

37 Johansson, *Bostadsvanor och bostadsnormer,* 50-51.

38 "Letter from Åke Stavenow to Brita Åkerman 18 Nov 1939". Centrum för Näringslivshistoria. Svensk forms arkiv. F1:96 Registraturen 1938-1970. 598a Möbelinventering i samband med Statens bosättningslån 1939-1940.

who served as photographer. The reason for having only one person conducting all interviews was both due to Larsson's qualifications as a designer, but also to the idea that it would be easier to compile a uniform material in this way. In the work with the material, the move between attention to details and a view of the whole was emphasised. The reason that it would be best to appoint only one person to process the material was to avoid that "the whole overall picture of the individual case thereby risks sliding out of view".[39]

Interviews with people in the 100 households in the study were conducted as open conversations based on a core set of questions, deemed as a format that would increase the reply rate as well as the depth of the questions posed.[40] Through a series of thirteen pre-defined questions, the interviewed persons' replies regarding their preferences and experiences of their homes were noted in some detail, which gave "a plenitude of information not only about the inhabitants' subjective views on the dwelling and its different details, but also of the actual existing dwelling habits, the use of the rooms, and the placement of the functions."[41]

While the interviews would yield a particular type of information, one of the significant draw-backs was that it would only focus on the persons' opinions, and not give any insights into the layout and type of dwelling in question, or of how it was fitted or furnished, and how these factors would play into people's opinions: "It is thus of little value to know how many percent wish to have larger or more bedrooms when one knows nothing of the rooms' or family's size."[42] In order to provide adequate input for designing, people's preferences and practices needed to be understood in relation to the contexts of the individual families' diverse situations.

The material gathered in the investigation was therefore of three different types: basic facts about the family and the apartment, including its use and furnishing, the inhabitants' comments about the dwelling including complaints and wishes, and also the investigator's impressions and evaluations. For each apartment, the placement of furniture was drawn on a 1:100 scale floor plan, using a series of specific annotations to specify both the different types of objects and if and how these were used. Specific codes were used to show if a sofa, for example, was used as a sleeping place, who in the family used it, where daily meals were taken and

39 Johansson, *Bostadsvanor och bostadsnormer,* 47.

40 Examples pointed to where of this type of method, what we today would call a semi-structured interview, were taken from where the 1100 interviews made in England in the "wartime surveys" by the British Mass-Observation institute, published in 1943 as *People's Homes*. Johansson, *Bostadsvanor och bostadsnormer*, 29. Also: Arkitekturmuseets arkiv. SAR & SSF bostadsutredning. Vol 63 100 fam I 100 läg: Familjer och lägenheter; Vol. 79 100 fam i 100 läg.; and 100 fam i 100 läg, enkät 1-50 & Enkät 51-100.

41 Johansson, *Bostadsvanor och bostadsnormer*, 29. "…en mångfald upplysningar inte bara om de boendes subjektiva inställning till bostaden som helhet och dess olika detaljer utan också om de faktiskt existerande bostadsvanorna, rummens användning och funktionernas förläggning."

42 Johansson, *Bostadsvanor och bostadsnormer*, 30. "Det är sålunda av föga värde att få veta hur många procent av de boende som önska större eller flera sovrum, när man varken får veta något om rummen eller familjens storlek."

so on. Nearly all apartments were also photographed, in order to document the types of furniture as well as the "general character" of the apartment. These photos were later attributed with a specific importance in the analysis of furniture, since "they often say more than lengthy descriptions, which especially for the analysis of furnishing and furniture, which during the course of the work have shown to own a more intimate connection with the dwelling habits than previously suspected."[43]

Documentation of the home interiors was extensive: floor plans noting the intended room functions, the marking of the exact position of different pieces of furniture and furnishings as well as the materials these were made of, detailed photographs showing the interiors, and short written descriptions by the interviewer, often also accompanied by value judgments or reflections regarding the quality of the interior or the atmosphere in the home.[44]

In order to obtain more substantial insights into the daily life of people, a method was adopted in which people were asked to keep diaries, or protocols, over their daily activities in which "the dwellers themselves keep protocols over their daily doings, where it is noted when each family member gets up and goes to bed, goes to work or school, comes home, eats, does their homework etc. and where for each function there is a note made of in which room it takes place."[45]

While the combination of interviews, protocols, photographs and furnishing plans would provide a certain insight into what people did in, and thought about, their home, one additional issue remained to be solved – that of how this connected to not only how the apartment was designed, but to the context of its furnishing. Methodologically, it was stressed, the questions of *what* one wants to know needs to be tightly linked to *how* one would go about planning and conducting the research.

The SAR and SSF dwelling investigation was a systematic study of furniture and floor plans, in which in-depth interviews and diaries over daily activities brought in the point of view of people living in the apartments in regard to how they used their things and their apartments. The aims with gathering both objective and subjective information and insights was to provide a foundation for the design of new types of dwellings. In order to be able to design apartments that suited not only contemporary but future needs, the ambition was to take account of people's ways of living and everyday dilemmas, as well as the material and financial conditions of city planning in relation to the things with which people furnished and equipped their homes.

43 Johansson, *Bostadsvanor och bostadsnormer*, 46. "…de ofta säga mer än långa beskrivningar, som alldeles särskilt för analysen a möblering och möbler, vilka under arbetets gång visat sig äga ett intimare samband med bostadsvanorna än man tidigare anat."

44 Arkitekturmuseets arkiv. SAR & SSF bostadsutredning. Vol 63 100 fam I 100 läg: Familjer och lägenheter; Vol 79 100 fam i 100 läg.

45 Johansson, *Bostadsvanor och bostadsnormer*, 30. "…de boende själva föra protokoll över sina dagliga sysselsättningar, där det antecknas när varje familjemedlem stiger upp och lägger sig, går till arbetet eller skolan, kommer hem, äter, läser sina läxor etc. och där det för varje funktion anges i vilket rum den försiggår."

The methods developed and applied in the dwelling investigation were quite thoroughly discussed in the 1964 publication, and are presented as a contribution more significant to future housing research than the actual results obtained. Many of the methods applied were inspired by the new sociology of the 1940s, but also by the research into everyday life made in organisations dedicated to the rationalisation and understanding of work in the home. In the overview of methods applied, references are made to techniques already well-known and used in other, previous, contexts of mapping everyday life. Questionnaires were deemed the simplest and cheapest way to gather material, but also too limited in regard to what could actually be known through them:

> The method is best adapted to opinion polls, where one wishes to find out the public's wishes and opinions in relation to certain types of dwellings or certain details of the dwelling's fittings and equipment. But even there, it has limited value.[46]

The use of questionnaires in gathering information about living conditions dated back to home reform initiatives adopting this method already around the turn of the century. In the early years of the 20th century, several different initiatives were taken to, through questionnaires, find out how people lived, in order to provide a basis for building better housing.[47] However, since the response rates to questionnaires were consistently so low that they were practically useless for founding any substantial conclusion on, the issues of the need for better methods and more systematic research into housing and dwellings was raised in relation to the need for more extensive information and "propaganda" for better housing.[48] The dwelling investigation was conducted as a way of trying to establish a different method for gaining insights into people's everyday needs and actions in their homes.

46 Johansson, *Bostadsvanor och bostadsnormer*, 29. "Mest lämpar sig metoden för opinionsundersökningar, där man vill utröna allmänhetens önskemål och inställning till exempelvis olika bostadstyper eller vissa detaljer i bostadens inredning och utrusning. Men även där är den av begränsat värde."

47 In 1906, for example, writer Elna Tenow published a series of books, the Solidar series, aiming to critically discuss issues of home life and the need for building better dwellings in order to address more pressing social issues. In connection to the publication of Tenow's books, a committee was formed with the intent of finding out how people in Stockholm would like to see future apartments organised, in order to provide better dwellings. But of 800 questionnaires, only 117 were answered. Johansson, *Bostadsvanor och bostadsnormer,* 46.

48 Similar problems with low response rates to written questionnaires was exemplified by a more recent dwelling survey, conducted in Gothenburg in 1939. With the aim of finding out if the newly built residential area of Johanneberg had provided dwellings that the inhabitants found satisfactory, 500 questionnaires were distributed. Only about half of these came back with replies. Gustaf Munthe, "Hur byggas våra bostäder? Röhsska museets bostadsundersökning", *Form* 1939:5, 99-106.

These methods for gathering material documenting furnishings and everyday life, from perspectives of users as well as from designers, seem to have spread relatively rapidly. They became applied in several subsequent Swedish dwelling investigations during the 1940s and 1950s, both within housing research, in studies of family sociology, and in the context of research into the ways work in the home was carried out.[49] Also in the other Nordic countries, initiatives were taken to map and understand daily life in the home building on the methodologies developed – at least according to the way the story was told by the SSF.[50]

The combined methods of systematic, measurable data with photographs, inventories of objects, and in-depth interviews focusing everyday habits as well as hopes and aspirations that Brita Åkerman had spearheaded, was still after 20 years deemed relevant to share with a national and international community in the publication of the survey in *Bostadsvanor och bostadsnormer.* The strength of this form of inquiry, it was stated, is that it is closer to family sociology than to technical housing research, in its attempt to provide "an objective account of how the family's daily life elapses in the dwelling."[51] How, then, did these methods come about, and become applied in the design of apartments and objects for the home?

49 The Industriförbundet (National Federation of Swedish Industries) conducted an extensive investigation into the dwellings of some 250 working class families in 1944, building on the same methods used in the SAR and SSF study. A few years later, when an evaluation of the state funded Swedish loans for young couples setting up home was made, the combination of diaries and interviews was once again used. Architects Åke Huldt and Ingeborg Waern Bugge, both affiliated to the SSF, then conducted a dwelling investigation among 164 families who had obtained dwelling loans in the late 1930s and early 1940s, with the intent to find out if and how everyday life had been affected by the chosen furnishings.

50 In Norway, a committee led by architect Carsten Boysen undertook a study similar to the SSF and SAR project, in which studies of dwelling habits were made and published in 1948 as *Mennesker og boliger* (People and dwellings) and in 1952 as *Livsform og boligform* (Life form and dwelling form). The work was conducted during the war, in 1942-43, as a team effort comprising some 50 investigators with a background in either architecture or psychology, combining field work and interviews with documentation of interiors. In Denmark, a study was made in 1943 by the Landsforeningen Dansk Kunstaandvaerk, in which architects and statisticians mapped dwelling habits in Copenhagen, through interviews and documentation of furnishings. Johansson, *Bostadsvanor och bostadsnormer*, 32.

51 Johansson, *Bostadsvanor och bostadsnormer*, 30. The research carried out by Åkerman was planned and conducted years before the British war-time surveys.

Sociology and research into everyday life

The dwelling survey conducted by SAR and SSF as part of the more general housing study was heavily influenced by an investigation conducted in 1937 of the living conditions of 214 families in Stockholm, published in 1941 by Brita Åkerman as *Familjen som växte ur sitt hem (The Family that Outgrew its Home)*. In the mid-1930s, the new American sociology, with a strong emphasis on empirical and systematic methods for measuring and comparing social phenomena in modern society, was introduced in Sweden.[52] In the years 1936 to 1937, a class in this new sociology was given at Stockholm University, and the author of the book, Brita Åkerman had been one of the students in that class.

Åkerman was well connected in the socio-political contexts of government committees and initiatives aiming towards legislation and policymaking in Sweden at this time. She was engaged in several groups and committees engaging in politically strengthen women's representation in society and politics by highlighting women's issues. As chair of the Hemkommittén (Home Committee) in the mid-1930s, she came to focus on the design of homes and their rational organisation. Her husband Alf Johansson was between 1933 and 1947 head of the *Bostadssociala utredningen* (the State commission on housing), of which Hemkommittén was a sub-committee. Their friends Alva and Gunnar Myrdal were involved in core political initiatives and many governmental committees, advocating making research the basis for socio-political reform through bridging social science settings and political arenas.[53] Other close family connections of Åkerman's were also engaged in establishing new ways of making change, not least through new types of research: Carin Boalt, Åkerman's sister, was a trained architect and had worked with the consumer cooperation and later with the Institute of Public Health, and whose husband Gunnar Boalt was a researcher in sociology.

Issues of gender and women's representation in politics as well as in forming the conditions for modern society were strong driving forces for Brita Åkerman and others in her network. Socio-political issues connected to the home, and to everyday life, became a way for "women's issues" to enter the political agenda, with women as experts. Dissatisfied with the information and analyses available about everyday life in the home, and the work in Hemkommittén, Åkerman enrolled in the sociology course at Stockholm University to learn more about the new sociological methods. As part of the course, the students were to plan and conduct a sociological investigation themselves.

52 Alva Myrdal and her husband Gunnar Myrdal, professor in economy and colleague with Alf Johansson who was married to Brita Åkerman, had both studied sociology in the US. They managed to found the Social institute, that educated social workers, that became the basis for the introduction of the new sociology. See Anna Larsson & Sanja Magdalenić, *Sociology in Sweden: A History* (Basingstoke, Hampshire: Palgrave Macmillan, 2015).

53 Per Wisselgren, "Reforming the science-policy boundary: the Myrdals and the Swedish tradition of Governmental Commissions" in Sven Eliaeson & Ragnvald Kalleberg (eds.) *Academics as Public Intellectuals* (Newcastle, UK: Cambridge Scholars Publishing, 2008), 173-195.

Åkerman and her classmates proposed a study of dwellings, where the ambition would be to find out how people actually lived in their homes and what differences modern and more traditional dwellings and furnishings played to how life was led.[54] Inspiration came from an influential book, *Middletown*, published in 1929 by two American sociologists who had studied an American mid-size town in the year 1925, and compared everyday life then to what life had been like some thirty years earlier, around the 1890s.[55] The method of that study included analyses of publications and statistics, and engaging people, mainly women, in in-depth interviews about how their lives had changed in comparison to that of their parents inspired the methods used in Åkerman's study.[56] Combining a gathering of quantitative data with in-depth interviews that targeted everyday practices and the qualities of real life in relation to statistical analysis, led to the development of a methodology that gave both objective 'facts' and a sense of people's real-life experiences of how to negotiate different situations of complexity in relation to making things work.

New methods, new insights

Through Åkerman's network, funding was secured for conducting the sociological study of women's situation in the home. The cooperative housing association HSB provided funding for the project, as well as access to the HSB housing registry. Through the statistical office of Stockholm, a random selection of families with children living in 1- to 3-room apartments were contacted about participation in the study.[57] The 214 families that accepted to participate received a visit from a social worker, who interviewed the woman in the household according to a list of pre-specified questions. How did they feel about the apartment, what they would like to change, and what were their habits? Where did they eat, sleep, work, and spend time together? Initially, the primary motivation for conducting personal interviews in Åkerman's study was to increase the response rate in comparison to using questionnaires and to make sure that there would be enough material to base an analysis on. Methodologically, this was not much of an innovation, as there were earlier examples of studies made by philanthropists and social workers visiting people's homes in order to provide input aiming to change housing conditions.[58]

What was innovative in this study, were the methods tested to explore how the actual use of the home played out in relation to how it was furnished, and how that affected what activities could take place there. The study group had decided that it would be interesting to document the material context of the homes visited, through

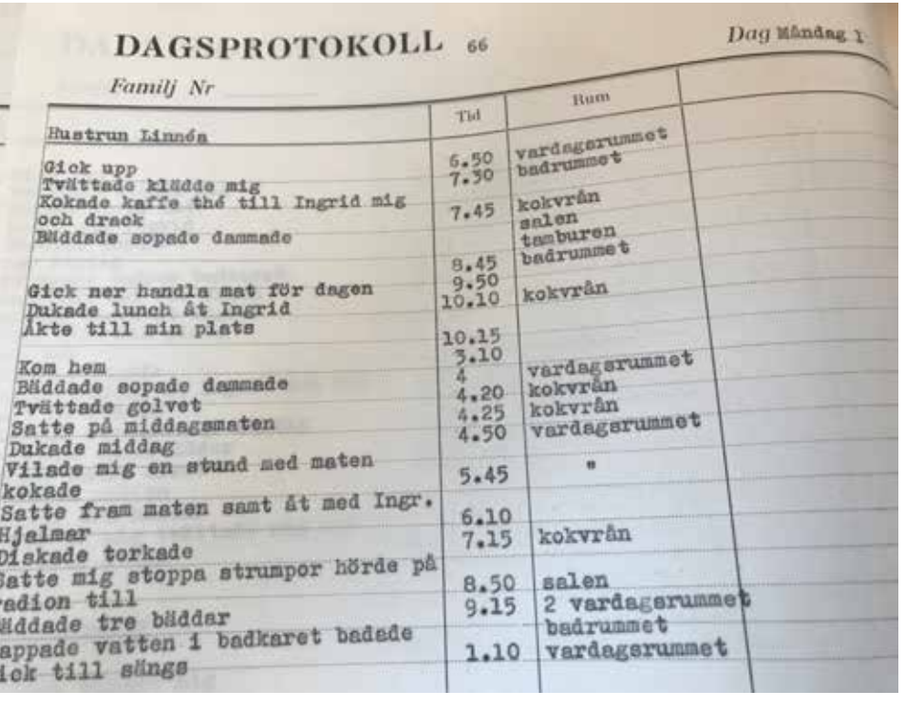
54 "Manuskript: I Familjen som växte ur sitt hem Hemkommittén, sociologi och bostadsboken", 6. Brita Åkerman's archive.

55 Robert Staughton Lynd & Helen Merrell Lynd, *Middletown: A Study in American Culture* (1929) (New York, 1956).

56 "Manuskript: I Familjen som växte ur sitt hem Hemkommittén, sociologi och bostadsboken", 19. Brita Åkerman's archive.

57 All of which lived in apartments except for 19 families who lived in small houses and who were included for comparison of eventual differences. Brita Åkerman, *Familjen som växte ur sitt hem*, 5.

58 Thörn, *En bostad för hemmet*, 32-133.



DAGSPROTOKOLL 66 — Dag Måndag 1

Familj Nr

Hustrun Linnéa	Tid	Rum
Gick upp	6.50	vardagsrummet
Tvättade klädde mig	7.30	badrummet
Kokade kaffe thé till Ingrid mig och dracK	7.45	kokvrån salen
Bäddade sopade dammade		tamburen
	8.45	badrummet
Gick ner handla mat för dagen	9.50	
Dukade lunch åt Ingrid	10.10	kokvrån
Åkte till min plats	10.15	
Kom hem	3.10	
Bäddade sopade dammade	4	vardagsrummet
Tvättade golvet	4.20	kokvrån
Satte på middagsmaten	4.25	kokvrån
Dukade middag	4.50	vardagsrummet
Vilade mig en stund med maten	5.45	"
kokade		
Satte fram maten samt åt med Ingr.	6.10	
Hjalmar	7.15	kokvrån
Diskade torkade		
Satte mig stoppa strumpor hörde på	8.50	salen
redion till	9.15	2 vardagsrummet
Bäddade tre bäddar		badrummet
appade vatten i badkaret badade	1.10	vardagsrummet
ick till sängs		

Protocol over a day's activities by one of the women in the 1937 study later published as *Familjen som växte ur sitt hem*. Brita Åkerman's archive, Umeå University library.

making notes of the way the apartments were furnished. Floor plans were provided through HSB, and during the interview, the social worker would use a pre-printed representation of the floor plan of the apartment, and schematically draw and make notes of how furniture was placed in the home. In some cases, photographs were also taken of the home, but not in a systematic and pre-defined way.

With the aim of gathering material to explore these different aspects of life in the home, every family was also asked to take notes about what they did on a daily basis, stating who did what, and how much time each activity took. This diary-keeping was something that, in later publications, is also described as a methodological novelty.[59] Not only did it give insights into the activities carried out in the home in relation to the layout of the dwelling. Besides the practical dimensions of what types of work and which actions took place, where, and who carried these out, the study also connected this to what that might mean for the relationship between wife and husband, and the children's place in the home and the family. The introduction of the furnishing drawings, photographs, and the diaries gave a new type of material to include in the sociological analysis of life and work in the home.

When the material gathering was concluded, the sociology course had

already ended, and the sociologists teaching it had left Sweden. Brita Åkerman was left with a massive amount of material to analyse more or less on her own.[60] For the quantitative data and the statistical analysis, she had some support from a colleague, but when it came to the material gathered through the new methods, and how to bring it all together to a coherent whole, she struggled with the analysis.

The methods of diary-keeping and furnishing drawings were novel. Since no similar study had been made previously, there were no established procedures or analytical tools that could be readily adapted to bring the different types of materials together, nor to base the analysis on. Instead, Åkerman worked on trying to analyse the material as best she could, between stepping into her new role as secretary of the Committee for increased women's representation (in 1937) and later as director of the organisation Aktiv hushållning (in 1940) and secretary for the women's delegation within the 1941 state Commission on population issues.

It was not until 1941 that the publication *Familjen som växte ur sitt hem* came out, and then in a format that both accounted for the "hard data" in the study, but that also foregrounded women's voices and experiences of everyday life and work in the home through examples, photographs and quotes from the interviews.

What stood out was that many families' living conditions were cramped, with the tiny apartments that people had in relation to the size of the families and also in relation to the types of furniture and norms for furnishing. Homes were furnished based on the types of furniture available to buy, rather than on what types of furnishings would support everyday activities. Between making and unmaking beds and sleeping places for everyone each evening and morning, the children had little room to play or do homework, and the parents had little time – and space – for their activities.

Moreover, as the study showed, women were trapped in a never-ceasing circle of domestic work and childcare, which led to that they had utterly slight possibilities to study, or work, outside the home. Meals were one of the most time-consuming chores in the home, since practically all meals were taken, or prepared, there for all family members. Packing lunch for the working fathers and schoolchildren, or preparing their lunches when they all came home to eat at midday, and providing snacks and meals throughout the day for the different family members, led to a minimum of six meals and sometimes up to 9-10 meals per day being prepared, served, with dishes being done afterwards.[61]

One conclusion in the study was that the living conditions – both standard of apartments and the furniture available on the market – as well as the difficulties for women to work and study, resulted in situations where couples hesitated to have children. This, in turn, was seen as a problem on state policy and national level, since diminishing numbers in the Swedish population had been identified as one

59 "Manuskript: I Familjen som växte ur sitt hem Hemkommittén, sociologi och bostadsboken", 7-8. Brita Åkerman's archive.

60 Riemer left for the US, and Sterner went to the US with Gunnar and Alva Myrdal as part of a research project, leaving Brita Åkerman to analyse the material on her own with some assistance in compiling graphs by Sam Ljungdahl.

61 "Manuskript: I Familjen som växte ur sitt hem Hemkommittén, sociologi och bostadsboken", 11. Brita Åkerman's archive.

of the most pressing issues to address if Sweden were to be able to develop into a strong and stable, modern, nation.[62]

The Family that Outgrew its Home came to inspire others to engage in systematic dwelling research through the adoption of these new methods. Methodologically, though, Åkerman meant that she was of little help to subsequent studies. She stated in her memoirs that

> I had so much to do at the beginning of 1941 that I did not have time to settle down and become an expert on dwellings. The sociology group that had been part of planning the study was dissolved. When people wanted to discuss with me how to plan new investigations, I could not provide them with any thoughts on the sociological or statistical method. I had no compilation of terms through which I could formulate our investigation methods.[63]

Despite that the publication had taken some time to finalise, the learnings and methods from the study had already begun to be spread and acknowledged through Åkerman's engagement within different committees and organisations. In these settings, the methods and findings presented as a result of the study resonated with the ambitions to fundamentally re-shape Swedish society, through the means of socio-political as well as through material reforms.

The attention towards the methods applied, as well as to the way that these gave insights into ordinary people's everyday situations, was picked up in three main areas. One was within social politics – hence the recruitment of Åkerman to the role as secretary in the Commission on population issues, given that issues relating to housing and family policies were at the core of this investigation.[64]

Secondly, as we shall return to later, the scientifically supported investigations focusing on the analysis of eating, cooking and on women's work in the home were picked up by women's organisations with different agendas and goals for increasing female representation in politics as well as raising women's issues in general. While the political organisation and representation in labour market matters were difficult for women to gain access to, issues of family politics, as well as domestic work and its professionalisation, provided spaces for women to engage politically as well as practically.[65]

Thirdly, the new methods were picked up by individuals and organisations searching for new ways to work with architecture and design, such as the Svenska

62 Alva Myrdal & Gunnar Myrdal, *Kris i befolkningsfrågan* (1934).

63 "Manuskript: I Familjen som växte ur sitt hem Hemkommittén, sociologi och bostadsboken", 11. Brita Åkerman's archive.

64 Nina Almgren, *Kvinnorörelsen och efterkrigsplaneringen: Statsfeminism i svensk arbetsmarknadspolitik under och kort efter andra världskriget*, diss. (Umeå, 2006); Johanna Overud, *I beredskap med Fru Lojal: Behovet av kvinnlig arbetskraft under andra världskriget*, diss. (Almqvist & Wiksell: Stockholm, 2005).

65 Ruth Schwartz Cowan, *More work for mother: The ironies of household technology from the open hearth to the microwave* (Basic books, 1984); Kjell Östberg, *Efter rösträtten: Kvinnors utrymme efter det demokratiska genombrottet* (Eslöv: Symposion, 1997).

slöjdföreningen, the National Association of Swedish architects, but also the consumer cooperation and the Industriens utredningsinstitut (the Research institute for industrial economics) engaged in adopting the new methods. Since these gave new potential insights into people's actual use of dwellings as well as furnishings and objects, as well as ways to holistically relate these insights to socio-economic and political initiatives and frameworks, it seems that the new methods opened up for other positions from where to approach the design of things and environments.

Domestic work and women's expertise

By the time Brita Åkerman's book was published in 1941, she had been recruited to a role as director of the Aktiv hushållning (AH, Active Housekeeping). This was a unit created under the Swedish government's information Office, as part of the crisis management of food, textiles and other resources set up due to the Second World War.[66] The work conducted at the Aktiv Hushållning aimed to spread information on how to make the most of rationed goods through advice on cooking, sewing, mending and caring for everyday things. Aktiv Hushållning did not develop any specific methods or advice of its own but functioned as a mediator of knowledge, information and advice that different women's organisations in their turn provided. The Husmödrarnas samarbetskommitté (the Housewives' collaboration committee) was formed with the purpose of networking and collaborating with Aktiv Hushållning. Represented were the Consumer cooperation's women's section, the Women's organisation of Sweden's rural areas, the National association of housewives' organisations, and the Social democratic Women's League.[67] Through the Husmödrarnas samarbetskommitté, contact was made with the so-called 'home consultants' that were tied to the regional household development organisations with the task of providing guidance and advice in housekeeping matters to rural homes, in order to activate them in the spreading of the Aktiv Hushållning advice.

The outreach of Aktiv Hushållning was – for its time – substantial, with average editions of its brochures of 100.000 copies and a strong presence in both radio and printed media.[68] As part of the organising of ways of collaborating, spreading information, and providing courses and education in domestic work, teachers in home economics formed a new national organisation – the Home

66 Aktiv Hushållning collaborated with the housewives' organisations through contacts with Husmödrarnas samarbetskommitté, but was run by women with expertise in other areas. The chair, Karin Kock, was an economist at Stockholm university, who when she was appointed to set up AH hired Brita Åkerman as director and Kaj Andersson, journalist, to work with producing media content. Brita Åkerman "Aktiv hushållning", in Brita Åkerman et al., (eds.) *Kunskap för vår vardag. Utbildning och forskning för hemmen* (Akademilitteratur: Stockholm, 1983), 78-79.

67 Kooperativa kvinnoförbundet, Svenska Landsbygdens Kvinnoförbund, Sveriges Husmodersföreningars Riksförbund, Socialdemokratiska kvinnoförbundet.

68 A brochure on conservation, for example, was printed in 200.000 copies in 1940, sold out almost immediately, and was re-printed in both 1942 and 1943. Some 6-8 articles

economics teachers' organisation – in order to have closer contact with AH.[69] In other words: through the Aktiv Hushållning, channels were established for collaboration, discussion, and for taking action in matters that regarded women's work and situation in the home in relation to a broader societal context.

With the affiliation to the government Information Office, the advice and information given in home matters through film, radio, brochures and courses had established women as experts in areas of domestic matters on the bases of both experience and professional knowledge. As stated in the first brochure, that introduced Aktiv Hushållning's mission as aiming towards "activating the experiences and knowledge that the Swedish housewives hold, to intensify research to find new paths toward increased collaboration between individuals and groups."[70]

Through the platform provided through Aktiv Hushållning, different Swedish women's organisations that in various ways, promoted a rational and scientific reform of housework gained a strong voice. While the work with coordinating and spreading information through Aktiv Hushållning also continued after the end of the war, mainly focusing then on textiles and clothes, several of the women who had been engaged in AH instead turned towards how it might be possible to establish ways of generating scientific knowledge about the home, in order to establish a solid foundation for making change on household as well as societal levels.

The feminist movement in Sweden had increasingly come to focus on issues of societal participation and equality in terms of bringing issues from the perceived private sphere of the home to the public sphere of socio-political discussion and reform.[71] In 1941, one of the leading debaters in the feminist movement, Elin Wägner, published the book *Väckarklocka* (*Alarm clock),* which emphatically criticised current societal structures. Wägner argued for the need to create a different form of society, built on radically different values as foundations for human life and culture, and proposed that women must enter into politics not within established power structures, but on terms of their own experience and knowledge in order for current systems to be dismantled.[72]

One of the ways to do this must be through women claiming more influence in society, in order to create a 'good life' that would not be based on ever-increasing consumption. Instead, the aim would be to find new ways of leading everyday life, that took into account a balance between collectivism and small-scale production, where a key was the proposal that quite small changes in patterns of everyday consumption could lead to a transformation of society as a whole.[73]

distributed each month through the AH press service, and Aktiv Hushållning also was allotted half an hour on national radio every Tuesday morning, to fill with content. Brita Åkerman, "Aktiv hushållning" in Åkerman, *Kunskap för vår vardag*, 87.

69 Hushållslärarinnornas samorganisation. Brita Åkerman, "Aktiv hushållning" in Åkerman, *Kunskap för vår vardag*, 79.

70 *Aktiv Hushållning nr 1 1940*, Statens informationsstyrelse (Stockholm, 1940), 13.

71 Helena Bergman & Peter Johansson (eds.), *Familjeangelägenheter: Modern historisk forskning om välfärd, genus och politik* (Eslöv: Symposion, 2002).

72 Katarina Leppänen, "At Peace with Earth—Connecting ecologial destruction and patriarchal civilisation", *Journal of Gender Studies,* 2004: 1 (vol 13), 37-47.

Brita Åkerman has later described that these ideas, in the early 1940s, became a turning point for her. Wägner's feminist philosophy led Åkerman to completely change her view on women, and on the world: "I perceived that Elin Wägner wanted to share a message of something new that ought to make us change our way of thinking and acting. For me, it meant a fundamental conversion."[74] The initiative to establish a research institute focusing on women's work in the home, to bring about societal change became – at least for Brita Åkerman – an attempt to bring Wägner's feminist ideas into practice.[75]

Hemmens forskningsinstitut

During the years at Aktiv Hushållning, Brita Åkerman and others had pointed towards the need for establishing not only sporadic but systematic and scientific studies of home life and housework that would be run based on women's expertise, and from feminist perspectives. The influence that women had gained during the war, also connected to the socio-political reforms that had begun to be initiated in different parts of society.

Establishing women's expertise in home matters as crucial to the (re) formation of industrial production, societal planning, and economic and political levels of decision-making, could be strategically linked to the expansion of social sciences as well as to political initiatives aiming towards improving and altering conditions for family life in Sweden.

State committees had stressed the need for building better housing, and funding was directed towards setting up research institutes, mainly focusing technical engineering research, such as Tekniska forskningsrådet (the Technical research council) and Statens kommitté för byggnadsforskning (the State committee for building research). Several research institutes that were part state and part industry-funded had also been initiated in the areas of wood, textile and nutrition research. However, despite suggestions from women's organisations to also set up a state-funded institute focusing on research about the home, this had not been done. Instead, the vast network of women's organisations took that matter into their own hands, through Husmödrarnas samarbetskommitté, and began planning for how to set up systematic research into everyday life in the home.

Planning for a research institute

In November of 1943, two sessions were convened by the Husmödrarnas samarbetskommitté, to discuss the possibilities of establishing research in the field of domestic work. Several representatives for different organisations and areas gathered to express the forms and directions that such an endeavour might

73 Elin Wägner, *Väckarklocka* (Stockholm: Bonnier, 1941); Katarina Leppänen, *Rethinking civilisation in a European feminist context: History, nature, women in Elin Wägner's Väckarklocka*, diss. (Gothenburg: Acta Universitatis Gothoburgensis, 2005).

74 Brita Åkerman, *88 år på 1900-talet: Bland vänner och ideer* (T. Fischer & co.: Stockholm, 1994), 99.

75 Åkerman, *88 år på 1900-talet,* 148.

take and the expectations and direction that could be set for the aims of this sort of work.[76] Many of the women engaged in this, were active in the various women's organisations and were also members of government committees and sub-committees – such as the Women's Committee under the Population Committee, in which Brita Åkerman was secretary. With professional training and experience in different areas of knowledge, ranging from nutritional science and architecture to economics and sociology, the participants in the discussion advocated that a research institute into matters of everyday domestic life must cover a variety of scientific disciplines. The questions of how to set up a research institute, organisationally as well as with which specific aims, led to intense discussions.

Architect Ingeborg Waern Bugge referred to the systematic and scientific studies of domestic work carried out in countries such as the US, Germany, England and Finland. Especially the activities at the US research institute under the department of agriculture, stemming from scientific home economics studies initiated by chemist Helen Richard in the 1870s, was mentioned as a format that could be interesting for a Swedish research institute to follow:

> The methods and formats for studies vary from case to case: they do not commit to a template, and do not confine themselves within their institutions (…) In regard to establishing standards based on the conducted studies, they proceed with utmost caution – especially in regard to the smallest and poorest homes: a tiny error can have such fatal consequences, the homes must in a sense be prepared for each adjustment, and an adjustment in one area must be tested in relation to other circumstances, the whole standard of the home, or else one does not move the home forward.[77]

Economist Margit Wohlin emphasised that the studies at a research institute would not benefit from spending too much time doing inventories and studies of the existing situation in actual homes since the quality of the dwellings and their standards differed much too much. Instead, the main question to be handled in relation to domestic

76 "Synpunkter på en kommande forskningsverksamhet på hemarbetets område. Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943". Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945.

77 "Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943", 4-5. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945. "Metoderna och formerna för studiearbetet växlar från fall till fall: man låser inte fast sig i schablon och låser inte in sig inom sina institutioner. (…) Då det gäller att på bas av utförda studier fixera standard går man fram med yttersta försiktighet – särskilt då det gäller de minsta och fattigaste hemmen: ett litet fel kan få så ödesdigra verkningar, hemmen måste liksom vara färdiga för varje förändring och en förändring på en punkt måste prövas i relation till andra omständigheter, hemmets hela standard, eljes för man inte hemmet framåt." Synpunkter på en kommande forskningsverksamhet på hemarbetets område.

research regarded what the actual aims were with engaging in research:

> We must clarify for ourselves what we want to achieve with new research, what we need to know. Not everything that is possible to know, is worth knowing. There is, for example, nothing interesting in knowing how many steps the women in Sweden walk each day, unless this is related to certain conditions, which could be improved. If we study how many steps one walks while making meatballs and lemon curd in one kitchen, and in another, and which appliances make the work easier, that does not mean that we are there. We must also know if it is actually rational to make meatballs and lemon curd from the viewpoint of full health of the nation, from a work savings point of view, and from an economic point of view. Which dietary habits are worth encouraging. We must not offhand assume anything permanent. Perhaps we should not knead and whisk to the extent that assisting appliances are worth having. – and besides, it is not certain that the preparation should take place in the individual home.[78]

The questions of what the research aims of the new institute should be were placed in the broader context of societal development as a whole. In the above quote, Wohlin's statement that nothing existing should be taken for granted speaks to that programmatic ambitions were not only embedded in the project of establishing a home research institute but were actually made very explicit.

Other speakers also stressed the importance of not taking a starting point in the current way of doing things as given. Writer Lilly Arrhenius, for example, pointed out that the research into housework must be as radical and as unprejudiced as possible:

> Such an investigation must not start from the home's and the housework's current outer forms as being eternal and unchangeable. It must with imagination

78 "Synpunkter på en kommande forskningsverksamhet på hemarbetets område. Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943", p. 9. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945. "Vi måste göra klart för oss vart vi vill nå med nya forskningar, vad vi behöver veta. Icke allt är vetvärt som är vetbart. Det har tex intet intresse att veta hur många steg kvinnorna i Sverige går om dagen, så vida detta inte sätts i relation till vissa förhållanden, som kan förbättras. Om vi studerar hur många steg man går för att laga köttbullar och citronkräm i det ena köket och i det andra och vilka maskiner som underlättar arbetet, så är vi därmed icke framme. Vi måste också veta om det verkligen är rationellt att laga köttbullar och citronkräm ur synpunkten av nationens fulla hälsa, ur arbetsbesparingssynpunkt och ur ekonomisk synpunkt. Vilka dietvanor är värda att uppmuntras. Vi får inte utan vidare godtaga något bestående. Vi kanske inte bör knåda och vispa så mycket, att assistentmaskiner lönar sig. – och för övrigt är det inte säkert at beredningen bör ske i det enskilda hemmet."

> consider how one will be able to find forms for solving the problems, that for many women in the current situation seem almost unsolvable.[79]

The ambition to bring about change in terms of what sort of future society might be possible, then, had to begin with finding ways to think differently. Developing new research methods would, it was hoped, provide the tools for not only solving problems but for making visible the assumptions and ideas underlying current social orders through a radical questioning of its foundations.

Two of the men invited to the discussion, Sigurd Erixon and Gotthard Johansson, both also supported and emphasised the need for research into domestic life from a systematic and scientific perspective. Gotthard Johansson, as the lead for the SAR and SSF investigation into housing norms and dwelling habits, had already earlier been involved in the preparatory discussions aiming to establish domestic research in relation to designing change.

Based on the SAR and SSF study, the view among many in the network of architects and designers in these organisations, was that the design of products, dwellings, behaviour and socio-political systems were all interconnected. Sigurd Erixon, professor of ethnology, advocated that if the studies were to aim towards reforming life and work in the home in general, also men's life in the home should be studied. The difficulty, then, would be where to draw the line between domestic work habits to be studied, and more general aspects of "leading life" in the home.

In order to be able to design new things as well as new methods of working, a core area of knowledge must be that of understanding people. In that context, Erixon advocated sociology with the addition of ethnographic methods to make sure that "methods must be to adapt inventions after the actually existing needs" that build on knowledge about people.[80]

The overall ambition with initiating domestic research was to find ways to shape a new type of society. Through understanding people's home life, and through radically questioning the current order of things, it seemed possible to make material interventions that could bring about a different everyday reality, in which perceived 'private' and 'public' issues were intertwined. Not taking for granted something so mundane as what food to cook, where it should be prepared, and how cooking should be done, and not only suggesting but stating that systematic knowledge about this could bring about a different way of living was the overall program of change associated with establishing domestic research.

The women, and men, who engaged in initiating the scientific and systematic study and redesign of domestic work, and its tools and environments, to a large

79 Lilly Arrhenius quoted in "Synpunkter på en kommande forskningsverksamhet på hemarbetets område. Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943", 15-16. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

80 "Synpunkter på en kommande forskningsverksamhet på hemarbetets område. Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943", 28. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

extent, saw this as a means to question the forms of social relations within the home as well as how these were mirrored in or connected to societal orders. Solving seemingly unsolvable problems was at the heart of the ambitions to use research to gain knowledge that could provide new ways of designing things, which in turn was linked to designing different ways of living. Designing new dwellings, new things to be used in them, and new methods for carrying out housework was not only a matter of rationalisation, but of radical societal redesign:

> Overall, when one asks how one can rationalise and simplify, and which the ideal ways of working are, and its methods and machines, one must also know what the goal is. To rationalise means to do things sensibly. What is sensible? Perhaps one, in general, can call that sensible which furthers the individual family's psychological and physical health and happiness. But it is not enough, it must also further the people's life as a whole and the mutual relations between families.[81]

In the continued discussion about the aims of initiating research into domestic work, the question of what should be an individual activity in the home, and what could be addressed on collective, societal, levels arose and almost led to an ideation-like situation. What could one imagine being done together, and could there be forms that combined collective solutions with individual choices?

One of the speakers sketched an idea where each block in a city could have collective facilities for childcare, supporting schoolchildren's homework, and facilities for preparing semi-finished food dishes that women could pick up on the way from work and finish cooking at home with the seasonings and additions of their choice. If one also had a system for hiring domestic work support by the hour, say for cleaning, mending, canning and other tasks, women could find ways to combine paid wage work outside the home with managing the everyday domestic work – that despite these outspoken ambitions to think in new ways was not associated with anyone else but the housewife.[82]

81 Margit Wohlin quoted in "Synpunkter på en kommande forskningsverksamhet på hemarbetets område. Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943", 11. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945. "Överhuvud, när man frågar, hur man kan rationalisera och förenkla och vilka de idealiska arbetsformerna, metoderna och maskinerna är, så måste man veta vilket målet är. Att rationalisera betyder att göra saker förnuftigt. Vad är förnuftigt? Kanske man allmänt kan kalla det förnuftigt som gagnar den individuella familjens psykiska och fysiska hälsa och lycka. Men det räcker inte, det måste också gagna folkets liv i dess helhet och de inbördes relationerna mellan familjerna."

82 "Synpunkter på en kommande forskningsverksamhet på hemarbetets område. Inlägg vid diskussioner anordnade av Husmödrarnas samarbetskommitté den 10 och 18 november 1943", 17-18. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

Developing domestic research methods

In 1944, the research institute Hemmens forskningsinstitut (HFI, the Home Research Institute) was officially founded, and located in the basement of the Statens institutut för folkhälsan (the National Institute of Public Health). There, the SSF and SAR housing and dwelling investigation had already previously, in collaboration with Aktiv Hushållning, set up a fully equipped test kitchen to prototype the planning of kitchens for new types of dwellings.

When the SAR and SSF dwelling investigation closed, the fact that there was a test kitchen to build home research around was one of the factors promoted to push the establishment of the HFI.[83] The investment in equipment was already made, the Institute for Public Health was keen on continuing supporting home research, and Aktiv Hushållning would initially handle the administration of the new research institute. Funding was secured from the consumer cooperation, from the Industry association, and also a government grant, which enabled the research institute to begin its work, in a relatively independent and flexible manner.[84]

The outspoken ambition at the HFI was to not begin with defining exactly what the forms of research at the institute should be. How its work should be conducted was instead left rather open, since the aim was to develop entirely new investigative methods. These should contribute to systematically building knowledge about domestic work and life in the home to bring about changes of how these were done, and led. As stated in the constituting meeting in April of the same year, the institute should:

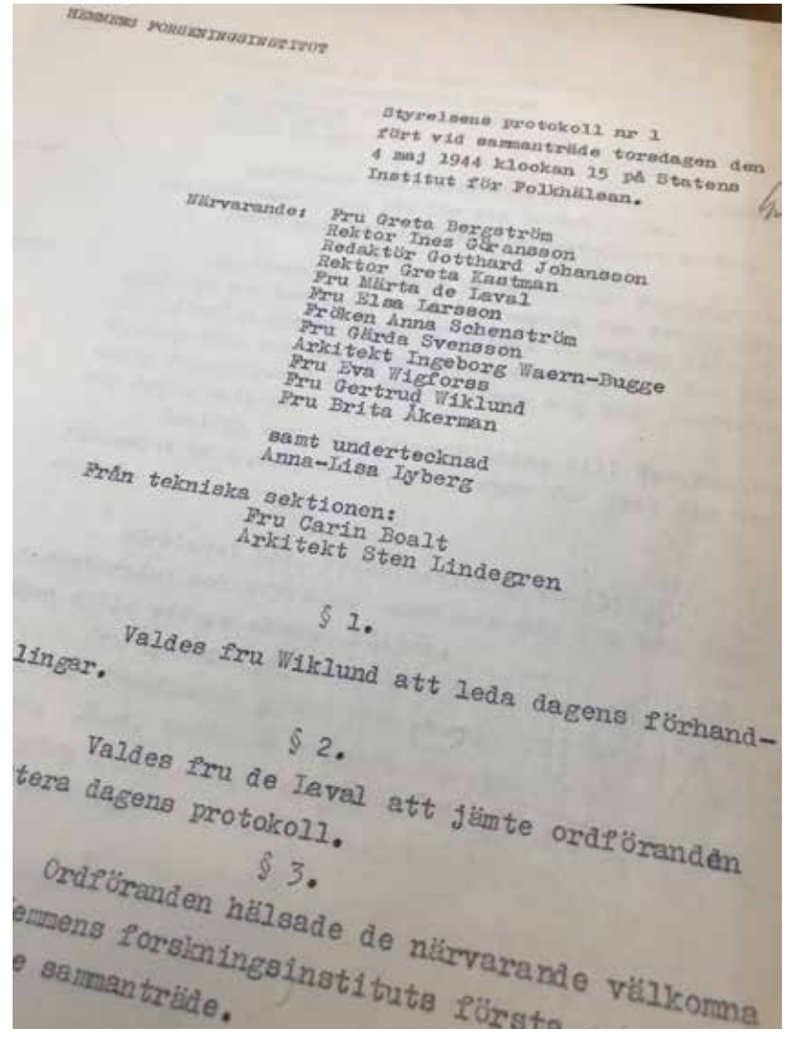
> Work for a systematic rationalisation of the working conditions in the Swedish homes through research concerning the technical and economic problems that are associated with the roles of homes as centres for consumption and work, considering the psychological, hygienic, and social problems that are linked to the general roles of the homes.[85]

In other words, the HFI inquiries aimed at understanding what happened when things were brought into people's lives in the home, and how activities defined as "work" were carried out, and what that meant from a broad range of perspectives. That the work needed to begin quickly was stressed in the planning and preparation work:

83 "Förslag till första etappen av undersökningsarbetet inom kommittén för hemhushållnings- och hemarbetsstudier" Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

84 Brita Åkerman, "Forskning för att förändra" in Åkerman, *Kunskap för vår vardag,* 131-139.

85 "Riktlinjer för verksamheten och aktuella arbetsuppgifter vid HFI", *HFI-meddelanden* 1948-49:1, 1-2



HEMMENS FORSKNINGSINSTITUT

Styrelsens protokoll nr 1
fört vid sammanträde torsdagen den
4 maj 1944 klockan 15 på Statens
Institut för Folkhälsan.

Närvarande: Fru Greta Bergström
Rektor Ines Göransson
Redaktör Gotthard Johansson
Rektor Greta Kantman
Fru Märta de Laval
Fru Elsa Larsson
Fröken Anna Schenström
Fru Gärda Svensson
Arkitekt Ingeborg Waern-Bugge
Fru Eva Wigforss
Fru Gertrud Wiklund
Fru Brita Åkerman

samt undertecknad
Anna-Lisa Lyberg

Från tekniska sektionen:
Fru Carin Boalt
Arkitekt Sten Lindegren

§ 1.

Valdes fru Wiklund att leda dagens förhand-
lingar.

§ 2.

Valdes fru de Laval att jämte ordföranden
tera dagens protokoll.

§ 3.

Ordföranden hälsade de närvarande välkomna
emmens forskningsinstituts förste
e sammanträde.

Thirteen women and two men formed the board of the Hemmens forskningsinstitut in 1944. Riksarkivet, Hemmens forskningsinstituts arkiv.

> Since many of the tasks to be investigated that this concerns are highly urgent due to the planning work that in other areas is conducted regarding the post-war production of dwellings and goods for consumption, regulations of educational issues for the female workforce, and for the societal construction in general, it is desirable that the investigative work be started as soon as possible.[86]

The measurability of domestic work

The initial studies carried out at the HFI in 1944 focused work physiological investigations into three different types of housework: baking, doing dishes, and cleaning floors. The committee planning the studies consisted of members of the HFI board, including director Carin Boalt, while several persons were hired for 4 to 7 months for some or all of the three studies.[87] Over the years, the number of permanent or temporary staff would vary, depending on the types of competencies needed in the different studies, and the available funding.

The three activities – baking, doing dishes and cleaning floors – were chosen since they were activities that would be carried out in all sorts of homes, on a daily or weekly basis. The study focused on finding ways to measure ergonomics as well as strain and energy consumption in working at different heights of workbenches and in different postures, with the aim to:

> investigate the possibilities of establishing an objective basis for assessing certain domestic work in relation to the work method and work posture. Such a method of assessment is necessary for the development of norms for fittings and equipment of the work stations in the home and for the development of efficient work methods in domestic work.[88]

86 "Förslag till första etappen av undersökningsarbetet inom kommittén för hemhushållnings- och hemarbetsstudier" Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

87 These were Boalt herself, for 7 months as operative secretary, a home economics teacher, a rural home economics teacher, two laboratory assistants, a time studies clerk, and an office clerk. Besides Boalt, the members of the committee were Greta Bergström, Greta Kastman and Sten Lindegren (chair). "P.M. angående tekniska sektionens verksamhet". Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

88 *Arbetsfysiologiska undersökningar över hemarbetet. Hur mycket tid och energi går åt vid bakning, diskning och golvrengöring med olika arbetsställning.* En undersökning utförd av Carin Boalt, Ing-Britt Hagberg och Yngve Zetterman., 1-2 "...undersöka möjligheten att ernå en objektiv bas för bedömning av vissa husliga arbeten med hänsyn till arbetsmetod och arbetsställning. En dylik bedömningsmetod är erforderlig för utarbetandet av normer för inredning och utrustning av arbetsplatserna i hemmet och för utarbetandet av effektiva arbetsmetoder inom hemarbetet. I denna undersökning har tidsåtgång och energiomsättning vid utförande av vissa arbeten studerats."



Time studies of potato peeling, in the series of HFI tests of potato peeling tools. Riksarkivet, Hemmens forskningsinstituts arkiv.

The aim was to find methods for the systematic study of housework, in order to in a later stage redesign the ways certain types of work in the home was carried out. In other words, to develop new methods for analysis as well as new methods for doing housework.

In a pamphlet describing the procedures and outcomes of the study, all aspects of the trials were accounted for in minute detail. In all of the studies, fatigue, strain and energy consumption were measured in relation to different methods of conducting the work, and various tools and equipment used. The studies were thoroughly documented, down to details of room temperature atmospheric pressure and at what time of day the test took place. Besides the measurable factors of the context of the test, the more qualitative experience and training of the "test subject" was also accounted for. In the laboratory studies of dishing, cooking and cleaning, the same person performed all of the work tasks. She was described as a woman thirty years of age, 164 cm and 58,9 kg, which was the equivalent of

> the normal measures of a Swedish woman according to professor G Dahlberg at the Institute for Racial biology, Uppsala. She was an experienced domestic worker, who for several years had worked in a large number of different homes, which had led to her not being tied to any specific working conditions such as height of workbench, etc.[89]

The time studies were conducted by "a trained time study clerk", who in detailed protocols noted the exact amount of time spent on each step in different activities, from the 110 seconds it took to mix yeast and milk when baking, to the 52 seconds needed to scrape the dough from the bowl onto the workbench.[90]

In the HFI studies, the thoroughness and detail to which every aspect of work was studies were reported with precision, in writing as well as with illustrations of photographs and graphs. In the study of floor cleaning, for example, posture (standing, kneeling) and equipment (broom, scrubber) were studied in relation to the floor material (linoleum, wood) and accounted for as to the time and the energy consumption for each method:

89 *Arbetsfysiologiska undersökningar över hemarbetet,* 5. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945. "denna typ motsvarade normalmåtten på svensk kvinna enligt professor G Dahlberg vid Rasbiologiska institutet, Uppsala. Hon var en van hemarbeterska, som i flera år arbetat i ett stort antal olika hem, vilket medfört att hon ej blivit bunden vid några särskilda arbetsförhållanden såsom bestämd bänkhöjd etc."

90 "Tidprotokoll för bakning, 20/6 kl 9.40". Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

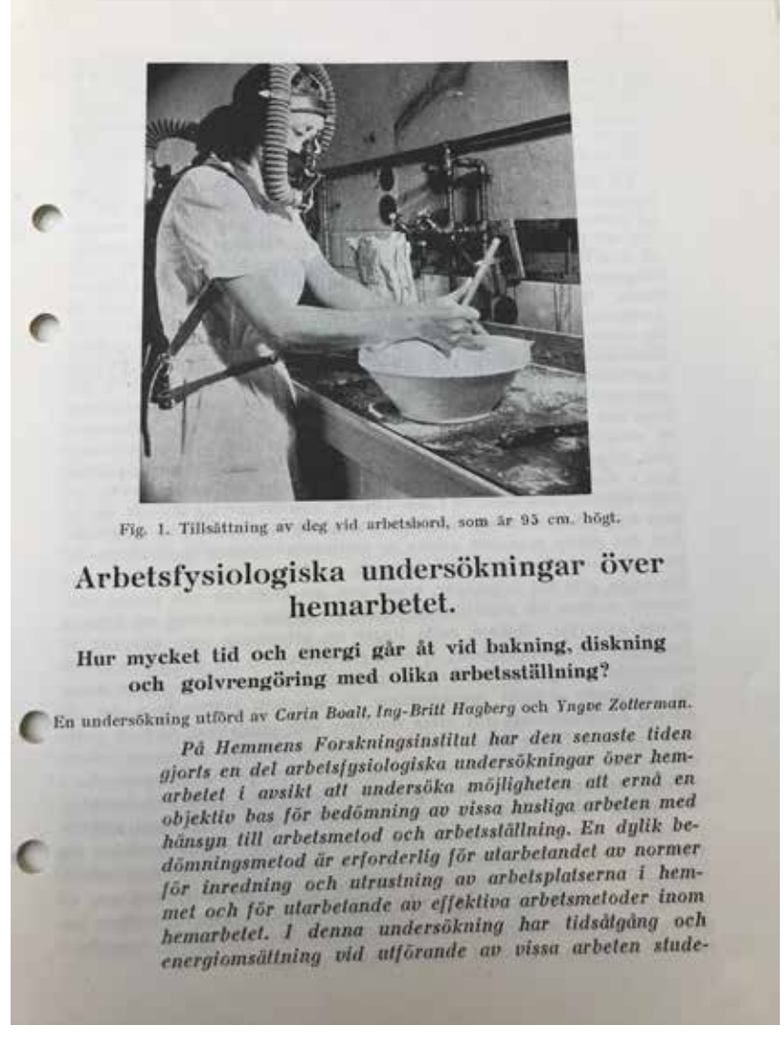


Fig. 1. Tillsättning av deg vid arbetsbord, som är 95 cm. högt.

Arbetsfysiologiska undersökningar över hemarbetet.

Hur mycket tid och energi går åt vid bakning, diskning och golvrengöring med olika arbetsställning?

En undersökning utförd av *Carin Boalt, Ing-Britt Hagberg* och *Yngve Zetterman.*

På Hemmens Forskningsinstitut har den senaste tiden gjorts en del arbetsfysiologiska undersökningar över hemarbetet i avsikt att undersöka möjligheten att ernå en objektiv bas för bedömning av vissa husliga arbeten med hänsyn till arbetsmetod och arbetsställning. En dylik bedömningsmetod är erforderlig för utarbetandet av normer för inredning och utrustning av arbetsplatserna i hemmet och för utarbetande av effektiva arbetsmetoder inom hemarbetet. I denna undersökning har tidsåtgång och energiomsättning vid utförande av vissa arbeten stude-

Measuring energy and fatigue in the activity of baking. *Work physiological investigations in the home,* the first HFI report in 1944. Riksarkivet, Hemmens forskningsinstituts arkiv.

> *Linoleum carpet.* The energy consumption during standing scouring was 4,04 kcal/min. and when lying considerably lower, 3,59 kcal/min. The time spent was on the other hand higher when kneeling than in standing posture, 9,41 resp. 6,88 min. For the after work, rinsing and drying there was no significant difference in regard to energy consumption. As with scouring of wooden floors, also in this case a lower score was measured for the total energy consumption in standing posture 27,8 than in lying posture 33,8 kcal. The difference here amounted to 22 percent.[91]

In the measurement of energy consumption for different tasks, the equipment adopted was "the Douglas bag method for indirect calorimetry".[92] The instrument used was a breathing mask connected with a tube to a rubber bag – at HFI carried by the test person on her back – and that measured oxygen uptake. While much attention has been given in historical accounts of these energy studies – indeed, the photographs of women wearing the Douglas bag while working are one of the most frequent illustrations of the HFI – the question of what was possible to measure, and what was actually important to find out, were foregrounded already in this first study:

> That a type of work is fatiguing can generally be seen to be synonymous with its being tiring, that one thus can keep going for long periods of time. One asks oneself, then, what it is that lies behind this subjective feeling of work being tiring.[93]

So, while the more long-term scope of the research would be to investigate the socio-psychological effects of certain types of work, a foundation of methods to gain measurable data would be needed to provide the basis for moving further towards more qualitative ways of studying work.

Collaboration with industry

From the viewpoint of the development of design methods, the way that the HFI in the 1940s took on the rethinking of seemingly taken-for-granted practices of women's work in domestic life, bears strong similarities to some of the central arguments brought forth in the design methods movement of the 1960s and 1970s. If the aim is to base designing on formulations of design problems or design situations on as broad a basis of information about situations and verifiable facts as possible, instead of making assumptions about what the design problem might be, there need to be ways of providing that broad collection of information.

The systematic ambitions at the Hemmens forskningsinstitut to bring about

91 *Arbetsfysiologiska undersökningar över hemarbetet,* 11. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

92 *Arbetsfysiologiska undersökningar över hemarbetet,* 3. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

93 *Arbetsfysiologiska undersökningar över hemarbetet,* 2. Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

changes in domestic work practices and its material manifestations included collaborating with Swedish manufacturing industries to assess, and to develop, the available tools for conducting domestic work as well as with architects and planners to redesign dwellings. Thus, already from the beginning of its research practices, the HFI cultivated collaborations with both individual companies and with different organisations, in a similar manner to how the networks of women's organisations had been established.

In order to be able to have a direct dialogue with different companies manufacturing household goods, the HFI set up a collaboration committee with representatives for many different types of Swedish manufacturing industries.[94] The chair was the CEO of Kockums ironworks, and represented were Husqvarna weapons, the association of iron manufacturers (SJM), Sweden's mechanical workshop industries' association, the electrical goods companies Elektriska ab Helios and Electrolux, and Separator. Through this collaboration committee, a number of more specialised technical sub-committees with engineers and other staff from companies were organised within different industry areas. After a few years, there were technical committees for cutlery, floors and flooring materials, knives, crockery, stoves and cooking vessels (pots and pans).[95] These would meet with the HFI staff to both discuss adequate testing methods, and to work together in different steps of the testing processes carried out at the research institute.

At the HFI, it was the 'technical section', led by architect Carin Boalt who planned and conducted all tests and studies of housework from technological, economic and practical aspects. The institute planned which studies to perform on an overall level – for example dishwashing, kitchens for disabled housewives, tools for cutting and slicing, vacuum cleaners etc. – and then developed ways to perform the investigations in collaboration with the technical committees where manufacturers were represented.

From 1948, the HFI also set up more permanent "housewives' committees" with the task of testing utensils and tools in their homes on an everyday basis. The HFI would also receive material in the form of objects for testing on the direct initiative from the producers, and in the process have a dialogue and collaboration with the specialised technical committees in regard to changing the design of the items based on the results. In some cases, certain manufacturers could discontinue a product that did not work well. In others, the tests could instead lead to joint initiatives towards agreements on standards for certain types of items – and, eventually, also to national norms and standards for environments and things.

Based on the work at HFI, a Swedish national standard for kitchen layouts was established in 1950, regarding heights, sizes of cupboards, and recommendations for placement of sinks, stoves and other fittings. Standards were also found, for example, for jars and containers for canning and preservation, and children's clothing.

94 *Samarbetskommitté mellan Hemmens forskningsinstitut och industriens branschföreningar 6/7 1944*. Riksarkivet, Hemmens forskningsinstituts arkiv, AIII:1 Fullmäktiges protokoll, Övriga protokoll 1943-1956.

95 Brita Lövgren, *Hemarbete som politik: diskussioner om hemarbete, Sverige 1930-40-talen, och tillkomsten av Hemmens forskningsinstitut* (diss., Stockholm: Almqvist & Wiksell), 148.

Understanding users

Besides the laboratory-based studies into domestic work, the HFI also stressed the importance of understanding what these practices and situations looked like in ordinary Swedish homes. Initially, the methods for this built on traditional questionnaires. In the studies carried out in the first year of HFI's practice, the national organisation Husmödrarnas samarbetskommitté, (the Housewives' Collaboration Committee) distributed a vast number of questionnaires through the local sections of different women's organisations.[96] One targeted floor materials and floor care, another dishwashing and dishwashing utensils.[97]

The questionnaires covered the ages and occupations of the household members, the annual income per year (husband, wife and children), the specifics of the dwelling, and the level of education of the housewife. Details about the kitchen infrastructure (indoor plumbing, hot water, how much water was used per time dishes were done) and the utensils used were asked for, along with assessments of pros and cons with these. The participants were also asked to draw a simple sketch of the kitchen sink and other areas used for dishes, from above, with notations about where clean and dirty dishes were placed. Also noted was the amount of time spent doing dishes, along with details of which member of the household had taken part in the work. Several sections were dedicated to more open questions regarding how doing dishes could be made easier and more efficient in regard to both utensils and other aspects, as well as if there were other measures to be taken to make doing dishes less straining or time-consuming.

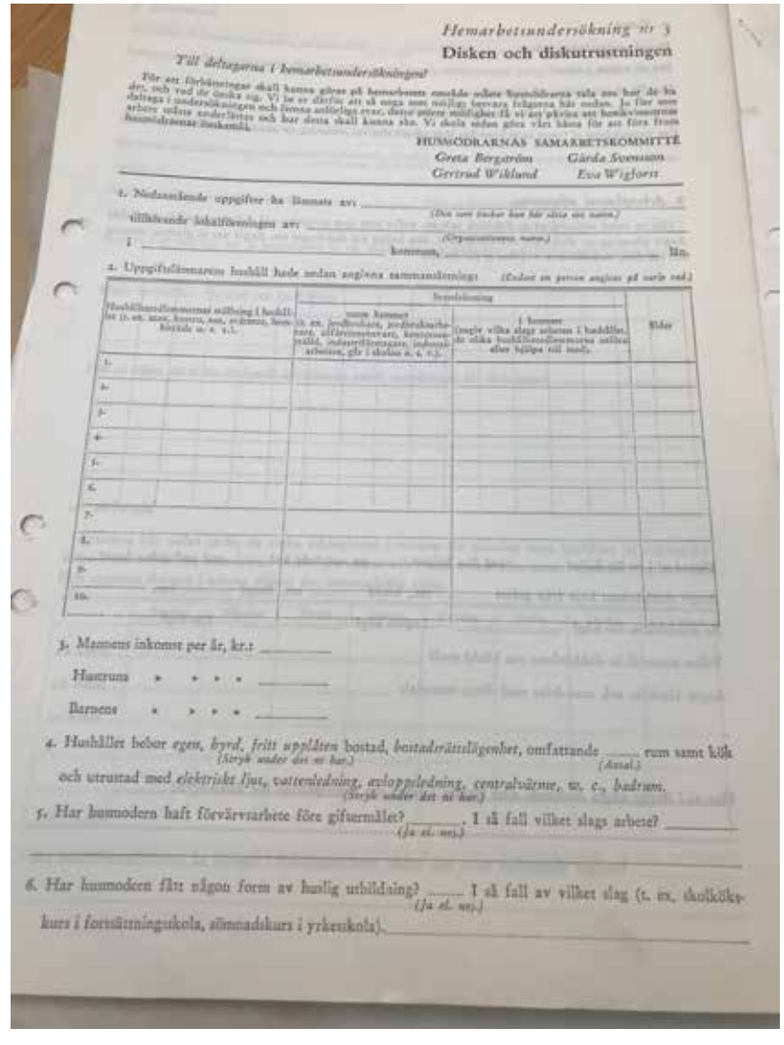
Engaging women to contribute with their experiences was explicitly framed as a means to both highlight the importance of domestic work, and that the aim was to find ways of making change happen based on the input from the participants. In the introduction to the questionnaires, this aim towards making change was stated clearly:

> To the participants in the domestic work survey! In order for improvements to be made in the area of domestic work the housewives must tell what their situation is, and what they wish for. We therefore ask you to answer the below questions as thoroughly as possible. The more who participate in the survey and leave detailed answers, the greater the possibility for us to show that the domestic women's work must be relieved and how this will be able to be done. We shall then do our best to promote the housewives' wishes.[98]

96 Aktiv Hushållning and Husmödrarnas samarbetskommitté coordinated, and the distributing organisations were Kooperativa kvinnogillesförbundet, Svenska Landsbygdens kvinnoförbund, Sveriges Husmödraföreningars Riksförbund and Sveriges Socialdemokratiska Kvinnoförbund. "Hemarbetsundersökning nr 5. Disken och diskutrustningen" (rapport). Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 1: Hemarbetsundersökningen 1944-1945

97 See, for example the questionnaire forms: "Hemarbetsundersökning nr 5. Disken och diskutrustningen," "Hemarbetsundersökning nr 6. Golvmaterial och golvvård". Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

98 "Hemarbetsundersökning nr 5. Disken och diskutrustningen". Riksarkivet. Hemmens forskningsinstitut. FII a



Hemarbetsundersökning nr 5

Disken och diskutrustningen

Till deltagarna i hemarbetsundersökningen!

För att förbättringar skall kunna göras på hemarbetets område måste husmödrarna tala om hur de bäst vill, och vad de önska sig. Vi be er därför att så noga som möjligt besvara frågorna här nedan. Ju fler som deltaga i undersökningen och ju mera utförliga svar, desto större möjlighet få vi att påvisa att hemarbetsområdet behöver ändras och har detta skall kunna ske. Vi skola sedan göra vårt bästa för att föra fram husmödrarnas önskemål.

HUSMÖDRARNAS SAMARBETSKOMMITTÉ
Greta Bergström *Gärda Svensson*
Gertrud Wiklund *Eva Wigforss*

1. Nedanstående uppgifter ha lämnats av: _____
(Det som önskar kan här sätta sitt namn.)
tillhörande lokalföreningen av: _____
(Organisationens namn.)
i _____ kommun, _____ län.

2. Uppgiftslämnarens hushåll både ordinarie sammansättning: (Endast en person angives på varje rad.)

Hushållsmedlemmens ställning i hushållet (t. ex. make, hustru, son, svärdotter, barn, föräldrar, hyresgäst m. fl.)	Sysselsättning: inom hemmet (t. ex. jordbruksarbete, fiskeriarbete, affärsmedhjälpare, kontorsbiträde, industriarbetare, fabriksarbetare, gift i skolan m. fl.)	Sysselsättning: i kommer (ingiv vilka slags sysslor i hushållet de olika hushållsmedlemmarna utföra eller hjälpa till med).	Ålder
1.			
2.			
3.			
4.			
5.			
6.			
7.			
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9.			
10.			

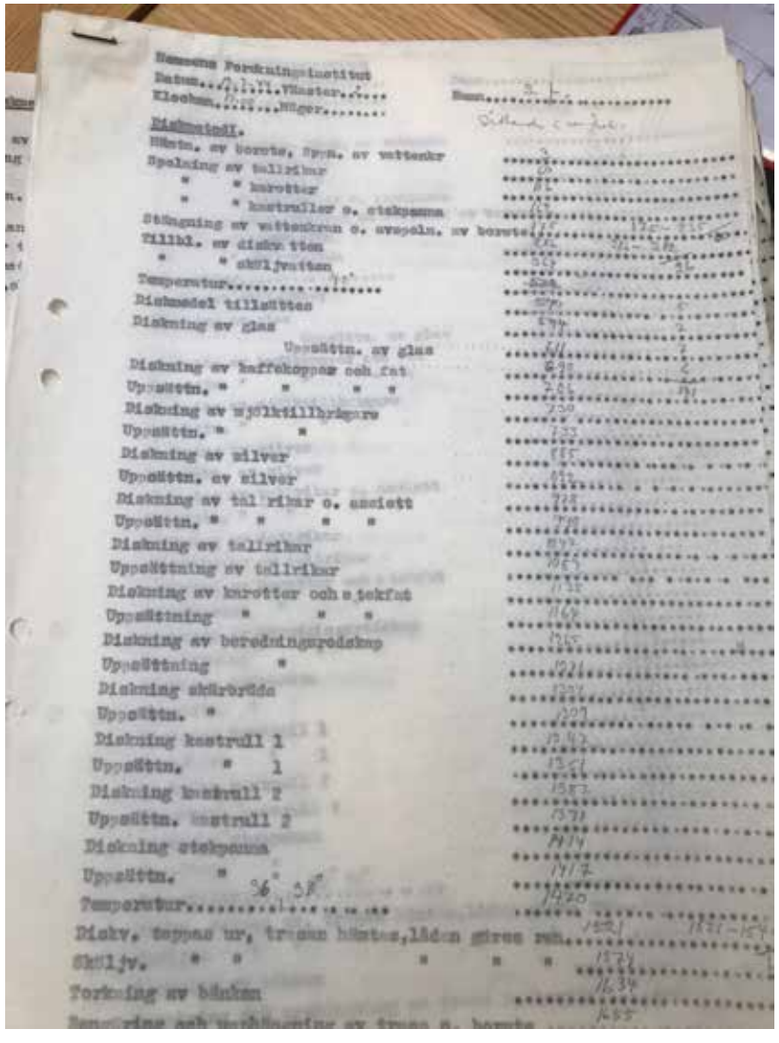
3. Mannens inkomst per år, kr.: _____
Hustrus » » » » _____
Barnens » » » » _____

4. Hushållet bebor *egen, hyrd, fritt upplåten* bostad, *bostadsrättslägenhet*, omfattande _____ rum samt kök (Antal.)
(Stryk under det ni har.)
och utrustad med *elektriskt ljus, vattenledning, avloppsledning, centralvärme, w. c., badrum.*
(Stryk under det ni har.)

5. Har hustrun haft förvärvsarbete före giftermålet? _____. I så fall vilket slags arbete? _____
(Ja el. nej.)

6. Har hustrun fått någon form av huslig utbildning? _____ I så fall av vilket slag (t. ex. skolköks-
(Ja el. nej.)
kurs i fortsättningsskola, sömnadskurs i yrkesskola). _____

Questionnaire about doing dishes, from the HFI housework investigation 1944-45. Riksarkivet, Hemmens forskningsinstituts arkiv.



Time study protocol, doing dishes. Riksarkivet, Hemmens forskningsinstituts arkiv.

The women asked to participate were indeed interested in making their voices heard. In the dishwashing study, a total of 2260 replies were gathered. Along with highlighting of the probable knowledge bias in terms of previous engagement with issues concerning domestic work, given that the women filling in the forms also all were members of different organisations engaged in promoting such matters, it was also considered probable that both equipment and work methods used by the respondents might be of a higher than average quality.[99]

The material gathered for analysis in these early domestic work studies ranged from the chemical study of components in detergents and cleaning products to the controlled conditions of experiments in a laboratory context, to the gathering of information about everyday practices in homes through questionnaires, and later also observations and interviews with people in their home environments.

Laboratory studies formed the basis for the more quantitative analysis of different aspects of housework, applying methods from the natural and technical sciences in both analysis and communication of the results. In diagrams, different relations were made visible, such as how the ratio between the time spent doing dishes and the time it took for dishes to dry changed in relation to how dirty the dishes had been, and what sort of food stuffs they had been used for.[100] In the lab-based time studies of dishwashing the time-study sheets divided the steps of the activity based on what category of objects were washed thus also setting not only a standard procedure for the test but for how dishwashing ideally should be carried out. Different varieties of sinks, and different ways of doing the dishes – different "methods of dishwashing" – were tried out in various combinations, clocking the time needed and monitoring the movements and procedures.

The timing of the total activity started with "[g]etting brush, opening of water tap" and the rinsing of plates, bowls, pots and pans, before the preparation of water for dishing and rinsing. After that followed the following actual washing of the dishes in order from glasses over cutlery and plates to utensils and pans, ending with the tapping out water, wiping the sink, and hanging up the dish brush.[101] Drying the dishes was monitored separately, noting both the variety of dish stand, and the type of towel used.[102]

Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

99 More than half came from women living in the countryside, and out of the 545 replies from women living in cities only a hundred or so were from the three largest Swedish cities of Stockholm, Gothenburg and Malmö. "Hemarbetsundersökning nr 5. Disken och diskutrustningen" (rapport). Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

100 Diagram "Sambandet mellan disktiden och intorkningstidens längd." Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

101 "Diskmetod I 19.9-44" Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

102 "Torkning o. nedsättning av glas 31.10-44"Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

The same equipment examined in the lab kitchen environment was also sent out to several households to be tried and assessed in a home environment throughout one to two months. A full inventory of all available dish brushes currently on the Swedish market had already been made, and the brushes had been analysed in regard to cost, materials and form. Many had been tried and tested in a laboratory setting to assess work efficiency. Now, the question was how the brushes worked in everyday environments over a longer period.

Dish brushes were distributed to women, through home consultants and organisations, along with a questionnaire asking the participants that "the enclosed dish brush be used regularly during a month in the household" and after that returned together with the filled-in form. Besides the number of family members in the household, the information provided by participants covered the number of meals per day and which time of day they took place, together with the approximate number of items washed each time. The brand of detergent was listed, along with details about how the dish brush was stored in between doing dishes, how long the total time spent each day doing dishes was, and if the test brush was satisfactory or not – and the reasons for the opinion.[103] The same was done with dish stands, to be used for two months before returning them and the form, stating their general comments along with opinions if the stand's size was "just right", "unnecessarily large" or "too small".[104]

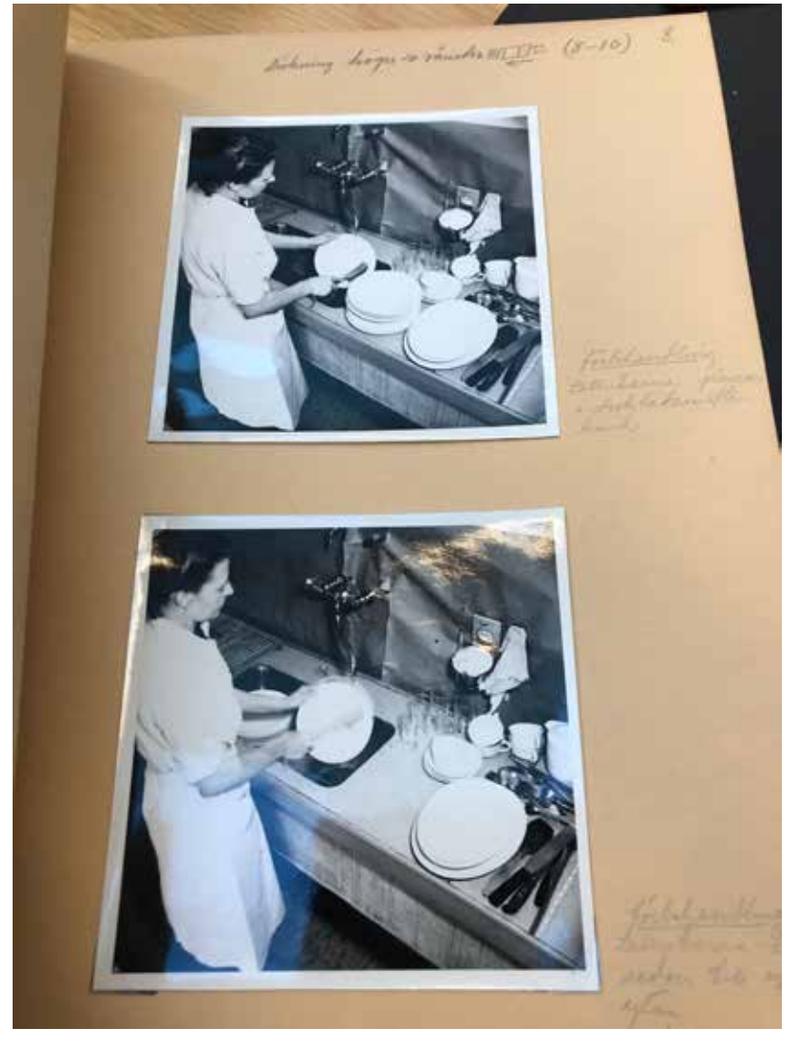
In the questionnaires, the participating women gave their opinions on the utensils used, often noting how materials used in the utensils made them difficult to use – some wooden stands didn't respond well to water, others left rust stains on dishes and sinks – and how the design of different items played out in actual use. Many found the dish stands difficult to use because of them being too large, or too small, or impossible to place certain types of items such as glasses or oven forms in.

The feedback from the women using the dishwashing equipment in their everyday environments was then compiled together with the lab-based studies of different sinks and equipment in relation to time, movements, strain and posture, the chemical analyses of different detergents and water temperatures, etc. and published in the first HFI report, *Diskning i hemmen* (*Dish-washing in the homes*).

Time, it was stated, could definitely be saved if adequate tools and properly planned sinks of a reasonable height were used. "Standard dishes" would be ready 8 minutes earlier in such a setting than if a too small and poorly planned workbench was used. Actions were also suggested in order to improve the "physical and mental well-being" as well as the efficiency of the person doing the dishes. In the conclusions, and the overview of tools and equipment included in the publication, the suggestions made by the HFI spanned over recommendations for the most efficient methods for doing dishes, over recommendations for producers of equipment for improvements, to suggestions for standards in regard to heights of workbenches and sinks. The work initiated in these studies eventually led to the

103 "Hemarbetsundersökningen. Utprovning av diskborstar i hushåll. 2.4504" Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945

104 "Hemarbetsundersökningen. Utprovning av disktorkställ i hushåll" Riksarkivet. Hemmens forskningsinstitut. FII a Handlingar rörande hemarbetsundersökningar. Volym 2: Hemarbetsundersökningen 1944-1945



Dishwashing study in the HFI test kitchen. Riksarkivet, Hemmens forskningsinstituts arkiv.



Dish brushes from the dishwashing study published in 1946. Riksarkivet, Hemmens forskningsinstituts arkiv.

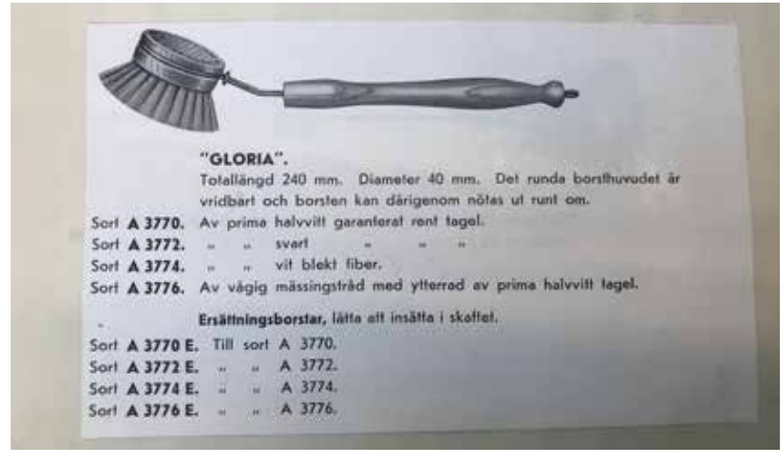
establishment in 1950 of a national standard for kitchen layout and fittings.

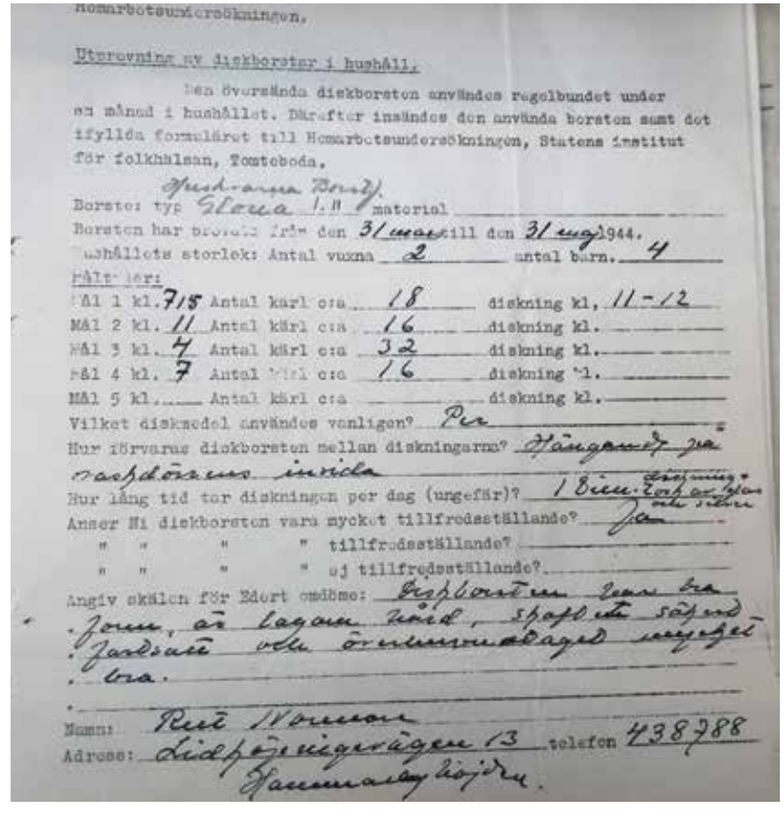
Taking the washing of dishes, an everyday activity that played out in all Swedish households multiple times on a daily basis, as the first study to conduct and publish was significant for how the HFI continued its work. As ambition was stated initially, the research at the HFI should focus on the home, but not take any of the current situations or ways of doing things for granted. The aim was to develop methods for investigating everyday practices in the home, through assessing and redesigning the tools used as well as investigating and developing the methods employed in housework. Ultimately, the scope was to provide a change in material as well as immaterial conditions for women's position and power in society.

The studies conducted during the first years of the institute's existence were set up in light of what could strategically be expected to be managed in terms of available people, their expert areas of knowledge and the funding and time frames at disposal. Even if time measurement was a central part of the assessment of how strenuous or time consuming a work task was, time studies alone would not provide a basis for creating possibilities for change.

Already from the beginning, the explicit ambition at HFI was to push methods of investigation and analysis beyond what was measurable – such as time studies, chemistry and statistics – to more qualitative methods that would support studying what was considered the crucial and complex issues: the individual and collective experiences in the home, from psychological as well as sociological perspectives. However, a fair amount of groundwork in regard to the more material and measurable aspects of housework had to be covered first, to lay a foundation for the more socio-psychological studies planned.

In descriptions of the methods used for evaluating different utensils for the home, the quantifiable and measurable properties of an object were reasonably easy to assess. However, equal weight and importance must be given to other factors determined through practical testing of these by actual users, "…such as shape of





Questionnaire evaluating a month of doing dishes with the dishbrush "Gloria". Riksarkivet, Hemmens forskningsinstituts arkiv.

handles, risk of causing damage during use, etc., [which] must of necessity be purely subjective evaluations".[105] As one of the founders of HFI stated years later:

> One wanted to see the different work processes in the home as a whole, where there are a variety of different factors to take into account. It is for example not enough to state that work is done in such or such a long time, or with a tool of this or that making. One must try to see to the whole at the same time as one studies the particularities of the work processes.[106]

In histories that touch upon the HFI, the main focus has been placed on the institute as a promoter of the rationalisation of housework, and its role in establishing of both consumer advice and women as expert consumers. Through its application of scientific methods aiming to change the ways women worked in the kitchen and its role in establishing consumer advice based on systematic comparison and studies, the HFI has figured as an important organisation in the history of consumer activism and empowerment.[107] However, very little attention has been turned to the practices of designing that were developed at the HFI. Many of the methods developed at the HFI for investigating and assessing the tools and equipment already for sale on the Swedish market, also became applied in processes aiming to determine how new things could be designed based on the insights given through the new methods.

Rational home management

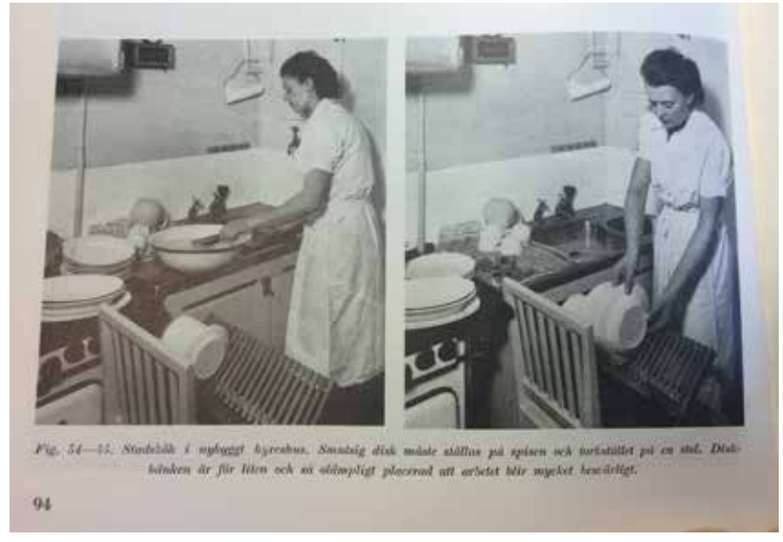
As such, some of the methods developed at the HFI fell back on a previously existing discourse of rational home reform that had been advocated in relation to the increasing industrialisation in the West since the mid-19th century. Household work should be considered a profession equal to salaried work outside the home, and improvements in terms of the work environment, tools and methods were needed in order for housewives to become established as true professionals.

Through a systematic study of the ways cooking and related work was done, the ambition was to create a labour-saving kitchen that would ease the burden of women's work as well as establish a professional role for women as experts on domestic work. The emphasis on rational and scientific methods for assessing best practices and the most efficient ways to carry out work in the kitchen placed the common everyday tasks of women in a context of potential societal change.

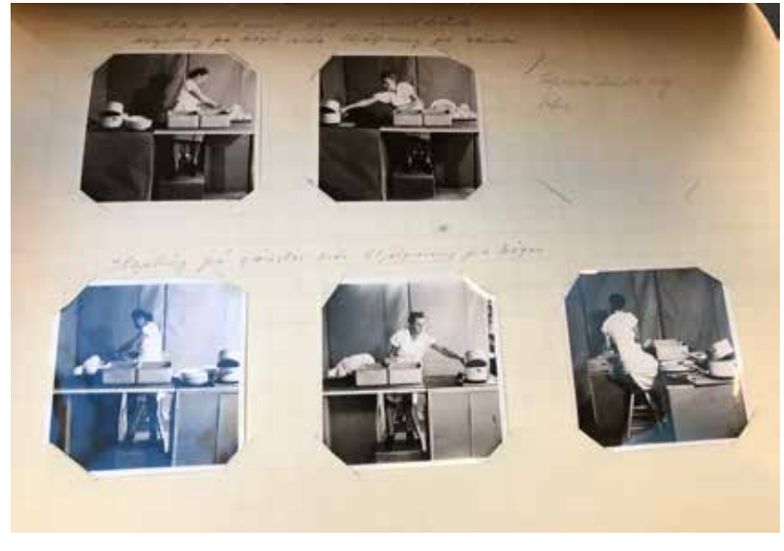
If work were to be carried out differently, and if kitchens were considered to be workplaces of equal importance to the workplaces men held outside the home, women's roles and influences in society would be strengthened. Using a rational and industrial logic to study and analyse domestic work became a legitimising factor used to argue for the value of housework.

These initiatives were connected both to the increasing lack of domestic

105 *Undersökningar av småredskap: Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden* 1946:4, 9.
106 Carin Boalt, "Hemmens forskningsinstitut", in Åkerman, *Kunskap för vår vardag,* 152.
107 Ulrika Torell, Jenny Lee, & Roger Qvarsell (eds.) *Köket: Rum för drömmar, ideal och vardagsliv under det långa 1900-talet* (Stockholm: Nordiska museets förlag, 2018).; Britta Lövgren *Hemarbete som politik: Diskussioner om hemarbete, Sverige 1930-40-talen, och tillkomsten av Hemmens forskningsinstitut,* diss. (Stockholm, 1993).



Dishwashing in a new apartment kitchen, with a sink so small the dish stand needs to be placed on a chair. *HFI-meddelanden 1944:1, Diskning i hemmen,* 94.



Studies of stacking dishes while sitting. Riksarkivet, Hemmens forskningsinstituts arkiv.

servants, as young women would rather work in industries and offices than in the homes of others. In the US, the advocacy of rational home management was also, for example in the case of Catharine Beecher, argued rationalisation of housework both as a method for professionalising the housewife's work and as a strategy towards the abolishment of slavery. With efficient methods for home keeping, a woman could run her own household in a modern way, also without enslaving other persons.[108] For early home economics and domestic work rationalisation advocates such as Catharine Beecher around the 1850s, the aim was thus to both change the ways work in the home was conducted, and how housework was valued, and how social structures and societal systems functioned.[109] Rationalising domestic work was one means towards the liberation of women, and enslaved persons, through the application of scientific methods and arguments.

Around the turn of the century 1900, Christine Fredericks and others argued for the introduction of scientific management and industrial engineering as means to increase efficiency and financial gain in the industry was quickly also advocated as methods suitable for adoption in the home.[110] The methods for motion studies conducted of bricklayers and other workers by Frank and Lilian Gilbreth were later introduced to study women working in kitchens. Emphasising efficiency in movement, the studies aimed to simplify housework and make it more rational. The movements, postures and procedures involved in everyday work were as thoroughly analysed as that of different types of professional work, using time-taking and motion studies, in combination with photography.

The Gilbreths worked with analysing motion studies, as well as fatigue, of the assembly and disassembly of small arms during world war one, as well as studies of cooking and other kitchen work. The way that these studies were made bear strong resemblance to the posture studies also carried out in what later became known as ergonomics. In a context of tracing the roots of where design methods, such as ergonomic studies, once emerged, it seems just as plausible to draw a connection to these motion and posture studies in everyday kitchen settings, as it is to draw them to military technological development situations.

That the kitchen became a location for modernisation, in terms of technological innovation as well as architectural layout and the rationalisation of housework in the early decades of the 20th century has been emphasised in several historical studies.[111] In a Western and global North context, the attention paid to re-thinking and

108 Catherine E. Beecher, *A treatise on domestic economy, for the use of young ladies at home, and at school* (New York, 1848).

109 Janice Williams Rutherford, *Selling Mrs. Consumer: Christine Frederick and the rise of household efficiency* (Athens & London: University of Georgia Press, 2003).

110 Christine Fredericks, *The New Housekeeping: Efficiency Studies in Home Management* (Garden City, N.Y., 1918); Mrs. Frank A. Pattison, "Scientific management in home-making", *The Annals of the American Academy of Political and Social Science* 1913:1 (vol. 48), 96-103.

111 Juliet Kinchin & Aidan O'Connor, *Counter Space: Design and the Modern Kitchen* (New York: Museum of Modern Art, 2011); Ulrika Torell, Jenny Lee & Roger Qvarsell (eds.) *Köket: Rum för drömmar, ideal och vardagsliv under det långa 1900-talet* (Stockholm: Nordiska museets förlag, 2018); Barbara Penner, "The Cornell Kitchen: Housing and Design Research in Postwar America", *Technology and Culture* 2018:1(vol.59), 48–94; Ruth Oldenziel & Karin Zachmann, *Cold War kitchen: Americanization, Technology, and European Users* (Cambridge, MA: MIT Press, 2009).

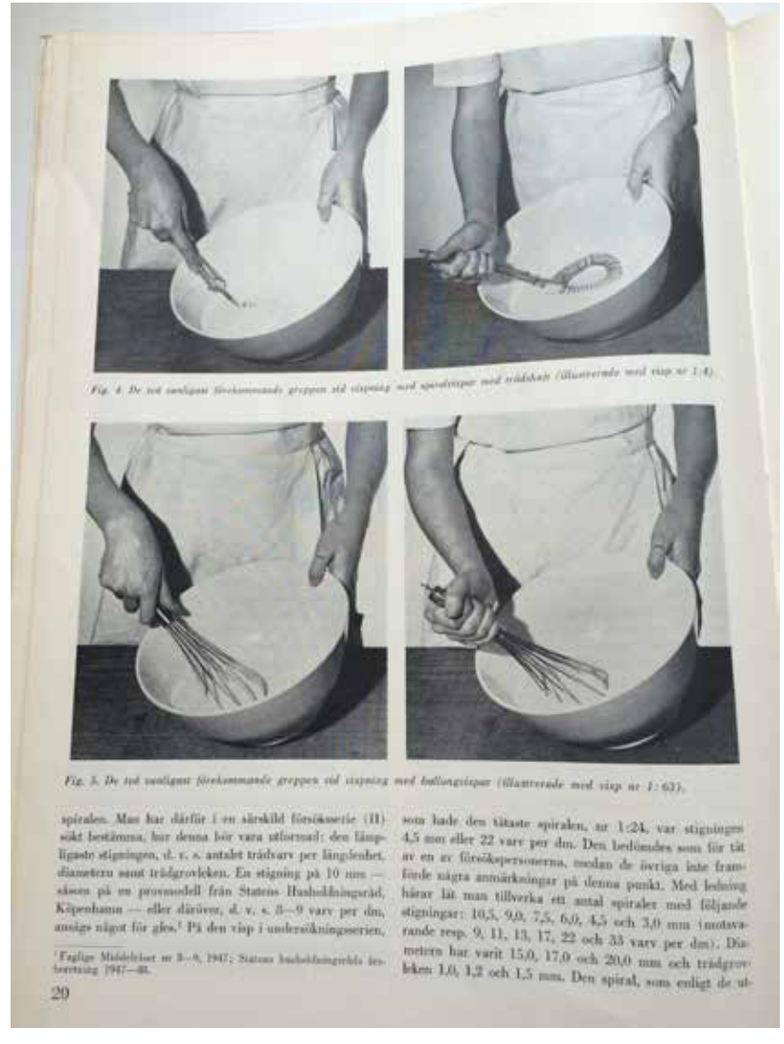


Fig. 4. De två enligt förekommande greppen vid vispning med spiralvisp med trådskaft (illustrerade med visp nr 1:4).

Fig. 5. De två enligt förekommande greppen vid vispning med ballongvisp (illustrerade med visp nr 1:63).

spiralen. Man har därför i en särskild försöksserie (II) sökt bestämma, hur denna bör vara utformad: den lämpligaste stigningen, d. v. s. antalet trådvarv per längdenhet, diametern samt trådgrovleken. En stigning på 10 mm — såsom på en provmodell från Stateins Hushållningsråd, Köpenhamn — eller däröver, d. v. s. 8—9 varv per dm, ansågs något för glest.[1] På den visp i undersökningsserien, som hade den tätaste spiralen, nr 1:24, var stigningen 4,5 mm eller 22 varv per dm. Den bedömdes som för tät av en av försökspersonerna, medan de övriga inte framförde några anmärkningar på denna punkt. Med ledning härur lät man tillverka ett antal spiraler med följande stigningar: 10,5, 9,0, 7,5, 6,0, 4,5 och 3,0 mm (motsvarande resp. 9, 11, 13, 17, 22 och 33 varv per dm). Diametern har varit 15,0, 17,0 och 20,0 mm och trådgrovleken 1,0, 1,2 och 1,5 mm. Den spiral, som enligt de ut-

[1] Tagliga Middelsbiet nr 3—9, 1947; Statens hushållningsrådsförhandlingar 1947—48.

20

The most common ways to hold a whisk. *HFI-meddelanden* 1948-49:1, 20.

redesigning the kitchen has played a quite significant role in attempting to reform everyday life. In Germany, as part of the Bauhaus-inspired modernist endeavours to produce rational and modern dwellings adapted to a new and democratic way of life – as well as rational and low-cost in production – included making the kitchen smaller and more laboratory-like. The Frankfurt kitchen, designed by Margarete Schütte-Lihotzky for the social housing project in Frankfurt in 1926 built on the ideas of labour-saving layout and efficiency in work as defined in works of scientific home management, and incorporated modernist ideals of the definition and separation of different functions in the floor plan of the home.[112]

The ideas about the rationalisation and scientific management of the home and housework came to be adopted in Sweden as well, already from the early years of the 20th century, by women's organisations as well as by the consumer cooperation.[113] Modernising women's work through adopting new technologies and through the reformation of ways of conducting everyday domestic work came to unite women with quite diverse social and political agendas.

The rationalisation of housework was seen as a means for giving women increased status as expert professionals, and thus a position from which to argue for the rights to full citizenship in a new type of society – with rights to vote, and new relations between men and women. For others, the focus on female caring and domestic abilities in relation to homemaking instead served as a way to strengthen and maintain existing societal structures based on ideas of women's specific abilities as wives and mothers.

Whatever the aim of the initiatives, the result did become that women actually gained influence in the public and political realm through establishing expertise in the domestic sphere. As experienced users, and professionals in the areas of cooking, cleaning and caring for children and family, the 'housewife' was established as an expert on equal terms with expert professionals from other areas. [114]

Motion studies and ergonomics

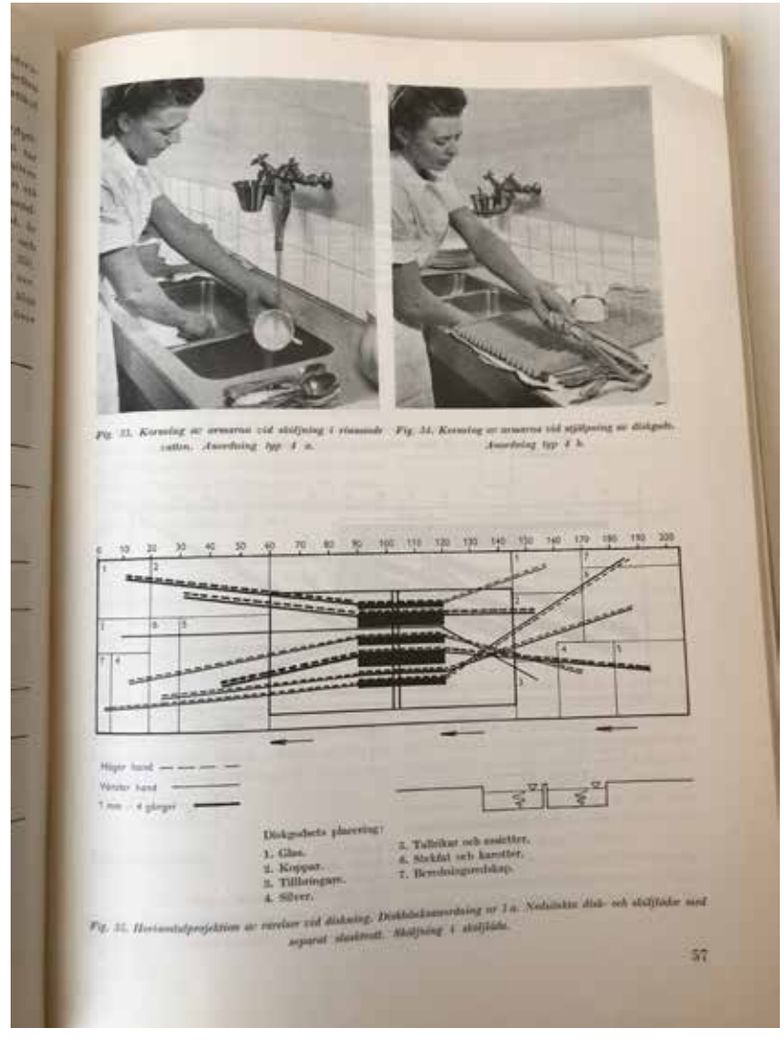
The research at HFI was conducted in multidisciplinary teams, building on bringing together different sorts of expertise. Teachers in home economics and cooking would work together with physiotherapists, chemists and engineers both in setting up and in evaluating various studies, and bring in ordinary housewives' views as well, which was a practice adopted from the very beginning of the HFI. And quite quickly, some methods of study became core to almost all the different domestic work areas and utensils investigated.

One area that became prominent, already from the first work studies conducted at the HFI, was that of posture, motion studies and ergonomics. Testing different grips and documenting movements established attention to the ways that women interacted with specific tools as well as with kitchen – or apartment – environments as

112 Finn Arne Jørgensen, "Frankfurt Kitchen, Germany (Grete Lihotzky, 1926)", in Grace Lees-Maffei (ed.) *Iconic designs: 50 stories about 50 things* (London: Bloomsbury, 2014), 167–164.

113 Peder Aléx, *Konsumera rätt: Ett svenskt ideal: behov, hushållning och konsumtion* (Lund: Studentlitteratur, 2003).

114 Boel Berner, "Housewives' Films and the Modern Housewife. Experts, Users and Household Modernization. Sweden in the 1950s and 60s" *History and Technology,* vol 18 (3), 2002, 155-179.



Horizontal projection and motion study of doing dishes using different sinks. *HFI-meddelanden 1944: 1, Diskning i hemmen,* 57.

a whole, which became an important factor for assessing and developing work tools as well as workflows.

In many of the tests conducted, studies of movement and grips stand out in photographs as well as in diagrammatic representations. In the dishwashing study, the arm movements were documented in relation to the placement of the sinks and drying stands. Pots and pans were assessed in relation to how the handles worked for different size hands, as well as how the grips worked when lifting or pouring from, them. Images documented and explained the differences materials and designs made for ease of handling objects from an ergonomic perspective.

As an example, we can take a closer look at the 1946 study of household equipment. Around 100 different utensils categorised as "small tools"— spatulas, potato peelers and can openers – were inventoried and investigated.[115] The survey studied the "main types" of each group of utensils, based on a systematic analysis of what was available on the market. The different tests and comparisons were structured so that the functions of each category were determined, and the utensils were then tested in regard to their usability:

> The material and construction has been determined and for each group a series of practical functional tests have been made, all with the purpose to shed light on the qualities that are essential to an assessment of the usability of each tool.[116]

The tests followed the same basic method, with slight variations where "one has wanted to try different ways". Durability and flexibility of the material in, for example, spatulas were tested using established quantitative methods. While it is relatively easy, the report stated, to make quantitative assessments of time used, materials and construction, it is more difficult to assess more subjective aspects of use, such as the form and suitability of grips. In order to address this challenge, groups of people were engaged in tests where the ways of working with a utensil in context could give insights into form and usability.[117]

Aspects of form and ergonomics when it came to grips and handles became a central aspect of consideration in the evaluation of tools. Function requirements were set up, sometimes in the form of flow charts, that then guided the evaluation in the practical user testing. In a description of the testing of potato-peelers, the function tests were accounted for in detail:

> The method for the *practical function tests* has been the following. Batches of 0,5 kg (6 pcs.) mid-sized, well-washed potatoes of homogenous quality were peeled and cleaned. The time consumed was measured, by which part-times were noted for peeling and cleaning, motivated ty that these operations were carried out with different parts of the utensil. The test batches were first peeled and then cleaned in context. The scale is in one hundreds of a minute. The peeled and polished potatoes were weighed with an accuracy of 5 g and the peeling loss has been calculated in percent of the weight of the starting material. Four test batches were peeled with each utensil and medians of time consumed and

115 *Undersökning av småredskap. Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden 1946:4.*
116 *Undersökning av småredskap. Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden 1946:4,* 3.
117 *Undersökning av småredskap. Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden 1946:4,* 3-4.



sig, finns för närvarande inte i marknaden. De typer, avsedda för hembruk, som finns, är för låga, har för små hjul och är för klent byggda för att kunna tjäna både som transportmedel för servisen och samtidigt tåla påfrestningen när man stöder sig på dem. De vagnar, som användes på sjukhus och restauranger, är för tunga och klumpiga för att vara riktigt lämpliga i föreliggande fall. Man efterlyser en stabilt byggd, men ändå inte för tung rullvagn med kraftiga och stadigt fastsatta hjul och i ett material, som inte fördyrar vagnen alltför mycket. En rullvagn som man kan förflytta utan att behöva gå alltför krokig bör vara försedd med ett stadigt handtag ca 90 cm över golv.

Förvaringsutrymmen.

Då hushållsutrustningens och livsmedelsförrådets storlek och förvaringsbetingelser är densamma i dessa hushåll som i andra finns det ingen anledning att frångå de krav på förvaringsutrymmenas utformning och antal som HFI kommit fram till och sedan utformats av Byggstandardiseringen.

Väggskåp.

Väggskåpen bör i båda typerna av kök sitta lägre än vad som är brukligt. I kök för rullstolsinvalider bör skåpen sitta 112 cm och i kök för käppinvalider 130 cm över golvet. Då man i det förra fallet inte från rullstolen kan nå mer än två hyllor i väggskåpen måste de två överskåpen över diskbänken utökas med ett s.k. skänkskåp vid sidan av diskbänken (bild 2). Detta skåp bör lämpligen utformas så att skåpets nedre del har samma höjd och djup som diskbänksskåpet och den övre delen är 20 cm grundare så att man får en avställningsyta framför överskåpet. Bredden bör vara ca 30 cm. Ett dylikt skåp fyller ingen funktion i ett kök för käppinvalider, där däremot ett högt skåp med svängbara hyllor är mycket lämpligt.

Vanliga slagluckor fungerar dåligt i båda köken. Vid skåpen, som sitter över diskbänken i köket för rullstolsinvalider, bör därför ha skjutluckor i stället, skänkskåpets nederdel likaså. Den övre delen av skänkskåpet kan emot ha uppåtgående s.k. motviktsbalanserad skjutlu Den lucktypen fungerade bättre än skjutluckor i s men även dylika kan användas. I köket för käppinv föredrog försökspersonerna uppåtgående skjutluckor även här kan skjutluckor i stället användas. Samtliga skåp bör förses med flyttbara hyllor och skänks underdel inredas med utdragbara backar i stället för då den som sitter i rullstol många gånger har sv höja sig djupt framåt.

Diskbänksskåp.

I ett kök för rullstolsinvalider bör utrymmet un bänken inredas med ett skåp på vardera sidan ningen för benen. I skåpet närmast spisen skall

Bild 3. Hörnskåpet med roterande hyllor är ett bra utnyttjande av ett utrymme i köket, som vanligen brukar byggas bort. Skåpet är i köken för rullstols- och käppinvalider avsett till förvaring av kokkärl och en del tyngre beredningskärl. I det slutliga utförandet kommer det att förses med tre hyllor.

Bild 2. Ett skänkskåp med hyllor placerade i god räckhöjd för den som sitter i en rullstol kompletterar väggskåpen över diskbänken. Det avbildade skåpet är en modell, vars slutliga utformning studerades under prövn.

Kitchens for 'disabled housewives'. *HFI-meddelanden* 1951:5.

> peeling loss have been calculated. These tests were carried out by a test person accustomed to household work, that during the test gave their assessment of the form of blade and handle from a perspective of work and of ease of cleaning. These qualities were tested by another two test persons, that each peeled and cleaned four batches with each utensil.[118]

Bringing in the voices and experiences of users into the work with assessing things and environments was something that happened from the early years. These users were brought in not only as subjects of study in time or motion studies but in their roles of being experts of their experiences, whether interviewed in a home or a lab environment. Working together with physiotherapists and with other HFI experts, the users' comments and suggestions on how equipment could be improved were taken into account and reported in the HFI publications.

The conclusions in regard to the design and production of household equipment thus took into consideration not only materials and construction but also form and use as factors determining what a usable and appropriate design would be. In the 1946 report, for example, it was stated that for spatulas "[t]he handle should be formed so that it lies well in the hand and have an adequate length to that the fingers do not come into contact with the, under use, hot blade."[119]

This attention to ergonomics, to grips, postures and movement in relation to work in the home contributed to bringing these considerations into the designing of household equipment. It could be possible that these studies also were foundational for the later practices of similar design methods in the 1960s and 1970s Sweden. By the end of 1947, the HFI annual report stated that the methods developed at the institute had now reached a level of maturity that would allow for a closer collaboration with other stakeholders, nationally as well as internationally:

> The methods for the investigation and the reporting of results have stabilised. The time of preparatory experiments has passed and the institute can now f.ex. when it comes to taking up a new area fairly well overview the test that need to be made and the different methods that should come into use. Simplifications of the methods have gradually been able to be bade without diminishing the demands on versatility and exactitude. The institute believes that it is desirable to bring about methods discussions with researchers from other areas and with those who in other countries earlier than us have picked up on home research according to the same guidelines. One area where one above all is interested in increased methodical efficiency is in regard to the collaboration between the tests made at the research institute and the tests and trials that housewives and teachers make in their respective workplaces.[120]

118 *Undersökning av småredskap. Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden 1946:4,* 16.
119 *Undersökning av småredskap. Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden 1946:*4, 15.
120 "Styrelsens protokoll nr 19 fört vid sammanträde den 26 maj kl 10 å HFI. Årsberättelse 1947 bilaga 1". Riksarkivet, Hemmens forskningsinstituts arkiv, AI:1 Styrelsens protokoll 1944-1957



Study of small tools: can openers, spatulas, and potato peelers. *HFI-meddelanden* 1946:4.

Designing at the HFI

The HFI had, in its first three years set up an extensive research practice, and developed new methods for investigating work and life in the home. The initial focus on quantitative methods was quite strong but always set in relation to both considerations and attempts to find methods to probe the less quantitively measurable aspects of home life and domestic work. The explicit aims of the HFI were to influence the production of household goods and have an impact on the things and tools used in the households, and on developing ways of understanding and investigating the use and the users of these.

As seen in the methods development activities sketched above, this is where we possibly can discern the early roots of a user-centered design process that incorporated ergonomics, ethnography, iterative prototyping, user testing and user experience considerations. Already in 1950, a study focusing only on work posture and ergonomics in different types of housework was inspired by the latest research and developments in industry-focused occupational science.[121] And a year later, a similar study was made of how homes and equipment would need to be designed to accommodate the needs of disabled housewives' in carrying out different types of work.[122]

In the HFI studies of objects, the materials, production, use and function were all investigated. HFI mapped work methods and equipment relating to the processes (conservation, food preparation) and tools (knives, whisks, spatulas, tin openers etc.) relating to cooking, laundry, cleaning and so on. In all cases, there were precise advice and guidelines given both about which tools were preferable, and also which improvements should be made to existing objects. Testing included user studies, ergonomic evaluations and often also prototyping other solutions for design and production. Suggestions for improvements were

121 *HFI-meddelanden* 1950:4, 52-58.
122 "Kök för invalidiserade husmödrar", *HFI-meddelanden* 1951:5, 13.

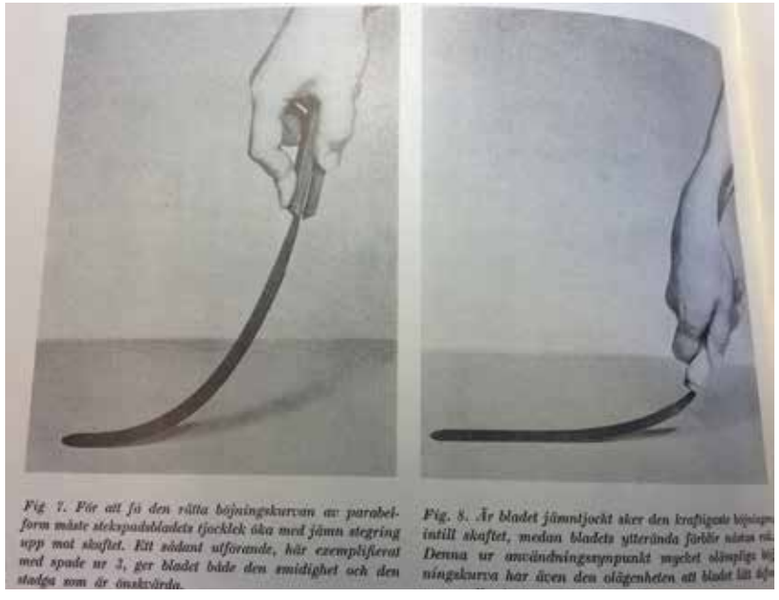


Fig. 7. För att få den rätta böjningskurvan av parabelform måste stekspadbladets tjocklek öka med jämn stegring upp mot skaftet. Ett sådant utförande, här exemplifierat med spade nr 3, ger bladet både den smidighet och den stadga som är önskvärda.

Fig. 8. Är bladet jämntjockt sker den kraftigaste böjningen intill skaftet, medan bladets ytterända förblir nästan rät. Denna ur användningssynpunkt mycket olämplige böjningskurva har även den olägenheten att bladet lätt böjer

Study of a variety of spatulas. *HFI-meddelanden* 1946:4.

mainly communicated in the technical committees, directly to the manufacturers, but sometimes also more publicly made evident in the HFI publications. At times, the HFI also prototyped and proposed designs of their own of different tools and equipment. The design could be either openly published in their reports, or proposed to all producers engaged in a technical committee.

In a study initiated in 1944, kitchen knives were evaluated in terms of construction and materials, as well as in use. Around 100 categories of knives were available on the market, and the 300 different designs of knives from various categories assessed in the study were of very varied quality. The user tests showed that none of the knives was satisfactory from an ergonomic perspective of use. In a later article describing the study, the authors note that "there was at that time very little literature that treated grip functions, but a few German work science studies that gave impulses".[123]

Methods for studying the ergonomics of knife handles in use situations, therefore, had to be developed at the HFI. Women with varying degrees of experience in domestic work were enrolled to take part in user testing of different knife grips in order to

> define the traits of the hand amongst others by taking imprints of the hands to determine the hand size. [---] In order to study the grips during work with cutting knives the test persons were asked to perform a number of ordinary kitchen tasks, like cutting meat in slices, fileting fish, cutting root vegetables. One could discern four main types of operations: 1) cutting straight slices vertically 2) cutting slanted slices vertically, 3) cutting horizontally – fileting 4) dicing or slivering.[124]

123 Åkerman, *Kunskap för vår vardag*, 173.
124 Åkerman, *Kunskap för vår vardag*, 173.

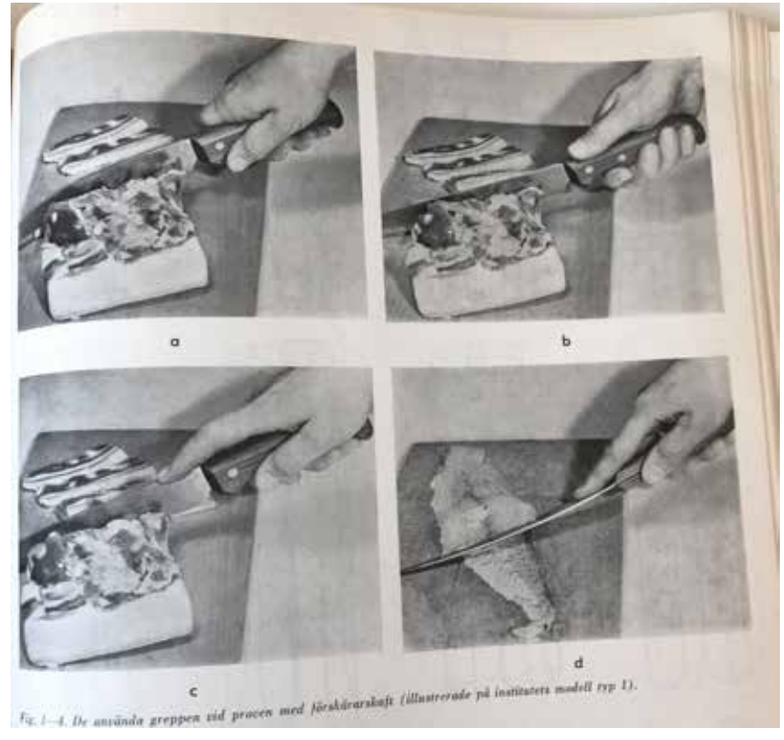


Fig. 1—4. De använda greppen vid proven med förskärarskaft (illustrerade på institutets modell typ 1).

The four different grips tested when prototyping different knife handles. *HFI-meddelanden* 1947:2, 119.

The 24 users were asked to give their comments on the knife grips, assigning scores from 1 to 5 and to also during the tests give "specified assessments about the form and dimensions of the knife shafts, which have been noted in the protocols and after that summarised."[125] At the same time, the grips that the users applied were studied by the HFI experts, who noted similarities and differences between test persons and types of cutting, in order to try to define the design space for a new type of handle.

Based on these studies, seven new forms for handles were prototyped in plastic clay and tested together with the users. These clay prototypes were adjusted so that the dimensions were increased, the edges more rounded, and the back of the shaft dented to make a support for the fingers. The manufacturers represented in the technical committee were asked to create a set of wood prototypes and send back to HFI for a new round of user testing in the HFI test kitchen.[126]

After a new round of testing, three refined models of kitchen knife handles were decided upon. A new round of testing took place to define the materials most suited for the handles, as well as the methods for fastening the blade. After different tests, including 300 rounds of dishwashing, the recommendation was to use the densest wood possible for the handles – ebony, bubinga, rosewood – and possibly, in the future, birch or beech if the development of synthetic dyes would eventually allow colouring processes that dyed the wood entirely through.

This iterative prototyping, with alternating qualitative user studies and more quantitative tests in which prototypes were increasingly refined in terms of both form and materials, resulted in the design of new kitchen knives that came into production the year after the HFI report was published. But besides the design of the physical tools, what was also designed at the institute was a new type of process for designing based on evaluating and developing form and usability in an iterative process.

The process of designing was itself also iterated immediately after defining the types of kitchen knives to take to production, this time prototyping smaller types of kitchen knives for peeling and trimming. The process with testing grips in relation to cutting techniques, blade length, materials, and prototyping different handles, led to that also two new models of smaller knives came into production.[127]

Another early HFI product design that built on extensive testing, both of materials, production methods and prototyping with users, was the launch of new types of pots and pans for cooking. That study was also initiated in 1944, based on the centrality and frequency of use in the kitchens of these products. The functional demands set as criteria for the study were formulated in terms resembling those of a function analysis set-up:

125 *HFI-meddelanden 1947,* 118.

126 "Protokoll fört vid tekniska sektionens konferens den 27 maj 1946". Riksarkivet, Hemmens forskningsinstituts arkiv, AIII:1 Fullmäktiges protokoll, Övriga protokoll 1943-1956.

127 Knives were manufactured by at least two companies; Berg manufaktur, and Karlsson & Nilsson knivfabrik (examples of these are, since 2017, part of the collection of industrial design artefacts at the Swedish national museum, Nationalmuseum: NMK 65/2017). However, in order to describe and analyse the details around the manufacturing of these knives more in detail, more extensive archival studies would be needed than have been made here.

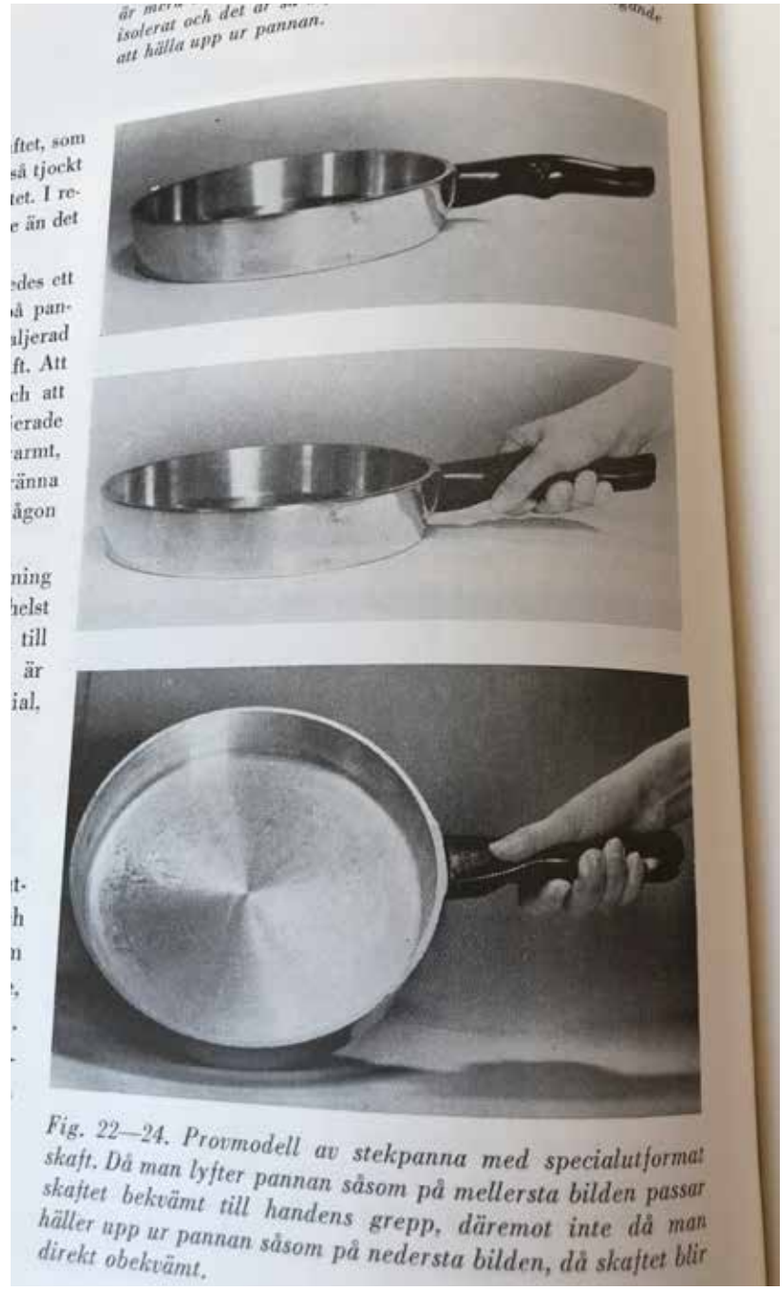


Fig. 22—24. Provmodell av stekpanna med specialutformat skaft. Då man lyfter pannan såsom på mellersta bilden passar skaftet bekvämt till handens grepp, däremot inte då man häller upp ur pannan såsom på nedersta bilden, då skaftet blir direkt obekvämt.

Prototype of frying pan tested for grip and weight. *HFI-meddelanden* 1950:1-2.

> They shall have a mechanical solidity and resist high temperatures and quick temperature changes. They shall further be resistant from a chemical point of view and not be affected by different ingredients used in cooking. They shall not either discolour the food or tinge the taste or contain poisonous substances that can precipitate. It must not be easy to burn the food in them and they shall be effective and economical from a heating perspective. The cooking vessels shall further be easy to lift and carry and easy to pour from. It shall be easy to stir in them and easy to flip when frying and baking. It shall be easy to clean them.[128]

Extensive material testing and inventories of the current production were done together with the HFI technical committee for cooking vessels and a committee for standardisation of cooking vessels appointed by the Swedish workshop industries (SMS), the Swedish electrical companies testing unit (KPA) and the Institute for public health. Together, these all developed methods for how to go about conducting the actual material testing, and made proposals for standards for how to test vessels for resistance to outer force, to chemicals and so on. Methods for how to assess criteria such as how easily food was burnt, led to descriptions of the methods for how to boil milk and flour in such a way that the whole bottom of the pan was burnt, and then methods for analysing the degree of burns on a scale:

> in each study a thorough inspection and assessment of the bottom surface was done after a 7-step scale produced with colour crayons. In each test, the bottom was drawn on white paper, the burnt area measured, drawn up and coloured with the according to the colour scale appropriate colour, after which it was numbered.[129]

From the user testing, it was concluded that the placement of handles on all available pots and pans was generally dissatisfactory from an ergonomic perspective. In the redesign of the cooking vessels, and in the material that also came to be proposed as standards for cooking vessels, the placement of the handles was moved lower on the pans and the sizes and shapes of the handle in relation to the size and function of the vessel specified. Additionally, to facilitate pouring from the pots, a new design for a fold or lining of the opening was designed, to ease the movement as well as to reduce dripping. In 1953, many of these innovations were established as the Swedish national standard (SMS 1536 66), and the studies subsequently also formed the bases for national standards of cooking utensils in both Finland, in 1967, and Denmark, in 1972.[130]

Yet another example of a prototyping-based design process was the redesign of a series of cutlery that was used in the military and in health care. Together with the technical committee on cutlery, a process was initiated in 1947, aiming to further improve the cutlery from both usability and manufacturing points of view. In a series of iterations – where the prototypes were produced by the Gense cutlery manufacturer – grip studies were made, as well as studies of which type of fork teething was most suitable for different sorts of foods, and ease of cleaning.

128 Quoted in Åkerman, *Kunskap för vår vardag,* 170.
129 Åkerman, *Kunskap för vår vardag,* 171.
130 Åkerman, *Kunskap för vår vardag,* 172.



Prototyping different knife handles in clay at HFI, based on user testing. Riksarkivet, Hemmens forskningsinstituts arkiv.

Clay knife handle prototypes. *HFI-meddelanden* 1947:2.

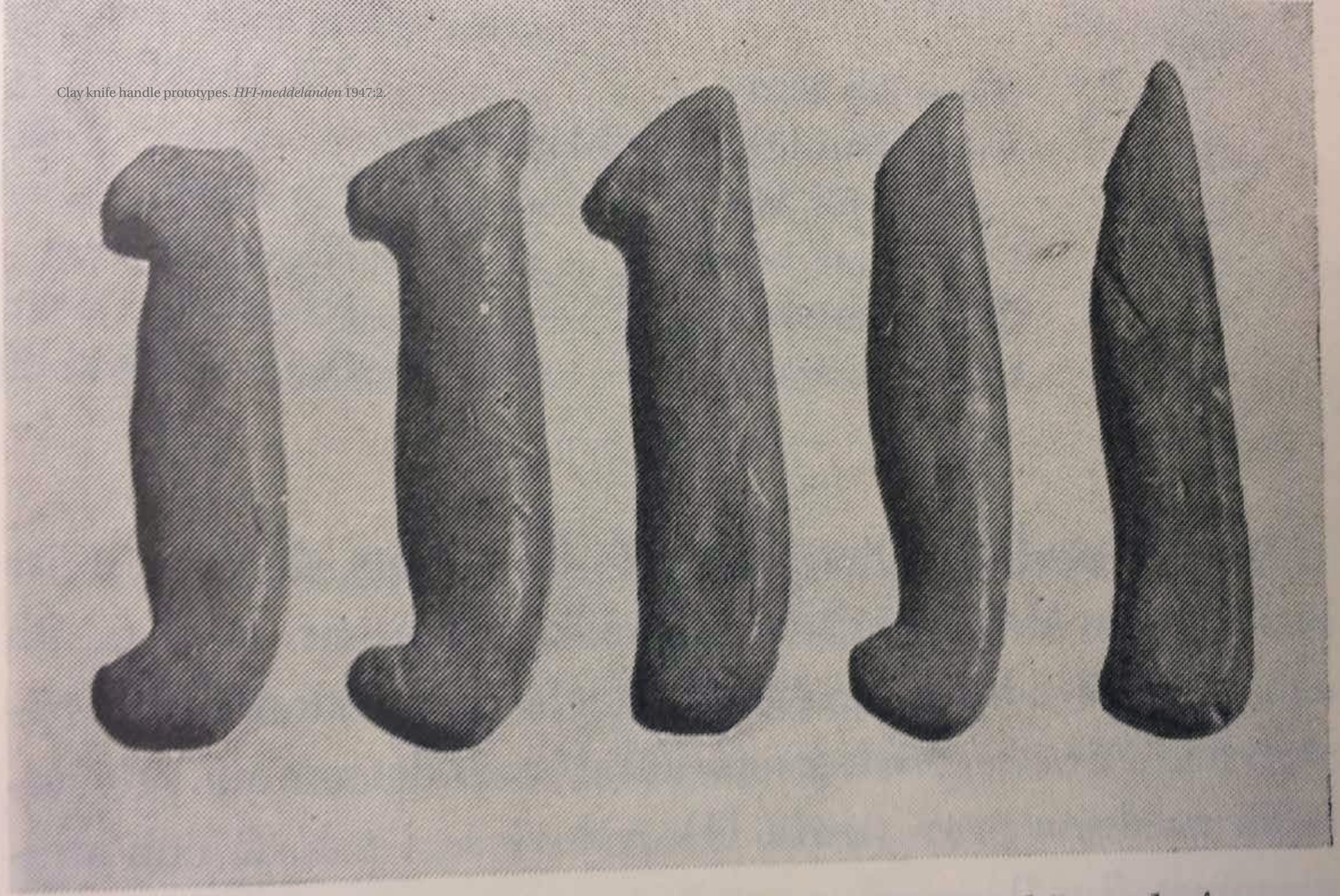


Fig. 7. Modeller, utförda i plastelina, till förskärarskaften nr 25, 27, 28, 30 och 31.



Old knives, HFI-designed knives, and clay handle prototypes. Riksarkivet, Hemmens forskningsinstituts arkiv.

When the results of the redesign were presented in the HFI-meddelanden in 1950, these were compared to the newly launched Gense series "Facette" designed by Folke Ahrström. In the comparison, the article stated that from a usability point of view, the two series were both very well designed. Still, while Ahrström's cutlery probably was "the most aesthetically pleasing cutlery to have been produced in Sweden", it was found lacking in comparison to the standard cutlery design when it came to the ergonomics of the grips.[131]

This comparison of on the one hand design as something "aesthetic", and on the other as something to do with usability and ergonomics, points to the shift in design process and foundations established at the HFI. The methods developed for defining functional requirements, the methods for assessing different aspects from a user perspective, and the collaborative processes in research as well as in developing the final designs were something else than that which was usually considered to be "design" at the time.

131 *HFI-meddelanden* 1950:2, 1p. 03.

Whose design?

Already in 1945, the question if the HFI should take credit for innovations and new designs had surfaced during a board meeting. An engineer had contacted the HFI, proposing that they should hire him to assist with innovation and patent processes, so that the HFI could begin producing designed objects and gain income from these.[132] After a few weeks of discussion, the conclusion from the HFI board came to be that the institute needed to be clear about how to handle any inventions that would be made by staff within the HFI:

> It was emphasised that the HFI should not seek patents, but publish developed proposals. Improvements that one wishes to suggest in tools or machines, which have been handed in by individual firms should be notified to said firms.[133]

The policy for the HFI was thereby set: There should be no individual claims to designs made by HFI staff, and any designs that were not specifically previously linked to a specific company should be openly published according to a sort of open access policy, for anyone to make use of. This reluctancy to credit individuals as designers of products at the HFI is not expanded on further in the archival material. Still, it might possibly have something to do with the strong emphasis on collective processes at the institute.

The work at the HFI was conducted in team constellations covering diverse disciplines and expertise. Methods for testing and developing tools thus came together from different fields and disciplines, and many times also further developed or re-invented in the iterative processes of testing and designing at the HFI. Quantitative methods were brought in from science and engineering, and qualitative methods from sociology and physiotherapy. These investigations, and the development of methods, took time. In the archival material, the board and steering group of HFI quite frequently discuss how to respond to the increasing demands from different stakeholders in the women's organisations to provide clear recommendations and advice on best practices in regard to methods for carrying out housework.

From notations in meeting minutes, it seems that the issue of publications in the HFI report series were up for discussion now and then. It seems these were considered to be much too academic, too scientific, and too extensive, to be accessible for the general public. For the purpose of spreading information, a press function was established that could translate the scientific findings into more accessible formats, providing articles for print and radio. Similarly, producers of different tools and equipment meant that the publications were too extensive, and asked that the HFI would launch a clear and straightforward system of approval, so that producers then could apply a statement of "approved by the HFI" to their products as part of their marketing work.

132 *Arbetsutskottets protokoll nr 21 fört vid sammanträde den 1 februari 1945.* Riksarkivet, Hemmens forskningsinstituts arkiv, AII:1 Arbetsutskottets protokoll, Arbetsutskottets protokoll 19-45 1945.

133 *Arbetsutskottets protokoll nr 26 fört vid sammanträde den 5 april 1945.* Riksarkivet, Hemmens forskningsinstituts arkiv, AII:1 Arbetsutskottets protokoll, Arbetsutskottets protokoll 19-45 1945.

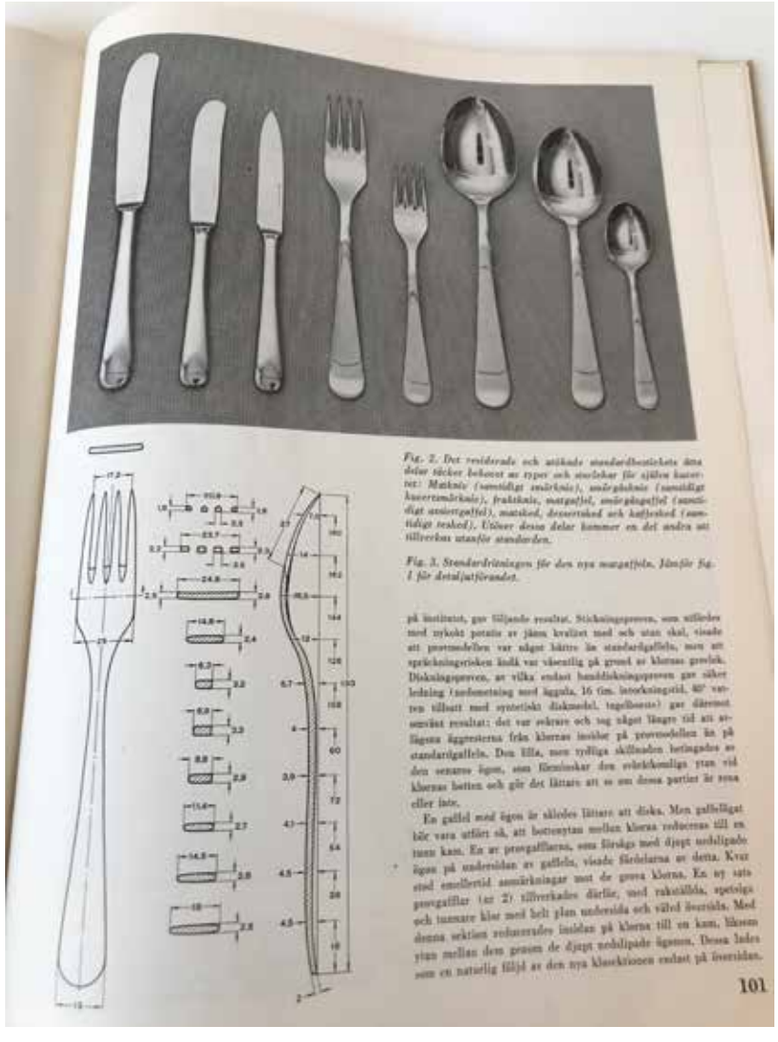


Fig. 2. Det reviderade och utökade standardbestickets åtta delar täcker behovet av typer och storlekar för sjutton kuvert: Matkniv (samtidigt smörkniv), smörgåskniv (samtidigt kuvertsmörkniv), fruktkniv, matgaffel, smörgåsgaffel (samtidigt ansiovisgaffel), matsked, dessertsked och kaffesked (samtidigt teasked). Utöver dessa delar kommer en del andra att tillverkas utanför standarden.

Fig. 3. Standardritningen för den nya matgaffeln. Jämför fig. 1 för detaljutförandet.

på institutet, gav följande resultat. Stickningsproven, som utfördes med nykokt potatis av jämn kvalitet med och utan skal, visade att provmodellen var något bättre än standardgaffeln, men att sprickningsrisken knappt var väsentlig på grund av klornas storlek. Diskningsproven, av vilka endast handdiskningsproven gav säker ledning (medelmetning med åtgång, 16 tim. inverkningstid, 40° vatten tillsatt med syntetiskt diskmedel, tegelbeats) gav därvmot samma resultat: det var svårare och tog något längre tid att avlägsna äggresterna från klornas insidor på provmodellen än på standardgaffeln. Den lilla, men tydliga skillnaden betingades av den smalare ägon, som förminskar den svårdiskbara ytan vid klornas botten och gör det lättare att se om dessa partier är rena eller inte.

En gaffel med ägon är således lättare att diska. Men gaffelägat bör vara utfört så, att botten mellan klorna reduceras till en tunn kam. En av provgafflarna, som försågs med djupt nedslipade ägon på undersidan av gaffeln, visade förtelarna av detta. Kvar stod emellertid anmärkningar mot de grova klorna. En ny serie provgafflar (nr 2) tillverkades därför, med rakställda, spetsiga och tunnare klor med helt plan undersida och välvd överdel. Med denna sektion reduceras insidan på klorna till en kam, likaväl ytan mellan dem genom de djupt nedslipade ägonen. Denna lades som en naturlig följd av den nya klassektionen endast på översidan.

101

HFI-designed cutlery. *HFI-meddelanden* 1950 1-2.

While similar discussions seem to have surfaced now and then, the HFI continued to resist invitations from manufacturers to provide some sort of label of recommendation. In the case of the HFI designed kitchen knives that came into production, the issue of recommendations is again surfaced at an internal HFI meeting when the manufacturer E.A. Berg had sent over their latest production examples of the knife design: "In the knife packaging had been placed a note with the following text 'Tried and recommended by the Hemmens forskningsinstitut.' Decision to discuss this item further with the technical committee for knives and raise it with the industry collaboration committee."[134]

While some of the designs for new tools and utensils developed by the HFI did come into production, several other proposals for new designs made at the HFI did not. In the *HFI-meddelanden*, reports on studies conducted, there were several examples of quite specific suggestions for new designs of different things that the HFI shared in an open-access format in the publication. The aim with this was to provide both inspiration – and in some cases also technical drawings and material specifications – that would allow producers to pick up on the item and start producing it. In the report on tools for mashing, for example, there were two different "HFI-types" of mashers designed and described, with the hope that someone would produce them. However, the companies were often reluctant to do this, which caused the HFI to also raise this issue during a meeting with the industry collaboration committee in 1948.

In the discussion that followed, the lead of the HFI technical section Carin Boalt raised the question of why the producers often did not readily alter their existing manufacture according to the results presented by the HFI. She took the example of the potato peelers studied early on by the HFI, and pointed to that the ones that had been deemed the best in the study were still hard to find on the market, and that none of the suggested improvements on the others had seemed to have been made.

The industry representatives pointed to that one problem was that any changes in design and production of for example a potato peeler would need to be renegotiated with the state price control committee, to adjust the price range that the company could place it in sales-wise. This committee, active between 1940 and 1956, was tasked with regulating the level of costs and prices for goods and services in Sweden. Negotiations were held with companies and organisations, but it was the committee that ultimately decided the reasonable price ranges for different products. From 1942 and for the duration of the war, there was a cap on almost all increases in prices.

This led to that any changes to existing designs that would make production more expensive, could often not result in the manufacturer increasing the price for the item unless the redesign was so substantial that it could be argued to be a new design altogether, which then could allow setting a higher price.[135] Making changes

134 "Arbetsutskottets protokoll nr 107 fört vid sammanträde den 20 januari 1948". Riksarkivet, Hemmens forskningsinstituts arkiv, AII:1 Arbetsutskottets protokoll, Arbetsutskottets protokoll 19-45 1945.

135 "Protokoll fört vid sammanträde med Samarbetskommittén mellan industrin och Hemmens forskningsinstitut onsdagen den 4 februari 1948." Riksarkivet, Hemmens forskningsinstituts arkiv, AIII:1 Fullmäktiges protokoll, Övriga protokoll 1943-1956

in existing production, therefore, seemed more difficult for the companies than working together with the HFI on developing new designs. For the HFI, already from the experiences gathered after the first year of its running research, it was the focus on developing new designs in collaboration with industry that seemed the most interesting and strategic route, rather than only comparing, assessing and approving existing products.[136]

Despite this close collaboration with industry, and the aims to engage in design for commercial production of goods, the institute aimed towards finding ways to balance this against its free-standing character. Being an independent research institute was important to the HFI, and any recommendations or suggestions that would lead consumers towards a particular brand or producer in the situation of acquisition had to be based on very objective criteria and evaluations, so as not to be seen as a bias from HFI towards a particular manufacturer. Instead, it was the independent work, with support and contacts with the manufacturers that the HFI believed would bring about better quality goods for people to buy and to use.

So, while the information shared through the HFI publications to a great extent focused comparisons of existing goods on the market, with as unbiased information as possible about strengths and weaknesses, the overarching ambitions with the work at HFI went well beyond providing a basis for a decision of what to buy. The main goal was to effect change on a level of production, so that tools provided were of better quality in general, and to change the position of women in society through professionalising housework. As the aim was to improve the overall work conditions in the home, through both repositioning the value of housework and the status of women's work through providing adequate tools and environments, the HFI wished that a range of producers would widely adopt the design suggestions or innovations developed in their processes.

However, these ambitions came to a halt in the mid-1950s. Due to changes in leadership, organisation and funding, the experimental practices were discontinued. In 1954, the HFI and Aktiv Hushållning merged into a single unit, which led to a restructuring of internal organisation as well as of work tasks. The staff of HFI was no longer allowed to continue designing products but were instead tasked to provide information for consumers.

The HFI was instructed to only work with the objective comparison and assessment of existing goods as a service to consumers to make informed choices of products. For this work, the institute was instructed by its new management to only use methods that had already been developed and proven efficient, and discontinue the development of any new methods or expansions into non-tangible areas of investigation. The once continuous exploration of methods thereby came to a halt and fossilised into defined assessment techniques and evaluation tools. When the HFI in 1957 once again was transformed and became the state-funded *Konsumentinstitutet* (Consumers' Institute), most of the team members who had been active in establishing the institute and developing its methods of working had by then either left or been fired.

136 "Protokoll 5 25 oktober 1945, på byggnadsföreningen Norrlandsgatan 11. Bilaga 2 PM rörande forsknings- och undersökningsverksamheten inom HFI 1945 och 1946 utarbetad i oktober 1945". Riksarkivet, Hemmens forskningsinstituts arkiv, AI:1 Styrelsens protokoll 1944-57.

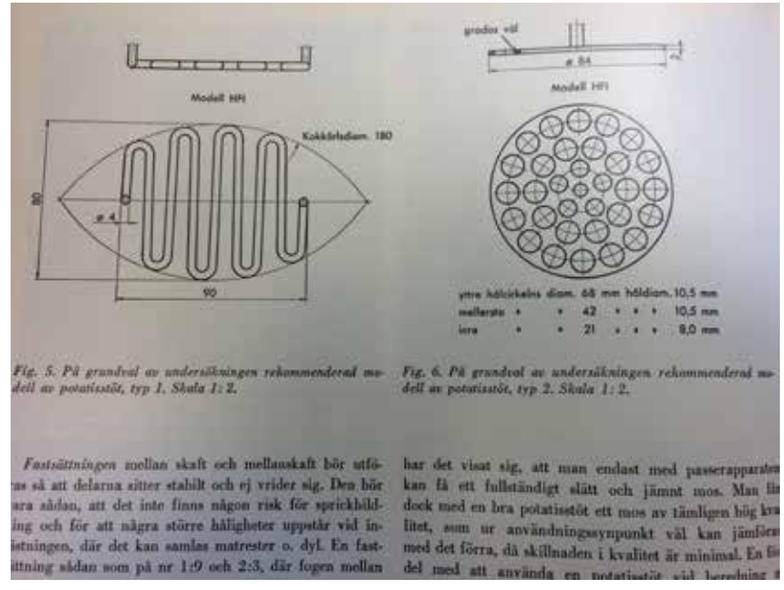


Fig. 5. På grundval av undersökningen rekommenderad modell av potatisstöt, typ 1. Skala 1:2.

Fig. 6. På grundval av undersökningen rekommenderad modell av potatisstöt, typ 2. Skala 1:2.

Fastsättningen mellan skaft och mellanskaft bör utföras så att delarna sitter stabilt och ej vrider sig. Den bör vara sådan, att det inte finns någon risk för sprickbildning och för att några större håligheter uppstår vid infästningen, där det kan samlas matrester o. dyl. En fastsättning sådan som på nr 1:9 och 2:3, där fogen mellan

har det visat sig, att man endast med passerapparat kan få ett fullständigt slätt och jämnt mos. Man får dock med en bra potatisstöt ett mos av tämligen hög kvalitet, som ur användningssynpunkt väl kan jämföras med det förra, då skillnaden i kvalitet är minimal. En fördel med att använda en potatisstöt vid beredning

HFI designs suggesting improved potato mashers. *HFI-meddelanden* 1948-49:1.

Making design methods

The aim with the brief history prototyped above has been to exemplify what a design history of design methods could look like when the development of methods for researching everyday life, and for designing products for the home at the Hemmens forskningsinstitut in the 1940s are approached as design. That these activities have not previously been included in mainstream histories of Scandinavian design – nor in broader sketches of histories of user-centered design inevitably brings up the question of what it is that qualifies as 'design,' and what as its history.

When we search for histories of design, we tend to look in places where others have looked before, and those places often seem very similar to those in which 'designing' takes place today. Much of design history has been narrated in formats and traditions close to art history, applying perspectives from material or visual culture. Design histories have come to have a relatively strong focus on design understood as outcomes, whether as design authorship and design objects or as the cultures and meanings these form and are a part of. Discussions of what qualifies as activities and things important to include in design history have expanded the areas in which to search for what could be important to understand in terms of histories. Still, in terms of highlighting different possibilities, these histories have often been approached as complementary or additive.

In histories where the work carried out at the HFI does surface, it is usually from the perspective of consumer perspectives. Feminist historians of technology have discussed HFI and similar women's organisations as significant for

establishing women as consumers, linking technological expertise to traditions of craft and domesticity. HFI has been seen either as a consumer rights organisation or as a social engineering initiative towards professionalising domestic work and a housewife ideal, rather than as an entity having to do with designing.

A recent general overview of Swedish modern history describes, with an ironically dismissive twist, the purpose of HFI as mainly aiming to rationalise domestic work, and that it "[d]uring its heyday, the end of the 1940s, research reports published shared important results, such as that the height of the kitchen sink affects the time consumed when baking sponge cake."[137] Swedish design historian Lasse Brunnström does point to the importance of HFI for developing standards, and quality tested approvals of domestic consumer goods. He mentions that the institute also designed certain objects, but seems to characterise the HFI products as "lacking aesthetic refinement" in comparison to the later ergonomic user-centered Swedish design of the 1960s and 1970s.[138]

However, since the aims of the design practices at the institute were not founded in exploring or developing form from an aesthetic foundation, this is hardly something that would make the HFI less interesting to include in histories of design. Rather the contrary: the work at the HFI explicitly aimed towards finding other foundations, and other methods, for the development of everyday tools for the home. It was precisely through the adoption of scientific methods and procedures that the HFI sought both foundations and legitimacy for their design work.

The HFI, during its first 8-10 years, worked according to an overall programmatic ambition, proposing that systematic research into everyday life in the home could provide a means for thinking differently about women's possibilities in society through reforming work practices in the home. The projects set in motion to explore if this could be the case, consisted of a series of explorations and experiments that took a starting point in tools, equipment, materials, leading to proposals of new designs for products, as a result of the projects, but also to suggestions for changes in practice.

But the explorations also led to suggestions for changes in practices that spoke to the establishment of other conceptual spaces, different ways of thinking, that in turn could re-shape what was considered as given or taken for granted. Was it really reasonable that women's time mainly was spent on cooking meals and doing dishes, hindering women from taking part in education, in salaried work outside the home? How could legislation, social services and public bodies provide ways of solving certain of these things on a collective level, opening the potential for women and men to lead lives in different ways than before? The development of new methods for understanding everyday life, and the development of design methods that followed at the HFI, fundamentally aimed towards changing the foundations for making things, in the context of an understanding of things as carriers of the potential for societal change.

The striving towards continuous development of methods, based on industrial and societal changes and a conscious approach towards not taking the current state of things for granted, was, therefore, an integral part of the HFI agenda. This approach can be compared to the statements regarding design methods, made by John Chris Jones in the foreword to the 1980s Spanish edition of *Design Methods*:

137 Yvonne Hirdman, Jenny Björkman & Urban Lundberg, *Sveriges historia 1920-1965: Välfärdens tid (*Nordstedt: Stockholm), 2012, 580.
138 Lasse Brunnström, *Swedish Design. A History* (Bloomsbury: London, 2019), 172.

> New methods cannot force us to do good, nothing can, but they do change the context of designing. There is a morality in widening the scope of life; what was before inevitable becomes now a conscious choice. And once this widening, this prior liberation, has occurred it is no longer realistic but insulting to refuse the ideals and purposes that are now possible. Industrial life HAS changed its nature in recent years and thus includes the means of pursuing aims never tried before. Of course the enlargement of what is possible includes the enlargement of what is bad as well as good, that is the risk, unavoidable. A new responsibility for which the institutions and professions and roles we inherit are surely unable by their form and nature to respond.[139]

Returning to the question of what design histories that take a starting point in methods might bring to designing, one question is what 'design' is understood to be, and where it is expected to take place. A reason why the HFI so far has not been foregrounded in design history, is probably that it has been considered as a place where designed things were evaluated rather than one where designing took place.

During the 1940s and 1950s, the work done at HFI was not spoken of as 'design' as such, so when looking for instances of that which is labelled 'design', it would not stand out. In Swedish, up until the mid-1940s, the term used was *formgivning*, 'form-giving'. And clearly, what the HFI focused on was something different than that. Designing understood as form-giving was mainly considered an individual artistic endeavour. Such a design process would begin with sketching and with considering qualities of material form and expression, rather than with user studies or a scientific questioning of why certain work was carried out in a certain way. The collaborative and investigative work at HFI was something different than that. The methods adopted, and invented, at the HFI came about through a systematic quest for knowledge and a desire to change socio-political situations through designing things as well as behaviours.

These methods, recognisable in later methodological descriptions of user-centered design, including both ergonomic studies and methods for probing user experience and expertise, have historically been attributed to a much later incorporation in design. In the 1950s, several industrial designers began advocating the inclusion of ergonomic and anthropometric studies in designing.[140] Since the early design methods conferences of the 1960s, the origins of ergonomics in design has been traced to military contexts and American industrial designers. Notably, Henry Dreyfuss has been highlighted as the first to bring human factors into design in the post-war years of the 1940s.[141] In one of the conference papers

139 John Chris Jones [1977], "now we are numerous. introduction to the Spanish edition of Design Methods" in John Chris Jones, *Designing Designing* (Architecture design and technology press: London, 1991), 33.
140 John Chris Jones, "Ergonomics, human data for design", *Design* 1954:66, 13-17; John Chris Jones, "Handles: The Ergonomic Approach", *Design* 1954:72, 34-38; Henry Dreyfuss, *Designing for People* (New York: Simon and Schuster, 1955)
141 Russell Flinchum, "Dreyfuss, design and human factors", *Ergonomics in Design* 2000:1 (vol.8), 18-24; B. Shackel, "Ergonomics and Design" in (ed.) S.A. Gregory, *The Design Method* (Butterworths: London, 1966), 50.

at the 1964 DRS conference on design method, for example, the relevance of ergonomic methods to design was introduced as a novel approach.

The studies made as part of home and housework rationalisation, such as movement studies or the methods developed by the HFI, have rarely been mentioned in terms of potential connections to the design methods movement, or to the emergence of user-centered design and usability. While not dismissing these situations as important for bringing about user-centered ways of designing, it also seems plausible that the appearance of user-centeredd design methods in a Scandinavian context were more likely to be linked to research and design contexts revolving around everyday life settings in the 1940s.

The emerging user-centered practices of designing at HFI did not resonate as something defined as 'design' neither then, nor in the later histories of Swedish design. However, when the term "industrial design" was introduced in a Swedish context, in the SSF journal *Form* in the mid-1940s, it is telling that objects in focus for the collaborative and iterative practices at HFI were included as examples of industrial design – and 'expert housewives' invited to discuss design in terms of form and function.[142] Linked to this newly emerging discipline of industrial design in Sweden, then, were not only ideas of a different type of process but also an inclusion of attention to user perspectives and an expanded notion of what (and who) could count as a designer. So, in a 1940s context, at least some of the work carried out by HFI could have been understood as "design".

In our contemporary outlooks, we might ask ourselves in which contexts new ways of designing are currently being developed, that we will not easily identify as 'design' given the ways we – both consciously and unconsciously – define what we consider designing to be. How could we go about recognising design when we quite literally may not even see it? Making design histories of designing, that highlight other narratives of what design could have been might also contribute to opening up conceptual spaces for thinking differently about what design might presently be becoming.

From the perspectives of what 'counts' as design in contemporary practices, and which types of histories that leads to, we might then consider what this does for views on what might be possible trajectories for design to take towards future practices. Would it make a difference to thinking and doing design in a Scandinavian user-centered and participatory 'tradition', if we understood some of its foundational methods and tools as emanating from a historical situation of feminist activism aimed towards rethinking societal norms through design interventions in the home rather than as coming from a trade-union driven context of agonistic power negotiations at work? Histories like these could contribute to raising awareness of the historical situations in which methods have come about, and what that has also meant in terms of the aims are with designing.

If we compare this to the approaches within the 1960s design methods movement where the ambition was to apply scientific methods to open up the design process to as broadly as possible bring in a variety of perspectives and information from which to draw design trajectories, the similarities are quite substantial.

142 "Bättre och vackrare men inte dyrare: Husmödrar diskuterar funktion och form", *Form* 1946:3, 65-67.

The new design methods proposed in the 1960s and 1970s aimed towards making visible and understandable the practices of designing that had until then been considered individual, experientially based, and difficult to formulate. Similar ambitions lay behind the attempts to develop objective methods for researching – and designing – everyday life in the 1940s. The introduction of systematic methods for describing, and understanding, at least certain stages of these processes did not only serve purposes of evaluating or defining opportunities for design or for improvement of methods for conducting housework. Ultimately, the goals with the practices developed at the Hemmens forskningsinstitut were founded in a feminist agenda of radical societal change, that aimed at changing ways of thinking and societal norms.

User-centered design tends to entail applying different methods and tools to understand various aspects of peoples' behaviours, actions, experiences, hopes and dreams. This is based on upholding a difference between 'designer' and 'user', where the agency and responsibility for designing lies with the 'designer'. Methods developed in contexts of participatory design question the distribution of responsibility and agency between 'designer' and 'user' and aim to propose different ways to support, enhance and collaborative and participatory practices in which designers and non-designers come together in different ways in design situations.

While there is continuously a vast amount of development work going on in the fields of user-centered and participatory design practices, considering the conceptual foundations of what a 'method' is and where it stems from might contribute to highlighting invisible assumptions embedded in the ways designing happens. In relation to how methods stabilise as core practices in certain ways of designing, we could perhaps question what happens if we begin to use certain methods as tools, or techniques, in design without asking what it then is that becomes cemented in terms of the perspectives and historicity built into these? As John Chris Jones pointed to, in relation to the developments in the design methods movement when 'methods' in many cases became used as 'tools':

> So the fault in method-making was that we made methods as 'products' and handed them on to the designers expecting them to use them, as 'tools', as means to an end. Which became a logical trap, turning the idea of process into its opposite?[143]

How and why methods are developed and applied in design differs in regard to what we expect the methods to be able to support. What is the difference between developing methods for participation, versus methods for designing together? What is the difference between developing methods for designing, in relation to developing methods for thinking differently, or for designing changes in societal norms? What happens if we consider methods developed in one context and historical situation in another time, as tools that can be put to work a contemporary time and context, with a different aim? Or without recognising how different levels of designing, for example, products, systems, and ontologies are entangled, and that what is done on one level also affects the others.

143 John Chris Jones [1978], "opus one, number two" in John Chris Jones, *Designing Designing* (Architecture design and technology press: London, 1991), 163.

Viewing design processes as a series of steps in which methods support divergence and convergence in a specific flow, has hitherto rested on an understanding of design as governed by a structure where collaborative work builds on designers bringing in expertise – other disciplines, users, stakeholders – from the 'outside' and 'into' the design situation. This is something more or less built into several of the participatory and user-centered methods commonly in use in designing.[144]

In an internalised understanding of design as something run by designers, whether as 'leaders' or 'coordinators' in a design process, the ways of thinking embedded in how the process comes to be and plays out perpetuates the very thing that many design research methods are trying to address: how people come together in other ways than before to take meaningful action through design. What is taken for granted, and what might be invisible to us in the ways we design, can be highlighted and brought into question through an attention to the historicity of design methods.

For the HFI, making new methods for researching everyday life ultimately aimed at changes that spanned the spectrum from designing dish brushes to changing the foundational understandings and structures of what society could be. As feminist debater Elin Wägner stated in her 1945 speech on the occasion of being elected member of the Royal Swedish Academy:

> It might seem insignificant, but one can begin with rejecting a dishbrush and end with liberating oneself from a whole form of civilisation.[145]

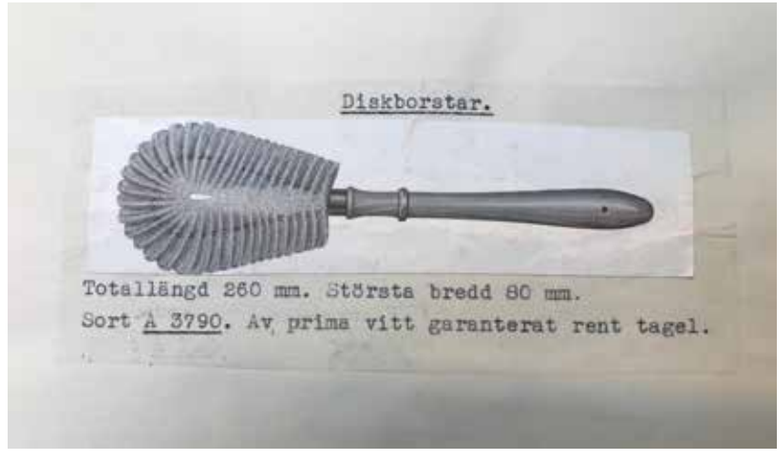
For change to become possible, there must be methods developed to support thinking in new ways about things that seemed so natural, and so embedded in everyday practice that they otherwise would be almost impossible to imagine being in any other way. At the core of the social design project that HFI was part of stood the making change in seemingly small and insignificant everyday material things. However, these changes were envisioned to promote different behaviours that would lead to shifting power relations in society, which in turn could open up for thinking differently about what society and civilisation could be.

In a current eco-political situation, with increasing attention to what it will take for the redirection of designing towards sustainable futures that will continue to include human civilisations, histories such as the one sketched here might not only contribute to questioning and expanding the conceptual spaces of designing. Histories of methods, rather than of things, can contribute to highlighting why certain ways of designing have come about – thus also contributing to the critical discussions of why, what, and how different ways of present-day design make or unmake possible futures. Making present the historicity of design also means recalling that many of the socio-material structures that we now take for granted or given, once seemed completely unrealistic or even unimaginable – and yet, we once brought them into being.

Making systemic change happens gradually, slowly, and with great effort, since it requires a constant movement between thinking differently and doing differently. If we view design as something that spans a spectrum ranging from

144 Monica Lindh Karlsson & Johan Redström, "Design Togetherness" in *Nordes 2015: Design Ecologies* (2015), no 6.
145 Elin Wägner quoted in Åkerman, *88 år på 1900-talet*, 148.

things to the foundational concepts that govern what becomes possible in, and understood as, designing, it follows that making new methods for designing differently today also will need to address the complexities of constantly shifting perspectives in order to see how designing dish brushes has something to do with rethinking civilisations.



One of the inventoried and tested dish brushes at HFI in 1944. Riksarkivet, Hemmens forskningsinstituts arkiv.



Dishbrushes as 'design objects' in the Nationalmuseum gift shop, Stockholm 2019.

7. Pasts/Presents

Returning to the program of transitional design histories, in light of the prototypical explorations presented in this thesis: what could its implications be for design, and for design history? After discussing what emerges in this first iteration of prototyping transitional design histories, I point to some potential trajectories for the continued exploration of how design and history could be made to matter more to each other. The prototyping of design histories *for* design is linked to the prototyping of practices *in* design, considering what activating aspects of temporality and historicity might mean for what design could become.

Program

The program of transitional design histories is formulated on the proposal that if design aims to develop new approaches and methods for responding to issues of complexity and sustainment, the historicity of designing must be made present in design.[1] In the opening chapter, I made the case that bringing history more actively into design can allow a reframing of the spaces in which to explore possibilities for how to go about designing differently.[2] This reframing has to do with how design sees its world and understands itself in processes of proposing changes in the present that aim towards certain futures. Recalling, again, the passage by Donna Haraway that I quoted in the opening chapter: this is about the things we think with in design, and about approaching design histories as stories that make worlds.[3]

In this thesis, I have presented a suggestion for how history could be made to matter more to design. Pointing to the a-historicity of design practice, and lack of practice-based outlooks in design history, I proposed that design history should be approached differently and that design practice should open up towards awareness of its historicity.[4] For design history to not only matter more to design,

1 Clive Dilnot "History, Design, Futures: Contending with what we have made" in Tony Fry, Clive Dilnot & Susan C. Stewart *Design and the Question of History* (Bloomsbury: London & New York, 2015), 149.

2 Eva Brandt and Thomas Binder, "Experimental Design Research: Genealogy – Intervention – Argument", *Proceedings of International Association of Societies of Design Research Conference* (Hong Kong, 2007).

3 "It matters what matters we use to think other matters with; it matters what stories we tell to tell other stories with; it matters what knots knot knots, what thoughts think thoughts, what descriptions describe descriptions, what ties tie ties. It matters what stories make worlds, what worlds make stories." Donna Haraway, *Staying with the trouble: Making kin in the Chthulucene*, (Durham: Duke University Press, 2016), 12.

4 Clive Dilnot, "Some Futures for Design History?" *Journal of Design History* 2009:4 (vol. 22).

but for it to also become a matter in designing, it is not enough to only bridge the current gap between design and design history. Closing the gap between design history and design requires that the outlook of design histories moves closer to that of contemporary issues in design, and from there make design histories that are positioned in design*ing*.

Transitional design histories are histories *for* design, not histories *of* design. I have argued that such design histories should be made experimentally as prototypes, addressing the historicity of designing rather than telling histories of design's past outcomes. In the design history prototypes presented in this thesis, I have sought to contribute to different ways of seeing and doing design through presenting – making present – the historicity of design concepts and methods. The intent with these histories was to shift perspectives on what design could be, by making visible the instability and historicity of things that are often taken for granted in designing. These transitional design histories thus aimed to present – to make present – historical tensions and assumptions embedded in the foundational concepts that form the frameworks for methods, outlooks, and approaches in contemporary designing.

Design does not only make and propose things that could make a difference; it also inherently makes and proposes ways that design itself could be different.[5] The transitional character of design typically is at its most visible in instances where established ways of doing design no longer suffice for handling the situations at hand. Design histories that can speak to changes in designing therefore need to support fluid and transitional understandings of design. I have proposed that such design histories would be made to offer re-positionings of how design can be understood as a changing practice that makes a difference, rather than aiming to provide explanatory narratives that consolidate specific definitions of what design and designing is, has been, or should be.

From this follows that transitional design histories would not only shift position from where they are made, but also continuously change in outlook, content, and narrative in response to changes in design's contexts and practices. I have suggested a methodology for the making of design histories as prototypes, that combines a programmatic approach from practice-based design research with research methods in history that focus on analysing concepts and ideas. Going on to explore the making of such prototypical design histories, I positioned the outlook for this exploration in practices of Scandinavian user-centered and participatory design.

Probing the historicity of concepts and internal world views that this type of designing is based on, I presented three prototypes of transitional design histories. Coming back to the questions that these prototypes set out to explore, what implications for design and for design history might we see emerging in the prototyping of transitional design histories? In the following, the design histories made and presented in this thesis are discussed in light of how they work, what becomes visible, and what is possible to say in terms of evaluating what they do for design, after this first iteration of prototyping. Considering the ways these histories work, some things are relatively easy to see, and to test, while others are more elusive and not easily evaluated at this point.

5 Søren Rosenbak, *The science of imagining solutions: Design becoming conscious of itself through design* (Umeå, 2019); Johan Redström, *Making Design Theory*, (Cambridge, MA: MIT Press, 2017) 129-130.

That design histories made from a position of designing do become different, and make visible other contexts relevant to design, is quite easy to see – which will be more elaborated here below. What is more difficult to say something about, and that in the following will need to be approached more in terms of speculation and potentiality, is if these types of histories can bring about different ways of approaching design. How would one know if different historical narratives could bring about individual or collective self-reflection in terms of what 'designing' or 'being a designer' might mean in terms of identity and historicity? Or if bringing forth historically formed instabilities in design concepts would make an impact on design situations and design methodologies? Although it is hard, perhaps even impossible, based on the prototypes presented here, to test or evaluate if people begin to think and do design differently after reading these historical representations, I will at least speculate about this, and point to some tendencies that could be discerned.

Seeing things differently

The design histories presented in this thesis are made from an outlook of present-day designing, though positioning the studies in design practices related to Scandinavian user-centered and participatory design.[6] This shift in outlook is not about moving away from a focus on design outcomes, such as products or things, to a focus on the physical places where those things are designed and made – like consultancies, design offices, or production lines. These are already phenomena well researched in design history, which also build on histories made from the point of view of design as outcome. The move made in these prototypes is instead one that builds on seeing design as becoming defined in moves along a spectrum that spans from what 'a design' is, to what 'designing' is.[7]



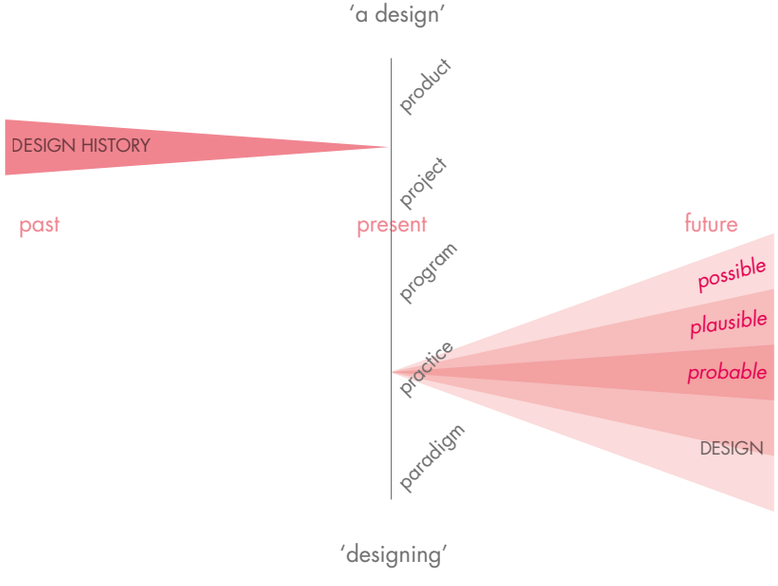
6 Jesper Simonsen & Toni Robertson (eds.), *Routledge International Handbook of Participatory Design* (Routledge: London & New York, 2013); Pelle Ehn, "Learning in Participatory Design as I Found It (1970–2015)", in Betsy DiSalvo, Jason Yip, Elizabeth Bonsignore & Carl DiSalvo (eds.), *Participatory Design for Learning: Perspectives from Practice and Research* (New York & London: Routledge, 2017).

7 Redström, *Making Design Theory*, 126-130.

Working with a more fluid understanding of how definitions of 'design' are made, making design histories from a vantage point of 'designing' does not mean that outcomes or products would be irrelevant or uninteresting. It means that 'design' is understood as defined not *only* through what it makes – products or outcomes – but through *how* and *why*, in which settings, with which ways of working, and with which understandings of the world designing takes place. One question guiding this research was: would this shift in outlook bring about other histories, and make visible other contexts and actors than previously have been visible in design history?

Position

Re-positioning design history and bringing it closer to designing entailed more than an idea of merely bridging the gap between these two fields of practice. Such bridging has frequently been attempted before, at least from the side of design history. This usually consists of providing studies into different versions of 'histories of […]'. The examples are many: this could, for example, be a history of a particular design subject (e.g. history of graphic design), a specific object (e.g. history of chairs), a certain material (e.g. history of steel), a geographical context (e.g. history of Scandinavian design) and so on, with an aim to explain or question outcomes in and meanings of design.[8]



8 Philip B. Meggs, *A History of Graphic Design*, 3. ed. (New York: John Wiley, 1998); Charlotte Fiell & Peter Fiell, *1000 chairs* (Köln: Taschen, 1997); Tony Fry & Anne-Marie Willis, *Steel: A Design, Cultural, and Ecological History* (London & New York: Bloomsbury Academic, 2015); Charlotte Ashby, *Modernism in Scandinavia: Art, Architecture and Design* (London: Bloomsbury Academic, 2017).

Design history from the outlook of what 'a design' is has undoubtedly brought about a broad design historical knowledge base, relating to design as not only products or outcomes, but as processes of production, consumption and mediation.[9] From the perspective of design historians, this has of course in many instances intended to be – and indeed also been – a contribution to design as a field today. In terms of addressing the a-historicity within design itself, however, this type of design history seems not to have made the difference it could or would have needed to make in presenting openings towards the historicity of contemporary designing. Also, while many of these histories indeed do bridge the divide between history and design in many areas, the fundamental disconnect in outlook has remained. This means that things that are important from the point of view of design have remained invisible in design history – and vice versa.

The prototypes presented in this thesis have called out this gap and the lack of connection between design and design history. Within present and emerging design practices that refer to their lineage in terms of belonging to a 'Scandinavian tradition' of user-centered and participatory design, the reference to 'tradition' points towards a claim towards a historical continuity between a particular, current practice and what is perceived as its historical origins and foundations.

Given that this way of referring to a 'Scandinavian tradition' is quite common in design research and practice building on collaborative ways of designing, often socio-politically oriented rather than commercial, it seems that this historical linkage is held as important from the viewpoint of design. From the point of view of design history, on the other hand, this practice of design has gone relatively unnoticed. In positioning a design historical outlook from the perspective of practices of designing, the prototypes in this thesis have made visible a blind spot in design history when it comes to histories of, and for, user-centered and participatory design.

In taking Swedish design history as an example (chapter 3), participatory practices of designing are seldom mentioned. If user-centered design is present, it is mainly described in terms of its outcomes. One could argue, of course, that the strong presence in design history of themes such as consumption and consumer practices, could point to certain aspects of user-centered design. But the idea of 'users' is quite different from that of 'consumers,' as are design practices primarily oriented towards 'use' versus design practices primarily targeting 'consumption.' However, going into distinctions and analyses of ideas of use in relation to ideas of consumption is not the topic here. What is the topic is the question of whether a shift in perspective towards design*ing* would bring other things, other practices, and other issues, into view when making design histories? Moreover, as I would argue is quite evident from what emerged in these first studies, this is the case in relation to participatory and user-centered design practices, at least in the examples presented here.

Making histories of participatory and user-centered design in this thesis did not, however, primarily aim to fill a gap in Swedish design history. Though that was, indeed, one of the potential workings envisioned with such a repositioning of the design historical outlook, the main mechanism intended with these prototypes was to activate other ways of seeing *within* design practice.

The prototypes in this thesis were made with the primary aim of being histories *for* design practices identifying with a Scandinavian user-centered and participatory tradition. Through applying different conceptual lenses from a perspective of

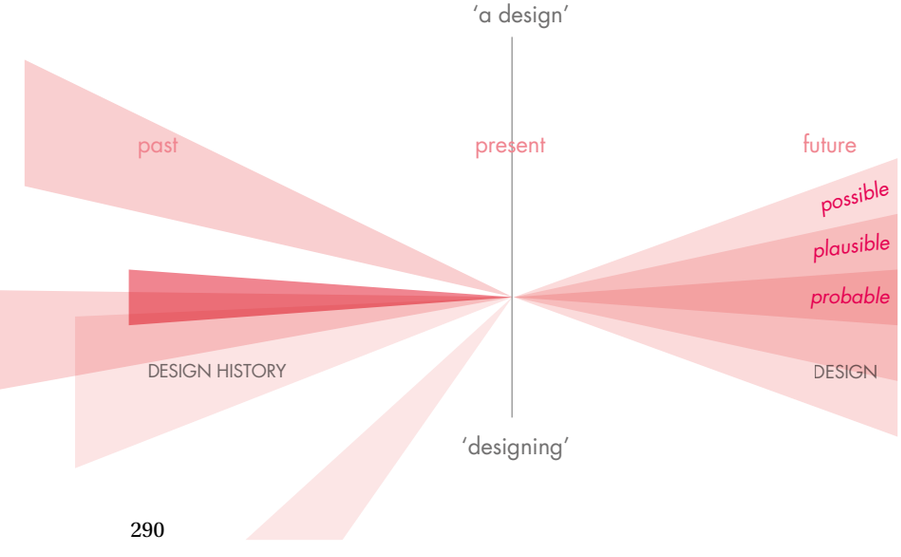
9 Grace Lees-Maffei, "The Production–Consumption–Mediation Paradigm", *Journal of Design History* 2009:4(22), 351–376.

participatory and user-centered designing, three different but partially overlapping design histories were presented as potentially relevant for making other things visible in this type of design practice. Each of the three histories took a different trajectory towards the past, depending on the concept it set out to explore, and depending on what became visible along the way of making the prototypes.

In the case of the study revolving around 'methods' (chapter 6), for example, an opening towards the search for ideas and practices that might have had to do with design methods carried over from the prototyping a design history of 'use.' In the texts and material studied in the 'use' prototype (chapter 5), something that came into view when applying the conceptual lens of 'use' was that there already in the 1930s and 1940s seemed to exist both experimentations and formulations aiming to establish new methods for designing. It stood out clearly that a search for methods with which to systematically investigate everyday life directly had to do with a search for methods for bringing insights from 'ordinary people' as well as from 'experts' and 'designers' into practices of designing. And this, in turn, was aimed towards making changes that would follow trajectories towards preferable futures that in their historical context might sometimes have seemed more or less utopian.

These three different historical prototypes began from the same position – Scandinavian user-centered and participatory designing – and resulted in proposals of what might be relevant histories for these design practices. Through an outlook towards the past from a position in present-day designing, the design histories that take shape do not aim to cover a coherent 'whole' in describing *the* history of collaborative and user-centered designing. As in the schematic visual below, the views that these perspectives open up towards overlap in some places, and diverge in others.

What they have in common is that they revolve around different aspects of design's conceptual foundations, thus making different contexts, relations, activities, and actors come into view. Even if the connections between these concepts are not explicitly accounted for here, they have over time come together and given shape to design practices. These concepts and their histories are related, even if it is maybe not clear exactly how, historically, those connections came to be.



Contexts and actors

One of the questions probed in relation to the positioning of design historical inquiries in designing was if an outlook from contemporary design practice would make other contexts and actors visible and relevant to consider in histories of design. In the three examples probed here, the shift in outlook resulted in bringing new perspectives on well-known phenomena in design history, as well as directing attention towards contexts previously not visible in histories of design.

One reason for making these prototypes based on printed and archival material was specifically to test what a different perspective would bring to a well-known material and context. Thus, the material studied in the prototype on 'use,' for example, sprang from a well-known context in Swedish design history. This prototype (chapter 5) revolved around design initiatives associated with home and housing reform in the years following the 1930 Stockholm exhibition, described in Swedish design histories as a watershed moment in introducing modernist design theory and practice in Sweden.

These are thoroughly researched topics and contexts in Swedish art and design history in particular, and in Swedish history in general. Since modernist approaches to housing and domesticity was a central aspect of the socio-political, as well as material, constructions of the Swedish welfare state, the *Folkhem,* there are many accounts already existing of these phenomena. The organisations I selected to study here were undoubtedly situated in contexts that, in the 1930s and 1940, already were closely associated with understandings of where design ordinarily would take place. As representatives for interests in design and architecture, the Svenska slöjdföreningen (SSF, still existing as Svensk Form) and the National Association of Swedish Architects (SAR, still existing merged with other architects' associations) would be expected to engage in questions of what design is, and how it should be done. This remains true in the present-day, as these contexts still are those where one would go looking for instances of 'design.' It would not, therefore, be surprising to find discussions on design methodology arising at some point in time in these contexts.

What might be slightly more surprising, in terms of initiatives directed towards developing new ways of designing to meet new contextual challenges, would be what came into view in the prototypes presented here. When applying the conceptual lenses of, respectively, 'users' and 'methods' to the known contexts of SSF and SAR, it became clear that the methodologies that these organisations began applying in studying the relationships between use and design were not invented or formulated there. Instead, they were directly brought into the SAR and SSF from contexts of women's organised initiatives to reform housework and redefine the roles of women in the home, as part of an agenda of gaining access to and influence in political and social arenas. In this, the Home Research Institute, HFI, stood out as a context that had not previously been directly considered as one where design would take place, but as one of importance for consumer-related influence on the production of

everyday things.[10] However, that the impetus towards searching for and formulating ways of use- and user-centered design*ing* actually was developed methodologically in the realm of inquiries into (women's) domestic work at the Home Research Institute (HFI) had not previously been clearly visible. In applying a perspective of 'methods,' the historical inquiry yielded insight into the work conducted at HFI as directly connected to developing what later became considered as 'design methods,' in the endeavour of both investigating and reshaping the conditions for everyday work in the home.

In making prototypical design histories, based on contextual readings of material with conceptual lenses and perspectives positioned in designing, somewhat different histories than the ones usually provided about 20th-century Swedish design have appeared. To give another example: previous design histories of Ellen Key's thinking have focused on specific publications of advice and examples on how to distinguish objects of 'good taste,' in which the aesthetic ideas and ideals advocated are described as normative and prescriptive.[11] The design history proposed here was shaped through the attention to participatory perspectives, placing more of an emphasis on Key's concern for issues of design reform and aesthetics against a context of programmatic ambitions to propose other possible ways of living. An understanding of the principles for creating new understandings of what would be a harmonious material environment – in the home as well as in public arenas in society – became an integral part of the ambition to activate design in making socio-political reform. New class relations, new means for influencing society, and new forms of democratic governance called for other material expressions and social behaviours.

Initiatives such as these called for practices that we might think of as a sort of infrastructuring:[12] providing situations for learning, through *folkbildning*, as well as initiatives such as Tolfterna (chapter 4) aimed at building relational foundations of mutual understanding that could lead towards collaborative efforts towards societal – and aesthetic – change. Without fully taking into account Key's and others' calls for domestic and production-directed material reform as situated within this broader programmatic socio-political design ambition, the normative aesthetic aspects easily tend to overshadow the basic proposal made at the time: that there was a real emancipatory potential in making life and society different –

10 Ulrika Torell, Jenny Lee & Roger Qvarsell (eds.), *Köket. Rum för drömmar, ideal och vardagsliv under det långa 1900-talet*, Nordiska museets handlingar 143 (Stockholm: Nordiska museet, 2018); Britta Lövgren, *Hemarbete som politik: diskussioner om hemarbete, Sverige 1930-40-talen, och tillkomsten av Hemmens forskningsinstitut*, diss. (Stockholm: Almqvist & Wiksell, 1993).

11 For example Zandra Ahl & Emma Olsson, *Svensk smak: Myter om den moderna formen* (Stockholm: Ordfront, 2001); Gunnela Ivanov, *Vackrare vardagsvara - design för alla? Gregor Paulsson och Svenska slöjdföreningen 1915-1925*, diss. (Umeå: Institutionen för historiska studier, 2004), 64-68.

12 Christopher A Le Dantec & Carl DiSalvo, "Infrastructuring and the formation of publics in participatory design" *Social Studies of Science* 2013:2 (vol.43), 241–264.

more egalitarian, more equal, more just – through designing everyday things and environments differently.

The methodological stance of moving design history's outlook to that of contemporary designing, then, has led to contributions also to histories *of* design. In terms of where we would go looking for histories of design, taking a perspective that is positioned in design*ing* does open up for other possible contexts, other actors, and other forms of agency than is otherwise often the case in design history. With a more fluid definition of design, along a spectrum spanning from what 'a design' is to what 'designing' is, moving design history towards the designing end of the spectrum allows for seeing other things in the past that can be relevant both for design history and for design.

These design history prototypes do not aim to give accurate and definitive descriptions of *the* history of participatory and user-centered design but are explorations of how histories of central concepts and methods in this field could speak to matters of concern in current design practices. What is in focus are why and how these ideas and concepts have come about in specific historical situations and what it might mean that these have transitioned into methods and approaches perceived as foundational for doing design. This takes us back to the main objective with making transitional design histories: that of activating an awareness of the historicity of design, and exploring whether prototypes such as these could expand understandings of core concepts in design.

Doing things differently

After establishing that shifting and making design histories from a perspective of designing did indeed in these prototypes bring forth other histories, it is, however, much more challenging to know if these histories could somehow support changes towards designing differently. Saying something more precise about *how* transitional design histories could be put to work in design, and potentially have an impact on redirecting or expanding contemporary design practice, is not possible after the prototyping carried out so far. Even if these design histories have been explicitly made *for* design, the ways in which they could be activated more directly *in* designing have not been tested.

It is not very straightforward to envision how these prototypical narratives would translate into actions leading to the activating of the historicity of design in actual design situations. Making history a matter for designing, one could speculate, could be something that happens as part of a design education, but it could also entail processes or methods introduced into design practices.

Perhaps, with time, design history might actually become an integrated part of designing, contributing with methodologies in design as well as with perspectives on design. Considering, for example, how the processes of gradually introducing of ethnographic perspectives and methodologies into design over the last thirty years have both formed and informed designing, as well as the field of design anthropology, perhaps the contributions of design histories made from

within design might take similar trajectories.[13] But how might this look, and what would be needed for bringing about other understandings of design's conceptual frameworks? How would one know if history had become a matter for design? These are all aspects that remain to be probed. Nevertheless, some projections for how this could be done, or at least begin to be explored, can be gleaned already.

Prototypes and practices

The program of transitional design histories merges methodologies and approaches from practice-based design research and historical research. I have proposed that design histories can be made and handled as prototypes, in a way similar to how prototyping is applied in design and design research. Following this, histories made as prototypes should be open, possible to adjust and change after trying them out, but still solid enough to be able to provide a certain functionality or experience that allows for specific aspects of an idea or a proposal to be investigated.

Prototypes can be made in different degrees of fidelity and finish, choosing materials and assemblies to make them look or work similarly or the same as a finished version would. The prototypes made in this study were made to look and work as histories, as historical representations. That these prototypes have worked as histories seems reasonable enough, but the question is if they work as *transitional* design histories? For that to be the case, these histories would have activated an awareness of design's historicity in designing, provided openings towards thinking and doing design differently, and also themselves be open to shifting and changing as designing changes.

With the examples presented here, trajectories have been drawn beyond the points in time when concepts such as 'use' and 'participation' at first glance might seem to have appeared in design. In the case of 'participation', which is usually taken to have entered design in the 1960s and 1970s and become more established from the 1990s onwards, the search for the ideas and practices that made it possible for this concept to come into designing were drawn to the turn of the century 1900 and a context of Swedish socio-political transformation.

In relation to practices of participation (chapter 4), I pointed to ideas and practices that emerged in and around *folkbildning* (adult learning) as central to creating new material as well as intellectual spaces that could support new types of participatory engagements aimed at setting up infrastructures for democratic reform. The emancipatory potential was linked to the conscious appropriation and sharing of different types of knowledge and experiences in situations that brought people of different backgrounds and social strata together. Nevertheless, while the aims explicitly were to empower people – defined as working-class women and men – these interventions also led to establishing frameworks and power relations founded in the middle class' normative and prescriptive ideas of propriety and education. This manifested difference between workers and middle class rather than eradicating it. Middle-class women were structurally in the position of holding knowledge that was to considered important to share with working-class women. Thus, the practices of 'inviting' working-class women to take part in pre-defined cultural and learning activities, was based in an imbalance of power already embedded in the setting up of situations intended to be egalitarian and participatory. While the overall vision, on a programmatic level, was the creation of an egalitarian society where class would not matter, the situation on project level – in Tolfterna – was still unequal and highly conditioned by the social structures of its time.

This example made visible that ideas embedded in concepts fundamental to design can carry historical tensions that would need more consideration in contemporary design practices. The core concept of 'participation', for example, carries both emancipatory ideas of equal say in processes, as well as ideas of making difference based on knowledge, expertise, and skill. Making these historical tensions more present in designing could contribute to other possible entry points for questioning the set-ups, processes and methods chosen in participatory design projects.

Practices in participatory design do already acknowledge issues of conceptually embedded aspects of power and agonism in negotiating co-design situations and explicitly aim to develop methods and approaches to support handling such issues in designing.[14] However, perhaps transitional design histories might support emerging practices in begining with intentional attempts to reshape and shift understandings of concepts, rather than aiming to make tools and methods based on existing conceptual frameworks?

Similarly, probing where ideas of 'use' spring from, and how they once came into designing speaks to inherent tensions between intended use and actual use that, in the design histories presented here, were noticeable already in the 1940s (chapter 5). For participatory and user-centered design, activating the historicity of design through pointing to tensions embedded in the concept of 'users' does not propose to bring solutions that could resolve complicated issues in contemporary collaborative design practices. There will still be many different ways one can go about prototyping practices of somehow finding ways for 'users' and 'designers' to come together in processes of designing, or to consider what 'use' could be in relation to 'design.' What transitional design histories *can* bring are widenings of the conceptual design spaces in which 'use' is considered as a stable foundation for designing.

Acknowledging the historicity of the concept of 'use', and its embedded tensions between ideal and real use might open up for ways of approaching designing differently through questioning what the concept in its current understanding actually allows for making and doing, and what it possibly counteracts, in situations of designing. An awareness of the historicity of how design practices have formed in relation to 'use' might then contribute to explorations of new ways of coming together in designing that could aim to question the conceptual division between 'use' and 'design' as a foundational starting point.

13 Alison J. Clarke, *Design Anthropology: Object Cultures in Transition*, revised edition (London: Bloomsbury, 2017); Rachel Charlotte Smith (ed.), *Design Anthropological Futures: Exploring Emergence, Intervention and Formation* (London: Bloomsbury Academic, 2016); Wendy Gunn, Ton Otto, and Rachel Charlotte Smith, *Design Anthropology: Theory and Practice* (London: Bloomsbury, 2013).

14 Thomas Binder et al., "What is the object of design", *CHI conference proceedings 2012,* (ACM, 2012), 21–30.

Making methods

In design, exploring and proposing the development of new design methods to support different ways of designing is closely related to changes in the contexts and situations where design takes place. In the 'methods' prototype (chapter 6), I explored how new practices developed at the Home Research Institute (HFI) already in the 1940s can be understood as proposals of user-centered design methods, responding to intended interventions into material and social practices in everyday life. From a perspective of designing, we saw that practices at the HFI included many of the methods that today are at the core of user-centered designing: ethnographic studies, user testing, iterative prototyping, etc. The development of these methods was actively pursued at the HFI as a feminist project of reforming women's conditions in the home, as well as in socio-political contexts.

This design history prototype was perhaps the one of the three presented here that most clearly can be identified as supporting – or pushing for – the most significant shift in regard to understandings of user-centered design's conceptual foundations. In making visible the work of developing user-centered design methods at the HFI, it stands to reason that many of the ideas underpinning the methods and concepts launched there were based in a feminist agenda for making change.

This is a radical re-positioning of outlook towards where the historical foundations of methods in user-centered design are usually sought. Whether one before has perceived user-centered designing as emerging in the early 1980s, attributing the formation of the concept to the introduction of ethnographic approaches in to commercial design research contexts,[15] or if one has seen it as originating from ergonomic usability and performance research in military contexts of the 1940s and 1950s,[16] the situated emergence of these methods in a feminist, domestic-research-oriented project in 1940s Sweden is something that brings quite different ideas to the table.

Understanding user-centered design methods as springing from a context fundamentally questioning norms and knowledge-making from a feminist and gender-critical foundation gives a different perspective than an understanding of user-centered design as emerging from aiming to improve the performance of military fighter-plane pilots through designing cockpit environments and controls based on studies in cognitive psychology and ergonomics. The mere fact that the conceptual foundations of developing user-centered design methods expand towards a radically different context of feminist activism opens up for thinking differently about ideas embedded in these design methods, and what these origins could mean for considering how these practices are conducted today.

The ways that methods were brought into designing at HFI built on experimentation, and borrowing or importing methods and approaches from different areas of practice and different fields of scientific study. Methods for investigating and designing the context of everyday housework did not exist, but were identified as needed for making solid foundations on which to base decisions for interventions into everyday practices. The processes of experimenting with

15 Lucy A. Suchman, *Plans and Situated Actions. The Problem of Human-Machine Communication* (Cambridge: Cambridge Univ. Press, 1987).

16 David Meister, *The History of Human Factors and Ergonomics* (Mahwah, NJ: Lawrence Erlbaum Associates, 1999).

how to find methods that could support making change – material and ideal – in everyday life, at the HFI, resonate strongly with how much of methods-oriented development work in design and design research takes place today.

In this example, it is also clear that what is retrospectively perhaps considered a peripheral design actor or context (HFI) in relation to what is generally defined as central actors or contexts (SSF & SAR) might at a certain point have been the more influential and central agent in terms of rethinking and making changes in how designing is done. From a contemporary perspective, what became visible in these prototypes, then, could support thinking around how to possibly reframe outlooks in contemporary initiatives that aim to develop new ways of designing. If radically new ways of user-centered designing once emerged in contexts that, at that time, were not identified as places where designers work or where designing is the core activity, where could one imagine to go looking in order to find the unexpected places for where new design approaches and methods might be taking shape today?

Positioning the outlook in core concepts central to participatory and user-centered design, the main aim was to draw forth and critically explore the historicity of its conceptual foundations: situations of participatory practices, concepts of use and users, the making of methods. These were chosen on basis of being foundational to practices of designing in the sense that they, as concepts, in some ways are more or less taken for granted, and in other ways conceived as deeply difficult.

The studies into the concepts of 'participation' and 'users' in prototypes presented here have traced some of the historical ideas that have been instrumental in forming these as foundations for doing design, highlighting that these methods and concepts do come from 'somewhere', and not from 'nowhere.' If, and how, this awareness of the historicity of methods and concepts would bring about differences to how designing is done is hard to say. But what is made reasonably clear is that methods and concepts in design are not neutral. The ways in which design is thought, and done, carry ideas that are still connected to the contexts in which they came about. Activating the historicity of concepts and methods in design, on the one hand means that these historical layerings will become visible and possible to address, but also that the changing nature of concepts will be actualised, opening up for broader understandings of what design's conceptual spaces might become.

Historicity and possibility

With the shift in position to making design histories from what 'designing' is, rather than what 'a design' is, the question to begin with is not what design makes, but what it is that makes design. Taking historical perspectives on concepts and methods at the core of designing today, it becomes clear that design's foundations are not all that stable as they sometimes might seem to be. What also comes across, is that ideas and practices have come into design at certain points in time that have contributed to forming embedded concepts and methods that design still uses, but without there being an awareness of what this historical layering implies for designing.

The prototypes that were made as part of this thesis project have investigated how to make design histories through inquiries into ideas and practices that have

formed some of the core concepts in design. As much as these design histories were inquiries, made in the form of prototypes, they were also interventions.[17] On the one hand, they were interventions in design history, proposing other stories than the usual. On the other hand, and perhaps more importantly in light of the main aims of this thesis, they were interventions on a conceptual level in design, calling out the instability and change inherent to the concepts and methods that designing rests on.

The proposal that design histories should be made as prototypes that are open and changing largely springs out of this attention to the conceptual foundations of designing and their inherent instabilities. Since design is directed towards change, and based in conceptual foundations that themselves are fluid and unstable, design histories that aim to support such change must also themselves be unstable and open to change. This means that transitional design histories will need to change in relation to designing, in response to what the conceptual foundations seem to be and how activating an awareness of design's historicity could open up for seeing certain situations and practices differently.

Instabilities and possibilities

Turning a historical attention towards designing, and using concepts as lenses for the analysis, central concepts in design come actively into view as not only 'being there', but actually 'having become' what they are at certain points in time, and over time. With time, however, they change form and shift meaning, as ideas proposing new understandings or practices play into defining the concept. These changes tend to happen only slowly, which can lead us to believe that concepts are more stable than they in practice are. Activating the historicity of designing thus also activates the instabilities that design necessarily has to work with, if the ambition is to not only replicate the existing but to make possible something different.

Approaching design histories as transitional aims to highlight what it could mean for design that several of its core concepts – use, participation, and even the concept of 'design' itself – are anything but stable, temporally as well as situationally.[18] In transitional design histories, what is emphasised is fluidity rather than stability – both when it comes to the concepts studied, and to the types of histories that can be made. Depending on what seems important, or difficult, to handle in design in the present, the design histories that could support addressing core issues will vary over time. As design moves in different directions, the outlooks from designing towards relevant histories also changes. In working with instability rather than solidity, questions rather than definitions can support in finding historical instances that shed light on why certain aspects of designing are difficult to handle given the concepts and methods we have at hand.

Through histories that address the historicity of designing, values and world views embedded in design's foundations can be questioned in terms of their capacity to respond to issues at hand. To what extent can current conceptual foundations support the handling of and responses to emerging challenges and complexities? In order to work towards doing design in ways that make other

17 Daniela K. Rosner, *Critical Fabulations, Reworking the Methods and Margins of Design* (Cambridge: MIT Press, 2018) 124.

18 Redström, *Making Design Theory,* 134-135.

futures possible than ones that are visible from our current perspective of practice, design's conceptual foundations will necessarily need to be different. In tackling issues of living together, sharing resources and making decisions in ways different from those that have been guided by the logics of industrial capitalism, not only does design need to change, but the frameworks and world views governing how it is understood and practiced also need to change. If an awareness of design's historicity can open up other understandings of what is made possible in design – in terms of proposing changes of how design could be done differently – could this, then, in actuality bring about real change through design? And if so, what would become different?

In contemporary and emerging design practices that emphasise the need for design-driven change towards more sustainable futures (eg. participatory design, transition design, design for social innovation), a foundational idea is that power needs to be redistributed and renegotiated in collaborative designing. As transition design is formulated, for example, the basic premise that design must play a substantial role in redirecting human practices towards sustainability on systemic as well as individual scale.[19] Transition design aims to change postures and mindsets, activating participatory design practices in new ways of designing that can support behavioural change on individual levels as well as systemic and values-based changes in order to create conditions for a sustainable and resilient society. The perspectives applied in transition design bring together multiple disciplines and practices, emphasising that transitions towards sustainment are complex processes that take time – and that fundamentally are about actively changing ways of thinking and behaving.

These formulations of what is expected of design in regard to making change echo the programmatic stances of early design theorists' calls for design's central role in bringing about behavioural and societal change. In aiming to bring about ways of designing that take on the challenges of change on systemic societal levels, such as in transition design, negotiating issues of balancing prescriptive and emancipatory stances in design are crucial.[20] These are by no means easy things to address. Bringing the historicity of design concepts to the fore will not in any way resolve these difficulties. But what it can contribute is an awareness of how such negotiations between prescribing and making possible, such as in the contexts described in the prototypes presented here, have been handled historically: and what that has meant in light of what sorts of futures – our current presents – those choices resulted in.

In the current situation, perhaps design interventions that prescribe certain actions over others are what is needed to bring about the sort of transitional change

19 Gideon Kossoff, Terry Irwin, & Anne-Marie Willis, "Transition Design", *Design Philosophy Papers: Transition Design* 2015:1 (vol.13), 1–2; Terry Irwin, "The Emerging Transition Design Approach", *Cuadernos del Centro de Estudios de Diseño y Comunicación* 2019:73; Cameron Tonkinwise, "Design's (Dis)orders: Mediating Systems-Level Transition Design", *Cuadernos del Centro de Estudios de Diseño y Comunicación* 2019:73.

20 Cameron Tonkinwise, "Design for Transitions – from and to What?", *Design Philosophy Papers* 2015:1 (vol.13), 85-92.

that can lead to more sustainable futures. However, bringing a prescriptive design intent to situations of participatory designing – which seems to be an aim several emerging design practices striving for sustainment – means bringing together quite different conceptual foundations for designing. Different complex design decisions and programmatic ambitions will necessarily bring about conflicting agendas on both practical and project levels when we try to design differently. How to go about negotiating these cannot be easily resolved – and will surely sometimes be seen as both normative and reductive by future historians in light of perspectives that we today cannot predict. However, the historical perspectives on what types of negotiations and considerations different conceptual design spaces allow for could probably contribute to seeing what needs to be part of the picture.

Over time, as other design practices emerge, the design histories that resonate with these will also need to change if they are to be meaningful for design. From what we see and where we stand, then, practices of designing will probably call for yet other histories. In some parts, they will build on previous design histories: in other aspects, the histories we make will need to be completely different in order to contribute something for design – and also to the histories of what becomes design. Activating historicity in design through the making of transitional design histories aims towards opening up conceptual spaces for thinking and doing design differently. If histories were made otherwise, the trajectories that lead from the past to the present might take other routes, and help us find other points of view that expand understandings of the present.

These other outlooks will in turn make it possible to think and see other potential futures. In this, prototypical design histories might also help us to understand how our present once might have been an unthinkable future, thus opening up perspectives that could make a difference for how possibilities in design are envisioned: what is reasonable to think of as possible, or as plausible; what that means for what design is called upon to do; and what could be expected for it to achieve.

Pasts, presents, futures

History is made by people. We make it through the way we choose to remember the past, and how we choose to tell stories of it – which is often that which we think of as 'history'. But we make history in many more ways than that. Everything we make and put in the world becomes history that shapes our ideas of the past as well as our understandings of the present. How we think, how we behave, how we relate to each other – in short, how we live our everyday lives and how we make sense of the world – is thoroughly conditioned by the historical materiality of what we have around us.

In a time when humanity's impact on the world has become so profound that we can speak of the world as completely artificial, we can also consider it as completely designed. As Clive Dilnot has pointed out, we now find ourselves in a situation where human activity has brought about a state where it is the artificial that conditions existence – human as well as non-human. Design thus becomes fundamental to approaching the artificial: the propositional is at the core of the artificial, and possibility is what conditions the world.[21]

21 Dilnot, "History, Design, Futures", 181.

But design is not only – or even primarily – about making things that take on material presence in our lives. Even more, design is about proposing that things could be otherwise. It is about proposing that we could do things differently: there could be other things that support us living our lives, but above all, there could be different ways to think about what it means to live life. What we value as important, what relationships we make with others, and what that means for the environments we create is at the core of designing. The ways of living that we can envision are dependent on where we stand, and what we can see from that point of view. If we are to make it possible to see other things, think other thoughts, propose other futures, we need to move to other places that allow for other lines of sight.

The propositional potential of transitional design histories could perhaps lead towards places of sheer wonder at imagining that designing could be seen otherwise. If this can contribute to laying new conceptual foundations for what designing could do for envisioning different futures, history would indeed become a matter of design. Perhaps rethinking and remaking the past in relation to the historicity of designing could spark something similar to Michel Foucault's reaction to a piece of literary fiction that, for him, opened up toward thinking completely differently about exploring the historicity of becoming human:

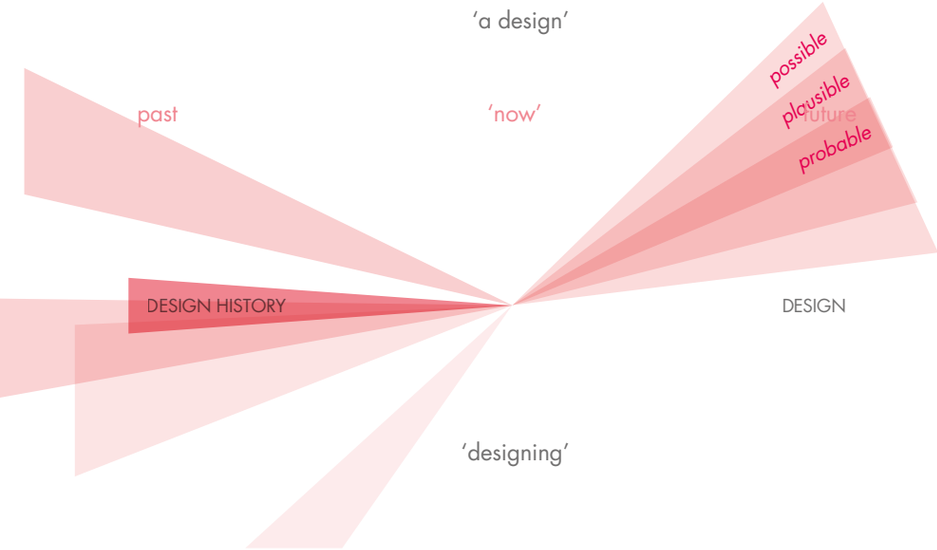
> This book first arose out of a passage in [Jorge Luis] Borges, out of the laughter that shattered, as I read the passage, all the familiar landmarks of my thought—*our* thought that bears the stamp of our age and our geography—breaking up all the ordered surfaces and all the planes with which we are accustomed to tame the wild profusion of existing things, and continuing long afterwards to disturb and threaten with collapse our age-old distinction between the Same and the Other. This passage quotes a 'certain Chinese encyclopaedia' in which it is written that 'animals are divided into: (a) belonging to the Emperor, (b) embalmed, (c) tame, (d) suckling pigs, (e) sirens, (f) fabulous, (g) stray dogs, (h) included in the present classification, (i) frenzied, (j) innumerable, (k) drawn with a very fine camelhair brush, (l) *et cetera,* (m) having just broken the water pitcher, (n) that from a long way off look like flies'. In the wonderment of this taxonomy, the thing we apprehend in one great leap, the thing that, by means of the fable, is demonstrated as the exotic charm of another system of thought, is the limitation of our own, the stark impossibility of thinking *that*.[22]

How to make forms of designing that could re-design the very concepts that set the limitations of our systems of thought require explorations of how to find ways of thinking the impossible as probable. If design histories, transitional and propositional, could contribute to the wonderment of imagining that design could be '*that*' – then we might also open up the conceptual design spaces that allow for designing different ways of being in the world.

Changing how and why design is done, then, is intrinsically linked to the enormous challenge of trying to think, and act, differently from within a framework that is so intricately embedded in most aspects of our lives that we hardly even notice its being there. Internalised categories and ways of thinking shape what is possible to envision through design. Purposely re-forming categories and concepts that often are taken for granted could open up new conceptual spaces for actually making different futures both visible and possible.

22 Michel Foucault, *The Order of Things* (New York: Pantheon, 1970), xv.

With this, then, the proposals for what to take action on in the present, given different trajectories possible to discern from the past to our 'now', would also be different. This view would also continue to change as design's contexts, outlooks, practices and histories change in relation to each other. Different pasts lead to different presents, from which the perspectives on potential futures can be turned in several different directions, depending on where we are able to find footing stable enough to provide a different outlook. Re-positioning the outlook of design history will also bring about a repositioning of the outlooks from presents to futures.

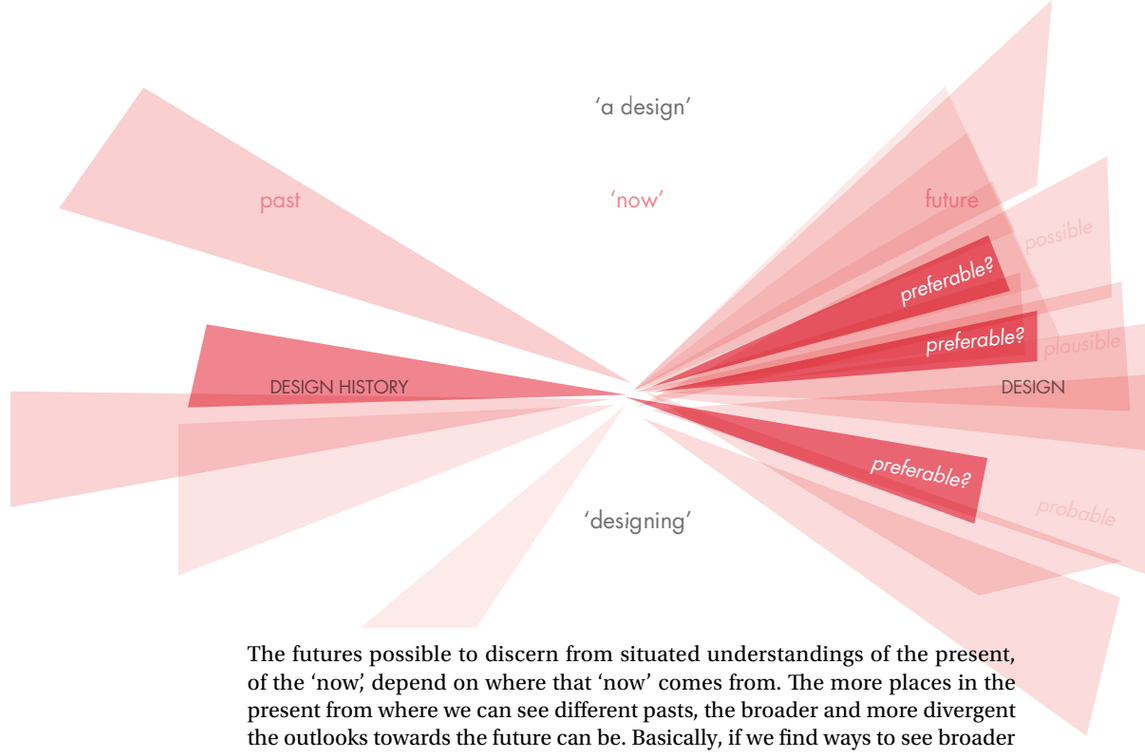


Design histories, therefore, need to be understood as multiple, situated, experiential, relational and contingent. As time passes, and things change, the histories that once were relevant from a certain point of view will no longer suffice to support probing questions and concerns in new contexts.

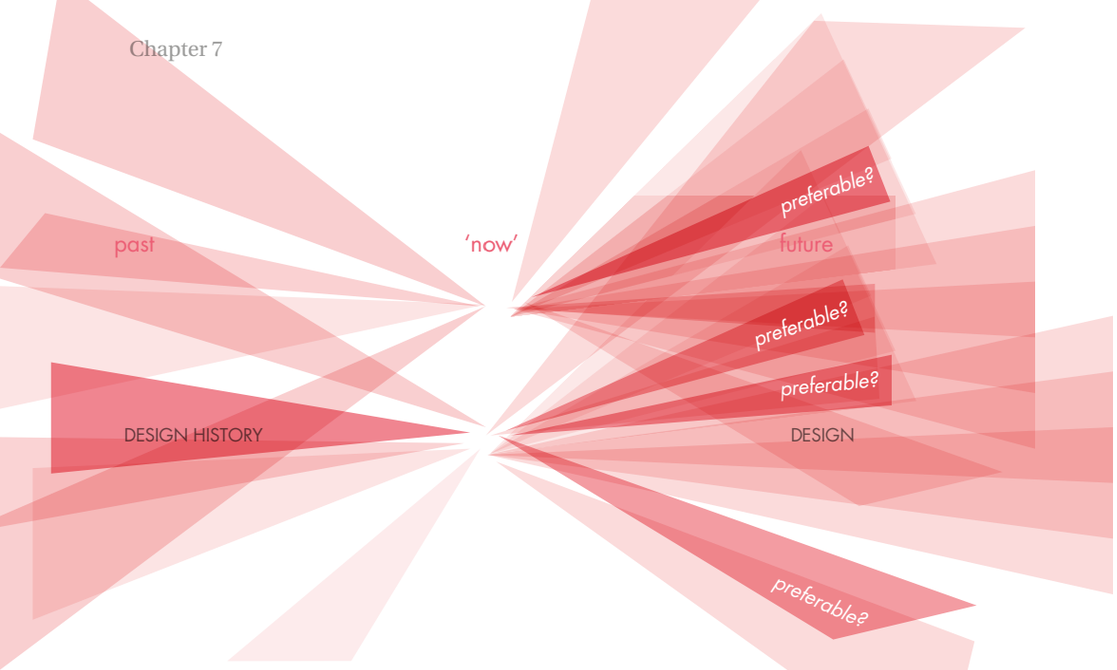
This speaks to the similarities between making design and making history. Proposals of what might be, or of what might have been, are always dependant on a specific situation, and are responses to the conditions of that particular context. A design proposal that becomes realised is only one of many potential things that could have been made, and a history is only one of many potential histories that could have been. Also, in making the choices that result in a certain design or a certain history, what has once been made cannot easily be unmade.

In choosing to make histories and represent them, and in choosing to make designs real, anything we make will inevitably affect other situations – other people, other contexts – than we might have intended and were able to perceive when making decisions. In activating design history in the drawing up of trajectories towards possible futures for design comes responsibilities of ensuring that the outlooks towards what could become encompass as many aspects as possible of how the presently visible is conditioned by the past. Even if we cannot unmake what has once been made, we can at least do our best to avoid repeating

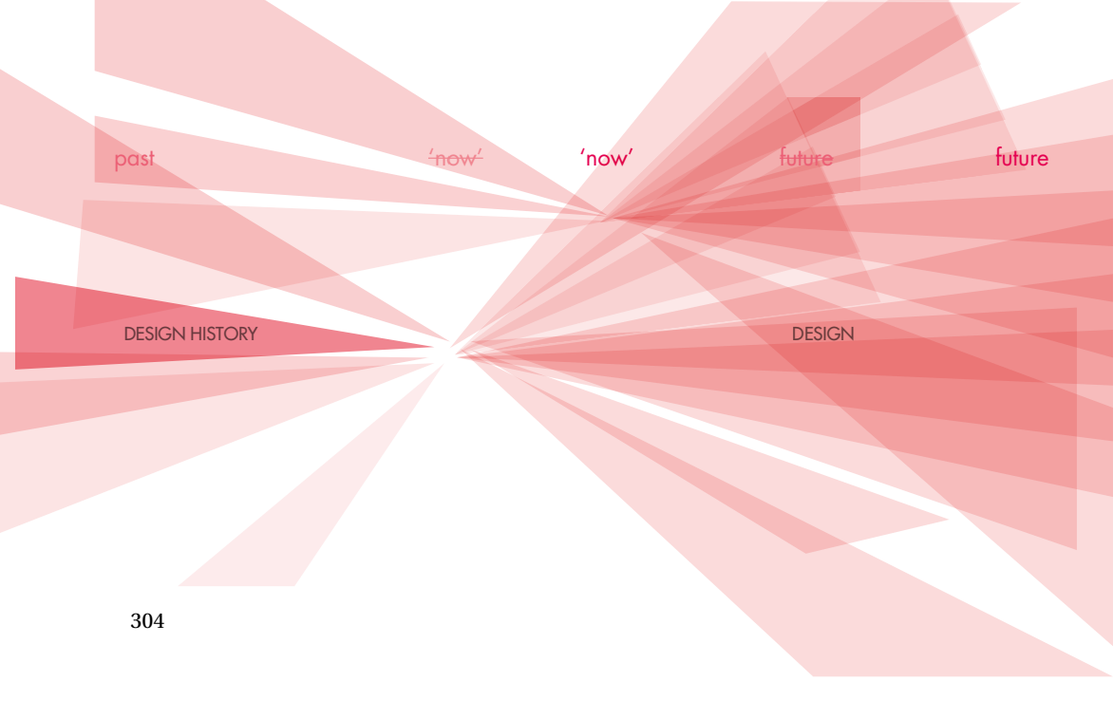
or reinforcing structures and attitudes that further ways of being we actually wish to leave behind. To not end up following trajectories that lead towards defuturing, increased unsustainment, or continued inequity and inequality, the past trajectories that point in those directions need to be challenged through finding other possible histories that point towards other potential ways of moving forward. It matters from where, and with what outlook, we trace our histories, since they are what make possible futures visible to us.



The futures possible to discern from situated understandings of the present, of the 'now', depend on where that 'now' comes from. The more places in the present from where we can see different pasts, the broader and more divergent the outlooks towards the future can be. Basically, if we find ways to see broader possibilities of what design could become in terms of things and situations it could engage with, the more we could articulate different preferable trajectories towards the future. Out of these ideas of what types of futures might be both possible and preferable, we can prototype practices in the present that might take designing in that direction. Of course, there would be not only one 'preferable future' that would open up through making transitional design histories. Depending on the outlook, the historical trajectories would point towards different potentially preferable trajectories that design could take towards the future. But at least there will be perhaps be active decisions that include perspectives of historicity and temporality in design, about what to aim for, to start with, and what to try to avoid. Activating different histories will expand and make a bigger 'now', needed to propose plural potential ways of moving towards other design practices.



If 'we' who design, and who hold accountability for processes and potential outcomes, are only designers or if we are many and diverse, the ways designing happens necessarily change, as do the outlooks towards what could be relevant histories for making preferable futures.[23] And as design always takes place in the present, in a 'now', the future previously envisioned will eventually become the present and the methods for designing that once were new will potentially be both questioned and changed. From there, what becomes visible – in the past, in the present, and as potential futures – as potentials for designing in other ways, will lead to needs for other probing into the conceptual foundations of designing.



Activating awareness of the historicity in design means paying attention to what it means to take action to make change not only in the world, but also in how we make it possible for us to think what the world is. Even if certain of the foundational concepts in design might seem stable and constant, when they are more taken for granted in an a-historical view of designing, they do change – and they *can* be changed. Through making present the assumptions and ideas that form these conceptual spaces for designing, it is also possible to address aspects of historicity of the very ways of thinking that guide the choices of what to do, and how to do it, in design.

What it is that we take for granted and what we challenge in design differs depending on the scale and scope of what we make visible in the process. If the conceptual foundations on which design methods and processes are built begin to increasingly be in conflict with emerging understandings governing situations in which design takes place, it is precisely this that might call for the need to explore this in terms of historicity, to potentially open up new understandings and call new practices of design into being.[24]

Unpacking the ideological contents and historical contexts embedded in current designing could support a more conscious and critical approach in rethinking and developing existing and emerging design practices. It is more than likely that the ways of doing design that have come about in accordance with concepts and ideas linking design to binary solutionism, financial gain, and industrial mass production and mass consumption will not be able to address complex contexts from perspectives of justice, sustainment, and multiplicity.

If ways of designing are to be developed that can build on understandings of the world different than the ones that brought design into being, it is crucial that an awareness of design's historicity can support unlearning and unmaking some of the methods, concepts and processes that designing historically has become built around.[25] This will unavoidably bring about other relationships, other priorities, and thus other dilemmas in designing.

Though history seemingly is about the past, it always has to do with what is relevant and meaningful in the present. The stories we make in the present – the enacted narratives about who 'we' are, what 'we' expect in life, and what futures 'we' aim for – are all shaped by the stories told about the past. Changing the stories we tell about what 'the present' is and where it comes from can support changing how and on what we choose to take action. This is why there needs to be a plurality of perspectives, of stories, and of understandings, that come together in negotiating what design could actually be making possible *now*, and not only in the future. Making things possible, however, is not the same as making things become a reality. The actions and choices that are made based on what could be possible are always anchored in particular ways of thinking and understanding the world – in certain concepts that guide our interpretations, that form our actions, and that make certain paths more likely to be taken than others.

23 Kristina Lindström & Åsa Ståhl, "Becoming response-able stakeholders: Participatory design in times of uncertainties", in *Proceedings of the 14th Participatory Design Conference* (ACM, 2016), 41–44.

24 Joanna Boehnert, "Design vs. the Design Industry", *Design Philosophy Papers* 2104:2 (vol.12), 133.

Making different histories is one way of shifting perspectives not only on, but in, the present. Making many, and other, potential futures come into sight requires creating spaces for a more multi-faceted and diverse 'now'. In making transitional design histories, many potential pasts speak to many potential understandings of what 'now' could be. This making of a bigger 'now' does not mean including as many perspectives as possible. Going to the etymology of the word, to 'include' originally means 'shutting in' or 'imprisoning'. Rather than shutting in diverse perspectives in a position where their outlooks converge into one, the ambition with prototyping multiple pasts is the drawing forth of many possible trajectories, through multiple presents, towards divergent potential futures.



25 John Chris Jones (1980), "...in the dimension of Time." in Jones, *Designing Designing* (London: Architecture design and technology press, 1991), 172

Archives

Arkitekturmuseets arkiv
ArkDes Archive
Svenska Slöjdföreningens arkiv
A10 Redogörelse för Bostadsutredningen, föredrag, artiklar, skrivelser
A50 Bostadsvaneundersökningen
A80 Andra utredningar I
A80II Andra utredningar II (Industriens utredningsinstitut)
Vol. 50 Andra utredningar.
Vol. 57 Plan för bostadsvaneundersökning
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Vol. 63 SAR & SSF Bostadsutredning I: 100 fam i 100 läg: Familjer och lägenheter
Vol. 79 SAR & SSF bostadsutredning I: 100 familjer i 100 lägenheter.
100 fam i 100 läg: Enkät 1-50
100 fam i 100 läg: Enkät 51-100

Gotthard Johanssons arkiv
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Centre for Business History
Föreningen Svensk Form
F1 Registraturen
F3 Övriga utställningar
F6 Handlingar rörande presstjänst
F9 Handlingar rörande kurser och konferenser

Kungliga biblioteket
National Library of Sweden
J10 Tolfternas samkväm för kvinnor ifrån olika yrkesområden
J10:1 Brev från Ellen Key
J10:3 Protokollböcker 1892-1964.
J10:5 "Tolfterna" Program 1901-1964
J10:9 Jubileumsskrift för samkvämet Tolfterna

Nordiska museets arkiv
The Nordiska Museet Archive
Lena Larssons arkiv
F3 Handlingar rörande Svenska slöjdföreningen 1933-1944

Svenska slöjdföreningens arkiv
F5 Bostadsvaneundersökningar
F6 Bäddundersökningen

Riksarkivet
The Swedish National Archives
Hemmens forskningsinstitut
FI a Uppdragsprovningar registrerade 1944-1956
FII a Handlingar rörande hemarbetsundersökningar.
FII b Handlingar rörande barnundersökning
FIV Handlingar rörande organisation
AI:1 Styrelsens protokoll 1944-1957
AII:1 Arbetsutskottets protokoll 1944-1952
A III:1 Fullmäktiges protokoll 1953-1957

Brita Åkermans arkiv [stray folder, kept at Umeå University Library for the time being]
Vol.7. Ämnesordnade handlingar.

Digitalt Museum
digitaltmuseum.se
Bohusläns museum
Grenna Museum– Andréexpeditionen Polarcenter
Nordiska museet
Stadsmuseet i Stockholm
Sundsvalls museum

References

A. Telier [Thomas Binder, Giorgio De Michelis, Pelle Ehn, Giulio Jacucci, Per Linde & Ina Wagner] *Design Things* (Cambridge, MA.: MIT Press, 2011).

Abrahamsson, Kenneth, Eriksson, Lisbeth, Myrstener, Mats, Svensson, Lena & Persson, Tore, *Folkbildning & forskning: Årsbok 2019* (Stockholm: Förening för folkbildningsforskning, 2019).

Adamson, Glenn, Riello, Giorgio & Teasley, Sarah, (eds.) *Global Design History*. (London: Routledge, 2011).

Ahl, Zandra & Olsson, Emma, *Svensk smak: Myter om den moderna formen* (Stockholm: Ordfront, 2001).

Ahmed, Sarah, *What's the Use? On the Uses of Use* (Durham & London: Duke University Press, 2019).

Aktiv Hushållning nr 1 1940, Statens informationsstyrelse (Stockholm, 1940).

Aléx, Peder, *Den rationella konsumenten: KF som folkuppfostrare 1899-1939* diss. (Umeå, 1994).

Aléx, Peder, *Konsumera rätt - ett svenskt ideal: Behov, hushållning och konsumtion* (Lund: Studentlitteratur, 2003).

Albers, Anni, Oral history interview with Anni Albers, 1968 July 5. Smithsonian Archives of American Art. https://www.aaa.si.edu/collections/interviews/oral-history-interview-anni-albers-12134, accessed 2020-06-10.

Almgren, Nina, *Kvinnorörelsen och efterkrigsplaneringen: Statsfeminism i svensk arbetsmarknadspolitik under och kort efter andra världskriget*, diss. (Umeå, 2006).

Ambjörnsson, Ronny, "Om möjligheten av en folkets idéhistoria", *Lychnos: årsbok för idé- och lärdomshistoria. 1983.* (Uppsala: Lärdomshistoriska samfundet, 1983), 158-165.

Ambjörnsson, Ronny, "*Ellen Key: En europeisk intellektuell* (Stockholm: Bonnier, 2012).

Ambjörnsson, Ronny, "Ellen Key and the concept of *Bildung*", *Confero* 2014:1 (vol.2), 133-160.

Andersen, Lars Bo, Danholt, Peter, Halskov, Kim, Hansen, Nicolai Brodersen & Peter Lauritsen, "Participation as a matter of concern in participatory design", *CoDesign* 2015: 3-4(Vol.11), 250-261.

Andersson, Norbert, *Designed in Umeå: Industrial Design at the Umeå Institute of Design,* Sweden (Stockholm: Infobooks, 2009).

Archer, Bruce, "The Three Rs" [1976], *Design Studies* 1979:1 (vol.1), 17-20.

Arendt, Hannah, *The Life of the Mind. One/Thinking. Two/Willing.* [1971/1978] (Harcourt: San Diego, New York, London, 1981).

Ashby, Charlotte, *Modernism in Scandinavia: Art, Architecture and Design* (London: Bloomsbury Academic, 2017).

Asplund, Gunnar, Gahn,Wolter, Markelius, Sven, Paulsson, Gregor, Sundahl,Eskil & Åhrén, Uno, *acceptera* (1931), in Lucy Creagh, Helena Kåberg & Barbara Lane Miller (eds.), *Modern Swedish Design: Three Founding Texts* (New York: The Museum of Modern Art, 2008).

"Att leva med handikapp", *Form* 1968:10.

Attfield, Judy, 'FORM/female FOLLOWS FUNCTION/male: Feminist Critiques of Design', in John A. Walker, *Design History and the History of Design* (London: Pluto Press, 1989), 199–225.

Attfield, Judy, *Utility reassessed: The Role of Ethics in the Practice of Design* (Manchester: Manchester University Press, 1999).

Attfield, Judy, *Wild Things: The Material Culture of Everyday Life* (Berg: Oxford, 2000).

Attfield, Judy, *Bringing Modernity Home: Writings on Popular Design and Material Culture* (Manchester: Manchester University Press, 2007).

Atzmon, Leslie & Boradkar, Prasad (eds.) *Encountering Things: Design and Theories of Things* (London & Oxford: Bloomsbury, 2017).

Aurell, Jaume, "*Rethinking History's* essential tension: between theoretical reflection and practical experimentation", *Rethinking History: The Journal of Theory and Practice* 2018:4 (vol.22), 439-458.

Avle, Seyram, Lindtner, Silvia & Williams, Kaiton, "How Methods Make Designers", in *Proceedings of the 2017 CHI Conference on human factors in computing systems* (ACM, 2017), 472–483.

Ball, Jonathan, "The Double Diamond: A universally accepted depiction of the design process" https://www.designcouncil.org.uk/news-opinion/double-diamond-universally-accepted-depiction-design-process, Retrieved 2020-06-10

Banham, Reyner, "Alternative networks for the alternative culture?" in Cross, Nigel (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972).

Banham, Reyner, *Theory and design in the first machine age (*London: Architectural Press, 1960).

Bannon, Liam J., "From Human Factors to Human Actors: The Role of Psychology and Human-Computer Interaction Studies in System Design" in J. Greenbaum & Morten Kyng (eds.), *Design at Work: Cooperative Design of Computer Systems* (Hillsdale: Lawrence Erlbaum Associates, 1991), 25-44.

Bannon, Liam J. & Ehn, Pelle, "Design: Design Matters in Participatory Design" in Jesper Simonsen & Toni Robertson (eds.) *Routledge International Handbook of Participatory Design* (London: Routledge, 2012).

Basballe, Ditte Amund, Halskov, Kim & Hansen, Nicolai Brodersen, "The Early Shaping of Participatory Design at PDC", in *PDC '16: Proceedings of the 14th Participatory Design Conference: Short Papers, Interactive Exhibitions, Workshops - Volume 2,* Aarhus, Denmark, August 2016, (New York: ACM, 2016) 21–24.

Beecher, Catherine E., *A treatise on domestic economy, for the use of young ladies at home, and at school* (New York, 1848).

Benktzon, Maria & Juhlin, Sven-Eric, *På lättare sätt: En rapport om utveckling av äta-drickaredskap* (Stockholm: Folksam, 1981).

Bergman, Helena & Johansson, Peter, (eds.) *Familjeangelägenheter: Modern historisk forskning om välfärd, genus och politik* (Eslöv: Symposion, 2002).

Berman, Marshall, *All that is Solid Melts into Air*: *The Experience of Modernity* (London: Verso, 1983).

Berner, Boel, "Housewives' Films and the Modern Housewife. Experts, Users and Household Modernization. Sweden in the 1950s and 60s" *History and Technology* 2002:3 (vol.18), 155-179.

Betänkande med förslag rörande lån och årliga bidrag av statsmedel för främjande av bostadsförsörjning för mindre bemedlade barnrika familjer jämte därtill hörande utredningar. Bostadssociala utredningen (Stockholm, 1935).

Binder, Thomas & Redström, Johan, "Exemplary design research", *Design Research Society Conference,* Lisbon, November 2006.

Binder, Thomas, De Michelis, Giorgio, Enh, Pelle, Jacucci, Giulio, Linde, Per & Wagner, Ina, "What is the object of design", *CHI conference proceedings 2012,* (ACM, 2012), 21–30.

Binder, Thomas, Brandt, Eva, Ehn, Pelle & Halse, Joachim, "Democratic design experiments: Between parliament and laboratory", *CoDesign*, 2015:3-4 (vol.11), 152-165.

Bjerknes, Gro & Bratteteig, Tone, "User Participation and Democracy: A Discussion of Scandinavian Research on System Development", *Scandinavian Journal of Information Systems*, 1995:1 (vol.7), 73-98.

Björgvinsson, Erling Bjarki, "Open-ended participatory design as prototypical practice", *CoDesign* 2008:2 (vol.4), 85-99.

Björgvinsson, Erling, Ehn, Pelle & Hillgren, Per-Anders, "Design Things and Design Thinking: Contemporary Participatory Design Challenges", *Design Issues* 2012:3 (vol. 28), 101-116.

Blomberg, Jeanette, Giacomi, Jean, Mosher, Andrea, Swenton-Wall, Pat, "Ethnographic Field Methods and Their Relation to Design" in (eds.) Douglas Schuler and Aki Namioka, *Participatory Design: Principles and Practices* (New York, 1993), 123-156.

Blythe, Mark A.,Monk, Andrew F., Overbeeke, K., Wright, P.C. (eds.), *Funology: From Usability to Enjoyment* (Dordrecht: Springer Science + Business Media, 2005).

Boalt, Carin, "Hemmens forskningsinstitut", in Brita Åkerman et al., *Kunskap för vår vardag. Utbildning och forskning för hemmen* (Akademilitteratur: Stockholm, 1983).

Boehnert, Joanna, "Design vs. the Design Industry", *Design Philosophy Papers,* 2014:2 (vol.12), 119–136.

Botsman, Peter, "Rethinking the CLASS struggle: Industrial democracy and the politics of production", *Economic and Industrial Democracy* 1989:1 (vol 10), 132-133.

Branzi, Andrea. *The Hot House: Italian New Wave Design* (London: Thames & Hudson, 1984).

Brandt, Eva & Binder, Thomas, "Experimental Design Research: Genealogy – Intervention – Argument", *Proceedings of International Association of Societies of Design Research Conference* (Hong Kong, 2007).

Braudel, Fernand, [1958] "Histoire et Sciences Sociales: La longue durée", *Réseaux* 1987:27 (vol.5), 7–37.

Braudel, Fernand, *Civilization and capitalism: 15th-18th century. Vol. 1, The structures of everyday life: The limits of the possible* [New ed.] (London: Collins, 1981).

Brochmann, Odd, (ed.) *Mennesker og boliger: Familieundersøgelsens resultater* (Oslo: J. Grundt Tanum, 1948).

Brochmann, Odd, *Livsform og boligform*, Oslo byes vel's boligundersøkelse (Oslo, 1952).

Brown, Tim & Katz, Barry, *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation* [2009] Revised and updated edition, (New York: HarperBusiness, 2019).

Brunnström, Lasse, (ed.) *Svensk industridesign: En 1900-talshistoria*, (Stockholm: Norstedt, 1997).

Brunnström, Lasse, "Hjälpmedel för ett säkrare och jämlikare liv" in Lasse Brunnström (ed) *Svensk industridesign: En 1900-talshistoria*, (Stockholm: Norstedt, 1997).

Brunnström, Lasse, *Swedish Design: A History* (London: Bloomsbury, 2019).

Buckley, Cheryl, 'Made in Patriarchy: Towards a Feminist Analysis of Women and Design,' *Design Issues,* 1986:2, 3-14.

Buckley, Cheryl, "Made in Patriarchy: Theories of women and design - a reworking" in Hazel Clark (ed.) *Design Studies: A Reader* (Oxford: Berg, 2009), 233-238.

"Bättre och vackrare men inte dyrare: Husmödrar diskuterar funktion och form", *Form* 1946, 65-67.

Bødker, Susanne & Buur, Jacob, "The Design Collaboratorium – a Place for Usability Design", ACM Transactions on Computer-Human Interaction 2002:2 (vol.9).

Bødker, Susanne, Dindler, Christian & Iversen Ole Sejer, "Tying Knots: Participatory Infrastructuring at Work", *CSCW* 2017:26, 245-273.

Bødker, Susanne, Ehn, Pelle, Romberger, Staffan & Sjögren, Dan (eds.), *The UTOPIA Project: An Alternative in Text and Images. Graffitti: Informationsblad från UTOPIA-projektet* nr. 7 (Aarhus & Stockholm, 1985).

Bødker, Susanne & Kyng, Morten, "Participatory design that matters – facing the big issues", *ACM Transactions on Computer-Human Interaction* 2018:1 (vol. 25), 1-21.

Carlsson, Allan, *The roles of Alva and Gunnar Myrdal in the development of a social democratic response to Europe's "population crisis", 1929-1938*, diss. (Ann Arbor, 1978).

Childs, Marquis, *Sweden: The Middle Way* (New Haven: Yale Univ. Press, 1936).

Clark, Hazel & Brody, David (eds.), *Design Studies: A Reader* (Berg: Oxford & New York, 2009).

Clarke, Alison J., *Tupperware: The Promise of Plastic in 1950s America* (Washington: Smithsonian Inst. Press, 1999).

Clarke, Alison J., "Design for Development, ICSID and UNIDO: The Anthropological Turn in 1970s Design", *Journal of Design* History 2015:1 (vol.29), 43-57.

Clarke, Alison J., *Design Anthropology: Object Cultures in Transition*, revised edition (London: Bloomsbury, 2017).

Conrads, Ulrich (ed.), *Programs and Manifestoes on 20th Century Architecture* (Cambridge: MIT Press, 1971).

Creagh, Lucy, Kåberg, Helena & Lane Miller, Barbara (eds.), *Modern Swedish Design: Three Founding Texts* (New York: The Museum of Modern Art, 2008).

Cross, Nigel, "Here comes everyman!", in Cross, Nigel (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972).

Cross, Nigel, (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972).

Cross, Nigel, "A History of Design Methodology", in M.J. de Vries et al. (eds.), *Design Methodology and Relationships with Science* (Kluwer Academic Publishers, 1993).

Cross, Nigel, "Designerly ways of knowing: design discipline versus design science", *Design Issues* 2001:3 (vol.17), 49-55.

Cross, Nigel, *Design thinking: Understanding how Designers Think and Work* (Oxford: Berg, 2011).

Curman, Jöran, "Industriens bostadsutredning" *Form* 1943:7-8.

Curman, Jörgen, *Industriens arbetarebostäder* (Stockholm: Industriens utredningsinstitut, 1944).

Design Research Society, Publications, Conference proceedings, https://www.designresearchsociety.org/cpages/publications-1, Accessed 2019-10-28.

Dikovitskaya, Margarita, *Visual culture: The study of the visual after the cultural turn* (Cambridge,MA: MIT Press, 2015).

Dilnot, Clive, "The State of Design History, part I: Mapping the Field," *Design Issues* 1984:1, 4-23.

Dilnot, Clive, 'The State of Design History, Part II: Problems and Possibilities", *Design Issues,* 1984:2 (vol.1), 3-20.

Dilnot Clive, "Some Futures for Design History?", *Journal of Design History* 2009:4 (vol. 22), 377-394

Dilnot, Clive, "History, Design, Futures: Contending with what we have made", in Tony Fry, Clive Dilnot & Susanne Stewart (eds.), *Design and the Question of History* (London/New York: Bloomsbury, 2015).

Dilnot, Clive, "The Matter of Design", *Design Philosophy Papers* 2015:2 (vol. 13), 115-123.

DiSalvo, Betsy, Yip, Jason, Bonsignore, Elizabeth & DiSalvo, Carl, (eds.) *Participatory Design for Learning: Perspectives from Practice and Research* (New York: Routledge, 2017).

DiSalvo, Carl, "The need for design history in HCI", *Interactions* 2014:6 (vol.21), 20–21.

DiSalvo, Carl, *Adversarial Design*, (Cambridge: MIT Press, 2012).

DiSalvo, Carl, Clement, Andrew & Pipek, Volkmar, "Communities: Participatory Design for, with, and by communities" in Simonsen, Jesper & Robertson, Toni, (eds.) *Routledge International Handbook of Participatory Design* (New York: Routledge, 2013).

Doordan, Dennis, "On history", *Design Issues* 1995:1 (vol.11), 76-81.

Dreyfuss, Henry, *Designing for People* [1955] new edition, (New York: Paragraphic, 1967).

Drain, Andrew & Sanders, Elizabeth B.-N, "A Collaboration System Model for Planning and Evaluating Participatory Design Projects", *International Journal of Design*2019:3 (vol. 15), 39-52.

Drew, Cat, "The Double Diamond: 15 years on", https://www.designcouncil.org.uk/news-opinion/double-diamond-15-years, accessed 2020-06-10.

Dunne, Anthony & Raby, Fiona, *Speculative Everything: Design, Fiction and Social Dreaming* (Cambridge, MA: MIT Press, 2013).

"Editorial policy", *Journal of Design History* 2019:4 (vol.32).

Edqvist, Samuel, "Det pragmatiska folkbildningbegreppet", in Kerstin Rydbäck & Henrik Nordvall (eds.) *Perspektiv på folkbildning. Fyra forskares tankar om folkligt bildningsarbete*, Mimer: Nationellt program för folkbildningsforskning (Linköping: Linköping University, 2015).

Ehn, Pelle, *Work-Oriented Design of Computer Artifacts,* diss. (Stockholm: Arbetslivscentrum, 1988).

Ehn, Pelle, "Participation in Design Things" in *Proceedings of Participatory Design Conference 2008* (Indiana University, 2008), 92-101.

Ehn, Pelle, "Learning in participatory design as I found it (1970-2015)" in Betsy DiSalvo, Jason Yip, Elizabeth Bonsignore & Carl DiSalvo (eds.), *Participatory Design for Learning: Perspectives from Practice and Research* (New York: Routledge, 2017).

Ehn, Pelle, Nilsson, Elisabet M. & Topgaard, Richard, (eds.) *Making Futures: Marginal Notes on Innovation, Design, and Democracy* (Cambridge, MA: The MIT Press, 2014).

Eliaeson, Sven & Kalleberg, Ragnvald, (eds.) *Academics as public intellectuals* (Newcastle, UK: Cambridge Scholars Publishing, 2008).

Eriksson, Eva, *Den moderna staden tar form: Arkitektur och debatt 1910-1935* (Stockholm: Ordfront, 2001).

Escobar, Arturo, *Designs for the pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds* (Durham: Duke University Press, 2018).

Eskilsson, Lena, "En arena för möten mellan människor och idéer: kvinnliga medborgarskolan på Fogelstad" in Erland Mårald & Christer Nordlund (eds.), *Topos: essäer om tänkvärda platser och platsbundna tankar* (Stockholm: Carlsson, 2006), 167-182.

Evans, Richard J., *Altered Pasts: Counterfactuals in History* (Waltham, MA: Brandeis University Press/Historical Society of Israel, 2013).

Fallan, Kjetil, *Design History: Understanding Theory and Method* (Oxford: Berg, 2010).

Fallan, Kjetil, "The 'Designer' – The Eleventh Plague": Design Discourse from Consumer Activism to Environmentalism in 1960s Norway", *Design Issues* 2011:4 (vol.27), 30-42.

Fallan, Kjetil (ed.), *Scandinavian Design: Alternative Histories* (London: Bloomsbury, 2012).

Fallan, Kjetil "De-tooling Design History: To What Purpose and for Whom Do We Write?", *Design and Culture*, 2013:1 (vol.5), 13-19.

Fallan, Kjetil, *Designing modern Norway: A history of design discourse* (London: Routledge, 2016).

Fallan, Kjetil & Jørgensen, Finn Arne, "Environmental Histories of Design: Towards a New Research Agenda", *Journal of Design History* 2017:2 (vol.30), 103-121.

Fallan, Kjetil & Lees- Maffei, Grace, (eds.) *Designing Worlds: National Design Histories in an Age of Globalization*. (New York & London: Berghahn Books, 2016).

Feo, Alberto, "Operational games applied to socio-technical problems", in Cross, Nigel (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972).

Fiell, Charlotte & Fiell, Peter, *1000 chairs* (Köln: Taschen, 1997).

Fiell, Charlotte & Fiell, Peter, *Scandinavian Design* (Köln & London: Taschen, 2002).

Flinchum, Russell, "Dreyfuss, design and human factors", *Ergonomics in Design* 2000:1 (vol.8), 18-24.

Floré, Fredie, *Lessen in modern wonen: Bronnenboek over woontentoonstellingen in België 1945-1958* (Gent: WZW, 2004).

Florin, Christina & Kvarnström, Lars, *Kvinnor på gränsen till medborgarskap: Genus, politik och offentlighet 1800-1950* (Stockholm: Atlas, 2001).

Floyd, C., Züllighoven, H., Budde, R., and Keil-Slawik, R., (eds.), *Software Development and Reality Construction* (Berlin: Springer-Verlag, 1992).

Forlano, Laura, Steenson, Molly Wright, Ananny, Mike, Mazé, Ramia, Turner, Fred, Ito, Joichi, Saltiel, Natalie, Uhl, Andre, Strohecker, Carol & Candy, Stuart, *Bauhaus Futures* (US: The Mit Press, 2019).

Forlizzi, Jodi, "Moving beyond user-centred design" *Interactions* 2018:5 (vol.25), 22-23.

Forty, Adrian, *Objects of Desire: Design and Society from Wedgwood to IBM* (Pantheon: New York, 1986).

Forty, Adrian, "DEBATE: a reply to Victor Margolin" *Design Issues* 1995:1 (vol 11), 16-18.

Foucault, Michel, *L'archéologie du savoir* (Paris: Gallimard, 1969).

Foucault, Michel, *The Order of Things* (New York: Pantheon, 1970).

Foucault, Michel, [1976] *Madness: The invention of an idea.* (New York: Harper, 2011).

Foucault, Michel [1975] *Discipline and punish: The birth of the prison* (New York: Vintage Books, 1979).

Frayling, Christopher, *Research in art and design, Royal College of Art Research Papers* 1993/94:1 (vol.1) (London: Royal College of Art, 1993).

Fredericks, Christine, *The New Housekeeping: Efficiency Studies in Home Management* (Garden City, N.Y., 1918);

Frick, Gunilla, *Svenska slöjdföreningen och konstindustrin före 1905*, Nordiska museets handlingar 91, diss. (Uppsala, 1978).

Fry, Tony, *A New Design Philosophy: An Introduction to Defuturing* (Sydney: UNSW Press, 1999).

Fry, Tony, *Design Futuring: Sustainability, Ethics, and New Practice* (Berg: Oxford, 2009).

Fry, Tony, "Whither design, whether history" in Tony Fry, Clive Dilnot & Susan C. Stewart *Design and the Question of History* (Bloomsbury: London & New York, 2015).

Fry, Tony, Dilnot, Clive & Stewart, Susanne, (eds.) *Design and the Question of History* (London/New York: Bloomsbury, 2015).

Fry, Tony & Willis, Anne-Marie, *Steel: A Design, Cultural, and Ecological History* (London & New York: Bloomsbury Academic, 2015).

Fröshaug, Anthony, "Visual Methodology", *ulm 4, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, 1959:4.

Fuad-Luke, Alastair, *Design activism: Beautiful strangeness for a sustainable world* (London: Earthscan, 2009).

Fuad-Luke, Alastair, "Design(-ing) for radical relationality: 'Relational Design' for Confronting Dangerous, Concurrent, Contingent Realities" in (eds.) Ma Jin & Lou Yongqi, *Emerging Practices: Professions, Values and Approaches in Design* (Shanghai: Tongji University Press, 2014), 42-72.

Gaziulusoy, İ dil A. & Ryan, Chris, "Shifting Conversations for Sustainability Transitions Using Participatory Design Visioning" *The Design Journal.* 2017 (vol.20, sup1), S1916–S1926.

Gedin, Per I., *När Sverige blev modernt: Gregor Paulsson, vackrare vardagsvara, funktionalismen och Stockholmsutställningen 1930* (Stockholm: Bonniers, 2018).

Gerrard, Victoria & Ricardo, Sosa, "Examining Participation", *PDC '14: Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1,* Windhoek, Namibia, October 2014 (New York: ACM, 2014), 111-120.

Gerritsen, Anne & Riello, Giorgio, *Writing Material Culture History* (London: Bloomsbury Academic, 2015).

Gerschenkron, Alexander, "On the Concept of Continuity in History", *Proceedings of the American Philosophical Society* 1962: 3 (vol. 106), 195-209.

Giaccardi, Elisa, "Metadesign as an emergent design culture", *Leonardo* 2005:4 (vol.38), 342-349.

Giaccardi, Elisa & Redström, Johan, "Technology and More-Than-Human Design", *Design Issues* 2020:4 (vol.36) [forthcoming].

Gideion, Sigfried, *Mechanization takes command: A contribution to anonymous history* (New York: Oxford University Press, 1948).

Ginzburg, Carlo, *Il formaggio e i vermi: il cosmo di un mugnaio del '500.* (Torino: Einaudi, 1982).

Gooding, Joanne, *Design history in Britain from the 1970s to 2012: Context, formation, and development,* diss. (University of Northumbria: Newcastle, 2012).

Goodman, Elizabeth, Kuniavsky, Mike & Moed, Andrea, *Observing the User Experience: A Practitioner's Guide to User Research*, 2nd edition (Waltham, MA: Morgan Kaufmann, 2012).

Greenbaum, Joan M. & Kyng, Morten (eds.*), Design at Work: Cooperative Design of Computer Systems* (Hillsdale: Lawrence Erlbaum Associates, 1991).

Gregory, Judith, "Scandinavian approaches to participatory design", *International Journal of Engineering Education* 2003:1 (vol 19), 62-74.

Gregory, Sydney A., *The Design Method* (Butterworths: London, 1966).

Gropius, Walter, "Principles of Bauhaus production [Dessau]" in Ulrich Conrads (ed.), *Programs and Manifestoes on 20th Century Architecture* (Cambridge: MIT Press, 1971).

Gunn, Wendy, Otto, Ton & Smith, Rachel Charlotte, *Design Anthropology: Theory and Practice* (London: Bloomsbury, 2013).

Göransdotter, Maria, "Smakfostran och heminredning: Om estetiska diskurser och bildning till bättre boende i Sverige 1930-1955", Johan Söderberg & Lars Magnusson (eds.), *Kultur och konsumtion i Norden 1750-1950* (Helsinki: Finska Historiska Samfundet 1997), 253-274.

Göransdotter, Maria, "Ting, tecken, text: Om semiotik och smakfostran", in Brita Brenna & Karen M. Fjeldstad (eds.), *Kollektive identiteter, ting og betydninger*(Oslo: Senter for teknologi og menneskelige verdier, 1997), 49-62.

Göransdotter, Maria, "Möbleringsfrågan: Om synen på heminredning i 1930- och 1940-talens bostadsvaneundersökningar", *Historisk tidskrift* 1999:3, 444-474.

Göransdotter, Maria, "Från moral till modernitet: Hemmet i 1930-talets svenska funktionalism", *Nordisk arkitekturforskning* 1999:2, 45-54.

Göransdotter, Maria & Redström, Johan, "Design methods and critical historiography: An example from Swedish user-centered design", *Design Issues* 2018:2 (vol.34), 20-30.

Halén, Widar & Wickman, Kerstin, (eds.) *Scandinavian design beyond the myth: Fifty years of design from the Nordic countries* (Stockholm: Arvinius, 2003).

Halskov, Kim & Hansen, Nicolai Brodersen, "The diversity of participatory design research practice at PDC 2002–2012", *International Journal of Human - Computer Studies*, 2015 (vol.74*)*, 81-92.

Hansson, Per Albin, "Folkhemmet, medborgarhemmet" in Per Albin Hansson, *Från Fram till Folkhemmet: Per Albin Hansson som tidningsman och talare* (Solna: Metodica, 1982).

Hansson, Per Albin, *Från Fram till Folkhemmet: Per Albin Hansson som tidningsman och talare* (Solna: Metodica, 1982).

Haraway, Donna, *Staying with the Trouble: Making Kin in the Chthulucene*, (Durham: Duke University Press, 2016).

Hatje, Ann-Katrin, *Befolkningsfrågan och välfärden: Debatten om familjepolitik och nativitetsökning under 1930- och 1940-talen*, diss. (Stockholm, 1974).

Hatje,Ann-Katrin, *Från treklang till triangeldrama: Barnträdgården som ett kvinnligt samhällsprojekt under 1880-1940-talen* (Lund: Historiska media, 1999).

Hedqvist, Hedvig, *Svensk form, internationell design* (Stockholm: Bokförlaget DN, 2007).

HFI-meddelanden, Hemmens forskningsinstitut, Stockholm, 1946-1956.

Highmore, Ben, (ed.) *The Design Culture Reader* (Routledge: New York, 2009).

Hillgren, Per-Anders, Seravalli, Anna & Emilson, Anders, "Prototyping and infrastructuring in design for social innovation", *CoDesign* 2011:3-4, vol.7, 169-183.

Hirdman, Yvonne, *Att lägga livet tillrätta: Studier i svensk folkhemspolitik*, Maktutredningens publikationer (Stockholm: Carlsson, 1989).

Hirdman Yvonne "Utopia in the Home: An Essay" *International Journal of Political Economy* 1992:22, 5-99.

Hirdman, Yvonne, Björkman, Jenny & Lundberg, Urban, *Sveriges historia 1920-1965: Välfärdens tid* (Stockholm: Nordstedt, 2012).

Holmes, Marcia E., *Performing proficiency: Applied experimental psychology and the human engineering of air defense, 1940-1965*, diss. (The University of Chicago, 2014).

Hort, Sven E.O., *Social policy, welfare state, and civil society in Sweden Vol. 1 History, policies, and institutions 1884-1988*, (Lund: Arkiv, 2014).

Hugoson, Rolf, "Clarifying Liquidity: Keynes and Marx, Merchants, and Poets", *Contributions to The History of Concepts*, 2019:2 (vol.14) 46–66.

Huldt, Åke H. & Waern Bugge, Ingeborg, *Bosättning* (Stockholm, 1944).

Hunt, Lynn, *Measuring Time, Making History* (New York: Central European University Press, 2008).

Huppatz, D.J., *Design: The Key Concepts* (London & New York: Bloomsbury, 2020).

Huppatz, DJ & Lees-Maffei, Grace, "Why design history? A multi-national perspective on the state and purpose of the field", *Arts & Humanities in Higher Education* 2012:12 (2–3), 310–330.

Huppatz, Daniel J., "Globalizing Design History and Global Design History", *Journal of Design History* 2015:2 (vol.28) 182–202.

Huppatz, D.J., "Revisiting Herbert Simon's 'Science of Design'", *Design Issues* 2015:2 (vol.31), 29–40.

Huppatz, D J, "Introduction to Methodology: Virtual Special Issue for the Journal of Design History 2018", *Journal of Design History*, 2020:1 (vol.33), e25–e40, https://doi-org.proxy.ub.umu.se/10.1093/jdh/epy021, Accessed 2020-05-31.

Huvudkatalog. Stockholmsutställningen 1930. (Stockholm, 1930).

Huybrechts, Liesbeth, Hendriks, Niels & Martens, Sarah, "Counterfactual scripting: challenging the temporality of participation" *CoDesign: Special issue: Participatory Design in an Era of Participation*. Guest editors: Rachel Charlotte Smith, Claus Bossen and Anne Marie Kanstrup. 2017:2 (vol.13), 96–109.

Huybrechts, Liesbeth, Tassinari, Virginia, Roosen, Barbara & Constantinescu, Teodora, "Work, labour and action: the role of participatory design in (re)activating the political dimension of work", in *Proceedings of the 15th Participatory Design Conference. i.* (ACM, 2018), 1–11.

Håkansson Petré, Lisbeth, *Tisdagar med Tolfterna: Nätverkande kvinnor i sekelskiftets Stockholm* (Stockholm: Stockholmia, 2019).

Hård, Mikael, "The good apartment: The social (democratic) construction of Swedish homes", *Home Cultures* 2010:2 (vol.7), 117-134.

IDEO, "Design thinking defined", https://designthinking.ideo.com, accessed 2020-06-10.

IDEO, "Human Centered Design Kit", https://www.ideo.com/post/design-kit, accessed 2020-06-10.

"Industrial Design Department", *ulm 1, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, 1958:1.

Irwin, Terry, "The Emerging Transition Design Approach", *Cuadernos del Centro de Estudios de Diseño y Comunicación* 2019:73.

Ivanov, Gunnela, *Vackrare vardagsvara – design för alla? Gregor Paulsson och Svenska slöjdföreningen 1915-1925*, diss. (Umeå, 2004).

Johansson, Gotthard, "Hur bor vårt folk? Bostadsvanorna utforskas" *Form* 1945:1, 8-16.

Johansson, Gotthard (ed.), *Bostadsvanor och bostadsnormer. Svenska Arkitekters Riksförbunds och Svenska Slöjdföreningens Bostadsutredning. Dwelling Habits and Housing Norms. The Housing Investigation of the National Association of Swedish Architects and The Swedish Society of Industrial Design* (Stockholm: Kooperativa förbundets bokförlag, 1964).

Jones, John Chris, "Ergonomics, human data for design", *Design* 1954:66, 13-17.

Jones, John Chris, "Handles: The Ergonomic Approach", *Design* 1954:72, 34-38.

Jones, John Chris, *Design Methods: Seeds of human futures* (London: John Wiley & Sons, 1970).

Jones, John Chris, *Design Methods* [1970/1981] 2nd edition (New York: Van Nostrand Reinhold, 1992).

Jones, John Chris, *Designing Designing* (London: Architecture design and technology press, 1991).

Jones, John Chris, [1973] "Love, hate and architecture. or what is a method? Memories of a seminar in Bologna" in John Chris Jones, *Designing Designing* (London: Architecture design and technology press, 1991).

Jones, John Chris, [1974] "how my thoughts about design methods have changed during the years" in John Chris Jones, *Designing Designing* (London: Architecture design and technology press, 1991).

Jones, John Chris, [1977] "now we are numerous. introduction to the Spanish edition of Design Methods" in *Designing Designing* (Architecture design and technology press: London, 1991).

Jones, John Chris, [1978] "opus one, number two" in John Chris Jones, *Designing Designing* (Architecture design and technology press: London, 1991).

Jones, John Chris, "1980 edition: a review of new topics" in John Chris Jones, *Design Methods* [1970/1981] 2nd edition (New York: Van Nostrand Reinhold, 1992).

Jones, John Chris [1980] "…in the dimension of Time." in John Chris Jones, *Designing Designing* (London: Architecture design and technology press, 1991).

Jones, J. Christopher & Jungk, Robert, "Closing comments" in Cross, Nigel (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972).

Jones, J. Christopher & Thornley, D.G., *Conference on design methods. Papers presented at the Conference on Systematic and Intuitive Methods in Engineering, Industrial Design, Architecture and Communications. London, September 1962.* (London: Pergamon Press, 1963).

Joost, Gesche, Bredies, Katharina, Christensen, Michelle, Conradi, Florian & Unteidig, Andreas, (eds.) *Design as Research: Positions, Arguments, Perspectives* (Basel: Birkhäuse, 2016).

Jørgensen, Finn Arne, "Frankfurt Kitchen, Germany (Grete Lihotzky, 1926)", in Grace Lees Maffei (ed.) Iconic designs: 50 stories about 50 things (London: Bloomsbury, 2014), 167–164.

Julier, Guy, Munch, Anders V., Folkmann, Mads Nygaard, Jensen, Hans-Christian & Skou, Niels Peter, (eds.) *Design Culture: Objects and Approaches* (London: Bloomsbury Visual Arts, 2019).

Kaufmann-Buhler, Jennifer, Pass, Victoria Rose & Wilson, Christopher S., (eds.) *Design History Beyond the Canon* (London & New York: Bloomsbury, 2019).

Keinonen, Turkka, *Designers, Users and Justice* (Bloomsbury: London & New York, 2017).

Kelly, Michael J. & Arthur, Rose, (eds.) *Theories of History: History Read Across the Humanities* (London: Bloomsbury, 2018).

Keshavarz, Mahmoud & Mazé, Ramia, "Design and Dissensus: Framing and Staging Participation in Design Research", *Design Philosophy Papers* 2013:1 (vol.11), 7–29.

Key, Ellen, "Skönhet i hemmen"[1897], in *Skönhet för alla* [1913] (Stockholm: Rekolid, 1996).

Key, Ellen, "Festvanor" [1896] in *Skönhet för alla* [1913] (Stockholm: Rekolid, 1996).

Key, Ellen, "Skönhet", in *Tankebilder I* (1898), 3.rd edition (Albert Bonniers förlag: Stockholm, 1922).

Key, Ellen, "Beauty in the Home" [1897], in Lucy Creagh, Helena Kåberg & Barbara Miller Lane (eds.), *Modern Swedish Design: Three Founding Texts* (The Museum of Modern Art: New York, 2008).

Key, Ellen, *Barnets århundrade* (Stockholm: Bonniers, 1900).

Key, Ellen, *Folkbildningsarbetet. Särskildt med hänsyn till skönhetssinnets odling: en återblick och några framtidsönskningar* (Uppsala: Appelberg, 1906).

Key, Ellen, *The Century of the Child* (New York & London: Putnam, 1909).

Key, Ellen, *The Morality of Woman and Other Essays,* (Chicago, 1911).

Key, Ellen, *Love and Ethics* (New York, 1911).

Key, Ellen, *The Woman Movement. Transl. by Mamah Bouton Borthwick. With an introd. by Havelock Ellis* (New York, 1912).

Key, Ellen, *War, Peace, and the Future: A Consideration of Nationalism and Internationalism, and of the Relation of Women to War*. *Transl. by Hildegard Norberg* (New York & London, 1916).

Key, Ellen, *Bildning: Några synpunkter*. Studentföreningen Verdandis småskrifter 67 (Bonniers: Stockholm, 1919).

Kirkham, Pat, & Weber, Susan, (eds.) *History of Design: Decorative Arts and Material Culture, 1400–2000.* (New Haven, CT and New York: Yale University Press and Bard Graduate Center, 2013).

Kinchin, Juliet & O'Connor, Aidan, *Counter Space: Design and the Modern Kitchen* (New York: Museum of Modern Art, 2011).

Kjellberg, Anders, "Från industriell demokrati till medbestämmande - fackliga utvecklingslinjer 1917-1980", *Arkiv för studier i arbetarrörelsens historia* 1981:21-22, 53-82.

Korsgaard, Ove, *Kampen om lyset: Dansk voksenoplysning gennem 500 år* (København: Gyldendal, 1997).

Korvenmaa, Pekka, *Finnish design: A concise history* (Helsinki: University of Art and Design Helsinki, 2009).

Koselleck, Reinhart, "A response to comments on the *Geschichtliche Grundbergriffe*", in H. Lehmann & M. Richter (eds.) *The Meanings of Historical Terms and Concepts: New Studies on Begriffsgeschichte*. (Washington, DC: German Historical Institute, 1996), 59–70.

Koselleck, Reinhart, "The temporalization of utopia" in Reinhart Koselleck, *The Practice of Conceptual History: Timing History, Spacing Concepts* (Stanford, CA: Stanford University Press, 2002), 84–99.

Koselleck, Reinhart& Presner, Todd Samuel, *The Practice of Conceptual History. Timing History, Spacing Concepts* (Stanford CA: Stanford University Press, 2002).

Koselleck, Reinhard, "Modernity and the planes of historicity", in Reinhard Koselleck, *Futures Past: On the Semantics of Historical Time* (New York: Columbia University Press, 2004), 9-25.

Koselleck, Reinhard, *Futures Past: On the Semantics of Historical Time* (New York: Columbia University Press, 2004)

Koskinen, Ilpo, Zimmerman, John, Binder, Thomas, Redström, Johan & Wensveen, Stephan, *Design Research through Practice: From the Lab, Field, and Showroom* (Waltham, MA: Morgan Kaufmann, 2011).

Kossoff, Gideon, Irwin, Terry & Willis, Anne-Marie, "Transition Design", *Design Philosophy Papers: Transition Design* 2015:1 (vol.13), 1-2.

Kries, Mateo, Klein, Amelia & Clarke, Alison J., (eds.) *Victor Papanek: The Politics of Design* (Weil am Rhein: Vitra Design Museum, 2018).

Kuenen, Stoffel, *Aesthetics of Being Together*, diss., (Umeå: Umeå University, 2018).

Kuijer, Lenneke & Giaccardi, Elisa,"Co-performance: Conceptualizing the Role of Artificial Agency in the Design of Everyday Life", *Proceedings of the 2018 CHI Conference on human factors in computing systems*. (ACM, 2018), 1–13.

"Kök för invalidiserade husmödrar", *HFI-meddelanden* 1951:5.

"The Lancaster Care Charter", *Design Issues* 2019:1 (vol.35), 73–77.

Larsson, Anna & Magdalenić, Sanja, *Sociology in Sweden: A History* (Basingstoke, Hampshire: Palgrave Macmillan, 2015).

Larsson, Lena, "På visit hos 100 familjer", *Form* 1945:1, 12-16.

Le Dantec, Christopher A & DiSalvo, Carl, "Infrastructuring and the formation of publics in participatory design", *Social Studies of Science* 2013:43(2), 241-264.

Le Goff, Jaques, "Les mentalités: Une histoire ambiguë" in Jaques Le Goff & Pierre Nora (eds.) *Faire de l'historie. III. Nouveaux objets* (Gallimard, 1974), 77-94.

Le Goff, Jaques & Nora, Pierre (eds.) *Faire de l'historie. I. Nouvelles Approches.* (Gallimard, 1974).

Le Goff, Jaques & Nora, Pierre (eds.) *Faire de l'historie. II. Nouveaux problemes* (Gallimard, 1974).

Le Goff, Jaques & Nora, Pierre (eds.) *Faire de l'historie. III. Nouveaux objets* (Gallimard, 1974).

Le Roy Ladurie, Emmanuel, *Montaillou: Cathars and catholics in a French village 1294-1324* (London: Scolar, 1978).

Leander, Sigfrid, *Folkbildningens födelse: Anton Nyström och Stockholms arbetareinstitut 1880-1980* (Stockholm: Nordiska museet, 1980).

Lees-Maffei, Grace, "From service to self-service: advice literature as design discourse, 1920-1970", *Journal of design history* 2001: 3 (Vol.14), 187-206.

Lees-Maffei, Grace, Modern living? Domestic advice literature and design discourse in post-war Britain, diss. (Portsmouth: University of Portsmouth, 2005).

Lees-Maffei, Grace, "The Production-Consumption-Mediation Paradigm", *Journal of Design History* 2009:4 (vol.22), 351-376.

Lees-Maffei, Grace & Houze, Rebecca, (eds.) *The Design History Reader* (Bloomsbury: London & New York, 2010).

Lees-Maffei, Grace, *Design at Home: Domestic Advice Books in Britain and the USA since 1945* (London: Routledge, 2014).

Lees-Maffei, Grace (ed.) *Iconic designs: 50 stories about 50 things* (London: Bloomsbury, 2014).

Lehmann, Hartmut & Richter, Melvin (eds.) *The Meanings of Historical Terms and Concepts: New Studies on Begriffsgeschichte*. (Washington, DC: German Historical Institute, 1996).

Leisure in the twentieth century. History of Design. Fourteen papers given at the Second Conference on Twentieth Century Design History 1976 (London: Design Council Publications, 1977).

Leppänen, Katarina, "At Peace with Earth—Connecting ecologial destruction and patriarchal civilisation", *Journal of Gender studies,* 2004: 1 (vol 13), 37-47.

Leppänen, Katarina, *Rethinking civilisation in a European feminist context: History, nature, women in Elin Wägner's Väckarklocka*, diss. (Gothenburg: Acta Universitatis Gothoburgensis, 2005).

Lie, Ida Kamilla, "'Make Us More Useful to Society!': The Scandinavian Design Students' Organization (SDO) and Socially Responsible Design, 1967–1973", *Design and Culture* 2016:3 (vol.8), 327-361.

Lichtman, Sarah A., "Reconsidering the History of Design Survey", *Journal of Design History,* 2009:4 (vol.20), 341-350.

Liedman, Sven-Eric, "A lesson for life", *Studies in Philosophy and Ecuation* 2002:21, 313-333.

Liedman, Sven-Eric, "Democracy, knowledge and imagination", *Studies in Philosophy and Education* 2002:21, 353-359.

Lindén, Claudia, "Förord av Claudia Lindén: Ellen Keys feministiska estetik", in Ellen Key, *Skönhet för alla* (Ellen Keyinstitutet: Ödeshög, 2006).

Lindh Karlsson, Monica & Redström, Johan, "Design Togetherness" in *Nordes 2015: Design Ecologies* (2015), no 6.

Lindinger,Herbert, *Ulm Design: The Morality of Objects* (Cambridge, MA: The MIT Press, 1991).

Lindström, Kristina & Ståhl, Åsa, *Patchworking publics-in-the-making: Design, media and public engagement*, diss. (Malmö: Malmö University, 2014);

Lindström, Kristina & Ståhl, Åsa, "Becoming response-able stakeholders – Participatory design in times of uncertainties", in *PDC '16: Proceedings of the 14th Participatory Design Conference: Short Papers, Interactive Exhibitions, Workshops - Volume 2,* Aarhus, Denmark, August 2016, (New York: ACM, 2016), 41-44.

Lindström, Kristina & Ståhl, Åsa, "Politics of inviting: Co-articulating issues in designerly public engagement", in Rachel Charlotte Smith et al., (eds.), *Design anthropological futures: Exploring emergence, intervention and formation* (London & New York: Bloomsbury Academic, 2016), 183-197.

Lindkvist, Anna, *Jorden åt folket: Nationalföreningen mot emigrationen 1907-1925*, diss. (Umeå, 2007).

Lovejoy, Arthur O., "Reflections on the History of Ideas", *Journal of the History of Ideas* 1940:1 (vol.1), 3–23.

Lundgren, Frans, *Den isolerade medborgaren: Liberalt styre och uppkomsten av det sociala vid 1800-talets mitt,* diss. (Hedemora: Gidlunds, 2003).

Lundh, Christer, *Nya perspektiv på Saltsjöbadsavtalet* (Stockholm: SNS, 2009).

Lynd, Robert Staughton & Lynd, Helen Merrell, *Middletown: A Study in American Culture* (1929) (New York, 1956).

Lövgren, Britta, *Hemarbete som politik: diskussioner om hemarbete, Sverige 1930-40-talen, och tillkomsten av Hemmens forskningsinstitut*, diss. (Stockholm: Almqvist & Wiksell, 1993).

Löwgren, Jonas, Larsen, Henrik Svarrer & Hobye, Mads, "Towards programmatic design research", *Designs for Learning* 2013:1-2 (vol.6), 80–101.

Maffei, Nic, *Norman Bel Geddes: American Design Visionary* (London: Bloomsbury, 2018).

Mainsah, Henry & Morrison, Andrew, "Participatory design through a cultural lens: Insights from postcolonial theory", *PDC '14: Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1*, Windhoek, Namibia, October 2014 (New York: ACM, 2014),

Maldonado, Tomás, "New Developments in Industry and the Training of the Designer", *ulm 2, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, October 1958.

Maldonado, Tomás, "Preliminary Note", *ulm 6, Quarterly bulletin of the Hochschule für Gestaltung, Ulm*, October 1962.

Maldonado, Tomás, *Design industriale: Un riesame* (Feltrinelli, 2008).

Manzini, Ezio, *Design, When Everybody Designs: An Introduction to Design for Social Innovation* (Cambridge: MIT Press, 2015).

Margolin, Victor, "A decade of design history in the United States 1977-87" *Journal of Design History* 1988:1 (vol.1), 51-72.

Margolin, Victor, "Design History or Design Studies: Subject Matter and Methods", *Design Issues* 1995:1 (vol 11), 4-15.

Margolin, Victor, "Design in History", *Design Issues* 2009:2 (vol.25), 94–105.

Margolin, Victor, *World History of Design, Vols. 1 & 2* (Bloomsbury: London, 2015).

Matchett, E. & Williams, K. G., "Strategic design for international healthcare", in Cross, Nigel (ed.) *Design Participation. Proceedings of the Design Research Society's Conference, Manchester, September 1971* (London: Academy editions, 1972).

Mattelmäki, Tuuli & Sleeswijk Visser, Froukje, "Lost in Co-X: Interpretations of Co-design and Co-Creation" in *Proceedings of the IASDR 2011, the 4th World Conference on Design Research* (Delft, 2011), 1-12.

Mattelmäki, Tuuli, Brandt, Eva & Vaajakallio, Kirsikka, "On designing open-ended interpretations for collaborative design exploration" *CoDesign: Special Issue: Design and Emotion*, 2011:2 (vol.7), 79–93.

Mazé, Ramia, "Politics of designing visions of the future", *Journal of Futures Studies* 2019:3 (vol.23), 23–38.

Medbestämmandelagen, Serien om arbetsrätt 5 (Stockholm: LO, 1977).

Meister, David, *The History of Human Factors and Ergonomics* (Mahwah, NJ: Lawrence Erlbaum Associates, 1999).

Meggs, Philip B., *A History of Graphic Design*, 3. ed. (New York: John Wiley, 1998).

Merriam-Webster dictionary, https://www.merriam-webster.com, accessed 2020-06-10.

Michl, Jan, "Taking Down the Bauhaus Wall: Towards Living Design History as a Tool for Better Design", *The Design Journal* 2014:3 (vol.17), 445-453.

Mitchell, Thomas, *Redefining Designing: From Form to Experience* (New York: Van Nostrand Reinhold, 1993).

Monö, Rune, "Produktdesign för arbetslivet" in Lasse Brunnström (ed.) *Svensk industridesign: En 1900-talshistoria* (Stockholm: Norstedt, 1997).

Movilla Vega, Daniel & Hallemar, Dan, *99 years of the Housing Question in Sweden* (Lund: Studentlitteratur, 2017).

Munthe, Gustaf, "Hur byggas våra bostäder? Röhsska museets bostadsundersökning", *Form* 1939:5, 99-106.

Mussari, Mark, *Danish Modern: Between Art and Design* (London: Bloomsbury, 2016).

Myrdal, Alva & Myrdal, Gunnar, *Kris i befolkningsfrågan* (Stockholm: Bonnier, 1934).

Möbelmarknaden. SAR:s och SSF:s bostadsutredning, (Stockholm, 1941).

"Nahman & Flanders DC Service Design Toolkit" https://www.servicedesigntoolkit.org, accessed 2020-06-10.

Nelson, Harold G,. & Stolterman, Erik, *The Design way: Intentional Change in an Unpredictable world* (Cambridge, MA: MIT Press, 2012).

Norman, Donald A., *The Design of Everyday Things* (MIT Press: London, 1998).

Nygaard, Kristen, "How Many Choices Do We Make? How Many Are Difficult?" in Floyd, C., Züllighoven, H., Budde, R., and Keil-Slawik, R. (eds.), *Software Development and Reality Construction* (Berlin: Springer-Verlag, 1992), 52-59.

Ruth Oldenziel & Karin Zachmann, *Cold War Kitchen: Americanization, Technology, and European Users* (Cambridge, MA: MIT Press, 2009).

Oudshoorn, Nelly, Rommes, Els & Stienstra, Marcelle, "Configuring the User as Everybody: Gender and Design Cultures in Information and Communication Technologies", *Science, Technology & Human Values* 2004:1 (vol.29), 30–63.

Overbeeke, Kees, Djajadiningrat, Tom, Hummels, Caroline, Wensveen, Stephan & Frens, Joep, "Let's make things engaging" in Blythe, Mark A.,Monk, Andrew F., Overbeeke, K., Wright, P.C. (eds.), *Funology: From Usability to Enjoyment* (Dordrecht: Springer Science + Business Media, 2005), 7-17.

Overud, Johanna, *I beredskap med Fru Lojal: Behovet av kvinnlig arbetskraft under andra världskriget*, diss. (Almqvist & Wiksell: Stockholm, 2005).

Pagold, Susanne, *Function rules: A Swedish approach to design glamour* (Stockholm: Arvinius, 2006).

Papanek, Victor *Miljön och miljonerna: Design som tjänst eller förtjänst* (Stockholm: Bonnier, 1970).

Papanek, Victor, *Design for the Real World: Human Ecology and Social Change*, (New York: Pantheon Books, 1971).

Pattison, Mrs. Frank A., "Scientific management in home-making", *The Annals of the American Academy of Political and Social Science* 1913:1 (vol. 48),96-103.

Paulsson, Gregor, *Better things for everyday life* [1919], in Lucy Creagh, Helena Kåberg & Barbara Lane Miller (eds.), *Modern Swedish Design: Three Founding Texts* (New York: The Museum of Modern Art, 2008).

Paulsson, Gregor, "Konstindustrien och den nya tiden" in *Huvudkatalog. Stockholmsutställningen 1930*. (Stockholm, 1930).

Paulsson, Gregor, "Demokratisk miljö. Hälsningsanförande vid S.S.F.:s konferens på 'Vår Gård' den 7 december", *Form* 1944:1.

Paulsson, Gregor, "Min väg till konstvetenskapen", in *Uppsala universitets konsthistoriska institution femtio år* (Uppsala, 1968).

Pawar, Aditya & Redström, Johan, "Publics, Participation and the Making of Umeå Pantry", *International Journal of Design* 2016:1 (Vol.10).

Penner, Barbara, "The Cornell Kitchen: Housing and Design Research in Postwar America", Technology and Culture 2018:1(vol.59), 48–94.

Pevsner, Nikolaus, *Pioneers of the Modern Movement: From William Morris to Walter Gropius* (London, 1936).

Pevsner, Nikolaus, *Pioneers of Modern Design: From William Morris to Walter Gropius* (New York: The Museum of Modern Art, 1949) [later editions: 1957, 1960, 1968, rev. 1974, 1986, rev. & exp. 2005].

Pevsner, Nikolaus, *The sources of modern architecture and design* (London: Thames and Hudson, 1968) [rev. ed. 1975].

Pred, Allan, *Recognizing European Modernities: A Montage of the Present* (London: Routledge, 1995).

Rampell, Linda, *Designatlas: En resa genom designteori 1845-2002* (Stockholm: Gabor Palotai, 2003).

Rampell, Linda, *En kritisk undersökning av det modernistiska projektet för design i Sverige*, diss. (Lund, 2004).

Redström, Johan, "Towards user design? On the shift from object to user as the subject of design", *Design studies* 2006:27, 123-139.

Redström, Johan, "RE:Definitions of use", *Design Studies* 2008:4 (vol.4), 410–423.

Redström, Johan, *Making Design Theory* (Cambridge, MA: MIT Press, 2017).

Redström, Johan, "Certain uncertainties and the design of design education", *She Ji: The Journal of Design, Economics, and Innovation* 2020:1 (vol. 6), 83-100.

Redström Johan & Wiltse, Heather, *Changing Things: The Future of Objects in a Digital World* (London: Bloomsbury Academic, 2018).

"Riktlinjer för verksamheten och aktuella arbetsuppgifter vid HFI", *HFI-meddelanden* 1948-49:1, 1-2.

Ringarp, Johanna & Carlsson, Karin, "Tolfterna – ett bildningsprojekt inför kvinnornas fullständiga medborgarskap?" in Peter Brubech, Ketil Knutsen & Jens Aage Poulsen (eds.) *Historie – didaktik, dannelse og bevisthed. Rapporter til det 29. Nordiske Historikermøde, Bind 1*. (Aalborg, 2017), 9-39.

Ringarp, Johanna & Anne Berg, "Borgerlig feminism eller klassolidaritet? Tolfternas bildningssamkväm vid förra sekelskiftet", in Kenneth Abrahamsson, Lisbeth Eriksson, Mats Myrstener, Lena Svensson & Tore Persson, *Folkbildning & forskning: Årsbok 2019* (Stockholm: Förening för folkbildningsforskning, 2019), 17-26.

Robach, Cilla, *Formens frigörelse: Konsthantverk och design under debatt i 1960-talets Sverige* (Stockholm: Arvinius, 2010).

Robertson, Toni & Simonsen, Jesper, "Participatory Design: An introduction" in (eds.) Jesper Simonsen & Toni Robertson, *Routledge International Handbook of Participatory Design* (Routledge: London & New York, 2013).

Robertson, Toni & Wagner, Ina, "Ethics: Engagement, representation and politics-in-action", in (eds.) Jesper Simonsen & Toni Robertsson (eds.), *Routledge International Handbook of Participatory Design* (Routledge: London & New York, 2013).

Rodgers, Paul A. & Yee, Joyce, (eds.) *The Routledge Companion to Design Research* (New York: Routledge, 2015).

Rosenbak, Søren, *The science of imagining solutions: Design becoming conscious of itself through design,* diss. *(*Umeå, 2019).

Rosner, Daniela K., *Critical Fabulations: Reworking the methods and margins of design* (Cambridge: MIT Press, 2018).

Routledge International Handbook of Participatory Design, (eds.) Simonsen, Jesper & Robertson, Toni, (New York: Routledge, 2013).

Rudberg, Eva, *The Stockholm exhibition 1930: Modernism's breakthrough in Swedish architecture* (Stockholm: Stockholmia, 1999).

Rutherford, Janice Williams, *Selling Mrs. Consumer: Christine Frederick and the rise of household efficiency* (Athens & London: University of Georgia Press, 2003).

Ryan, Deborah S., *Daily Mail – Ideal Home Exhibition: The Ideal Home Through the 20th Century* (London: Hazar, 1997).

Salvo, Michael J., "Ethics of Engagement: User-Centered Design and Rhetorical Methodology", *Technical Communication Quarterly* 2001:10 (vol 3), 273-290.

Sanders, Elizabeth B.-N., Brandt, Eva & Binder, Thomas, "A framework for organizing the tools and techniques of participatory design" *Proceedings of the 11th Biennial Participatory Design Conference* (Sydney, 2010), 195-198.

Sanders, Elizabeth B.-N & Stappers, Pieter Jan, "Co-creation and the new landscapes of design", *Co-Design* 2008:1 (vol.4), 5-18.

Sanders, Elizabeth B.-N. & Stappers, Pieter Jan, *Convivial Toolbox*: *Generative Research for the Front End of Design* (Amsterdam: BIS, 2012).

Sanders, Liz & Stappers, Pieter Jan, "From Designing to Co-Designing to Collective Dreaming: Three Slices in Time", *Interactions* Nov.-Dec. 2014.

Shackel, RB., "Ergonomics and Design" in (ed.) S.A. Gregory *The Design Method* (Butterworths: London, 1966).

Schuler, Douglas, & Namioka, Aki, *Participatory Design: Principles and Practices* (New York, 1993).

Schwartz Cowan, Ruth, *More work for mother: The ironies of household technology from the open hearth to the microwave* (Basic books, 1984).

Scott, Joan W., "Gender: A Useful Category of Historical Analysis", *The American Historical Review* 1986:5 (vol. 91), 1053-1075.

Schnyder, Gerhard, "Like a phoenix from the ashes? Reassessing the transformation of the Swedish political economy since the 1970s", *Journal of European Public Policy* 2012:8 (vol.19),1126-1145.

Shove, Elizabeth, Matthew, Watson, Martin, Hand & Ingram, Jack, *The Design of Everyday Life* (Berg: Oxford & New York, 2007).

Simon, Herbert, *The Sciences of the Artificial* [1969] 3rd edition. (Cambridge, MA: MIT Press, 1996).

Simonsen, Jesper & Robertson, Toni, (eds.) *Routledge International Handbook of Participatory Design* (New York: Routledge, 2013).

Smith, Rachel Charlotte, Vangkilde, Kasper Tang, Kjaersgaard, Mette Gislev, Otto, Ton, Halse, Joachim & Binder, Thomas, (eds.) *Design Anthropological Futures* (London: Bloomsbury Academic, 2016).

Singles, Kathleen, *Alternate History: Playing with Contingency and Necessity* (Berlin: De Gruyter, 2013).

Slann, Peter A., "Foreword" in Jones, J. Christopher & Thornley, D.G., *Conference on design methods. Papers presented at the Conference on Systematic and Intuitive Methods in Engineering, Industrial Design, Architecture and Communications. London, September 1962.* (London: Pergamon Press, 1963).

Snow, C.P., *The Two Cultures* [1959] (Cambridge: Cambridge University Press, 1993).

SOU 1943:18 Utredning och förslag angående planmässigt sparande för familjebildning och statens bosättningslån (Stockholm, 1943),

Sparke, Penny, "Design History: Fad or Function? Some Afterthoughts", in *Design History Society Newsletter number three*, December 1978, 14-16.

Sparke, Penny, *As Long as it's Pink: The Sexual Politics of Taste* (London: Pandora, 1995).

Sparke, Penny, *Designing the Modern Interior: From the Victorians to Today* (Oxford: Berg, 2009).

Sparke, Penny & Fischer, Fiona, (eds.) *The Routledge Companion to Design Studies* Routledge: New York, 2016).

Spitz, René, *HfG Ulm: The View Behind the Foreground. The Political History of the Ulm School of Design (1953-1968*) (Stuttgart & London: Edition Axel Menges, 2002).

Spivak, Gayatri Chakravorty, "Can the subaltern speak?" [1985], in Spivak, *A Critique of Postcolonial Reason: Toward a History of the Vanishing Present* (Cambridge, Mass: Harvard University Press, 1999).

Spivak, Gayatri Chakravorty, *A Critique of Postcolonial Reason: Toward a History of the Vanishing Present* (Cambridge, Mass: Harvard University Press, 1999).

Steenson, Molly Wright, *Architectural Intelligence: How Designers and Architects Created the Digital Landscape* (Cambridge, MA: MIT Press, 2017).

Stewart, Susan C., "And so to another setting", in Tony Fry, Clive Dilnot & Susan C. Stewart *Design and the Question of History* (Bloomsbury: London & New York, 2015), 275-297.

Stockholm: Stockholmsutställningen 1930 (broschure), Svenska Trafikförbundet & Stockholmsutställningen 1930 (Stockholm: Centraltryckeriet, 1929).

Ståhl, Åsa & Lindström, Kristina, "Politics of Inviting: Co-Articulating Issues in Designerly Public Engagement", in Rachel Charlotte Smith et al., (eds.) *Design Anthropological Futures* (London: Bloomsbury Academic, 2016).

Suchman, Lucy A., *Plans and Situated Actions. The Problem of Human-Machine Communication* (Cambridge: Cambridge Univ. Press, 1987).

Suchman, Lucy," Working Relations of Technology Production and Use", *Computer Supported Cooperative Work* 1994:1, (vol.2), 21–39.

Sundblad, Yngve, "UTOPIA - Participatory Design from Scandinavia to the World", *History of Nordic Computing* 2010:3, 176–186.

Så ska vi ha't: Bo-cirkel propagerar för bättre hem (Stockholm 1946),

Söderberg, Johan & Magnusson, Lars, (eds.) *Kultur och konsumtion i Norden 1750-1950* (Helsinki: Finska Historiska Samfundet 1997).

Tassinari, Virginia, "The Wunderkammer of the Future: Philosophical Notes on the Idea of History in Speculative Design Practices", *z33Research*, https://z33research.be/2018/04/virginia-tassinari (2018-04-13), accessed 2020-05-25.

Teasley, Sarah, "Methods of Reasoning and Imagination", in eds. Michael J Kelly & Arthur Rose, *Theories of History: History Read Across the Humanities* (London: Bloomsbury, 2018).

Thörn, Kerstin, *En bostad för hemmet: Idéhistoriska studier i bostadsfrågan,* Idéhistoriska skrifter 20, diss. (Umeå, 1997).

Tie, Warwick John, "The subject supposed to design", *CoDesign* 2017:1 (vol. 13), 16-31.

Tonkinwise, Cameron, "Always Historicise Design. Review of Bruce Sterling Shaping Things", *Design Philosophy Papers* 2006:1 (vol.4), 5-20.

Tonkinwise, Cameron, "Designing transitions – from and to What?", *Design Philosophy Papers* 2015:1 (vol.13), 85-92.

Tonkinwise, Cameron, "Design's (Dis)orders: Mediating Systems-Level Transition Design", *Cuadernos del Centro de Estudios de Diseño y Comunicación* 2019:73.

Torell, Ulrika, Lee, Jenny & Qvarsell, Roger (eds.) *Köket: Rum för drömmar, ideal och vardagsliv under det långa 1900-talet* (Stockholm: Nordiska museets förlag, 2018).

Twemlow, Alice, *Sifting the Trash: A History of Design Criticism* (Cambridge, MA: MIT Press, 2017).

Undersökningar av småredskap: Stekspadar, potatisskalare, konservöppnare. HFI-meddelanden 1946:4.

"Ungdomen behöver vägledning vid hembildningen" *Form* 1943:1, 22-26.

Valtonen, Anna, *Redefining industrial design: Changes in the design practice in Finland*, diss. (Helsinki: University of Art and Design, 2007).

Vardouli, Theodora, "*User Design*: Constructions of the 'user' in the history of design research", P. Lloyd & E. Bohemia, eds., *Proceedings of DRS2016: Design + Research + Society - Future-Focused Thinking,* (Brighton 2016).

Vaughan, Laurene (ed.), *Practice Based Design Research* (New York: Bloomsbury Academic, 2017).

de Vries, M.J. et al. (eds.), *Design Methodology and Relationships with Science* (Kluwer Academic Publishers, 1993).

von Wachenfeldt, Paula & Nyberg, Klas, D*et svenska begäret: Sekler av lyxkonsumtion* (Stockholm: Carlssons, 2015).

Walker, John A., *Design History and the History of Design* (London: Pluto, 1989).

Wensween, Stephan & Matthews, Ben, "Prototypes and prototyping in design research", in Paul A. Rodgers & Joyce Yee (eds.) *The Routledge Companion to Design Research* (New York: Routledge, 2015).

Westerlund, Bo, "The use of the absent and Othering in design and critical analysis of PD activities", *PDC '16: Proceedings of the 14th Participatory Design Conference: Short Papers, Interactive Exhibitions, Workshops - Volume 2,* Aarhus, Denmark, August 2016, (New York: ACM, 2016), 29-32.

"What is the framework for innovation? Design Council's evolved Double Diamond", https://www.designcouncil.org.uk/news-opinion/what-framework-innovation-design-councils-evolved-double-diamond, accessed 2020-05-30.

Whittle, Jon, "How much participation is enough? A comparison of six participatory design projects in terms of outcomes", *PDC '14: Proceedings of the 13th Participatory Design Conference: Research Papers - Volume 1,* Windhoek, Namibia October 2014 (New York: ACM, 2014),121-130.

Wickman, Kerstin, *Formens rörelse: Svensk form genom 100 år* (Stockholm: Carlsson, 1995).

Wickman, Kerstin, *A&E Design: The Book* (Bromma: Business History Publishing, 2018).

Wilkins, Bridget, "Introduction", *Leisure in the twentieth century. History of Design. Fourteen papers given at the Second Conference on Twentieth Century Design History 1976* (London: Design Council Publications, 1977).

Willis, Anne-Marie, "Ontological designing", *Design Philosophy Papers* 2006:2 (vol 4), 69-92.

Wisselgren, Per, *Samhällets kartläggare: Lorénska stiftelsen, den sociala frågan och samhällsvetenskapens formering 1830-1920*, diss. (Umeå, 2000).

Wisselgren, Per, "Reforming the science-policy boundary: the Myrdals and the Swedish tradition of Governmental Commissions" in Sven Eliaeson & Ragnvald Kalleberg (eds.) *Academics as public intellectuals* (Newcastle, UK: Cambridge Scholars Publishing, 2008), 173-195.

Wisselgren, Per, *The Social Scientific Gaze. The Social Question and the rise of Academic Social Science in Sweden* (Farnham & Burlington: Ashgate, 2015).

Wittgenstein, Ludwig, *Philosophical Investigations* (1953), revised 4th edition (Blackwell: Oxford, 2009).

Witz, Leslie & Rassool, Ciraj, “Making histories”, *Kronos: Southern African Histories* 2008:34, 6-15.

Woodham, Jonathan M., “Resisting Colonization: Design History Has Its Own Identity”, *Design Issues* 1995:1 (vol 11), 22-37.

Woodham, Jonathan M., *Twentieth Century Design* (Oxford: Oxford University Press, 1997).

Wägner, Elin, *Väckarklocka* (Stockholm: Bonnier, 1941).

Zetterlund, Christina, “Just Decoration? Ideology and Design in Early-Twentieth-Century Sweden”, in Kjetil Fallan (ed.) *Scandinavian Design: Alternative Histories* (London: Berg, 2012), 103-116;

Åhrén, Uno, “Brytningar”, in Gregor Paulsson (ed.), *Svenska slöjdföreningens årsbok 1925. Moderna strömningar inom Europas konstindustri* (Stockholm, 1925), 7-36.

Åkerman, Brita, *Familjen som växte ur sitt hem* (Stockholm, 1941).

Åkerman, Brita, “Aktiv hushållning”, in Brita Åkerman et al., (eds.) *Kunskap för vår vardag. Utbildning och forskning för hemmen* (Akademilitteratur: Stockholm, 1983), 78-79.

Åkerman, Brita, “Forskning för att förändra” in Brita Åkerman et al., (eds.) *Kunskap för vår vardag: Utbildning och forskning för hemmen* (Stockholm, 1984), 131-139.

Åkerman, Brita et al., *Kunskap för vår vardag: Utbildning och forskning för hemmen* (Akademilitteratur: Stockholm, 1983).

Åkerman, Brita, *88 år på 1900-talet: Bland vänner och idéer* (T. Fischer & co.: Stockholm, 1994).

Östberg, Kjell, *Efter rösträtten: Kvinnors utrymme efter det demokratiska genombrottet* (Eslöv: Symposion, 1997).

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Utopia is a transitional serif typeface, designed by Robert Slimbach for Adobe in 1989. A transitional typeface is one that – stylistically – builds on typefaces that carried over from the older renaissance style typefaces (such as Bembo, Garamond or Caslon) to the more modern and high contrast typefaces (such as Bodoni or Didot) designed towards the late 18th century.

Futura is a sans-serif typeface designed by Paul Renner in 1927 as part of the New Frankfurt project, which aimed to propose rational and affordable housing according to modernist ideas about design and everyday life. Futura is based on geometrical forms – circles, triangles and squares – and was presented as 'the typeface of our time'.



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